

**CIMS**

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## Welcome

CIMS<sup>1</sup> is an integrated set of economic, energy, and materials models designed to provide information to decision makers on the likely response of firms and households to policies that influence their technology acquisition and use decisions. Technology is widely defined to include not just equipment but also buildings and major infrastructure such as transit networks. Thus, it is sometimes described as a technology simulation model, which seeks to reflect how people actually behave rather than how they ought to behave. CIMS is able to provide policy makers with information on: (1) the ability of their policies to achieve specific objectives (like greenhouse gas (GHG) emissions reductions), (2) the likely costs of achieving these objectives, and (3) the uncertainties associated with the model's simulations. CIMS is maintained and operated by the Energy and Materials Research Group at the School of Resource and Environmental Management at Simon Fraser University.

CIMS is based on energy flows through a country's economic system. Like its predecessor, the ISTUM family of models, which form the individual sub-components of CIMS, it tracks the flow of energy, beginning with energy services and production processes through to eventual end-use by individual technologies. Unlike the partial equilibrium ISTUM models, which compete technologies against each other to serve pre-specified demands for end-uses, CIMS is nearly a full equilibrium system that allows the practitioner to include price driven energy and goods and services supply and demand equilibrium, energy trade, and some indirect macroeconomic effects, specifically disposable income and savings multiplier effects.

The model's focus on energy flows through technologies makes it ideal for modelling policies intended to affect energy efficiency, greenhouse gas emissions, and air quality. Energy consumption and emissions of all pollutants are technology specific; unless a model operates on a detailed technology basis, as CIMS does, energy consumption and emission estimates can only be approximated by economic activity.

---

<sup>1</sup>CIMS originally stood for Canadian Integrated Modelling System. The acronym is now used as a proper name, since over its history the model has been applied to several other countries and regions.

**Part I**

**About**

# 1 History and Development

CIMS was built on top of, and partly out of, a body of technology-simulation work that had been accumulating at the Energy and Materials Research Group (EMRG) at Simon Fraser University since the late 1980s. Understanding how CIMS arrived at its current shape helps explain why some of its modelling conventions are the way they are — the strong emphasis on technology-by-technology accounting, the partial-equilibrium sub-models that still organize the code, the discrete-choice competition equations, and the iterative supply–demand feedback loop all carry forward decisions made over the model’s now thirty-plus year history.

## 1.1 ISTUM origins (late 1980s – 1990s)

CIMS’s direct ancestor was the **Industrial Sector Technology Use Model**, ISTUM, a “bottom-up” engineering simulation model of industrial energy demand. ISTUM was initially developed by the US Department of Energy and later maintained at EMRG, working closely with the Canadian Industrial Energy End-Use Data Analysis Centre (CIEEDAC) which provided much of the empirical technology and stock data that the model needed. The first published application is the trial of ISTUM-PC to the British Columbia pulp and paper industry by Jaccard and Roop (1990), which demonstrated the central modelling idea that EMRG would carry forward for the next several decades: represent each end-use service explicitly (process heat, mechanical drive, lighting, etc.) and let candidate technologies compete to provide it, tracking the resulting energy demand and capital stock evolution rather than inferring them from sectoral aggregates.

Over the next decade ISTUM was extended in several directions. Nyboer (1997) worked out the simulation framework in greater detail and applied it to additional sub-sectors. By the early 2000s ISTUM had been built out into a family of sector-specific sub-models covering most of the Canadian economy: primary energy supply (natural gas, coal, crude oil), energy processing (refining, electricity), and end-use demand for energy services (commercial/institutional, transportation, residential, iron & steel, pulp & paper, metal smelting, chemicals, mining, industrial minerals, “other manufacturing”). Each ISTUM sub-model was, on its own, a partial-equilibrium technology simulation — given exogenous demand for the relevant service and exogenous energy prices, it simulated what technology stock would emerge over time. The pieces did not yet talk to each other.

## 1.2 Construction of CIMS (early 2000s)

The hybrid energy–economy model that became CIMS was led by Chris Bataille as the central work of his PhD at SFU under the supervision of Mark Jaccard and John Nyboer (Bataille 2005). The project’s animating question was a methodological one: bottom-up models like ISTUM had strong technological realism but treated demand and prices as exogenous; top-down macro-econometric and CGE models had economic completeness but were largely silent on technology. A “hybrid” model — bottom-up in its technology representation, top-down in its market-clearing behaviour — promised to combine the strengths of both (Rivers and Jaccard 2005b). Bataille’s dissertation established design criteria for what such a hybrid should do, surveyed the prior attempts (MARKAL, MESSAGE-MACRO, NEMS), and then built the integration framework on top of the existing ISTUM sub-models (Bataille 2005). Other key members of the EMRG team at SFU who contributed to this phase of the model’s development included: Jotham Peters, Rose Murphy, Bryn Sadownik, Alison Laurin, and Nic Rivers.

The principal additions to ISTUM that produced CIMS were:

- **Energy supply–demand equilibration:** the previously isolated sub-models were connected via an iterative loop in which demand-side energy requirements drive supply-side production, supply-side cost-of-production drives the energy prices fed back to the demand side, and the system is iterated until energy prices stabilize within a threshold.
- **Goods-and-services demand feedbacks:** Armington-style price elasticities allow domestic demand for traded goods to respond to changes in domestic production costs (relative to imports), and an empirically estimated relationship between industrial value-added and commercial/residential building permits closes the loop between manufacturing activity and floorspace.
- **Behavioural parameters in the competition:** the technology choice within each sub-model was reformulated as a probabilistic competition with a variance parameter  $v$  controlling cost-sensitivity, alongside intangible-cost parameters that capture non-financial preferences and perceived risk. This is the equation now described in Chapter 4.
- **Forward-looking and learning behaviour:** declining-capital-cost and declining-intangible-cost functions were added to represent learning-by-doing and the “neighbour effect” of falling perceived risk as market share grows (Mau et al. 2008); the foresight functions (myopic, average, and discounted expectations of future emissions charges) allow expected policy costs to enter today’s investment decisions.

The discrete-choice basis for the technology-competition equation was formalized by Rivers and Jaccard (2005b), which sets out the family of choice models the CIMS competition function belongs to and discusses calibration of  $v$  against empirically estimated behavioural parameters. The empirical-parameter programme — collecting stated- and revealed-preference data for personal vehicle and home heating decisions and using it to constrain the  $v$  values

in those technology competitions — was developed in parallel through several papers in the mid-2000s (Bataille et al. [2006](#); Mau et al. [2008](#)).

### 1.3 Evolution

CIMS has been re-implemented several times since its initial construction. The first integrated version used APL for the sub-model calculations and feedback dynamics, with Microsoft Excel/Access for data tables. That generation was flexible but had a difficult user interface. A second version layered Visual Basic and a C++ front-end on top of the APL engine — easier to use but more rigid because data flowed through four different programs in four different languages. A third generation, kept APL as the calculation engine but moved the interface and data storage into Java.

The current implementation has continued this trajectory: the model engine and surrounding extensions are organized as a Python project, with model inputs as CSV and text files, and operator interactions through a marimo notebook layer. The shift in implementation languages over time has tracked broader research-software trends — APL → Visual Basic → Java → Python — while the underlying simulation logic (the partial-equilibrium technology competitions, the iterative supply–demand loop, the discrete-choice competition function, the lifecycle-cost formulation) has remained recognizable from one version to the next.

The Python redevelopment of CIMS has been led by Brad Griffin, Jillian Anderson, and Matt Sniatynski, with considerable support provided by Thomas Budd and Emma Starke as part of their respective PhD research.

### 1.4 Current status

CIMS continues to be maintained at EMRG and is used by a network of researchers and policy clients across Canada (Energy Modelling Hub [2026](#)). It has been registered in the Energy Modelling Hub’s open-models inventory and is part of the Open Insights suite of open-source models, where it sits among the country’s small set of national-scope hybrid energy-economy models. Recent extensions include the disaggregation of the Atlantic provinces and territories, expanded coverage of low- and high-temperature industrial heat (see Chapter [34](#)), updated demand and price-elasticity calibrations, and continued evolution of the calibration workflow described in Chapter [39](#).

## 1.5 Selected applications

A representative — not exhaustive — list of areas where CIMS has been used to inform policy and academic discussion:

- Evaluation of bottom-up cost curves of US abatement at a range of carbon prices (Murphy and Jaccard 2011).
- Comparisons of bottom-up and top-down estimates of marginal abatement cost for various Canadian sectors and regions.
- Analyses of provincial climate plans and the federal carbon-pricing benchmark.
- Behaviourally realistic assessments of the decarbonization potential of innovative technologies.
- Studies of consumer adoption of new vehicle technologies through discrete-choice analysis (Mau et al. 2008; Horne 2003).
- A book-length academic treatment of the energy-economics research programme in Jaccard (2005).

For a deeper academic introduction to the CIMS modelling philosophy and its place in the broader hybrid-modelling literature, the most useful single reference is still Bataille (2005); Rivers and Jaccard (2005b) is the methodological companion piece on the discrete-choice competition, and Jaccard (2005) gives a book-length statement of the energy-economics research programme which produced CIMS.

## 2 Model Architecture

### 2.1 Scale and Scope

CIMS covers the entire Canadian economy and can connect to aggregated representations of other regions (e.g., United States, Europe, or Rest of World). Table 2.1 illustrates the sectors and regions currently covered in CIMS.<sup>1</sup>

**Table 2.1:** Sectors represented by region

| Sectors                  | BC | AB | SK | MB | ON | QC | AT <sup>2</sup> |
|--------------------------|----|----|----|----|----|----|-----------------|
| Coal Mining              | +  | +  | +  |    |    |    | +               |
| Natural Gas              | +  | +  | +  | +  | +  | +  | +               |
| Petroleum Crude          | +  | +  | +  | +  | +  |    | +               |
| Petroleum Refining       | +  | +  | +  |    | +  | +  | +               |
| Electricity              | +  | +  | +  | +  | +  | +  | +               |
| Ethanol                  | +  | +  | +  | +  | +  | +  | +               |
| Biodiesel                | +  | +  | +  | +  | +  | +  | +               |
| Hydrogen                 | +  | +  | +  | +  | +  | +  | +               |
| Mining                   | +  |    | +  | +  | +  | +  | +               |
| Industrial Minerals      | +  | +  |    |    | +  | +  | +               |
| Chemical Products        | +  | +  | +  |    | +  | +  |                 |
| Pulp and Paper           | +  | +  |    |    | +  | +  | +               |
| Metal Smelting           | +  |    |    | +  | +  | +  | +               |
| Iron and Steel           |    |    | +  |    | +  | +  |                 |
| Light Industrial         | +  | +  | +  | +  | +  | +  | +               |
| Construction             | +  | +  | +  | +  | +  | +  | +               |
| Residential              | +  | +  | +  | +  | +  | +  | +               |
| Commercial               | +  | +  | +  | +  | +  | +  | +               |
| Transportation Passenger | +  | +  | +  | +  | +  | +  | +               |
| Transportation Freight   | +  | +  | +  | +  | +  | +  | +               |
| Waste                    | +  | +  | +  | +  | +  | +  | +               |
| Agriculture              | +  | +  | +  | +  | +  | +  | +               |
| Forestry                 | +  | +  | +  | +  | +  | +  | +               |

<sup>1</sup>Canadian territories are not currently represented in CIMS.

TODO: Could include major metro regions like Toronto, Montreal, and Vancouver to represent city building and transportation policies.

While the model is simple in operation, it can appear complex because it is technologically explicit and covers the whole economy. CIMS represents a detailed list of technologies (fridges, light bulbs, cars, industrial motors, steel furnaces, buildings, power plants, etc.), and includes their linkages to each other. The model is especially large for the industrial sectors because industry has the largest variety of technologies.

CIMS highlights the interplay of energy supply and demand because energy-related GHG emissions are currently a key policy concern. Thus, the model focuses on the interaction between sectors that use energy (industrial, residential, commercial / institutional, and transportation sectors) and sectors that produce or transform energy (electricity generation, fossil fuel supply, oil refining, natural gas processing, biofuels, and hydrogen). This interaction is shown in Figure 2.1. A policy that seeks to influence energy supply and demand may also influence the cost of producing goods and services, thus affecting their price and final demand. To reflect these interactions, CIMS includes a goods and services final demand feedback loop, composed of Armington elasticities for traded goods and empirically driven relationships between the traded and non-traded sectors.

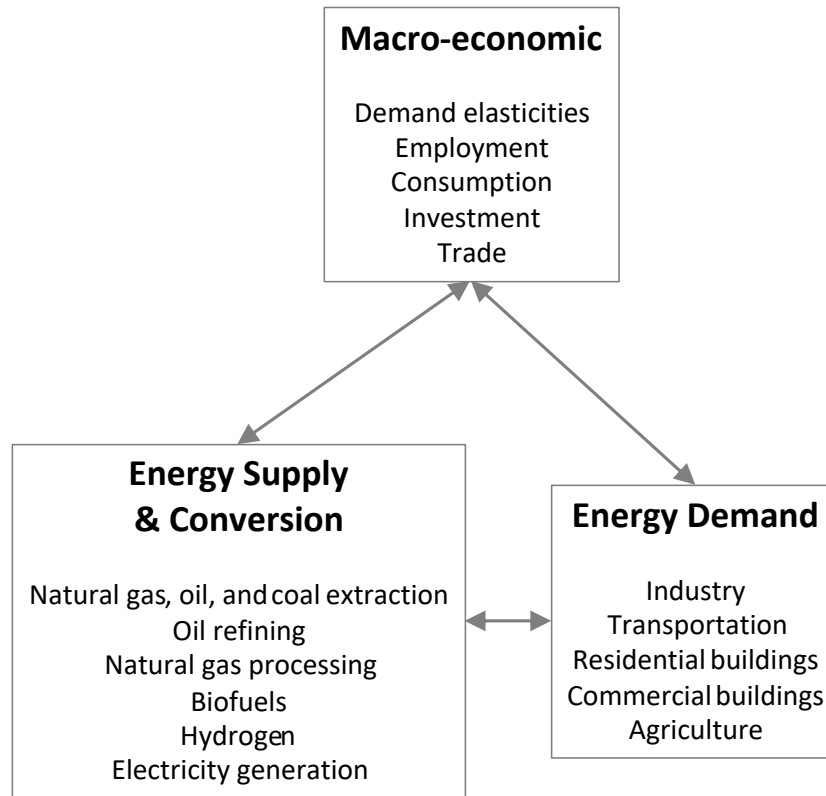
## 2.2 Simulation: Six Steps

As a simulation model, CIMS simulates the evolution of technologies acquired to meet energy services, and it uses both engineering and economic information. The engineering information details the unit energy consumption of each energy service technology and explains, using flow models of technologically heterogeneous sectors, how final energy service demands, intermediate products, and energy supply are related. The economic information determines which technologies are preferable in terms of cost minimization. While CIMS is not an optimization model, it can be adapted to produce results like those of an optimization model by calling on certain built-in functions. These functions are known as winner-take-all functions, and they give all new market shares in a given competition to the technology or process with the least cost.

CIMS is designed to mimic a sector or industry by representing a number of industrial establishments, residential or commercial buildings, and delivery of goods or personal travel within a defined region. The model focuses its analysis on the aggregate energy end-use requirements of that industry or sector. For example, in industry the major end-uses are process heat, motive force, space heat, and lighting. In the pulp and paper sector, the major end-uses are steam heat or mechanical energy to digest wood fibre into pulp, mechanical energy to convey and form pulp into paper, mechanical energy to press moisture out of formed pulp or

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<sup>2</sup> Atlantic includes New Brunswick, Nova Scotia, Prince Edward Island, and Newfoundland and Labrador.



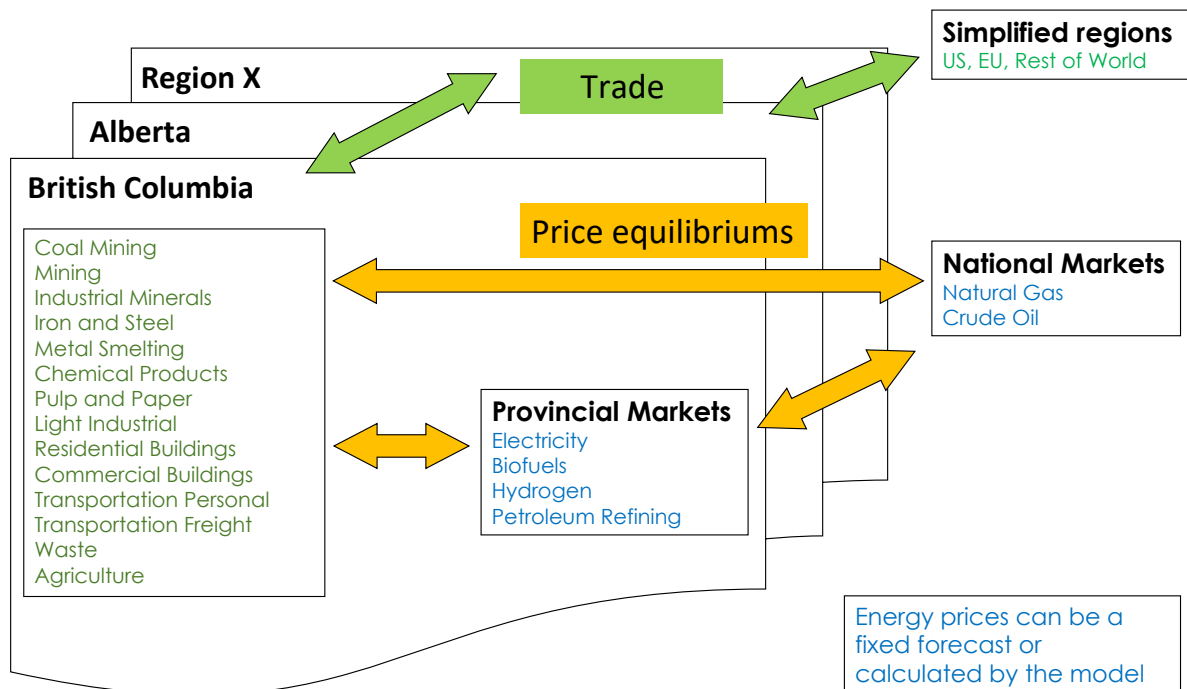
**Figure 2.1:** General model structure

paper, steam heat to dry pulp or paper, space heat, and lighting. In residential and commercial buildings, the primary end-uses include space heating and cooling, hot water production, refrigeration, and other assorted appliances.

**A CIMS simulation comprises six basic steps:**

1. The baseline macroeconomic drivers, energy supply drivers, and energy service technology characteristics are set for the desired time frame (15 years, 20 years, etc.) in five-year increments.
  - a. Technologies are represented in the model in terms of the quantity of energy service they provide. This could be, for example, vehicle kilometers traveled, tonnes of paper, or square metres of floor space heated and cooled. Then, a forecast of growth in energy service demand is used to drive future demand for energy services, usually in five-year increments.
  - b. Macro-economic drivers include income, economic output, structural evolution of sectors, interest rates, employment rates, regulations, etc. They account for all energy service demands in the energy demand module.

- c. Energy supply drivers include international commodity prices, domestic market developments, forecasts of costs and availability of marginal energy supplies, and the resulting forecasts of domestic prices.
  - d. Energy service technology characteristics include the date a technology or process will be available, technology energy efficiencies and capital / operating costs, firm and household time preferences (discount rates) for discretionary and non-discretionary expenditures (new and retrofit), and other behavioural parameters (perceived risks and consumer surpluses of different technologies, interdependence of certain technology choices that serve different energy services, such as domestic water heating and domestic space heating).
2. The energy demand module (industry, commercial, residential, and transportation) is run for the initial driver values (including energy supply prices).
- a. In each future period, a portion of the initial year's stock of technologies is retired. Retirement depends only on age. The remaining technology stocks in each period are subtracted from the forecast energy service demand and this difference determines the amount of new technology stocks in which to invest.
  - b. Prospective energy service technologies compete to fill the new service demand. The objective of the model is to simulate this competition so that the outcome approximates what would happen in the real world. Here the model relies heavily on market research of past and prospective firm and household behavior. Technology costs depend on recognized and financial costs, but also on identified differences in non-financial preferences (e.g., differences in the quality of lighting from different light bulbs) and risks (one technology is seen as more likely to fail than another). Even the determination of financial costs is not straightforward as time preferences (discount rates) can differ depending on the decision maker (household vs. firm) and the type of decision (non-discretionary vs. discretionary). Consequently, the competition is simulated probabilistically; new stock purchases are distributed among prospective technologies with financial costs being just one of several factors in determining market share.
  - c. In each period, a similar competition occurs with remaining technology stocks to simulate retrofitting (if desirable and likely from the firm or household's perspective). The same financial and non-financial information is required, except that the capital costs of remaining technology stocks are excluded, having been spent earlier when the remaining technology stock was originally acquired.
3. The demand models then send their requirements for energy to the energy supply models. The supply models send back corresponding energy prices. If the changes in energy prices are over a set threshold value for equilibrium (usually 5%), the demand models are then rerun with the new energy prices. This process is repeated until all energy price changes fall under the threshold value. Figure 2.2 describes this process:



**Figure 2.2:** Energy Supply and Demand

4. Policy or market driven energy price changes may change the lifecycle cost of some energy services, thereby changing the end-use price for these services. If the change in lifecycle cost from the business-as-usual case is beyond a pre-set threshold price elasticities for these services are applied, thereby changing the demand for these services. This new demand is sent back to the supply and demand models for re-calculation.
5. The model reiterates step 2, 3 and 4 until all variables stabilize. It then proceeds to the next time period. Steps 1 through 5 are repeated for all time periods.
6. Output is provided for new and total capital stock, energy use, and trade flows, GHG emissions, levelized lifecycle costs by service and technology,

## **Part II**

# **Methodology**

## 3 Theory

### 3.1 CIMS in the energy-economy modelling landscape

CIMS sits in a family of integrated energy-economy models that policy analysts use to study questions like the cost of greenhouse gas abatement, the energy-sector consequences of carbon pricing, or the long-run response of capital stocks to regulation. These models broadly fall into three traditions, and CIMS is built as a deliberate hybrid of them. Locating CIMS within this landscape — and being explicit about what it borrows from each tradition and what it deliberately rejects — is the easiest entry point into the conceptual machinery that the rest of this chapter develops in detail.

#### 3.1.1 Top-down general-equilibrium models

The top-down tradition is dominated by *computable general equilibrium* (CGE) models, which represent the entire economy as a system of interconnected production and consumption functions calibrated to a social accounting matrix. Producers and consumers respond to price changes through elasticities of substitution; markets for goods, labour, and capital clear endogenously; and the model returns a new equilibrium price-quantity vector consistent with the imposed policy shock. CGE models are well suited to questions about how a carbon price reshapes terms of trade, sectoral value-added, household income distribution, or the inter-sectoral reallocation of capital — questions whose answers depend on circulating effects across the whole economy.

What CGE models do not do well is *technology*. An entire production sector is typically represented as a Cobb-Douglas or nested CES production function over a handful of aggregate inputs, and “switching to a cleaner technology” appears, if at all, as movement along an aggregate substitution curve rather than as the retirement of one furnace and the purchase of another. This makes a pure CGE model awkward for analysing policies whose mechanism runs through specific pieces of equipment — vehicle fuel-economy standards, equipment efficiency regulations, capital subsidies for heat pumps, retrofits to industrial boilers. The aggregate elasticities can be made to *mimic* the average response, but the model cannot tell the analyst *which* technologies turned over to produce it.

### 3.1.2 Bottom-up optimization models

The bottom-up optimization tradition — the MARKAL / TIMES family, MESSAGE, OSeMOSYS, and similar — takes the opposite stance. Each represents the energy system as an explicit graph of technologies with engineering-realistic capital costs, conversion efficiencies, and stock dynamics, then solves a (typically linear) programme that chooses the least-cost mix of technologies to satisfy exogenous service demands subject to constraints. This delivers full technological resolution and a clean, defensible objective function: the answer is the cost-minimising solution under the modeller’s assumptions.

The price of that clarity is a strong normative assumption about decision-making. Investment is made by an omniscient, infinitely patient social planner using a single social discount rate; near-identical alternatives are split arbitrarily by the solver rather than smeared across decision-makers; and there is no formal place for non-financial preferences, perceived risk, or the heterogeneity of actual firms and households. Real adoption patterns are observably smoother and more dispersed than the corner solutions that linear-programme formulations tend to produce, and the reasons — uncertainty, information frictions, financing constraints, brand loyalty, perceived reliability — are precisely the factors a least-cost solver does not represent.

### 3.1.3 Bottom-up simulation models

The third tradition, *bottom-up simulation*, is exemplified by the US Energy Information Administration’s National Energy Modeling System (NEMS) and by CIMS’s own ancestor, the ISTUM family. These models keep the explicit technology representation of the optimization tradition but drop the social-planner assumption. Technology choice is modelled instead as it is observed to occur: probabilistically, on the basis of decision-maker-specific perceived costs, with the share of new investment going to each option depending on how that option compares to its rivals on both financial and non-financial grounds. There is no claim that the resulting mix is socially optimal — only that it is a plausible representation of what real decision-makers would actually do.

This is the tradition CIMS most directly belongs to.

### 3.1.4 CIMS as a hybrid

What distinguishes CIMS from a pure bottom-up simulation is that it grafts a partial-equilibrium feedback structure onto the simulation core, producing what its designers called a **hybrid energy-economy model** (Bataille 2005; Rivers and Jaccard 2005b). The bottom-up half is a network of sector-specific simulation sub-models in which candidate technologies compete to provide energy services — process heat, motive force, space conditioning, freight

movement, and so on — using a discrete-choice market-share equation whose variance parameter  $v$  is calibrated against empirically observed adoption patterns rather than assumed (see Chapter 4). The top-down half is the iterative price-quantity equilibration that connects the sub-models: aggregate energy demand from the demand-side sub-models drives the energy-supply sub-models, the supply sub-models return prices that feed back into the next iteration’s lifecycle-cost calculations, and Armington-style elasticities allow demand for traded goods to respond to the relative cost of domestic versus imported production. The system is iterated until energy prices and goods demands stabilise within a tolerance.

This positioning gives CIMS a different mix of strengths and weaknesses from its neighbours. Compared with a CGE model, CIMS sees individual furnaces, vehicles, and building shells, and so can speak directly to technology-specific policies; but it is silent on labour markets, the full inter-industry circulation of value-added, and most of what a CGE practitioner would call “general equilibrium.” Compared with a MARKAL-style optimization model, CIMS produces smoother, more behaviourally realistic adoption patterns and can be calibrated against revealed-preference data, but it makes no claim that the resulting technology mix is socially optimal — only that it is what real decision-makers, with their actual heterogeneity and risk perceptions, would plausibly choose.

The two framings are not mutually exclusive in practice. Where the analyst wants an optimization-style answer — the lowest-cost technology mix that satisfies a constraint — CIMS can be coaxed in that direction by setting  $v$  very high in the relevant competitions, which collapses the probabilistic share equation toward a winner-take-all rule. This is discussed in its own right below and formalised in Chapter 4. The point for now is that the probabilistic-versus-deterministic framings of technology choice sit on a continuum in CIMS rather than being separate model paradigms, and which end of the continuum is appropriate is a calibration question, not a structural one.

The remainder of this chapter develops the conceptual basis that makes this hybrid structure work: the choice of energy *services* as the unit of analysis, the discrete-choice / Weibit family of competition equations, the iterative price-quantity loop, the distinction between behavioural and financial discount rates, and the role of intangible costs in calibrating non-financial preferences.

## 3.2 The discrete-choice family of technology competition models

The new-stock competition equation in Chapter 4 is, mathematically, a normalised inverse-power function on lifecycle cost: the market share of each technology is proportional to its lifecycle cost raised to  $-v$ , divided by the same quantity summed over all competitors. That short description hides a deliberate methodological choice. CIMS is not the only model that converts an array of candidate technology costs into a share split, and the form it uses sits inside a well-developed family of *discrete-choice* models drawn from microeconometrics. This

section unpacks where that family comes from, where the CIMS competition equation sits inside it, and why that placement matters when calibrating or interpreting model output.

### 3.2.1 Random utility theory and the multinomial logit

The discrete-choice literature begins with the observation that an analyst trying to predict which option a decision-maker will pick from a small menu of alternatives never has all the information available to that decision-maker. There are preferences, situations, and constraints that are invisible to the model but visible to the chooser. The standard response, formalised by McFadden (1974), is to model the latent utility of each option as a sum of a deterministic component the analyst can compute — a function of observable attributes like price, efficiency, and capacity — and a stochastic component representing everything the analyst cannot see. The chooser is assumed to pick the option with the highest realised utility; the analyst can then predict only the *probability* that each option is chosen, given the distribution of the unobserved component.

The functional form of the choice probability depends entirely on what distribution one assumes for that unobserved error. If the errors are independent and identically distributed Gumbel (extreme-value type I), a tidy closed-form expression drops out: the probability that option  $k$  is chosen equals  $e^{V_k} / \sum_j e^{V_j}$ , where  $V_k$  is the deterministic utility of option  $k$ . This is the *multinomial logit* model, and it is the mainstay of applied discrete choice in transportation, marketing, and energy. Its appeal is partly empirical — it fits well in many settings — and partly computational, since the likelihood functions for logit models are globally concave and easy to estimate.

### 3.2.2 The Weibit and the CIMS power form

The multinomial logit’s tractability comes from one specific distributional assumption. Change the assumption and a different choice probability falls out. If the unobserved component enters utility *multiplicatively* rather than additively, and is distributed Weibull rather than Gumbel, the closed-form result is a power expression rather than an exponential one: the probability that option  $k$  is chosen equals  $V_k^{-v} / \sum_j V_j^{-v}$ , where  $V_k$  is now a *cost* rather than a utility and  $v > 0$  plays the same scale role the scale parameter plays in logit. This is the **Weibit** model, derived in closed form by Castillo et al. (2008), and it is this form that CIMS uses for technology competition.

The two forms are related: taking logs of the cost terms in the Weibit equation produces an expression that is algebraically identical to a multinomial logit on  $\ln V_k$  with coefficient  $-v$ . So an equivalent way to describe the CIMS competition is “a multinomial logit on log-lifecycle-cost.” The choice between describing it as Weibit-on-cost or logit-on-log-cost is largely a matter of which side of the algebra the analyst finds more natural. The empirical content — that

share splits depend on cost *ratios* rather than cost differences, and that the dependence is parameterised by a single scale parameter  $v$  — is the same either way.

CIMS adopted the power form partly for historical reasons — the equation was inherited and refined from the ISTUM family of models that predated CIMS — and partly for its behavioural properties. Working in cost-ratio space makes the share split scale-free: doubling all lifecycle costs in a competition (because, say, currency units changed) leaves the split unchanged, which is the behaviour one wants from a competition equation. The same is not true of a logit applied directly to costs in dollar units. The power form does come with one awkwardness:  $V_k^{-v}$  is only well-defined for strictly positive  $V_k$  (or strictly negative), so when a competition includes technologies whose lifecycle cost is negative — for example because intangible-cost adjustments offset the financial cost — a separate treatment has to be applied. TODO: cross-reference the section where the negative-lifecycle-cost handling is described.

### 3.2.3 The heterogeneity parameter $v$

The parameter  $v$  deserves attention in its own right, because it is the single most consequential scalar in the technology competition. Algebraically,  $v$  is the scale of the underlying error distribution: small  $v$  corresponds to large unobserved variance — the analyst’s deterministic cost calculation captures relatively little of what the chooser is actually responding to — and large  $v$  corresponds to small unobserved variance, with the deterministic cost calculation explaining almost everything.

The two limits are instructive. As  $v \rightarrow 1$ , the share split flattens toward an even split across all available technologies regardless of cost, and the model effectively concludes that lifecycle-cost ratios contain no information about decision-maker behaviour. As  $v \rightarrow \infty$ , the share split sharpens onto the single lowest-cost option, and the competition becomes a winner-take-all rule equivalent in outcome to a linear-programme corner solution. Real-world adoption patterns sit between these extremes, and the practical calibration question is *how far* between — which is exactly what  $v$  is asked to encode.

A second, often-missed point about  $v$  is that it varies across competitions. The “right”  $v$  for a decision over residential furnaces is not the same as the right  $v$  for a decision over industrial process-heat technologies, because the underlying decision-makers are different (households versus capital-budgeting committees), the stakes are different, and the set of unobserved factors driving the decision is different. CIMS exposes  $v$  as a per-competition parameter so that the model can carry different behavioural assumptions for different decisions, rather than imposing a single global “rationality dial.”

### 3.2.4 Empirical interpretation and calibration

Because  $v$  has a microeconomic interpretation as a scale parameter on revealed preferences, it can in principle be estimated from data. The CIMS programme has pursued this most

aggressively for personal-vehicle and home-heating decisions, where stated- and revealed-preference surveys are tractable and where the choice sets are small enough to estimate cleanly (Rivers and Jaccard 2005b; Bataille et al. 2006; Mau et al. 2008). For competitions where direct estimation is not feasible — for example, deep inside an industrial process where the “decision-maker” is a capital-allocation committee whose criteria are not surveyable —  $v$  is calibrated indirectly against observed historical market shares as part of the broader workflow described in Chapter 39.

### 3.2.5 Limitations worth flagging

The Weibit equation, like the multinomial logit it is closely related to, inherits the *independence of irrelevant alternatives* (IIA) property: the ratio of choice probabilities between any two options depends only on those two options’ costs and is unaffected by what other options are in the choice set. This is convenient algebraically but unrealistic when the choice set contains close substitutes — adding a second “green” option to a menu should mostly steal share from the existing green option, not from the unrelated brown options, yet IIA implies proportional substitution from all incumbents. Nested-logit and mixed-logit variants address this in the broader literature; CIMS does not currently use them, and the practical mitigation is to be careful when structuring the technology hierarchy so that close substitutes are not all thrown into the same competition node.

## 3.3 Iterative price-quantity equilibration

A model that represents technology choice probabilistically inside individual sub-models still needs a way to make those sub-models behave consistently with one another. The lifecycle cost of an electric arc furnace depends on the price of electricity; the price of electricity depends on how much of it the steel sector and others actually buy; how much they buy depends in turn on which furnace technologies won their respective competitions — which depends on the price of electricity. This is a fixed-point problem, and CIMS solves it by iteration rather than by simultaneous solution. The mechanism deserves its own conceptual treatment because it is what gives CIMS its “partial-equilibrium” character without committing it to the full general-equilibrium machinery of a CGE model.

### 3.3.1 The fixed-point structure of the problem

At any given simulation year, the system contains two coupled blocks. The *demand-side* sub-models (residential, commercial, transportation, industrial) each take energy prices as exogenous inputs, run their technology competitions, and produce a vector of energy demands (so

much electricity, so much natural gas, so much refined petroleum, etc.) as output. The *supply-side* sub-models (electricity generation, refining, natural-gas processing, biofuels, hydrogen) each take energy demands as exogenous inputs, run their own technology competitions to decide how the demand is met, and produce a vector of energy prices as output. Each block, in isolation, is a partial-equilibrium calculation: it treats the other block's outputs as given. Coupling the two creates a circular dependency that the partial-equilibrium calculations do not themselves resolve.

CIMS resolves the circularity by *iteration to a fixed point*. The model uses an initial estimate of the energy-price vector, runs the demand-side at those prices, feeds the resulting demands into the supply-side, observes the new prices the supply block produces, and compares them to the previous input estimate. If the two vectors agree within a tolerance — historically 5% on every energy carrier — the system is declared equilibrated for that year and the simulation advances to the next time step. If they disagree, the new prices replace the old input and the demand-supply pass is repeated. In the latest version of CIMS, this price comparison is not only completed for energy carriers, but for every service lifecycle cost in the model. This better ensures all levels of the model are in equilibrium.

The iteration process is illustrated schematically by the demand-supply feedback figure in Chapter 4, and the same logic governs the goods-demand feedback described in the next paragraph.

The current implementation adds a structural wrinkle to this picture. CIMS now represents the model as a directed graph (using Python's [NetworkX](#) package), and each demand-supply pass is realised as a pair of graph traversals — top-down and bottom-up — that require the underlying graph to be acyclic (that is, no loops are present in the graph). Deliberate cycles do, however, exist in the model design: for example, a biodiesel-fuelled tractor consumes biodiesel as an input, and the biodiesel sub-model in turn calls back on tractor service as part of its own production. To run the traversal at all, CIMS applies a small set of loop-resolution heuristics that pick one node in each cycle to evaluate first, using estimated values from its parents on that first pass. Because that estimate is only an approximation of the true equilibrium value, the outer iteration is required to converge several times in succession rather than just once, giving the values inside the broken loop a chance to stabilise alongside the price vector.

### 3.3.2 Goods and services demand feedbacks

A second product feedback loop sits alongside the energy iteration. When a policy shifts production costs in one of the trade-exposed sectors — pulp and paper, primary metals, chemicals — the price of the domestically produced good moves relative to its imported substitute. Armington-style elasticities then shift the demand for the domestic good, which in turn changes the activity level driving the demand-side technology competitions in the affected sector, which can change energy demand and so on. CIMS treats this loop in the same iterative spirit: the goods-price-vs-quantity relationship is evaluated, the activity vector is updated,

and the energy-price iteration restarts under the new activity levels. Convergence in practice depends on having well-behaved elasticities and a finite-amplitude policy shock; the elasticities themselves are documented in the goods-and-services demand feedbacks chapter.

### 3.3.3 Why use partial-equilibrium rather than general

A full CGE model would close *every* circular dependency in the economy simultaneously — energy prices and demands, but also labour markets, capital allocation, terms of trade, exchange rates, household consumption baskets — typically by feeding a single nonlinear system of equations to a solver and asking for the price-quantity vector that zeros out all the market-clearing residuals at once. CIMS does not do that. It closes the energy market and the trade-in-goods market against the technology sub-models, and treats everything else (labour, capital cost, exchange rates, aggregate macroeconomic activity) as exogenous drivers set at the start of the run. Equilibrium is reached over the markets the model considers endogenous, but not over the whole economy.

The trade-off is the same one that runs through the rest of the chapter. Closing fewer markets keeps the technology sub-models tractable and lets them carry the engineering and behavioural detail that distinguishes CIMS from a CGE model, at the cost of being silent on the macroeconomic feedbacks a CGE model would capture. Closing more markets would push the model toward CGE territory and probably toward greater aggregations of technology, which would decrease the model's utility for evaluating regulatory policies. Further, it is sometimes advantageous to disconnect a sector or region from the wider economy and explore policy effects in isolation. This flexibility of scenario design and modelling approach is one of the core strengths of CIMS.

### 3.3.4 Convergence and tolerance

A couple of practical points are worth noting about the iteration itself. First, convergence is not guaranteed in general — a competition equation with high  $v$  and tightly coupled supply and demand can in principle exhibit limit cycles, where the share of, say, electric vehicles oscillates between iterations because each side over-corrects to the other. In practice CIMS rarely encounters this regime, partly because the smoothing imposed by the Weibit equation (intermediate  $v$ ) damps oscillations and partly because the historical 5% tolerance is loose enough to accept small residual oscillations rather than chase them indefinitely. Second, the iteration is performed independently in each time step; convergence achieved in one period is not assumed to carry over to the next, since the technology stock has rolled over and the cost surface has shifted.

### 3.4 Discount rates

Discount rates appear in two distinct roles in CIMS, and conflating them is one of the easier mistakes to make when reading the equations in Chapter 4. The roles correspond to different *questions* about future cash flows. Being explicit about which question is being asked is what motivates CIMS to carry both a *financial* discount rate and a *behavioural* (sometimes called *revealed* or *empirical*) discount rate, and to use them in different places.

#### 3.4.1 Financial rate

The financial discount rate is the rate at which a competently advised firm or household, applying standard cost-benefit logic, would discount the future cash flows associated with a piece of equipment. In a textbook treatment it is the weighted-average cost of capital for a firm, or the appropriate risk-adjusted social discount rate for a public project; in practice it is the rate that converts an undiscounted stream of future operating and energy costs into a present value comparable to today's capital outlay.

CIMS uses a discount rate, denoted  $r$ , inside the capital recovery factor:

$$CRF = \frac{r}{1 - (1 + r)^{-N}}$$

which annualises a one-time capital cost over the equipment's lifespan  $N$  (see Chapter 4). Conceptually, the CRF is asking "what flat annual payment, discounted at  $r$  over  $N$  years, has the same present value as the lump-sum capital cost?" In many financial situations, the right answer to that question turns on what the analyst thinks the *time-value of money* is, and so the value of  $r$  should reflect a defensible cost-of-capital estimate. Typical values are single-digit annual percentages — a few percent above the prevailing real interest rate, with adjustments for the riskiness of the investment.

#### 3.4.2 Behavioural rate

A behavioural discount rate, by contrast, is not a question about the time value of money at all. It is a question about *how decision-makers actually behave*. Empirical studies of consumer choice over energy-using equipment — refrigerators, light bulbs, water heaters, vehicles — find again and again that the implied trade-off between higher up-front capital cost and lower future operating cost is far steeper than any defensible financial discount rate would predict. Consumers, in the language of the literature, act as though they are discounting future energy savings at rates of 20%, 30%, sometimes higher. Firms making smaller, more reversible equipment decisions exhibit something similar, if usually less extreme.

This wedge between “what the financial calculation says they should do” and “what they actually do” is real, and it is not just irrationality. It encodes risk aversion in the face of uncertain future fuel prices, liquidity constraints, information costs, the implicit option value of not committing capital to a long-lived piece of equipment, transaction costs of switching, and decision-maker-specific tastes that the analyst cannot observe. A behavioural discount rate is the device CIMS uses to summarise this entire wedge, along with the financial time-value of money, in a single parameter. When it appears in the lifecycle-cost calculation, it is the rate at which a *plausibly modelled* decision-maker, not a textbook one, is treating future costs.

### 3.4.3 How the two rates are used in practice

In ordinary use, CIMS applies the *behavioural* rate throughout the lifecycle-cost calculation — the capital recovery factor, the foresight expectations of emissions charges, and the other places where future costs are weighed against present ones. This is the default for the reason given above: if CIMS evaluated those quantities at a textbook financial discount rate, it would predict consumers eagerly switching to high-capital-cost / low-operating-cost technologies — heat pumps, LED lighting, electric vehicles — in volumes that decades of empirical observation tell us they do not choose. The behavioural rate corrects the model toward observed behaviour without requiring the analyst to encode every individual reason for that behaviour separately.

The financial rate is still available to use in the model, but as a sensitivity and a normative reference point rather than as the default driver of technology choice. An analyst can substitute it in to test alternative views on the cost of capital, to explore how technology choice would look if decision-makers were behaving textbook-rationally, or to produce an after-the-fact cost-benefit appraisal at a defensible social discount rate.

One caveat that comes with this distinction is that model output should be read with the behavioural rates in mind. A scenario that holds the behavioural rate fixed across a counterfactual is implicitly assuming that the policy in question does not change the underlying barriers to adoption — and many of the policies CIMS is asked about (information campaigns, financing programmes, dealer-incentive schemes) are precisely policies that target those barriers. In such cases, the behavioural rate itself could be considered an object of policy, not a fixed parameter.

### 3.4.4 Behavioural rate versus intangible cost

The behavioural discount rate is not the only device in CIMS that absorbs non-financial preferences — intangible costs (described in the next section) do similar work. In an ideal world the two would carry conceptually distinct loads: the behavioural rate would capture the *time-preference* component of the wedge — impatience, liquidity constraints, the implicit option value of deferring an investment — while the intangible cost would capture the *state-preference*

component — risk aversion to a new technology, perceived quality, social signalling, switching costs unrelated to time. In this idealisation the two are independent and separately identifiable from the data.

In practice they are not. Most observable patterns of technology adoption can be rationalised by any of several combinations of high discount rate and high intangible cost, and the experiments that would in principle separate the two — varying payback horizon while holding perceived risk constant, say — are difficult to run cleanly for real durable-goods decisions. CIMS handles this empirical muddiness pragmatically: it adopts behavioural discount rates from the published literature, where they have been estimated for specific decision contexts and survive across studies, and uses intangible costs as the *adjustable* calibration handle that closes the remaining gap to observed market shares (see Chapter 39). Drawing the line in this place is a choice, not a statement about which device is more “real” — it reflects the practical fact that the literature is a better source for one and historical market-share data the better source for the other.

### 3.5 Intangible costs as a calibration device

The lifecycle cost of a technology in CIMS is, in its short form, the sum of an annualised capital cost, an operating-and-maintenance cost, and a service cost — each of which has an obvious engineering interpretation. The full expression in Chapter 4 adds two further terms inside the capital cost: a *fixed intangible cost* ( $FIC$ ) and a *declining intangible cost* ( $DIC_t$ ). These are not engineering quantities, and they are not measured from invoices or fuel bills. They are calibration handles, and their role inside the model is best understood in those terms.

#### 3.5.1 What intangible costs are doing

An intangible cost is the model’s way of expressing, in dollars, everything that drives a decision-maker’s preference for or against a technology *other than* the financial costs the engineering data already covers. The list of factors any one of these terms can be standing in for is long: the perceived reliability of a new technology relative to an incumbent, the quality or aesthetic of the service it provides (the warmth of incandescent light, the cargo flexibility of a pickup), the inconvenience of switching, the worry about resale value, the social signal of choosing one option over another, the perceived risk of fuel or technology lock-in. Each of these is real and each affects market share, but none of them shows up in a capital-cost or operating-cost spreadsheet. The intangible-cost terms give the model a place to put them.

Mechanically, intangible costs enter the lifecycle-cost calculation through the same channel as the engineering-derived capital cost (see Chapter 4). They are denominated in the same units, they affect the Weibit competition through the same lifecycle-cost ratio, and a high intangible

cost on a technology has the same modelled effect as a high financial cost: it lowers that technology's market share in the competition. The difference is only in interpretation — financial costs are measured externally, intangible costs are inferred internally.

### 3.5.2 Why this is a calibration device

The defining feature of a *calibration device* is that the analyst sets its value not from independent measurement but from the requirement that the model reproduce something observable. For intangible costs, the observable target is the empirical history of market shares: in any given historical year, the actual stock-share split among competing technologies is known, and the value of the intangible-cost parameters can be adjusted until the model's competition produces the same split.

In effect, the intangible costs absorb whatever wedge remains between the engineering cost ratios — capital plus operating plus service — and the cost ratios that, run through the Weibit equation, would actually produce the observed shares. The smaller this wedge is, the closer the engineering data alone comes to explaining adoption behaviour. The larger it is, the more of the adoption story is being carried by the things engineering data cannot see. Either outcome is informative, and one of the secondary purposes of the calibration exercise is to expose where the engineering story is and is not sufficient.

CIMS treats intangible costs as a *per-competition* parameter for the same reason it treats  $v$  that way: the non-financial preferences attached to a residential furnace decision are not the same as those attached to an industrial process-heat decision. The calibration workflow that fits these parameters jointly with  $v$  against historical data is described in Chapter 39.

### 3.5.3 Fixed vs. declining intangible costs

CIMS distinguishes two intangible-cost terms because the underlying behaviour has two distinct timescales.

The *fixed intangible cost* ( $FIC$ ) represents the time-invariant component of non-financial preference — the wedge that does not go away with experience or market exposure, because it reflects a genuine and durable feature of the technology. A user who finds gas-stove cooking irreducibly preferable to induction for reasons that have nothing to do with the price of either is not going to change their mind because more of their neighbours bought induction stoves.  $FIC$  is the calibration handle for that kind of preference.

The *declining intangible cost* ( $DIC_t$ ), by contrast, represents the time-varying component — the wedge that *does* shrink as the technology becomes more familiar, more widely available, and less perceived-risky. New technologies tend to attract a risk premium that fades as adoption climbs, both because direct experience accumulates and because adopters can observe their neighbours' experience. CIMS captures this via the declining intangible-cost function

(in Chapter 4), which lets  $DIC_t$  shrink as a function of new market share. This formalises the *neighbour effect* documented in the empirical literature on consumer adoption of new vehicle technologies (Mau et al. 2008) and is a key channel through which policies that boost early adoption — pilot programmes, fleet procurement, financing assistance — produce dynamic feedback in the model.

### 3.5.4 Limitations of intangible costs

Calibrated parameters can be made to fit almost any historical observations, and intangible costs are a flexible enough device to absorb a great deal of model error if used without discipline. Practically, the calibrated values should be *interpretable*: an intangible cost large enough to dominate the lifecycle cost should correspond to a story the analyst can tell about why the technology actually has that much non-financial baggage. A calibrated value that has no narrative is more likely to be picking up a misspecification elsewhere in the model — a wrong elasticity, a missing technology, an unrepresented constraint — than a genuine non-financial preference. Further, the values should be *stable across reasonable variations in the calibration target*: an intangible cost that swings wildly between two equally defensible historical fits is also more likely to be model error in disguise than a real preference.

Intangible costs are CIMS’s primary device for expressing the gap between what engineering data says decision-makers should do and what observed adoption tells us they actually did. The calibration chapter (Chapter 39) discusses the practical workflow and the diagnostic checks that try to keep intangible costs realistic.

## 4 Technology Competition

Sub-model technology choice occurs according to the following order:

- Demand Assessment
- Retirement
- Assessment of capital stock availability
- Retrofit & new capital stock competition to meet demand not supplied by existing capital stock

### 4.1 Demand Assessment

All of the CIMS energy demand models (industry, transportation, commercial / institutional, and residential) start with an exogenously determined demand for physical goods and services in each run period. These demands are mostly taken from Natural Resources Canada forecasts. The energy supply models may either use exogenously determined forecasts or they can be run using energy demand calculated from the demand model plus net exports.

#### 4.1.1 Request-Provide demand

#### 4.1.2 Distributed supply

### 4.2 Capital Stock Retirement

CIMS begins each simulation with an existing base stock of technologies. Equation 4.1 describes how the existing stock of unknown age already in use is retired. Each year a constant amount of the base stock is retired until there is no more base stock remaining (Figure 4.1).

$$BS_t = BS_0 \left( 1 - \frac{T_t - T_0}{L} \right) \quad (4.1)$$

$BS_t$  base stock at time  $t$

$BS_0$  initial base stock

$T_t$  current simulated year

$T_0$  initial year  
 $L$  lifespan of the technology

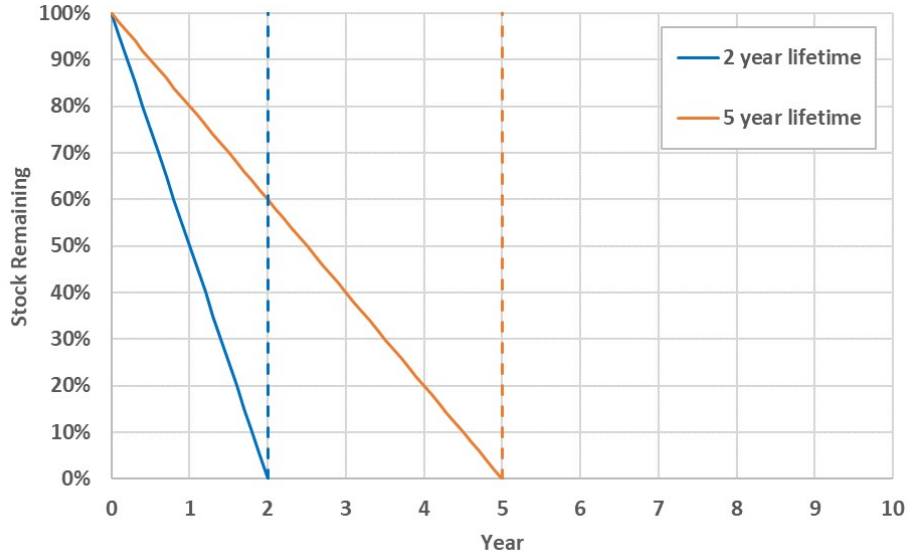


Figure 4.1: Existing Stock Retirement

Retirement of capital stock purchased during a CIMS simulation is calculated using a different methodology since the vintage of new stock is known and an explicit lifetime (and retirement schedule) can be applied to the stock purchase.

$$NS_t = \frac{NS_p}{1 + e^{\frac{11.513}{L}((T_t - T_p) - L)}} \quad (4.2)$$

$NS_t$  new stock not yet retired at time t

$NS_p$  new stock acquired at purchase time p

$T_t$  current simulated year

$T_p$  year at purchase time p

$L$  lifespan of the technology

11.513 intercept value from ISTUM model for average technology retirement rates

This function uses an s-shaped, logistic curve to simulate the retirement of technologies. Figure 4.2 shows examples of the retirement function for technologies with different lifespans. Half of the stock is retired before the average lifespan of the technology due to early replacement and failure and half is retired afterward to simulate technologies that last longer than the average. The default intercept value of 11.513, which comes from market research on product failure rates, is used in CIMS for many technologies that are purchased and installed. For

other longer-lived technologies, like building shells, lower intercepts are used that flatten the logistic curve.

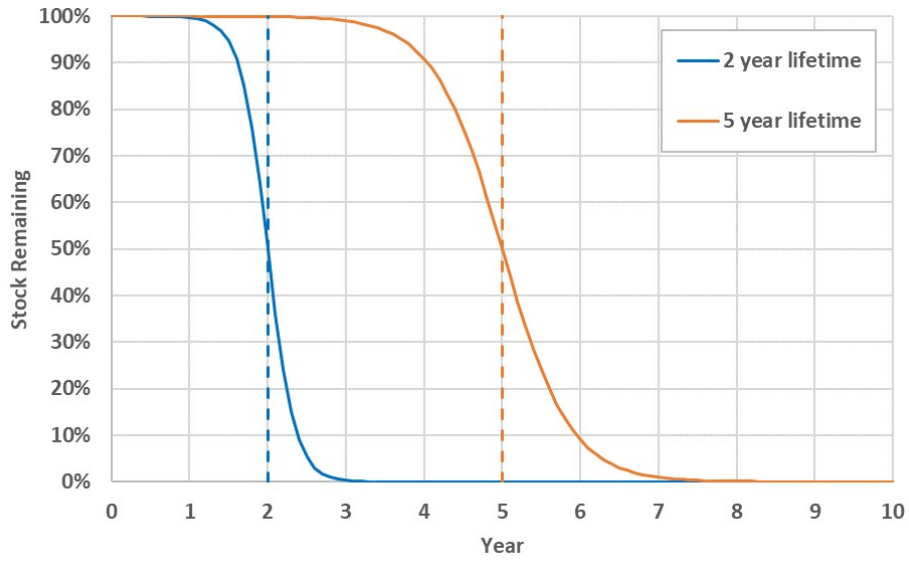


Figure 4.2: New Stock Retirement

The total stock of a technology in any given year  $t$  is the sum of the unretired base stock and the unretired new stock remaining from each previous year (Equation 4.3).

$$Stock_t = BS_t + \sum_{p=0}^t NS_t \quad (4.3)$$

### 4.3 Retrofit and New Stock Competition

While retrofitting occurs before new stock competition, the retrofitting equations are a subset of the new stock equations and are therefore presented after them. The market shares for individual technologies are determined using the equations below.

#### 4.3.1 New Stock Competition

Equation 4.4 is a normalized inverse-power function: the market share of each technology is proportional to its lifecycle cost raised to  $-v$ , divided by the sum of that quantity over all competitors. It is the multinomial choice analogue of softmax, but with a power-law transform

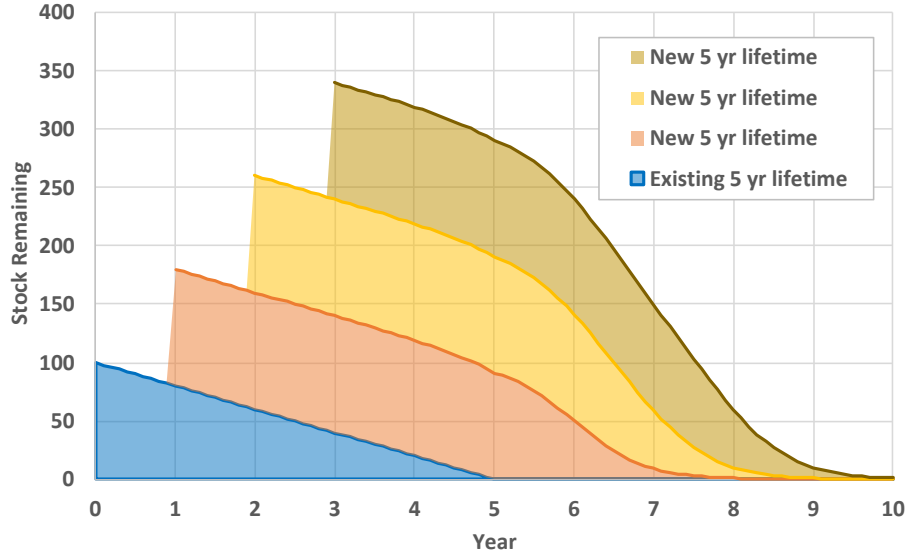


Figure 4.3: Total Stock

of cost instead of an exponential<sup>1</sup> — which means lifecycle-cost *ratios*, rather than differences, drive the share split. The variance parameter  $v$  controls how sharply the competition responds to those ratios: a high value, such as  $v = 30$ , makes the lowest-cost technology capture almost all new equipment stocks (approaching the winner-take-all rule of a linear program), while a low value, such as  $v = 1$ , spreads new equipment shares almost evenly across competing technologies even when their lifecycle costs differ significantly. Equivalently, in log-cost space the equation has the form of a multinomial logit on  $\ln LCC$ .

$$MS_k = \frac{LCC_k^{-v}}{\sum_{j=1}^J LCC_j^{-v}} \quad (4.4)$$

$MS_k$  market share of technology  $k$  for new equipment stocks

$LCC_k$  annual lifecycle cost of technology  $k$

$v$  variance parameter representing cost homogeneity

<sup>1</sup>The CIMS choice rule belongs to the family of log-linear utility, multiplicative-error discrete choice models sometimes called the **Weibit** model. Standard multinomial logit (McFadden 1974) assumes a Gumbel-distributed additive error term in utility and yields choice probabilities of the form  $e^{V_k} / \sum_j e^{V_j}$ ; the Weibit instead assumes a Weibull-distributed multiplicative error and yields the power-form expression used here (Castillo et al. 2008). The distinction occasionally matters when calibrating  $v$  against empirical data, because the implied stochastic interpretation of the lifecycle cost differs. For background on how the logit-style framework is used in CIMS more broadly, see Rivers and Jaccard (2005a).

$$LCC_{kt} = CC_{kt} + OM_k + SC_{kt} \quad (4.5)$$

$LCC_{kt}$  annual lifecycle cost of technology  $k$  at time  $t$

$CC_{kt}$  capital cost of technology  $k$  at time  $t$

$OM_k$  operating costs of technology  $k$

$SC_{kt}$  service cost of technology  $k$  at time  $t$

Note somewhere that technology capital costs should be included as CAPEX where possible to ensure apples-to-apples comparison across the competition.

$$CC_t = \max \{DCC_t, CC_{2000} \cdot DCC_{limit}\} \quad (4.6)$$

$CC_t$  annual capital cost of a technology at time  $t$

$DCC_t$  declining capital cost at time  $t$

$CC_{2000}$  capital cost of a technology in the year 2000 (before applying the declining capital cost function)

$DCC_{limit}$  lowest value to which a technology's capital cost can be reduced by the DCC function (as a percentage of the capital cost in year 2000)

See the section below for the declining capital cost functions.

$$CC_{2000} = \left( \frac{CC_{overnight} + FIC + DIC_t}{Output} \right) \cdot CRF \quad (4.7)$$

$CC_{2000}$  capital cost of a technology in the year 2000 (before applying the declining capital cost function)

$CC_{overnight}$  overnight capital cost

$FIC$  upfront fixed intangible cost, representing non-financial costs of investment

$DIC_t$  upfront intangible cost that can decline over time

$Output$  material output of a technology

$CRF$  capital recovery factor

$$CRF = \frac{r}{1 - (1 + r)^{-N}} \quad (4.8)$$

$CRF$  capital recovery factor of a technology

$r$  financial discount rate

$N$  lifespan of the technology

$$SC_{kt} = \sum_j P_{jt} \cdot R_{kjt} \quad (4.9)$$

$SC_{kt}$  service cost of technology  $k$  at time  $t$

$P_{jt}$  price of service type  $j$  at time  $t$  (can be either an energy price or the weighted-average life-cycle cost of a service, based on the market shares of technologies competing to provide that service)

$R_{kjt}$  amount of service type  $j$  requested by technology  $k$  at time  $t$

Note that market shares used to calculate the weighted-average LCC of a service are from the previous iteration of the demand-supply equilibrium process (i.e. the final market shares from the previous year if calculating the first iteration or the previous demand-supply iteration if equilibrium prices have not yet been found).

## Subsidies

### 4.3.2 Retrofit

The retrofit option allows functioning equipment to be converted to a different type of equipment before its retirement date. For example, a boiler could be retrofitted into a cogenerator before its lifetime-based retirement. After existing stock is retired, CIMS treats the initial purchasing price of the incumbent technology as a sunk cost and uses the operating and maintenance (O&M), energy, service, and emissions costs as the only costs of the incumbent technology. The costs of the retrofit options are then calculated as the sum of O&M, energy, service, and emissions costs plus the annualized retrofit capital cost less any tax or other credit. CIMS then determines, using a similar competition calculation to those already described, how many incumbent technologies will be retrofitted.

The analyst can place some bounds on this retrofit option. A maximum (R-MAX) or minimum (R-MIN) market share can be established such that only a certain amount of the old stock could (maximum) or must (minimum) be retrofitted in any year. In an unconstrained market, maximum should be set to 1 and minimum to 0.

## 4.4 Modifications to the technology choice equations

Several functions have been added to the competition process to reflect both real world conditions and policy concerns. These currently include functions to:

- Reduce capital costs as total production increases to reflect learning by doing and efficiencies of scale;
- Reduce intangible costs as new market share climbs to reflect reduced perceived risk (i.e. the neighbour effect); and
- Eliminate obsolete but still functioning technologies.

#### 4.4.1 Declining Capital Cost

The declining capital cost function allows the capital costs of technologies to decline as a result of learning by doing and improving economies of scale. In CIMS, capital costs decline as a result of the cumulative production of a technology across Canada and an exogenous rate of decline which represents production in other countries.

**Need to align this description and function below with Thomas' update of DCC function**

$$DCC_t = GCC_t \cdot \left[ \left( \sum_1^p \frac{CumulNS_{2000,p} + \sum_{j=2005}^{t-5} NS_{jp}}{CumulNS_{2000,p}} \right)^{\log_2(PR)} \right] \quad (4.10)$$

$DCC_t$  declining capital cost of a technology at time  $t$

$GCC_t$  capital cost of a technology adjusted for cumulative stock in other countries

$NS_{jp}$  new stock of a technology added from 2005 to the previous time period in each province  $p$

$CumulNS_{2000,p}$  cumulative new stock of a technology for all years up to and including the year 2000 in each province  $p$  (exogenous)

$PR$  progress ratio

Increased stock in other countries reduces the capital costs of a technology in CIMS. Therefore, even if there is no change in stock of a technology in Canada, its capital costs may still decline. The rate at which the capital costs decline as a result of global stock change is set exogenously.<sup>2</sup>

$$GCC_t = GCC_{t-5} \cdot (1 - AEEI)^5 \quad (4.11)$$

$GCC_t$  capital cost of a technology after it has been adjusted for cumulative stock in other countries

$AEEI$  exogenous rate of decline of a technology's capital costs

In order to prevent capital costs from declining indefinitely, an analyst may specify the percentage of the initial capital costs below which capital costs may not decline.

Research has shown that investment in one technology (e.g., hybrid cars) can lower costs of another technology (e.g., battery cars). This can be achieved by setting technologies to the same DCC category so that the sum of all installed stock for technologies within that category is used when determining capital costs.

---

<sup>2</sup>This is equivalent to the autonomous energy efficiency index (AEEI).

#### 4.4.2 Declining Intangible Cost Function

The declining intangible cost function allows non-financial costs to fall as a result of reduced perceived risk as market exposure increases. The following general formula applies.

$$\text{(Equation 20 — placeholder for the declining intangible cost function)} \quad (4.12)$$

$I_t$  intangible cost in time  $t$

$I_0$  initial intangible cost (column N of the economic file for capital, column R for operating costs)

$A$  constant in column O of the economic file for capital, column S for operating costs

$K$  constant in column P of the economic file for capital, column T for operating costs

$NMS$  new market share

$NMS_{t-5}$  new market share from the previous run, expressed between 0 and 1 (i.e. between 0% and 100%, not 0 and 100).  $NMS$  is the technology's share among all other technologies that provide the same service.

#### 4.4.3 Forward looking behaviour

The foresight functions simulate business and consumer behaviour for considering future costs. The functions calculate the expected emissions cost of technologies, considering different lifespans and discount rates.

CIMS has three options for generating expectations: 1) myopic, where people do not have foresight into their future emissions costs; 2) foresight based on the average emissions costs of a technology; and 3) foresight based on the discounted value of emissions costs of a technology.

##### Myopic

The emissions charge used to calculate the lifecycle cost of technology is the current tax rate.

$$E(\text{EmissionsCharge})_{kt} = \text{EmissionsCharge}_t \quad (4.13)$$

##### Average Expectations

The second option generates expectations of future emissions costs based on the average emissions charge over the lifespan of a technology.

$$\text{(Equation 22 — placeholder for the average-expectations function)} \quad (4.14)$$

**EmissionsCharge<sub>n</sub>** emissions charge in year  $n$ , set in CIMS's Simulation Creation Centre  
*base* year in which technology  $k$  is purchased

$N$  lifespan of technology  $k$

### Discounted Expectations

The third option generates expectations of future emissions costs based on the present value of emissions charges. After the present value of emissions charges has been calculated, it is annualized using the capital recovery factor.

$$\text{(Equation 23 — placeholder for the discounted-expectations function)} \quad (4.15)$$

**EmissionsCharge <sub>$n$</sub>**  emissions charge in year  $n$ , set in CIMS's Simulation Creation Centre

*base* year in which technology  $k$  is purchased

$N$  lifespan of technology  $k$

$r_k$  revealed discount rate of technology  $k$ , set in the Techno Economic file

## 5 Energy Supply and Demand

### 5.1 Energy Pricing

#### 5.1.1 Primary Energy

##### 5.1.1.1 Fixed Price

##### 5.1.1.2 Cost Curve – Annual

##### 5.1.1.3 Cost Curve – Cumulative

#### 5.1.2 Secondary Energy

#### 5.1.3 Carbon pricing

TODO: Add section on biomass supply modelling

In CIMS's integrated energy supply and demand system, the demands of final demand sectors directly drive the activities of electricity production, refining, coal mining and natural gas and crude oil extraction. Inter-regional transfers as well as net exports are added, as demand for Canada's energy supply sector includes US demand. The importance of domestic demand varies by energy commodity. Most electricity is produced for Canadian consumers, more than half of natural gas produced is shipped to the US, and there is a heavy trade in crude oil and refined petroleum products. Half our domestic crude oil production is shipped to the US, while we import from both OPEC and non-OPEC suppliers.

**Need to revise this description to reflect the current setup, but should preserve a more complete version as a comment for later upgrades. Adjust the descriptions below to detail the fully endogenous options and note that many can be switched to exogenous function.**

Each of the energy markets in CIMS has the option of fixed or dynamic pricing, and they are described below:

- Regional electricity markets are included with endogenous internal pricing based on the cost of production. The amount of electricity produced is the sum of endogenous domestic demand and net extra-regional exports, which may be adjusted using an export own-price elasticity.
- One national natural gas market is included, with dynamic pricing using a supply curve based on a combination of Canadian and US market feedbacks. The amount of gas produced is the sum of endogenous domestic demand and net exports. Net exports may be adjusted to reflect changes in the domestic cost of production using an export price elasticity.
- One national crude oil market, based on an exogenously specified world price of crude oil. The amount of crude oil produced is the sum of endogenous domestic demand and net exports. Net exports may be adjusted to reflect changes in the domestic cost of production using an export elasticity. The world crude oil price is assumed to be unresponsive to Canadian demand.
- Petroleum refining regions divide their production amongst demand regions and net exports. All can use endogenous internal pricing based on the cost of production. The amount of refined petroleum products produced is the sum of domestic regional demand and net extra-regional exports.
- Coal producing regions divide their thermal coal production amongst endogenous domestic regional demand and exogenous exports (i.e. BC exports 100% of its production while Alberta exports about 30% of its production). Due to the differences between the ways each non-producing region and sector gets its coal, these regions are assumed to buy coal at a fixed market price or according to a local supply curve – for example, some electricity plants in Alberta and Saskatchewan are sited at coalfields and are the sole customers. British Columbia, Alberta, Saskatchewan, and the Atlantic provinces all buy their coal at a fixed market price.
- **-> Revise coal description to separate thermal coal (exogenous) and metallurgical (endogenous).**

CIMS divides energy flow in the economy into demand for energy end-use services, intermediate processing, and primary supply (Figure 5.1). It begins with an initial demand for four general end-use energy commodities: electricity, natural gas, coal, and refined petroleum products. The electricity needs of the domestic energy demand models (transportation, residential, commercial, and industry) plus net electricity transfers (inter-regional transfers and exports) are used to drive electricity generation. The refined petroleum products sector is driven by the net demand from the energy demand models, electricity, inter-regional transfers, and net exports. The primary energy supply models, crude extraction, coal mining and natural gas extraction and transmission are driven by the net demand for their products from the final demand sectors, electricity and refined petroleum.

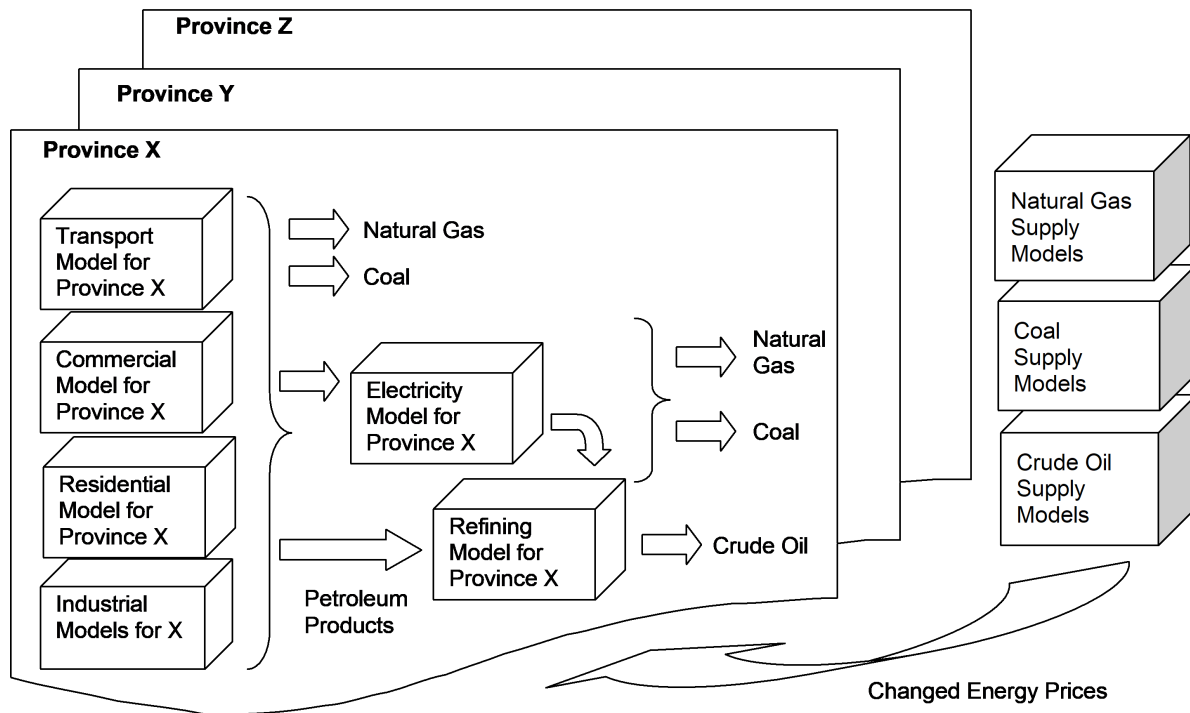


Figure 5.1: final\_intermediate\_primary\_flow

The electricity demands of petroleum refining, the refined petroleum product demands of the primary energy supply models, and the interdependencies among the primary energy supply models create feedback loops within the system. Because CIMS operates as a simulation model rather than a computable general equilibrium model, these feedback loops cannot be resolved within a single iteration of the demand–supply algorithm. To address this limitation, CIMS is run iteratively until lifecycle costs throughout the model converge below a specified threshold criterion (typically less than a 5% difference between iterations). Additional iterations may also be performed to ensure that the effects of the feedback loops have stabilized.

Canada trades a significant amount of energy with the United States and, to a lesser extent, with both OPEC and non-OPEC suppliers. Given the possible effect of Canadian energy policy on the price of producing these commodities, and hence their marketability to the US, electricity, crude oil, and natural gas export feedbacks are included in CIMS using price elasticities calculated for the Canadian Federal Department of Finance (Table 5.1).

change this table to GTAP? talk to Thomas

**Table 5.1:** Price Elasticities for Energy Exports, Wirjanto (1999)

| Energy Commodity         | Long-Run Own-price Elasticity for Export Demand |
|--------------------------|-------------------------------------------------|
| Electricity              | -0.54                                           |
| Natural gas              | -0.67                                           |
| Coal                     | -0.51                                           |
| Gasoline and fuel oil    | -0.92                                           |
| Other petroleum products | -0.82                                           |

These elasticities follow the Armington specification, whereby the demand for an imported good rises as its price falls relative to the price of its domestic substitute. At all times, consumers buy at least some of the domestic good and some of the imported good, even when the price difference is large (Iorwerth et al. 2000, Wirjanto 1999). See Bataille (2005) for further elaboration. The Armington specification assumes that goods produced by different countries are not perfect substitutes. The energy trade elasticities are represented in Equation 5.1, which applies the elasticity of the ratio of import volumes to domestic production in relation to a change in the relative price of imports to domestic prices.

$$\text{(Equation 25 — placeholder for the energy trade Armington elasticity)} \quad (5.1)$$

$FLCC_t$  current financial lifecycle cost for producing the energy commodity

$FLCC_{t-1}$  previous financial lifecycle cost for producing the energy commodity

$COP$  cost of production covered by the energy-producing sub-model, as a percentage of total cost

$\sigma$  price demand elasticity for the energy commodity

$IT$  initial trade volume of the energy commodity

$Adj_{prod}$  additive adjustment to production of the energy commodity

Western Canadian exports of crude oil are separated from eastern Canadian imports, as there are currently no large cross-country shipments. While electricity trade is currently a very small portion of our energy trade, electricity import substitution elasticities are included to cover the possibility of a policy driven increase or decrease in ele

## 6 Goods and Services Demand Feedbacks

The economy is commonly divided into: households, who provide labour and financial capital and who consume goods and services; firms, who consume capital and labour and provide goods and services; government, which consumes goods and services, provides public goods, and regulates the market; and finally foreign producers and consumers of goods, services, and capital (Dornbusch et al. 1989). It is common to assume that Canada is a price taker for capital (i.e., the interest rate is set at the world price of capital).

Canada is assumed to be a small open economy... the world interest rate is assumed to be the US prime rate, and foreign savings are assumed to be available elastically. This means that the foreign sector automatically clears any surplus or deficit in the domestic savings market each period, so the domestic interest rate is effectively exogenous. (McKittrick 1998) p.552

The following suite of options face the consumer or producer when the price of a good or input rises, modelled after the consumer's utility map in CASGEM (Iorwerth et al. 2000):

- Substitute an import
- Buy something else
- Substitute current consumption for savings
- Work less and enjoy more leisure, which is the same as reducing output

These options were used as the primary guide for adding demand feedbacks in CIMS.

### 6.1 Manufactured products

Most Canadian industrial products are highly traded; their sensitivity to foreign competition is of prime importance. To reflect the sensitivity to foreign competition, CIMS employs the same Armington specification for price elasticities as used for energy trade, whereby the demand for a foreign good rises as its price falls relative to the price of a domestically produced substitute. Using the Armington specification, consumers always buy at least some of the domestic good and some of the imported good. The price elasticities are applied with the general formula in Equation 6.1.

(Equation 26 — placeholder for the manufactured-products Armington elasticity) (6.1)

$FLCC_t$  current financial lifecycle cost for the product

$FLCC_{t-1}$  previous financial lifecycle cost for the product

$COP$  cost of production covered by the sub-model that produces the product, as a percentage of total cost

$\sigma$  price demand elasticity for the product

*Multiplier* value by which demand for the product is multiplied

To accommodate the differences between products in industrial sectors, especially Industrial Minerals, Chemical Products, Metal Smelting, and Light Industrial, demand feedbacks are applied at the level where production of individual products is determined. Table 6.1 provides the elasticities used for sectors and their products. See Appendix E of Chris Bataille's PhD dissertation for a full breakdown of their sources and how they were mapped to CIMS' sub-models and their sub-products.

**Table 6.1:** : Price elasticities used from Wirjanto (1999)

| Sector                     | Product                                                            | Elasticity |
|----------------------------|--------------------------------------------------------------------|------------|
| <b>Metal Smelting</b>      | Lead, Zinc, Aluminium, Nickel                                      | -0.59      |
|                            | Copper                                                             | -0.57      |
| <b>Chemical products</b>   | Chlor-alkali, Sodium Chlorate, Hydrogen Peroxide, Ammonia Methanol | -0.44      |
|                            | Polymers                                                           | -0.83      |
| <b>Industrial Minerals</b> | Cement                                                             | -1.75      |
|                            | Lime, Glass, Bricks                                                | -0.53      |
|                            | Pulp export                                                        | -1.72      |
| <b>Pulp and Paper</b>      | Coated                                                             | -0.79      |
|                            | Uncoated, linerboard, tissue & newsprint                           | -1.54      |
| <b>Iron and Steel</b>      | Slabs, blooms, billets & molten steel                              | -0.60      |
| <b>Mining</b>              | Open pit, underground                                              | -1.38      |
|                            | Potash                                                             | -0.58      |
| <b>Light manufacturing</b> | Food, tobacco, beverage                                            | -0.32      |
|                            | Leather, textile, clothing; Wood products                          | -0.86      |
|                            | Rubber and plastics product                                        | -1.30      |
|                            | Electrical and electronic equipment                                | -0.81      |
|                            | Transportation equipment                                           | -1.06      |
|                            | Furniture, printing, machinery                                     | -0.74      |

## 6.2 Commercial and residential

Check this section because I don't think we do this! Review with Thomas how much of Chris Bataille's work is actually included. Also need Thomas to document his macroeconomic work.

There is no import substitution for non-traded goods and services such as residential and commercial floorspace. CIMS uses a log-linear econometric relationship to estimate the relationship between manufacturing activity – used dynamically as valued added but proxied using the value of shipments – and residential / commercial activity – measured by the value of building permits. The resulting relationship is used to allow commercial and residential activity to respond to changes in manufacturing activity.

(Equation 27 — placeholder for the commercial/residential demand feedback) (6.2)

$BP_{c,r}$  natural log of activity (value of building permits) in the commercial or residential sectors in a given year

$IVA$  natural log of industrial value-added in a given year

$e$  error term

$m$  coefficient of the relationship (elasticity)

$b$  intercept

This equation was estimated, using SHAZAM v.9, by running ordinary least squares regressions between manufacturers' shipments and residential building permits between 1961 and 1991 and manufacturers' shipments and commercial, institutional, and governmental building permits between 1961 and 1991. All values were deflated using a standard consumer price index. We tested various permutations of the regressions, including lagging and the use of population and year as an independent variable. Table 9 provides the parameter estimates.

Table 7. Parameter Estimates Used to Provide Demand Feedbacks for the Residential and Commercial/Institutional Sectors

The process for calculating manufacturing valued-added, the driver for residential and commercial activity in CIMS, is complex. The sub-model's outputs are designed to capture changes in energy, new capital, and labour expenditures engendered by policy. These costs are a sub-component of the total cost of production, which also includes materials and supplies, as well as capital investment, labour, and repairs not related to energy using equipment.

Value-added is defined as the total value of shipments and other revenue, minus the cost of intermediate inputs (energy, materials, and supplies) (Lipsey et al. 1998, p.175). Returns to capital include both amortization and profits. Using values for wages, energy expenditures, costs of material, total value of shipments and capital expenditures from Statistics Canada, we found that while the ISTUM sub-models do not capture expenditures for materials or capital unrelated to energy consumption, they capture most capital expenditures in the sectors for which sub-models exist. Supplemental to this, the demand feedback function of CIMS, when faced with a reduction in demand, removes first new growth, then the necessity for replacing retired technology stock, and finally useful stock that is no longer required. Given that most sectors in CIMS grow in most time periods, a simplifying assumption was made that all the productive investment being removed was new or replacement investment, not useful

stock. We then compared CIMS' new investment in 2000 with figures from Statistics Canada, and derived a set of multipliers to scale up CIMS' investment figures to reflect the ratio of coverage (the multipliers affect the demand for all stock, not just the portion of the cost of production covered by ISTUM). Finally, we added normal profits as a percentage of revenue, having ascertained that normal profits are included in value-added and that these are not normally included in CIMS. These were calculated as a percentage of total costs from Statistics Canada's "Financial and taxation statistics for enterprises, 2000, 61-219".

### 6.3 Freight transportation

Freight transportation is important for energy and emissions policy because of the large amount of energy consumed and the amount of GHGs and criteria air contaminants produced. Activity in this sector is directly related to the amount of activity in the rest of the economy. To provide feedback relating freight and economic activity, a similar relationship to that established between manufacturing value-added and residential and commercial activity was incorporated into CIMS. It proved difficult to find an equivalent statistic for freight activity to manufacturing activity as used for the commercial and residential sectors, so instead of an estimated relationship a rule-of-thumb value was temporarily used instead (0.95 – a 1% fall in manufacturing value-added causes a 0.95% fall in freight activity). This relationship allows changes in manufacturing value-added to adjust freight transportation activity. This relationship will be estimated once statistics are available.

### 6.4 Personal transportation

Demand feedbacks for personal transportation operate in a similar fashion to the demand feedbacks for manufacturing and industry, except that in the place of an Armington price elasticity an elasticity for personal kilometers traveled (pkt) in response to the cost of travel is used instead (price elasticity = -0.02) (Michaelis and Davidson 1996). Transportation is also a bit different from the other sub-models in that some demand adjustments occur within the sub-models, specifically the choice between urban modes of transport (single occupancy vehicles, high occupancy vehicles, transit, and walking and cycling). The upshot is that much of the structural change possible in transportation is directly included in the sub-sector models.

### 6.5 Limits to demand adjustments

Experimentation with early versions of CIMS showed that the demand adjustments had to be regarded critically. Elasticities are by nature log-linear measurements, while demand re-

sponses are very often non-linear. It may be hypothesized that there will always be a demand for some commodities, no matter the price. A function for limiting their effects was installed; 50% was chosen as the maximum demand reduction as a rule-of-thumb for the maximum extent to which CIMS' substitution elasticities were still applicable. This demand limiting adjustment is somewhat simplistic, and in the future some form of function that applies limits to demand adjustments in a progressive

## **Part III**

# **Sector Descriptions**

## 7 Sector Overview

This part of the documentation sets out how CIMS represents each part of the economy. Every sector is a self-contained sub-model — a flow model of the energy services and products that drive its energy use and emissions — but all of them are built from the same handful of ideas: hierarchical nodes linked by engineering ratios, technologies competing on lifecycle cost at the lowest nodes, and a split between the things a sector produces and the auxiliary services it consumes. The chapters that follow take each sector in turn; this overview sets out the shared structure they rest on, the conventions they have in common, and a map of which sectors are covered and how they relate.

### 7.1 How a sector is modelled: the flow model

In CIMS, the product and energy-service demands of a sub-sector are linked in a **flow model** that traces the sequence of activities needed to generate the sector's output. A CIMS flow model is geared toward representing technology evolution and energy consumption, rather than the monetary flows of an econometric model (where units are typically in dollars) or the literal mechanical detail of the blueprints and process-flow diagrams an engineer would use. Because the emphasis is on energy rather than material flow, each node represents a process stage at which energy consumption can be distinctly estimated. Nodes are arranged hierarchically and linked by engineering ratios — the physical quantity of input each stage requires per unit of its output — and technology competitions take place at the lowest-level nodes in the hierarchy.

### 7.2 Nodes and competition rules

Each sector sub-model is a hierarchical tree, implemented in the model code as a NetworkX directed tree. Every node carries a competition rule that determines how the demand passed down to it is allocated among its children:

- **Fixed-ratio** nodes split demand among their children in fixed, exogenously specified proportions — the engineering ratios that capture the unavoidable sequence of process steps required to produce the sector's output.

- **Tech-compete** nodes host a competition among alternative technologies that can provide the same service. Each technology carries a capital cost, a fixed operating and maintenance cost, an output (node-unit) basis, service requests for the energy and upstream services it consumes, and availability and lifetime years; these are annualised and combined with service costs into a lifecycle cost. Market shares are then assigned by the standard CIMS technology-choice rule of Chapter 4, on the basis of those lifecycle costs and a heterogeneity (variance) parameter that governs how sharply the competition responds to cost differences.

The interplay of these two rules is what gives each sector its shape: a fixed-ratio backbone fixes the process chain, while tech-compete leaves decide which equipment and fuels actually serve each step — and it is at those leaves that policy, whether a carbon price, a regulation, or a subsidy, changes the outcome.

## 7.3 Product and Service sub-trees

At the top of each sector tree, the nodes are organised into two sub-trees, a **Product** sub-tree and a **Service** sub-tree:

- The **Product** sub-tree is the physical production chain: the sequence of process stages that converts raw inputs into the sector’s saleable output — tonnes of steel, square metres of floor space, vehicle-kilometres, and so on. It is driven by the sector’s exogenous output or activity forecast, and its shape is specific to each sector’s process flow.
- The **Service** sub-tree holds the auxiliary energy and utility services that production technologies request rather than producing themselves — for example space heating and cooling, lighting, machine drive, steam and cogeneration, fuel blending, and carbon capture where applicable. Factoring these shared loads into their own sub-tree avoids re-specifying the same equipment inside every process stage, and lets efficiency and fuel-switching measures compete once and serve the whole sector.

### 7.3.1 Auxiliary services and the MECS structure

For the industrial sectors, the auxiliary services follow the end-use structure of the U.S. Energy Information Administration’s Manufacturing Energy Consumption Survey (MECS) — in particular, **Machine Drive** is disaggregated into compressed air, fans, materials handling, materials processing, and pumps. Each service is represented at a general level, as a competition between existing and more efficient equipment with fuel-switching options where relevant, rather than as a detailed equipment catalogue. The specific set of services varies by sector; the buildings and transportation sectors, for instance, carry their own service definitions, which are described in their respective chapters. The cross-cutting services that many sectors share — machine drive, low-temperature heat and steam, cogeneration, carbon capture, and energy

storage — are documented once in the [General Services](#) part rather than repeated in every chapter.

TODO: Several shared services are still being consolidated into the General Services part — in particular renewable-fuel blending, cogeneration, and waste-heat recovery — and some sector files still carry older, sector-specific versions of these structures. Migrate them to the shared definitions and cross-reference them here.

## 7.4 The sectors represented

The sub-models fall into a few natural groups, reflecting how energy and materials move through the economy.

Energy supply and transformation covers the industries that produce and convert the fuels everything else burns: [Coal Mining](#), [Natural Gas](#), [Petroleum Crude](#), and [Petroleum Refining](#) on the fossil side; [Electricity](#) as the central energy carrier; and the low-carbon fuels [Ethanol](#), [Biodiesel](#), and [Hydrogen](#). Because these sectors supply the fuel blends and electricity that every demand sector draws on, their emissions intensity propagates throughout the model.

Extraction and heavy industry covers the material-producing sectors: [Mining](#), [Industrial Minerals](#) (including cement and lime), [Chemical Products](#), [Pulp and Paper](#), [Metal Smelting](#), [Iron and Steel](#), and the catch-all [Light Industrial](#). These carry the richest Product sub-trees and the largest process-heat and process-emission loads, and they are where carbon capture and deep fuel-switching matter most.

Buildings and construction covers the built environment: [Construction](#), the activity of building itself, and the [Residential](#) and [Commercial](#) sectors, which represent the heating, cooling, lighting, and plug loads of the building stock once it is in use.

Transportation covers the movement of people and goods, split into [Transportation \(Passenger\)](#) and [Transportation \(Freight\)](#), each resolving its vehicle fleet by mode and powertrain.

Emissions-driven and land-based sectors are those whose footprint is dominated by something other than energy use, and whose activity is therefore often measured directly in tonnes of CO<sub>2</sub>-equivalent: [Waste](#) (landfill methane), [Agriculture](#), and [Forestry](#), together with the negative-emissions and land-accounting sectors [Carbon Dioxide Removal](#) and [LULUCF](#). Several of these are deliberately simplified placeholders today, and their chapters flag what a fuller representation would add.

## 7.5 What each sector chapter contains

The sector chapters follow a common template, so that the same questions can be asked of every part of the economy. Each opens with the sector's **scope** — what it does and does not

include, and where adjacent activities are handled to avoid double-counting — and a **real-world context** section that grounds the model in the actual industry: its scale in Canada, its emission sources, and the levers available to reduce them. A **model structure** section then walks through the sector’s tree, anchored by a hand-drawn flow figure, distinguishing the fixed-ratio production chain from the tech-compete leaves and naming the services it draws on.

A **decarbonisation technologies** section surveys the low- and zero-carbon options for the sector, usually with two tables: a maturity table that places each option on the spectrum from commercial to emerging, and an indicative techno-economic cost table in 2025 Canadian dollars. These cost figures are order-of-magnitude values drawn from public literature, included for relative comparison rather than budgeting, and kept deliberately separate from the calibrated values the model carries internally. An **economics** section explains how the competition resolves for that sector and records its emissions accounting; a **regional coverage** section notes which regions carry the sector and how they differ; and a **limitations and planned extensions** section, with embedded TODOs, records the known gaps and the priorities for future work.

## 7.6 Cross-cutting conventions

A few conventions hold across all the sector chapters and are worth stating once.

The trees are written as a **region-agnostic master tree** and then instantiated for each of the seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — which share a structure but differ in their activity forecasts, fuel prices, and grid emissions intensity. Not every sector appears in every region; coal mining, for instance, is carried only where coal is actually mined.

Technology choice everywhere uses **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital. These higher rates capture the short payback horizons and adoption barriers that govern real investment decisions, and they are applied throughout the lifecycle-cost calculation, as set out in the [Theory](#) chapter and in Chapter 4.

Emissions accounting combines combustion CO<sub>2</sub> from the fuels each technology requests with **process emissions** released by chemical reactions independent of energy use — calcination in cement, for example — and, in a few sectors, fugitive methane, all expressed on a CO<sub>2</sub>-equivalent basis so that high-warming gases enter at their full weight. Fuels reach the demand sectors as blends, so a rising renewable content in a blend lowers the lifecycle carbon of every technology that draws on it, and **carbon capture** is represented as a shared service that sectors can call on where it applies.

Taken together, these shared structures are what let CIMS treat a landfill, a steel mill, and a passenger-vehicle fleet within one consistent framework: each is a flow model of competing technologies, driven by an activity forecast, discounted behaviourally, and accounted on

a common emissions basis. The chapters that follow apply that framework, one sector at a time.

## 8 Coal Mining

Coal mining is one of CIMS's energy-supply and extractive sub-models: it produces coal for metallurgical use (coking coal for steelmaking) and for thermal use (power and industrial heat). The emphasis, as in the other extractive sectors, is on the structure of the energy-service flow model — the production chain, the technologies that compete at each stage, and the engineering ratios that link them. The tree is region-agnostic. Coal mining has a distinctive emissions signature: alongside the diesel its mobile equipment burns, it releases fugitive methane liberated from the coal seam during extraction, and this coal-mine methane is the sector's largest greenhouse-gas source — yet, as discussed below, the current model carries no technology to abate it.

### 8.1 Scope

CIMS represents the extraction, size reduction, and finishing of coal — from the mine face through crushing, milling, washing, and drying to a saleable product. The driving variable is coal output in tonnes. Both metallurgical (coking) coal and thermal coal are within scope; the chain includes a Metallurgical Finishing stage (washing, cleaning/drying, tailings) that prepares coking coal to the specification the steel industry needs.

Adjacent activities are handled elsewhere to avoid double-counting: the combustion of thermal coal for power is in [Electricity](#) and for industrial heat in the relevant industrial chapters; the conversion of coking coal to coke and its use in ironmaking are in [Iron and Steel](#); and other (non-coal) mining is in the [Mining](#) sector. This chapter covers the mining of the coal itself.

### 8.2 Real-world context

Canada produced about 42.6 Mt of coal in 2024, roughly two-thirds of it metallurgical (Natural Resources Canada [2025a](#)). The geography is concentrated in the West: British Columbia accounts for around 69% of tonnage (the Elk Valley coking-coal mines), Alberta about 19% (metallurgical and thermal), and Saskatchewan about 11% (lignite/thermal). The two coal types are on opposite trajectories. Thermal coal is in steep decline as Canada phases out unabated coal-fired electricity by 2030 and moves to end thermal-coal exports, while metallurgical coal — for which there is no current large-scale substitute in conventional steelmaking —

has held up and is increasingly framed as an export commodity (Natural Resources Canada 2025a). Canadian coal production has fallen about 31% over the past decade, almost all of it thermal.

The sector’s emissions come from two places. Diesel powers the trucks, shovels, and other mobile equipment, as at any large mine. More distinctively, methane trapped in the coal seam is released during extraction — fugitive coal-mine methane (CMM), much of it carried out in the ventilation air (VAM). Coal-mine methane is the dominant greenhouse-gas source from coal mining, on the order of 60% of the total, and mature technologies exist to drain and destroy it — gas drainage, and ventilation-air-methane oxidation that can destroy more than 98% of the methane in mine ventilation air (International Energy Agency 2023; CSIRO 2026). The CIMS structure carries the fugitive-methane factor on the extraction node and the diesel emissions on the mobile equipment, but — as noted below — it does not yet offer a methane-capture option.

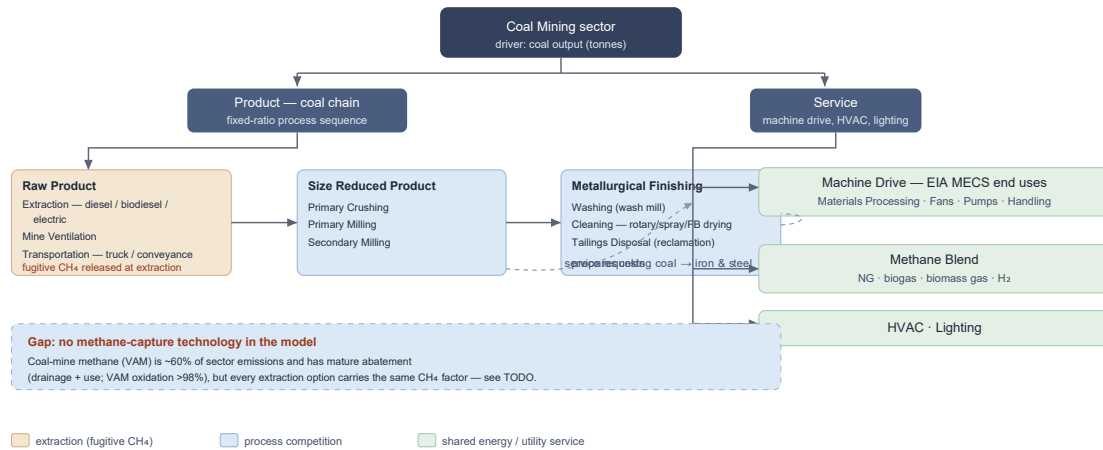
### 8.3 Model structure

The sector divides into a **Product** sub-tree — the coal production chain — and a **Service** sub-tree of auxiliary energy services. Figure 8.1 shows the structure. Each node carries a competition rule (see Chapter 4); the production chain is a fixed-ratio sequence and the technology choices occur at the tech-compete leaves.

#### 8.3.1 Products: the coal production chain

- **Raw Product** — the mine itself: Extraction (competing diesel, biodiesel, and electric equipment), Mine Ventilation, and Transportation (diesel and biodiesel trucks, or conveyance). The Extraction node carries the fugitive-methane emission factor.
- **Size Reduced Product** — Primary Crushing, Primary Milling, and Secondary Milling of the run-of-mine coal.
- **Metallurgical Finishing** — preparing coking coal: Washing (wash mill), Cleaning (rotary, spray, or fluidized-bed drying, on coal, natural gas, or electricity), and Tailings Disposal (reclamation, with diesel, biodiesel, or electric options).

The master tree also defines Simplified Extraction and Simplified Production nodes — lower-resolution alternatives to the full chain for regions or runs where the detailed structure is not warranted.



**Figure 8.1:** Energy-service flow model of the coal mining sector (region-agnostic master tree). The Product sub-tree runs from Raw Product (extraction, mine ventilation, transportation) through Size Reduced Product (crushing and milling) to Metallurgical Finishing (washing, cleaning/drying, tailings); the Service sub-tree holds the MECS-aligned machine drive and building services. Fugitive coal-mine methane is released at extraction.

### 8.3.2 Services

The services follow the standard pattern and are documented in their own chapters in the General Services part: **Machine Drive** (see [Industrial Machine Drive Systems](#)) carries the large motor loads (the model's current files still use the older per-horsepower and separate compression/conveyance/pumping/ventilation structure), together with **HVAC**, **Lighting**, and the **Methane Blend** gaseous-fuel node.

TODO: The current regional input files still implement the earlier, more detailed auxiliary structure (six per-horsepower machine-drive size classes; separate Compression, Conveyance, Pumping, Ventilation, and Direct Drive sub-trees) and predate the MECS-aligned Machine Drive grouping. Migrate them to the region-agnostic master tree and the shared General Services chapters.

## 8.4 Decarbonisation technologies

Coal mining's emissions split into fugitive methane (the larger share) and diesel combustion in mobile equipment, with smaller process-heat (drying) and electricity loads. The decarbonisation levers therefore are methane capture, equipment electrification or fuel-switching, and efficiency. Table 8.1 summarises maturity and Table 8.2 gives indicative costs. A crucial caveat: the single largest lever — methane capture — is not yet represented in the model.

**Table 8.1:** Indicative performance and maturity of key coal mining decarbonisation technologies

| Technology                                        | Indicative benefit                           | Maturity                                         |
|---------------------------------------------------|----------------------------------------------|--------------------------------------------------|
| Coal-mine methane drainage + use                  | Captures seam methane for energy             | Commercial — <i>not in model</i>                 |
| Ventilation-air-methane (VAM) oxidation           | Destroys >98% of dilute vent methane         | Commercial / demonstration — <i>not in model</i> |
| Battery-electric / biodiesel extraction & haulage | Cuts mobile-equipment diesel CO <sub>2</sub> | Commercial / emerging                            |
| Electric coal/concentrate drying                  | Removes drying-fuel CO <sub>2</sub>          | Commercial                                       |
| Ventilation-on-demand & efficient fans            | Lower ventilation energy                     | Commercial                                       |
| Electric conveyance & efficient motors/drives     | Lower auxiliary energy                       | Commercial                                       |

The estimates in Table 8.2 are order-of-magnitude figures from public literature (International Energy Agency 2023; Mungyeko Bisulandu et al. 2026; SRK Consulting 2024), expressed in 2025 Canadian dollars (US sources converted at roughly 1.37 CAD/USD and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 8.2:** Indicative techno-economic characteristics of coal mining decarbonisation technologies, 2025 CAD

| Technology                     | Cost / capex                        | Operating effect                           | Energy / emission                 |
|--------------------------------|-------------------------------------|--------------------------------------------|-----------------------------------|
| Methane drainage + utilisation | moderate capex; gas revenue offsets | often low net cost per t CO <sub>2</sub> e | captures high-GWP CH <sub>4</sub> |
| VAM oxidation (e.g. CataVAM)   | moderate capex                      | low cost per t CO <sub>2</sub> e abated    | destroys >98% of vent methane     |

| Technology                      | Cost / capex                               | Operating effect            | Energy / emission                   |
|---------------------------------|--------------------------------------------|-----------------------------|-------------------------------------|
| Battery-electric haul truck     | ~CAD 4–7 M per truck (+100–150% vs diesel) | ~65% lower cost per t moved | electricity                         |
| Biodiesel in existing equipment | none (drop-in fuel)                        | fuel-cost dependent         | lifecycle CO <sub>2</sub> cut       |
| Electric drying                 | ~150–400 CAD/kWth                          | electricity-priced heat     | removes drying-fuel CO <sub>2</sub> |

#### 8.4.1 Capturing coal-mine methane

The most important lever is the one the model is missing. Because methane has a high global-warming potential and is released in large quantities during mining, draining and using it (where concentrations allow) and oxidising the dilute methane in ventilation air (VAM) together address the majority of the sector’s emissions, with mature technology that can destroy more than 98% of vent methane (CSIRO 2026; International Energy Agency 2023). The model currently applies the same fugitive-methane factor to every extraction option, with no capture alternative.

TODO: Add coal-mine methane drainage and use and ventilation-air-methane (VAM) oxidation technologies at the Extraction / Mine Ventilation nodes. Fugitive methane is the largest single source of coal-mining emissions and has commercially available abatement, but the current tree offers no option to reduce it — every extraction technology carries the same methane factor.

#### 8.4.2 Electrifying and fuel-switching mobile equipment

As in metal mining, the diesel load is addressed by battery-electric and biodiesel extraction and haulage (already options at the Extraction and Transportation nodes), electric coal drying, ventilation-on-demand, and efficient motors, fans, and conveyors in the machine-drive service. These are the same levers detailed in the [Mining](#) chapter.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

### 8.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output

basis, service requests for the energy it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Coal mining technologies are discounted at a high rate — 35% at most process nodes in the current data, with some auxiliary nodes at 50%. These values are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the short pay-back horizons and adoption barriers typical of resource industries; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 10.

The emissions accounting combines combustion CO<sub>2</sub> from diesel and drying fuel with the distinctive fugitive CH<sub>4</sub> factor on extraction (entering on a CO<sub>2</sub>-equivalent basis at methane's high warming potential). Because there is no methane-capture option and no CCS service, the model's only channels for emissions reduction are equipment electrification, fuel-switching to biodiesel, and efficiency — which reach the combustion emissions but leave the dominant fugitive methane untouched. This is the most consequential representational gap in the sub-model, and it means the model understates the abatement available to the sector.

## 8.6 Regional coverage and differences

Four regions carry coal mining — Alberta, the Atlantic region, British Columbia, and Saskatchewan — reflecting where Canadian coal is actually mined:

- British Columbia dominates, with metallurgical (coking) coal from the Elk Valley; this production is export-oriented and durable.
- Alberta has both metallurgical and thermal coal, with thermal declining as coal power is phased out.
- Saskatchewan mines lignite/thermal coal, the most exposed to the power phase-out.
- The Atlantic region has a small coal presence.

Ontario, Québec, and Manitoba are absent, consistent with the absence of coal mining there.

TODO: Represent the diverging trajectories of metallurgical versus thermal coal explicitly — durable, export-driven coking-coal demand versus a sharp thermal-coal decline under the coal-power phase-out and export restrictions — since lumping them understates the structural change underway in the sector.

## 8.7 Limitations and planned extensions

The coal mining sub-model captures the production chain and its diesel and electricity loads in good detail, but it has one large gap and several smaller ones. The large gap is the absence

of any coal-mine-methane abatement technology, which omits the sector's single most important emissions lever. The smaller debts are the older auxiliary structure (to be migrated to MECS machine drive), the lack of an explicit metallurgical-versus-thermal split, and reliance on inherited cost data. The TODOs above record these priorities.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled coal output and fugitive-methane emissions against Natural Resources Canada coal statistics and Environment and Climate Change Canada's National Inventory Report.

## 9 Natural Gas

The natural gas supply sector is one of CIMS's energy-supply sub-models, covering the **upstream and midstream** chain — extracting raw gas, processing it, and moving it through transmission and distribution to the point of use. The emphasis is on the structure of the energy-service flow model: the production chain, the technologies that compete at each stage, and — distinctively for this sector — the emission-source nodes where methane and CO<sub>2</sub> escape. The tree is region-agnostic. Natural gas is defined, for climate purposes, by methane: the gas is itself almost pure methane, and the leaks, vents, and flares along the supply chain make fugitive and vented methane the sector's largest controllable emission — and, fortunately, one of the cheapest to abate anywhere in the economy.

### 9.1 Scope

CIMS represents the supply of natural gas: extraction (conventional, shale, tight, and coalbed methane), field and plant processing, transmission, and distribution, together with the venting, fugitive, flaring, and formation-CO<sub>2</sub> emission sources that occur along the way and the energy the sector consumes for compression, drilling, and heat. The driving variable is gas throughput. The model also carries the oil-and-gas fuel blends (diesel, gasoline, methane) the sector burns for its own operations.

Adjacent activities are handled elsewhere: the combustion of the delivered gas by end users is accounted in the demand sectors (residential, commercial, industry, electricity); crude oil production is in [Petroleum Crude](#); and downstream gas use as a chemical feedstock is in [Chemical Products](#) and as a hydrogen feedstock in [Hydrogen](#). This chapter covers getting the gas out of the ground and to market.

### 9.2 Real-world context

Canada is one of the world's largest natural gas producers, with output concentrated in the Western Canadian Sedimentary Basin — above all the Montney play straddling Alberta and British Columbia. The sector's defining climate issue is methane. Because methane has a high global-warming potential and the product is itself methane, leaks and intentional releases along the chain — fugitive equipment leaks, vented gas from pneumatic devices and

tanks, flaring, well blowdowns, and liquid unloading — together make up the sector’s largest greenhouse-gas source, and one of the largest controllable methane sources in the economy.

The decisive fact is that this methane is cheap to abate. Canada has set a target to cut oil-and-gas methane by 75% below 2012 levels by 2030, and the measures to do it — leak detection and repair, vapour-recovery units, zero-bleed (electric) pneumatic controllers, vent and blow-down capture, plunger lifts, and flare reduction — are among the lowest-cost greenhouse-gas abatement options available, on the order of \$11–48 per tonne of CO<sub>2</sub>-equivalent, with much of the captured gas recoverable for sale (Environmental Defense Fund 2023; Environment and Climate Change Canada 2024c). Alberta accounts for roughly 68% of the sector’s methane, while British Columbia has already cut its oil-and-gas methane about 51% from 2014 levels while increasing production (Environmental Defense Fund 2023). The CIMS structure is built precisely around this: it represents each emission source as its own node with a competition among abatement levels, so the model can buy down methane the way the real policy does.

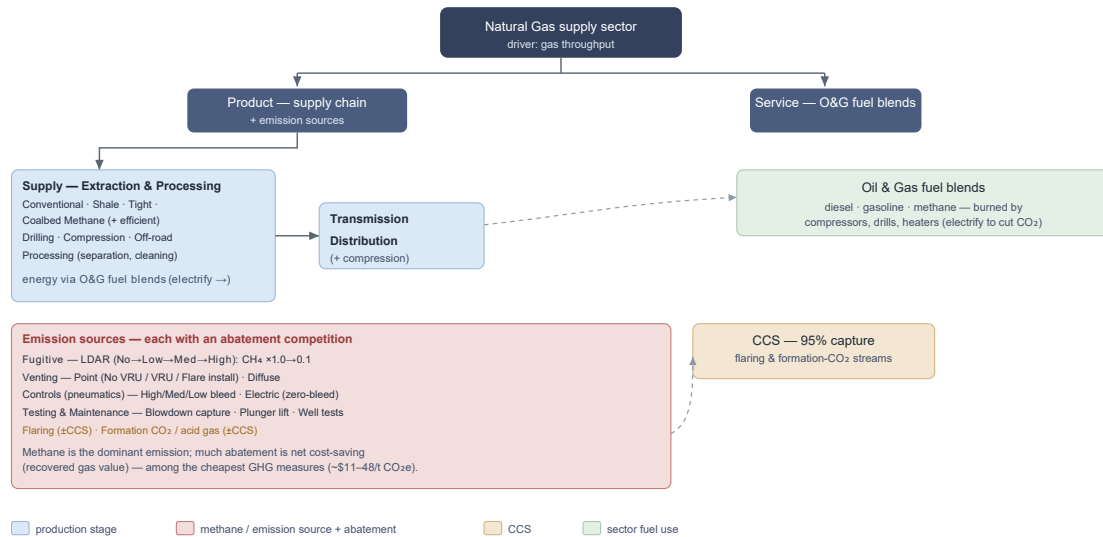
### 9.3 Model structure

The sector divides into a **Product** sub-tree — the gas supply chain, including its emission sources — and a **Service** sub-tree of the oil-and-gas fuel blends. Figure 9.1 shows the structure. Each node carries a competition rule (see Chapter 4); the emission-source nodes host competitions among abatement options.

#### 9.3.1 Products: the supply chain and its emission sources

The Supply node is the heart of the model. It combines the productive steps with the emission sources:

- **Extraction** — conventional, shale, tight, and coalbed methane, with drilling, compression, off-road transport, and efficient variants; the energy of getting gas out of the ground.
- **Processing** — separating and cleaning the raw gas (with compression).
- **Emission sources, each with an abatement competition:**
  - **Fugitive** — equipment leaks, competing leak-detection-and-repair (LDAR) intensities (No / Low / Med / High LDAR, cutting the methane factor from 1.0 to 0.75, 0.25, and 0.1).
  - **Venting** — Point Venting (no vapour-recovery unit, VRU, or flare install) and Diffuse Venting.
  - **Controls** — pneumatic devices, competing high-, medium-, and low-bleed and electric (zero-bleed, instrument-air or electric controllers).
  - **Flaring** — with or without CCS.



**Figure 9.1:** Energy-service flow model of the natural gas supply sector (region-agnostic master tree). The Product sub-tree runs Supply (extraction, processing, and the venting/fugitive/flaring/formation-CO<sub>2</sub>/testing emission sources, each with an abatement competition, plus a CCS service) → Transmission → Distribution; the Service sub-tree holds the oil-and-gas fuel blends the sector burns. Methane is the dominant emission, and most nodes carry an explicit abatement choice.

- **Formation CO<sub>2</sub>** — the CO<sub>2</sub> separated from the raw gas (acid gas), with or without CCS.
- **Testing and Maintenance** — blowdowns (capture high/low), liquid unloading (plunger install), and well tests.
- **CCS** — a shared capture service (95% capture) for the flaring and formation-CO<sub>2</sub> streams.

Downstream, **Transmission** (with compression) and **Distribution** move the gas to market, each carrying its own (smaller) emission factors.

### 9.3.2 Services: the sector's own fuel use

The **Service** sub-tree holds the oil-and-gas fuel blends — diesel, gasoline, and methane — that the sector's own engines, compressors, and heaters consume, allowing the model to track (and

decarbonise) the energy the gas industry uses on itself, including electrification of compression and drives.

TODO: The model carries a separate Natural Gas Market node (Supply / Transmission / Distribution) alongside the production tree, and a Rest-of-World region for trade. Document and, where appropriate, consolidate the market/trade representation, and add an explicit LNG export pathway (the LNG Compression node exists but the export chain is not fully built out).

## 9.4 Decarbonisation technologies

Natural gas decarbonisation is, overwhelmingly, methane abatement — stopping the leaks, vents, and flares — supplemented by electrifying the sector’s compression and heat and capturing concentrated CO<sub>2</sub> streams. Table 9.1 summarises maturity and Table 9.2 gives indicative costs. The striking feature is how cheap, even profitable, much of this is.

**Table 9.1:** Indicative performance and maturity of key natural gas decarbonisation technologies

| Technology                                  | Indicative benefit                                                    | Maturity                   |
|---------------------------------------------|-----------------------------------------------------------------------|----------------------------|
| Leak detection and repair (LDAR)            | Cuts fugitive methane up to ~90%                                      | Commercial                 |
| Zero-bleed / electric pneumatic controllers | Eliminates pneumatic vent methane                                     | Commercial                 |
| Vapour-recovery units (VRU)                 | Captures vented methane for sale                                      | Commercial                 |
| Vent / blowdown capture                     | Recovers methane from maintenance events                              | Commercial                 |
| Plunger lift (liquid unloading)             | Avoids unloading vent methane                                         | Commercial                 |
| Flare reduction / flare install             | Converts vented methane to (lower-GWP) CO <sub>2</sub> or captures it | Commercial                 |
| Electric compression & drives               | Removes combustion CO <sub>2</sub> from operations                    | Commercial                 |
| Flaring / formation-CO <sub>2</sub> + CCS   | Captures concentrated CO <sub>2</sub> streams (95%)                   | Demonstration / commercial |

The estimates in Table 9.2 are order-of-magnitude figures from public Canadian sources (Environmental Defense Fund 2023; Environment and Climate Change Canada 2024c), expressed in 2025 Canadian dollars. Costs are given as cost of abatement per tonne of CO<sub>2</sub>-equivalent, since most measures are equipment changes whose benefit is avoided methane (often partly offset by the value of the recovered gas). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 9.2:** Indicative cost of natural gas methane and CO<sub>2</sub> abatement, 2025 CAD

| Measure                                    | Cost of abatement<br>(2025 CAD/t CO <sub>2</sub> e) | Note                         | Effect                        |
|--------------------------------------------|-----------------------------------------------------|------------------------------|-------------------------------|
| Leak detection and repair (LDAR)           | often <i>net-negative</i> to ~30                    | recovered gas offsets cost   | up to ~90% fugitive cut       |
| Electric (zero-bleed) pneumatics           | often <i>net-negative</i>                           | recovered gas; simple swap   | eliminates device venting     |
| Vapour-recovery unit                       | ~10–50                                              | captures saleable gas        | cuts point-source venting     |
| Blowdown / vent capture                    | ~10–60                                              | event-based                  | cuts maintenance methane      |
| Plunger lift                               | low; often <i>net-negative</i>                      | recovered gas                | cuts liquid-unloading venting |
| Sector-wide 75% methane cut                | ~11–48 average                                      | among cheapest GHG abatement | ~\$2B recoverable gas value   |
| CCS on flaring / formation CO <sub>2</sub> | ~50–150                                             | concentrated streams         | 95% capture                   |

#### 9.4.1 Methane abatement: the dominant, low-cost lever

The overwhelming priority is methane. Leak detection and repair finds and fixes fugitive leaks; zero-bleed electric pneumatics replace the gas-driven controllers that vent by design; vapour-recovery units capture vented gas; blowdown and vent capture and plunger lifts address maintenance and liquid-unloading releases; and flare reduction converts otherwise-vented methane to CO<sub>2</sub> or captures it. The model represents each of these as an abatement competition at the relevant emission node. Because the captured methane is saleable gas, many of these measures are net cost-saving — which is why the sector-wide 75% reduction is achievable at an average cost of only around \$11–48/tCO<sub>2</sub>e (Environmental Defense Fund 2023).

#### 9.4.2 Electrifying operations and capturing CO<sub>2</sub>

Beyond methane, the sector burns gas for compression, drilling, and heat; switching these to electric drives and compression (the Controls “electric” and the fuel-blend electrification options) removes that combustion CO<sub>2</sub>. Where the raw gas contains formation CO<sub>2</sub> (acid gas) or where gas is flared, the concentrated CO<sub>2</sub> streams can be routed to CCS at 95% capture — among the lower-cost capture opportunities because the CO<sub>2</sub> is already concentrated.

TODO: Replace the indicative literature ranges above with the abatement costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files, and align the abatement levels with the trajectory of Canada's updated federal oil-and-gas methane regulations ( $\approx 75\%$  reduction by 2030).

## 9.5 Economics

Technology choice at every node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, operating cost, an output basis, service requests for the energy it consumes, any recovered-gas credit, and availability and lifetime years; these are combined into a lifecycle cost and converted to market shares.

Natural gas technologies are discounted at a high rate — about 35% at most nodes, with some at 20% in the current data. These values are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the payback horizons and adoption barriers typical of the industry; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** govern how sharply the abatement competitions respond to cost.

The emissions accounting is unusually detailed on the source side: the model carries explicit  $\text{CH}_4$  factors on the fugitive, venting, control, blowdown, liquid-unloading, and distribution nodes (entering on a  $\text{CO}_2$ -equivalent basis at methane's high warming potential),  $\text{CO}_2$  factors on flaring and formation gas, and a **CCS** removal factor on the equipped routes. Combustion  $\text{CO}_2$  from the sector's own fuel use enters through the oil-and-gas fuel blends. A distinctive economic feature is that abatement often carries a recovered-gas value that partly or fully offsets its cost, so a carbon price (or a methane regulation) can drive large reductions at very low or negative net cost — which the model reproduces through the abatement competitions.

## 9.6 Regional coverage and differences

The sector is represented for the producing regions and for trade. Alberta and British Columbia dominate production (the Montney and related plays) and therefore the sector's emissions and abatement opportunity; Saskatchewan adds production; Ontario, Québec, Manitoba, and the Atlantic carry mainly transmission and distribution. The model also includes a national (CAN) aggregate and a Rest-of-World region to handle trade and, prospectively, LNG export. Regional differences are large: Alberta holds most of the methane-abatement opportunity, while British Columbia has already moved further on methane controls.

TODO: Calibrate the regional production mix (conventional / shale / tight / coalbed methane) and the baseline methane-source intensities against Canada's National Inventory Report and

provincial methane data, since both the emissions and the abatement potential depend heavily on them.

## 9.7 Limitations and planned extensions

The natural gas sub-model is unusually well suited to its central question — methane abatement — because it resolves the individual emission sources and gives each an abatement competition. Its main limitations are the incomplete build-out of the LNG export chain, the somewhat separate market/trade representation, and reliance on inherited methane-source intensities that should track the rapidly evolving federal and provincial methane regulations. The TODOs above record these priorities.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled methane and CO<sub>2</sub> emissions against Environment and Climate Change Canada’s National Inventory Report and the federal methane-regulation projections.

## 10 Petroleum Crude

Crude oil and bitumen production is one of CIMS's energy-supply sub-models, covering the upstream oil industry — conventional crude oil and, dominating both output and emissions in Canada, the oil sands (surface mining, in-situ recovery, and upgrading to synthetic crude). The emphasis is on the structure of the energy-service flow model: the production routes, the technologies that compete at each, and the energy services — above all steam — that drive them. The tree is region-agnostic. The defining feature of the sector is that the energy used to *produce* the oil — particularly the natural gas burned to raise steam for in-situ bitumen recovery — is itself a very large emissions source, so the sector's decarbonisation is mostly about how that production energy is supplied and whether its CO<sub>2</sub> is captured.

### 10.1 Scope

CIMS represents the production of crude oil and bitumen: conventional light, medium, and heavy crude (onshore and offshore), and the oil sands in their three forms — surface mining, in-situ recovery, and upgrading of bitumen to synthetic crude oil. The driving variable is oil/bitumen output. The model carries the supporting energy services — steam, process hydrogen (for upgrading), industrial engines, and a **CCS** service.

Adjacent activities are handled elsewhere: refining crude into fuels and products is in [Petroleum Refining](#); natural gas production is in [Natural Gas](#); the combustion of the refined fuels is in the demand sectors; and the hydrogen the sector uses can be drawn from the [Hydrogen](#) sub-model. This chapter covers getting the oil and bitumen out of the ground and to a refinery-ready state.

### 10.2 Real-world context

Canadian crude production is dominated by the Alberta oil sands, the largest source of both output and emissions in the sector, supplemented by conventional crude (Saskatchewan, Alberta) and offshore production in the Atlantic (Newfoundland). The oil sands come in two forms: surface mining and upgrading, which digs out near-surface bitumen and upgrades it to synthetic crude; and in-situ recovery, which uses steam to mobilise deeper bitumen and now rivals mining as the main production route.

The sector’s emissions profile is distinctive. For in-situ production, the single largest source is steam generation — natural gas burned to make the steam that lowers bitumen viscosity in steam-assisted gravity drainage (SAGD) and cyclic steam — so the carbon intensity of a barrel depends heavily on how much steam it needs and how that steam is produced (“[Canadian Oil Sands Industry GHG Emissions Intensity and Mitigation Potential of Key Emerging Technologies Towards its 2050 Net-Zero Commitment](#)” 2025). Upgrading adds the emissions of coking, hydrotreating, and the hydrogen it consumes. The decarbonisation levers follow directly: solvent-assisted in-situ processes that cut steam use (with potential to lower intensity by up to ~50%), low-carbon steam (electric, hydrogen, nuclear, cogeneration), and carbon capture — the focus of the Pathways Alliance, whose proposed oil sands CCS network would be among the world’s largest (Pathways Alliance 2024; “[Canadian Oil Sands Industry GHG Emissions Intensity and Mitigation Potential of Key Emerging Technologies Towards its 2050 Net-Zero Commitment](#)” 2025). The CIMS structure captures all of these as competing technologies at the production and steam nodes.

### 10.3 Model structure

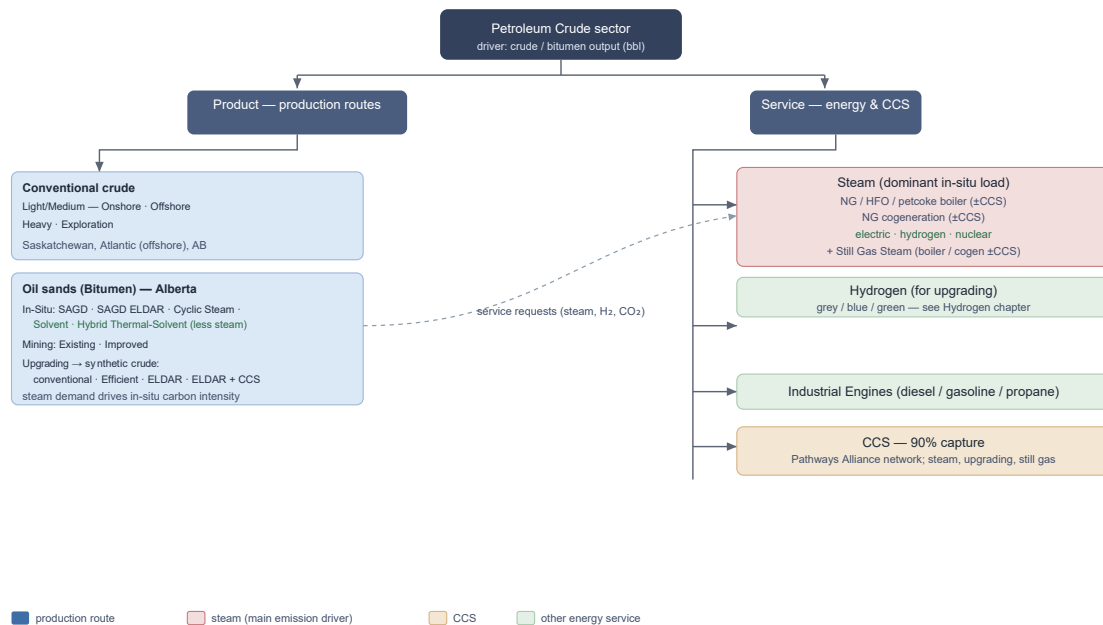
The sector divides into a **Product** sub-tree — the production routes — and a **Service** sub-tree of the energy services that supply them. Figure 10.1 shows the structure. Each node carries a competition rule (see Chapter 4).

#### 10.3.1 Products: conventional crude and the oil sands

- Light/Medium crude — Onshore and Offshore conventional production.
- Heavy crude — conventional heavy oil.
- Exploration — the upstream drilling and appraisal activity.
- Bitumen — the oil sands, in three linked nodes:
  - In-Situ — competing SAGD, SAGD with enhanced leak detection (ELDAR), Cyclic Steam, Solvent, and Hybrid Thermal-Solvent recovery. These differ in steam intensity (and therefore CO<sub>2</sub>) and in methane (the solvent and hybrid routes cut both).
  - Mining — surface mining (Existing and Improved).
  - Upgrading — converting bitumen to synthetic crude, competing conventional Upgrading, Efficient, ELDAR, and ELDAR + CCS routes.

#### 10.3.2 Services: steam, hydrogen, engines, and capture

- **Steam** — the most important service, because in-situ recovery is steam-intensive. It competes natural-gas, heavy-fuel-oil, and petcoke boilers (with CCS variants), natural-gas



**Figure 10.1:** Energy-service flow model of the petroleum crude sector (region-agnostic master tree). The Product sub-tree holds conventional crude (light/medium onshore and offshore, heavy) and the oil sands (in-situ recovery, surface mining, and upgrading to synthetic crude); the Service sub-tree supplies the steam, hydrogen, industrial engines and CCS the production routes draw on. Steam generation is the dominant emission source for in-situ bitumen.

cogeneration (with and without CCS), and electric, hydrogen, and nuclear steam. The choice here largely determines an in-situ barrel's carbon intensity.

- **Still Gas** — steam raised from refinery/upgrader still gas, with boiler and cogeneration options (with and without CCS).
- **Hydrogen** — for upgrading (can be sourced from the [Hydrogen](#) sub-model's grey, blue, and green routes).
- **Industrial Engines** — diesel, gasoline, and propane mobile equipment (mine trucks, and the like).
- **CCS** — the shared capture service (90% capture), central to deep decarbonisation of the sector.

TODO: Link the sector's hydrogen demand explicitly to the [Hydrogen](#) sub-model's production competition, so that upgrading hydrogen carries the cost and carbon intensity of the chosen (grey, blue, or green) route rather than an assumed value.

## 10.4 Decarbonisation technologies

Oil-sands and crude decarbonisation works through reducing steam demand (solvent processes), decarbonising the steam and process energy (electric, hydrogen, nuclear, cogeneration), capturing CO<sub>2</sub> (the Pathways CCS network), and controlling methane. Table 10.1 summarises maturity and Table 10.2 gives indicative costs.

**Table 10.1:** Indicative performance and maturity of key petroleum-crude decarbonisation technologies

| Technology                              | Indicative benefit                                 | Maturity                             |
|-----------------------------------------|----------------------------------------------------|--------------------------------------|
| Solvent / hybrid thermal-solvent SAGD   | Up to ~50% lower in-situ CO <sub>2</sub> intensity | Pilot / demonstration                |
| Electric / hydrogen / nuclear steam     | Decarbonises SAGD steam                            | Emerging                             |
| Cogeneration (gas, with or without CCS) | Combined heat and power; surplus export            | Commercial                           |
| CCS on steam / upgrading (Pathways)     | Captures ~90% of treated CO <sub>2</sub>           | Demonstration (large-scale proposed) |
| Efficient / ELDAR upgrading             | Lower upgrading energy and methane                 | Commercial                           |
| Enhanced leak detection (ELDAR)         | Cuts fugitive and vented methane                   | Commercial                           |
| Mining electrification (trucks)         | Lower mine-fleet diesel CO <sub>2</sub>            | Emerging                             |

The estimates in Table 10.2 are order-of-magnitude figures from public Canadian sources (Pathways Alliance 2024; “Canadian Oil Sands Industry GHG Emissions Intensity and Mitigation Potential of Key Emerging Technologies Towards its 2050 Net-Zero Commitment” 2025), expressed in 2025 Canadian dollars. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 10.2:** Indicative techno-economic characteristics of petroleum-crude decarbonisation technologies, 2025 CAD

| Technology              | Cost / capex                 | Effect                                   | Energy / input          |
|-------------------------|------------------------------|------------------------------------------|-------------------------|
| Solvent / hybrid SAGD   | process retrofit / new wells | up to ~50% lower steam CO <sub>2</sub>   | solvent + reduced steam |
| Electric steam (boiler) | ~150–400 CAD/kWth            | removes steam combustion CO <sub>2</sub> | electricity             |

| Technology                             | Cost / capex             | Effect                                     | Energy / input                  |
|----------------------------------------|--------------------------|--------------------------------------------|---------------------------------|
| Hydrogen steam / fuel                  | retrofit + hydrogen cost | low-carbon if blue or green H <sub>2</sub> | hydrogen                        |
| Gas cogeneration (with or without CCS) | ~1,800–3,000 CAD/kWe     | CHP; exports surplus power                 | natural gas                     |
| CCS on steam / upgrading               | +30–50% plant equipment  | ~CAD 70–150 / t CO <sub>2</sub>            | steam / electricity for capture |
| Enhanced leak detection (ELDAR)        | low; often net-negative  | cuts fugitive methane                      | — (recovered gas)               |

#### 10.4.1 Reducing and decarbonising steam

For in-situ bitumen, the game is steam. Solvent and hybrid thermal-solvent processes inject light hydrocarbons (often propane) in place of, or alongside, steam, cutting the steam needed per barrel and lowering carbon intensity by up to about 50% ([“Canadian Oil Sands Industry GHG Emissions Intensity and Mitigation Potential of Key Emerging Technologies Towards its 2050 Net-Zero Commitment” 2025](#)). Where steam is still needed, the **Steam** service can switch from gas boilers to cogeneration, electric, hydrogen, or even nuclear steam — the choice that most determines an in-situ barrel’s footprint.

#### 10.4.2 Carbon capture and methane control

Because the oil sands produce large, relatively concentrated CO<sub>2</sub> streams (from steam generation, upgrading, and hydrogen production), **CCS** is the central deep-decarbonisation lever, and the proposed Pathways Alliance network would capture CO<sub>2</sub> from a cluster of oil-sands facilities at large scale (Pathways Alliance [2024](#)). The model represents this through the CCS-equipped steam, upgrading, and still-gas routes and the shared **CCS** service. Enhanced leak detection (ELDAR) cuts the sector’s fugitive and vented methane, often at net-negative cost.

TODO: Replace the indicative literature ranges above with the capital costs, steam-to-oil ratios, efficiencies, availability years, and emission factors actually used at these nodes in the model input files, and refresh the solvent-SAGD and CCS options against the Pathways Alliance project and current pilots.

### 10.5 Economics

Technology choice at every node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, operating cost, an output (per-barrel) basis, service requests

for the steam, hydrogen, and engine energy it consumes, any recovered-gas credit, and availability and lifetime years; these are combined into a lifecycle cost and converted to market shares.

Petroleum-crude technologies are discounted at a high rate — about 35% at most nodes, with some at 20–40% in the current data. These values are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the payback horizons and risk typical of the industry; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4.

Emissions are tracked as **CO<sub>2</sub>** (chiefly from steam, upgrading, and engine combustion) and **CH<sub>4</sub>** (fugitive and vented, with the ELDAR and solvent routes carrying lower factors), and the **CCS** service applies a 90% removal factor to the equipped routes. Because in-situ carbon intensity is dominated by steam, the carbon price acts most powerfully through the **Steam** competition — making solvent processes, electric/hydrogen/nuclear steam, cogeneration, and CCS progressively more attractive as it rises — while methane controls add cheap, often profitable, abatement.

## 10.6 Regional coverage and differences

Six regions carry crude production, but it is highly concentrated:

- **Alberta** dominates, with essentially all of the oil sands (mining, in-situ, and upgrading) plus conventional crude — and therefore most of the sector’s energy use and abatement opportunity.
- **Saskatchewan** carries conventional heavy and light/medium crude.
- The **Atlantic** region carries offshore light/medium production (Newfoundland).
- **British Columbia, Manitoba, and Ontario** carry smaller conventional volumes.

Because the oil sands are so steam- and energy-intensive, Alberta holds the overwhelming majority of the sector’s decarbonisation challenge and opportunity.

TODO: Calibrate the in-situ versus mining production split and the steam-to-oil ratios against current Alberta Energy Regulator data, since the sector’s emissions and the value of steam decarbonisation depend heavily on both.

## 10.7 Limitations and planned extensions

The petroleum-crude sub-model captures the essential structure — conventional and oil-sands routes, the dominant steam service with its full range of low-carbon options, and CCS — which makes it well suited to analysing oil-sands decarbonisation. Its main limitations are the loose

coupling of upgrading hydrogen to the hydrogen sub-model, reliance on inherited steam-to-oil and cost assumptions, and the need to keep the solvent-SAGD and CCS options aligned with fast-moving industry projects. The TODOs above record these priorities.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled crude and bitumen production energy use and emissions intensity against the Alberta Energy Regulator and Environment and Climate Change Canada's National Inventory Report.

## 11 Petroleum Refining

Petroleum refining is one of CIMS's energy-supply and processing sub-models: it converts crude oil and bitumen into refined products — gasoline, diesel, jet fuel, fuel oils, petrochemical feedstocks, asphalt, lubricants, and coke. The emphasis is on the structure of the energy-service flow model: the process units that turn crude into products, the technologies that compete at each, and the energy services — above all process heat, steam, and hydrogen — that drive them. The tree is region-agnostic. The defining feature of refining is that almost all of its emissions come from burning fuel for process heat and steam (much of it the refinery's own still gas) and from on-site hydrogen production, so its decarbonisation is about electrifying and decarbonising that energy and capturing the concentrated CO<sub>2</sub> streams — against a backdrop of long-run demand decline as transport electrifies.

### 11.1 Scope

CIMS represents the refining of crude oil and synthetic crude into finished products, through the standard process units: distillation (atmospheric and vacuum), cracking (coking and hydrocracking), hydrotreating (desulphurisation and sulphur recovery), reforming, alkylation, and isomerisation, plus by-products such as asphalt and lubes. The driving variable is refinery throughput/product output. Lubricating-oil, grease, and asphalt manufacture are included where carried out within refineries. The model carries the supporting energy services — direct process heat, steam, still gas, hydrogen, machine drive, and a CCS service.

Adjacent activities are handled elsewhere: crude oil and bitumen production (including oil-sands upgrading) is in [Petroleum Crude](#); natural gas processing is in [Natural Gas](#); petrochemicals made from refinery streams are in [Chemical Products](#); and the combustion of the finished fuels is in the demand sectors. This chapter covers the refinery itself.

### 11.2 Real-world context

Canada has a modest fleet of refineries spread across the country, supplying gasoline, diesel, and other products to domestic and export markets. A refinery's energy use and emissions depend on its complexity — the type of crude it runs and how much cracking, hydrotreating, and reforming it does to make high-value products. The bulk of refinery emissions come from

burning fossil fuels for process heating and steam generation, with a further significant share from hydrogen manufacturing (steam methane reforming for hydrotreating and hydrocracking) (World Resources Institute 2023). Much of the process fuel is the refinery's own still gas (off-gas from the process units).

The decarbonisation levers follow from that emissions structure, and Canada's [Roadmap for the Decarbonization of the Oil and Gas Sector](#) names them directly: electrifying low- and medium-grade process heat, fuel-switching to low- and zero-carbon hydrogen for high-temperature heat, energy efficiency, and CCUS — and refining is regarded as one of the lower-cost CCS opportunities, with concentrated CO<sub>2</sub> available from the hydrogen plant and other units (Natural Resources Canada 2024f; World Resources Institute 2023). Over the longer term, refinery throughput is expected to decline as road transport electrifies, shifting the product slate toward aviation, petrochemicals, and exports. The CIMS structure captures the decarbonisation levers as competing technologies at the heat, steam, still-gas, and hydrogen services, with a shared CCS service.

### 11.3 Model structure

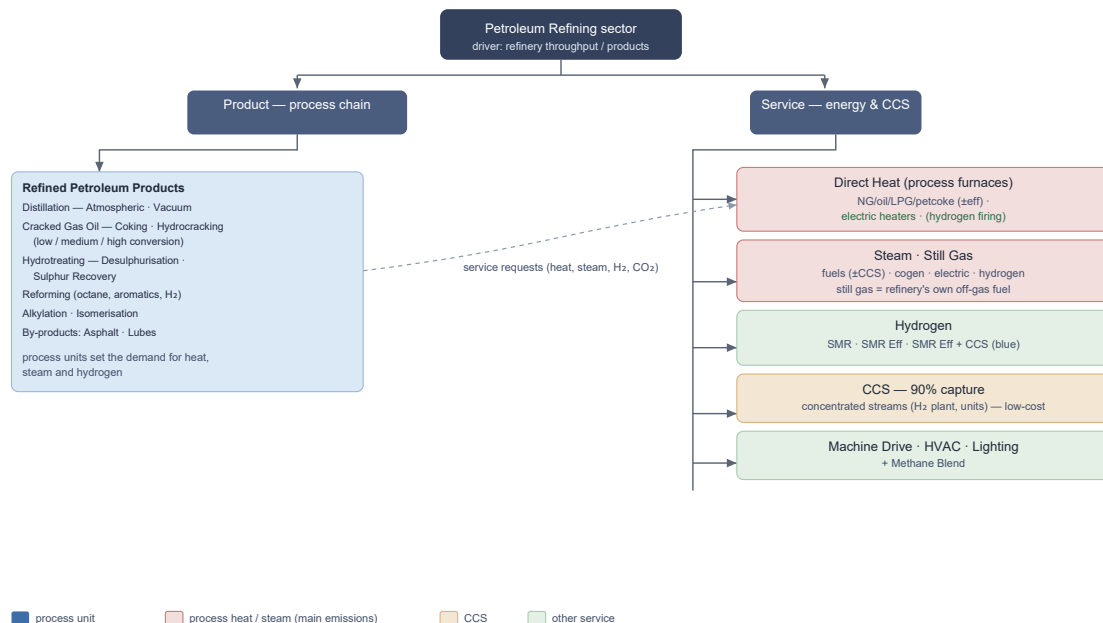
The sector divides into a **Product** sub-tree — the refinery process chain — and a **Service** sub-tree of the energy services. Figure 11.1 shows the structure. Each node carries a competition rule (see Chapter 4); the process chain is largely fixed-ratio, and the technology choices occur in the energy services.

#### 11.3.1 Products: the refinery process chain

The **Refined Petroleum Products** node holds the process units:

- **Petroleum Fractions** — the main chain: distillation (atmospheric and vacuum) separates crude into fractions; cracked gas oil upgrades heavy fractions via coking and hydrocracking (at low, medium, and high conversion); hydrotreating (desulphurisation and sulphur recovery) cleans the products; and asphalt and lubes are by-product streams.
- **Alkylates** — alkylation and isoparaffins (isomerisation), which make high-octane gasoline blendstocks.
- **Reforming** — catalytic reforming for octane and aromatics (and a hydrogen by-product).

These units are the demand for the energy services; their relative throughput sets how much heat, steam, and hydrogen the refinery needs.



**Figure 11.1:** Energy-service flow model of the petroleum refining sector (region-agnostic master tree). The Product sub-tree is the refinery process chain — distillation, cracking (coking, hydrocracking), hydrotreating (desulphurisation, sulphur recovery), reforming, alkylation/isomerisation, and by-products (asphalt, lubes); the Service sub-tree supplies the direct process heat, steam, still gas, hydrogen, machine drive, and CCS the process units draw on.

### 11.3.2 Services: heat, steam, still gas, hydrogen, and capture

- **Direct Heat** — the process furnaces, competing natural-gas, heavy- and light-fuel-oil, LPG, and petcoke firing (with efficiency variants) and electric heaters (high and low efficiency). This is the central electrification lever for low- and medium-grade heat.
- **Steam** — boiler- and cogeneration-raised steam, competing the full range of fuels (NG, oil, LPG, petcoke, with efficiency and CCS variants), electric and hydrogen steam, and gas cogeneration (CCGT/SCGT/back-pressure turbines, with and without CCS).
- **Still Gas** — the refinery's own off-gas, used in heaters (with efficiency and CCS variants) and boilers; the dominant self-generated fuel.
- **Hydrogen** — on-site steam methane reforming (standard, efficient, and efficient + CCS for blue hydrogen), for hydrotreating and hydrocracking.
- **CCS** — the shared capture service (90% capture), well suited to the concentrated CO<sub>2</sub> from the hydrogen plant and other units.
- **Machine Drive, HVAC, Lighting, Methane Blend** — the shared utilities.

TODO: The current regional input files still implement the earlier, more detailed auxiliary structure (six per-horsepower machine-drive size classes; separate Compression, Pumping, and Ventilation sub-trees) and predate the MECS-aligned Machine Drive grouping. Migrate them to the region-agnostic master tree and the shared General Services chapters. Also link the **Hydrogen** service to the [Hydrogen](#) sub-model so refinery hydrogen can draw on green and merchant blue supply, not just on-site SMR.

## 11.4 Decarbonisation technologies

Refinery decarbonisation works through electrifying and decarbonising process heat and steam, low-carbon hydrogen, efficiency, and CCS on the concentrated streams. Table 11.1 summarises maturity and Table 11.2 gives indicative costs.

**Table 11.1:** Indicative performance and maturity of key petroleum refining decarbonisation technologies

| Technology                                  | Indicative benefit                                         | Maturity                             |
|---------------------------------------------|------------------------------------------------------------|--------------------------------------|
| Electric process heaters (low/medium heat)  | Removes combustion CO <sub>2</sub> from electrifiable heat | Commercial (low/med grade)           |
| Hydrogen firing (high-temperature heat)     | Decarbonises hard-to-electrify furnaces                    | Emerging                             |
| Blue hydrogen (SMR + CCS) for hydrotreating | Captures hydrogen-plant CO <sub>2</sub>                    | Commercial / demonstration           |
| Green hydrogen (electrolytic)               | Zero-carbon hydrogen                                       | Emerging                             |
| Cogeneration (gas, with or without CCS)     | Combined heat and power; surplus export                    | Commercial                           |
| Energy efficiency (heat integration, units) | Lower fuel use per barrel                                  | Commercial                           |
| CCS on hydrogen plant / heaters / still gas | ~90% capture of concentrated CO <sub>2</sub>               | Demonstration (low-cost in refining) |
| Electrified drives + efficient motors       | Lower auxiliary electricity                                | Commercial                           |

The estimates in Table 11.2 are order-of-magnitude figures from public sources (World Resources Institute 2023; Natural Resources Canada 2024f), expressed in 2025 Canadian dollars (US sources converted at roughly 1.37 CAD/USD and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 11.2:** Indicative techno-economic characteristics of petroleum refining decarbonisation technologies, 2025 CAD

| Technology                             | Capex / cost                    | Effect                                                 | Energy / input        |
|----------------------------------------|---------------------------------|--------------------------------------------------------|-----------------------|
| Electric process heater                | ~150–400 CAD/kWth               | removes combustion CO <sub>2</sub> (low/med heat)      | electricity           |
| Hydrogen-fired furnace                 | retrofit + hydrogen cost        | decarbonises high-temp heat                            | hydrogen              |
| Blue hydrogen (SMR + CCS)              | +30–50% on H <sub>2</sub> plant | captures ~90% of H <sub>2</sub> -plant CO <sub>2</sub> | natural gas + capture |
| Gas cogeneration (with or without CCS) | ~1,800–3,000 CAD/kWe            | CHP; exports surplus power                             | natural gas           |
| CCS on concentrated streams            | +30–50% plant equipment         | ~CAD 50–120 / t CO <sub>2</sub> (low-cost in refining) | steam / electricity   |
| Energy-efficiency / heat integration   | unit-specific                   | lower fuel per barrel                                  | —                     |

#### 11.4.1 Decarbonising process heat and steam

Process furnaces and boilers are the largest emission source, so the central lever is how their heat is supplied. The model competes electric heaters (viable for low- and medium-grade heat), hydrogen firing (for high-temperature duty that is hard to electrify), and efficiency-improved fossil firing, at the **Direct Heat**, **Steam**, and **Still Gas** services, alongside cogeneration. Electrifying the electrifiable heat and switching the rest to hydrogen is the backbone of a low-carbon refinery (World Resources Institute 2023).

#### 11.4.2 Low-carbon hydrogen and CCS

Hydrogen manufacturing is the second-largest emission source. The model already competes blue hydrogen (SMR with CCS) at the **Hydrogen** service; linking it to the [Hydrogen](#) sub-model would also allow green (electrolytic) hydrogen. Because the hydrogen plant and several process units produce concentrated CO<sub>2</sub> streams, **CCS** is comparatively low-cost in refining, and the model routes the CCS-equipped heat, steam, still-gas, and hydrogen options to the shared capture service.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

## 11.5 Economics

Technology choice at every node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, operating cost, an output basis, service requests for the fuel, steam, hydrogen, and electricity it consumes, and availability and lifetime years; these are combined into a lifecycle cost and converted to market shares.

Petroleum-refining technologies are discounted at a high rate — 35% at most process nodes, with auxiliary and capital-intensive nodes at 40–50% in the current data. These values are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the short payback horizons and risk aversion typical of the industry; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** govern how sharply the energy-service competitions respond to cost.

Emissions are overwhelmingly combustion CO<sub>2</sub> from process heat, steam, still gas, and the hydrogen plant, entering through each technology’s fuel service requests and the carbon price; the CCS service applies a 90% removal factor to the equipped routes. Because so much of the fuel is self-generated still gas, the carbon price acts on the choice of how to supply heat and hydrogen — electrify, switch to hydrogen, or capture — rather than on a single dominant process emission. A structural consideration the model can represent is that refinery throughput is expected to decline over the long run as transport electrifies, which reduces absolute emissions independently of the per-barrel decarbonisation choices.

## 11.6 Regional coverage and differences

Six regions carry refining — Alberta, the Atlantic region, British Columbia, Ontario, Québec, and Saskatchewan — corresponding to where Canada’s refineries are located. Regional differences show up in refinery complexity and product slate, in the fuel mix available for process heat (cheaper gas in the West), and in grid carbon intensity, which determines how much electrifying process heat actually reduces emissions (large in hydro-rich Québec and British Columbia, smaller in Alberta and Saskatchewan until their grids decarbonise).

TODO: Calibrate the regional refinery configurations (complexity, product slate, hydrogen demand) and the long-run throughput trajectory against Canada Energy Regulator and Statistics Canada data, since both the energy use and the decarbonisation economics depend on them.

## 11.7 Limitations and planned extensions

The petroleum refining sub-model captures the process chain and, in detail, the energy services that dominate its emissions — direct heat, steam, still gas, and hydrogen — with a full set of electrification, hydrogen, cogeneration, and CCS options. Its main limitations are the older auxiliary structure (to be migrated to MECS machine drive), the on-site-only hydrogen representation (to be linked to the hydrogen sub-model), and reliance on inherited refinery-configuration and cost data. A broader consideration is the sector’s long-run decline: as transport electrifies, the most consequential question for refining emissions may be how fast throughput falls, which is driven by the demand sectors rather than within this sub-model. The TODOs above record the priorities.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled refining energy use and emissions against Canada Energy Regulator and Environment and Climate Change Canada (NIR) data.

## 12 Electricity

Electricity generation is the central energy-supply sub-model in CIMS: it converts fuels and renewable resources into the power that every other sector draws on, and as the rest of the economy electrifies, what happens here increasingly determines system-wide emissions. The emphasis, as elsewhere, is on the structure of the energy-service flow model — how demand is split across the load curve, the generation technologies that compete to meet each part of it, and the services (storage, transmission, capture) that support them. The model tree is region-agnostic, but electricity is the sector where regional differences are largest, because each province’s grid is built on a different resource base.

### 12.1 Scope

CIMS represents utility-scale electricity generation and the supporting storage, transmission, and carbon-capture services. Electricity co-generated within industry is accounted for in the industrial sub-models (see, for example, the cogeneration options in [Pulp and Paper](#) and [Low-Temperature Industrial Heat](#)) and is not double-counted here — with the deliberate exception that all renewable generation is accounted in the electricity model regardless of whether a utility or an industrial site operates it. Final electricity demand across the economy drives the sub-model; imports and exports are handled through the macroeconomic module, with an own-price elasticity for export demand of about  $-1.07$ .

The combustion of the electricity once delivered, and the end-use equipment that consumes it, belong to the demand sectors. This chapter covers only how the power is generated and delivered.

### 12.2 Real-world context

Canada has one of the cleanest electricity systems in the world. In 2024 renewables supplied about 65.5% of generation — hydropower 55.3% (with Québec alone producing about half of national hydro), wind 7.2%, and solar 1.3% — nuclear 13.7% (almost entirely Ontario’s CANDU reactors), and fossil fuels 20.8% (natural gas 15.8%, coal 4.1%) (Natural Resources Canada [2024c](#)). More than four-fifths of recent generation investment has gone to non-emitting sources (Natural Resources Canada [2024c](#)).

The system is, however, deeply regional. British Columbia, Québec, and Manitoba run on hydro and are almost emissions-free; Ontario combines nuclear, hydro, and a growing renewable share; the Atlantic provinces are mixed (Prince Edward Island is essentially all wind); and Alberta and Saskatchewan are the fossil-heavy outliers, historically coal and now increasingly natural gas. Federal regulation is phasing out conventional coal generation and pushing toward a net-zero grid, so the model’s central question is how each province’s mix evolves from its very different starting point. The CIMS structure is built to answer exactly that: it competes a full set of generation technologies — variable renewables, firm low-carbon resources, and flexible balancing resources — within a load-duration framework, and adds the storage, transmission, and capture services that a high-renewables grid needs.

## 12.3 Model structure

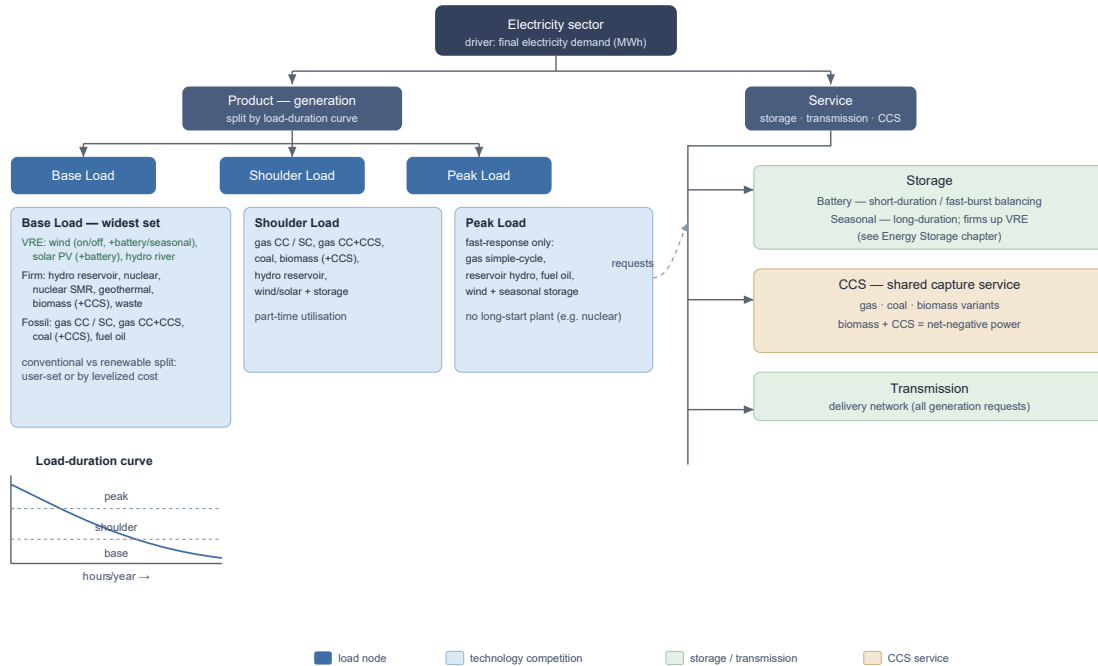
Following the standard CIMS pattern, the sector divides into a **Product** sub-tree — generation, organised by position on the load curve — and a **Service** sub-tree holding **CCS**, **Storage**, and **Transmission**. Figure 12.1 shows the structure.

Total generation is split across **Base Load**, **Shoulder Load**, and **Peak Load** nodes using ratios derived from historical load-duration curves (the user-specified `fixed_matrix_ratio` decision rule). A load-duration curve re-orders the year’s hourly demand from highest to lowest; its height gives the capacity required and its width the utilisation, so the three nodes correspond to power that is needed almost all the time (base), part of the time (shoulder), and only at peak. Within each node, technologies compete on the basis of their annual levelized cost (see Chapter 4).

### 12.3.1 Products: base, shoulder, and peak generation

The traditional baseload/load-following/peaking classification is, as the model’s own framing notes, less meaningful in a system with large wind and solar shares, where dispatch is driven by renewable variability and most low-carbon options are capital-intensive with near-zero variable cost. CIMS therefore keeps the base/shoulder/peak split as the demand-side framework but populates it with three functional classes of technology:

- Variable renewables (VRE) — wind (onshore, offshore), solar PV, run-of-river hydro: zero fuel cost, output set by the resource, valuable when available.
- Firm low-carbon resources — reservoir hydro, nuclear (including small modular reactors, SMRs), geothermal, biomass and biogas, and fossil plants with CCS: dispatchable over long durations.
- Fast-burst balancing resources — short-duration battery storage and flexible demand: energy-limited, used to balance supply and demand over short, high-value periods.



**Figure 12.1:** Energy-service flow model of the electricity sector (region-agnostic master tree). Total demand is split by load-duration-curve ratios into Base, Shoulder, and Peak Load nodes, each hosting a competition among generation technologies; a Service sub-tree provides CCS (coal, gas, biomass), Storage (battery and seasonal), and Transmission.

In the model the **Base Load** node hosts the widest set — wind (with battery or seasonal storage), solar PV (with battery), hydro reservoir and river, nuclear and nuclear SMR, natural gas combined-cycle (CC) and simple-cycle (SC), gas CC with CCS, coal and coal with CCS, biomass and biomass with CCS, geothermal/geoexchange, and waste-to-energy. **Shoulder Load** hosts a similar but slightly narrower set, and **Peak Load** is limited to fast-responding plant (gas simple-cycle, reservoir hydro, oil, and wind-plus-seasonal-storage) — long-start plant such as nuclear cannot serve peak. The split between conventional and renewable generation within a node can be user-specified over time or determined endogenously by relative leveled cost.

### 12.3.2 Services: storage, transmission, and capture

- **Storage** — Battery (short-duration, the fast-burst balancing resource) and Seasonal (long-duration storage that lets variable renewables serve firm demand). Storage is what allows high-VRE grids to remain reliable.
- **Transmission** — the delivery network that all generation requests.

- **CCS** — the shared carbon-capture service, with coal, natural gas, and biomass variants; biomass-with-CCS generation is net-negative.

TODO: The current regional input files predate the full Product/Service split (generation sits under a `Utility Generation` node rather than the master tree's `Product`). Migrate them to the region-agnostic master tree and harmonise the storage and transmission representation with the [Energy Storage](#) chapter.

## 12.4 Decarbonisation technologies

Electricity decarbonisation is largely about building clean firm and variable capacity to displace unabated coal and gas, and adding the storage and transmission that let a high-renewables system stay reliable. Because Canada starts so clean nationally, the action is concentrated in the fossil-heavy provinces. Table 12.1 summarises maturity and Table 12.2 gives indicative costs.

**Table 12.1:** Indicative role and maturity of key electricity generation and storage technologies

| Technology                          | Indicative role                     | Maturity                                   |
|-------------------------------------|-------------------------------------|--------------------------------------------|
| Onshore wind / solar PV             | Lowest-cost new energy; variable    | Commercial                                 |
| Offshore wind                       | Higher capacity factor; higher cost | Commercial<br>(emerging in Canada)         |
| Reservoir hydro                     | Firm, flexible, storage-capable     | Commercial<br>(site-limited)               |
| Nuclear — large & SMR               | Firm low-carbon; SMRs modular       | Commercial (large);<br>first-of-kind (SMR) |
| Battery storage<br>(short-duration) | Fast-burst balancing                | Commercial                                 |
| Seasonal / long-duration<br>storage | Firms up variable renewables        | Emerging                                   |
| Gas CC + CCS                        | Firm low-carbon dispatchable        | Demonstration                              |
| Coal + CCS                          | Retrofit of coal fleet              | Demonstration<br>(e.g. Boundary Dam)       |
| Biomass + CCS                       | Firm and net-negative               | Demonstration                              |
| Geothermal                          | Firm low-carbon                     | Emerging in Canada                         |

The estimates in Table 12.2 are order-of-magnitude figures from public Canadian and North American sources (Independent Electricity System Operator 2024; Natural Resources Canada 2024c; World Nuclear News 2025), in 2025 Canadian dollars (US sources converted at roughly

1.37 CAD/USD and adjusted to 2025). Overnight capital is per kW; levelized cost of electricity (LCOE) is per MWh and is highly site- and grid-dependent. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 12.2:** Indicative techno-economic characteristics of electricity technologies, 2025 CAD

| Technology                         | Overnight capex<br>(2025 CAD/kW) | LCOE (2025<br>CAD/MWh) | Notes                             |
|------------------------------------|----------------------------------|------------------------|-----------------------------------|
| Onshore wind                       | ~1,800–2,600                     | ~45–80                 | variable; falling costs           |
| Solar PV (utility)                 | ~1,300–2,400                     | ~45–85                 | variable; falling costs           |
| Natural gas<br>combined cycle      | ~1,500–2,200                     | ~65–100                | + carbon cost on<br>emissions     |
| Natural gas simple<br>cycle (peak) | ~1,100–1,600                     | ~140–220               | low utilisation,<br>peaking       |
| Gas CC + CCS                       | ~3,000–4,500                     | ~95–150                | capture +CAD<br>~25–45/MWh        |
| Reservoir hydro                    | ~3,500–7,000                     | ~50–110                | very site-specific;<br>long-lived |
| Nuclear (large)                    | ~7,000–11,000                    | ~90–170                | firm; high capital                |
| Nuclear SMR                        | ~5,500–9,000                     | ~120–190               | first-of-kind<br>premium          |
| Battery storage<br>(4-hour)        | ~1,800–3,100 (per<br>kW)         | ~110–200               | balancing, not bulk<br>energy     |
| Coal + CCS<br>(retrofit)           | retrofit-dependent               | ~100–160               | ~90% capture                      |

#### 12.4.1 Variable renewables plus storage

The cheapest new energy in most of Canada is now onshore wind and solar PV, but their variability means that beyond modest shares they need balancing. The model represents this directly: wind and solar appear both alone and paired with battery (short-duration) or seasonal (long-duration) storage, and the **Storage** service is what lets a high-renewables node meet firm demand. As battery costs fall — utility four-hour storage is approaching US\$100/MWh — the VRE-plus-storage combination displaces more gas.

#### 12.4.2 Firm low-carbon resources

Where firm capacity is needed over long durations, the model competes reservoir hydro (Canada's traditional firm clean resource, though site-limited), nuclear including small

modular reactors (Ontario is building Canada’s first SMR), geothermal, biomass, and gas or coal with CCS. These are the technologies that can replace unabated fossil baseload while keeping the grid reliable in low-wind, low-sun periods.

### 12.4.3 Carbon capture and net-negative power

In the gas- and coal-heavy provinces, CCS on combined-cycle gas, on the existing coal fleet (Saskatchewan’s Boundary Dam is a pioneer), or on biomass plants captures most of the combustion CO<sub>2</sub>; biomass with CCS goes further and delivers net-negative generation. The shared CCS service and its costs are detailed in the [Carbon Capture, Utilization, and Storage](#) chapter.

TODO: Replace the indicative literature ranges above with the capital costs, capacity factors, availability years, and emission factors actually used at these nodes in the model input files, and reconcile the storage representation with the [Energy Storage](#) chapter.

## 12.5 Economics

Technology choice within each load node follows the lifecycle-cost competition of Chapter 4, here on an annual levelized-cost basis: each technology carries an overnight capital cost, fixed and variable operating costs, a capacity factor, fuel service requests where applicable, and availability and lifetime years, which are annualised and combined into a levelized cost per MWh and converted to market shares.

Electricity technologies are discounted at lower rates than most demand sectors — around 12.5–25% in the current data — reflecting the regulated, capital-intensive, long-horizon nature of generation investment. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than pure financial costs of capital, but for a regulated utility they sit closer to the financial end of the range than the 35%+ used in the demand sectors; CIMS applies them throughout the levelized-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 10–25.

Emissions are tracked as CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O and arise from fuel combustion in the fossil and biomass plants, entering through each technology’s fuel service requests and the carbon price; the CCS service applies a removal factor to the captured routes, which can drive biomass-CCS generation net-negative. Because electricity is an input to every other sector, the carbon price acts here with unusual leverage: it raises the cost of unabated coal and gas relative to renewables, nuclear, and CCS, shifting the generation mix and, through the price of delivered electricity, the economics of electrification everywhere else in the model. Electricity imports and exports respond to relative prices through the macroeconomic module’s trade elasticities.

## 12.6 Regional coverage and differences

Electricity is the sector where the region-agnostic tree resolves to the most different outcomes, because each of the seven modelled regions starts from a distinct resource base:

- Québec, British Columbia, and Manitoba are dominated by hydro and are already almost emissions-free; their decarbonisation question is mainly how to add capacity (hydro, wind) for growing electrified demand.
- Ontario runs on nuclear and hydro with growing renewables, and faces choices about nuclear refurbishment, SMRs, and wind/solar/storage.
- Alberta and Saskatchewan are the fossil-heavy provinces, where the coal phase-out and the competition among gas, gas-plus-CCS, wind, solar, and storage is most consequential.
- The Atlantic region is mixed, with significant wind potential.

TODO: Confirm the regional capacity factors, resource limits (especially hydro and wind siting), and existing-stock vintages, since the regional generation mix and its evolution are highly sensitive to them.

## 12.7 Limitations and planned extensions

The electricity sub-model captures the essential structure of a modern power system — a load-duration split, three functional technology classes, and the storage and transmission services a high-renewables grid needs — but it remains a simplified representation of a system whose real behaviour is hourly and spatial. Its main limitations are the coarse (base/shoulder/peak) time resolution relative to the sub-hourly dynamics that actually drive VRE integration, the simplified storage and transmission representation, and reliance on inherited cost and load-curve assumptions. The TODOs above record the priorities, of which keeping the cost and capacity-factor data current and reconciling storage with the dedicated [Energy Storage](#) chapter are the most important.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark the modelled generation mix and grid emissions intensity by province against Canada Energy Regulator and Canada Energy Future data.

## 13 Ethanol

Ethanol production is one of CIMS's energy-supply sub-models: it produces a low-carbon liquid fuel that the transport sector blends with gasoline. The emphasis is on the structure of the energy-service flow model — the production routes, the technologies that compete at each service, and the engineering ratios that link them. The tree is region-agnostic. Two features define the ethanol sub-model: it carries two distinct production routes — first-generation corn (grain) fermentation and second-generation cellulosic conversion — that differ markedly in carbon intensity; and, because fermentation releases a stream of nearly pure, biogenic CO<sub>2</sub>, it is one of the cheapest places in the whole energy system to apply carbon capture, which the sub-model represents through a CCS service.

### 13.1 Scope

CIMS represents the production of fuel ethanol by two routes: corn (and, in Western Canada, wheat) starch fermentation, and cellulosic conversion of crop residues and other lignocellulosic material. The sub-model covers the conversion plant and its inputs — agricultural feedstock, process steam, electricity, and a gaseous process fuel — and produces ethanol as an output that flows into the gasoline blend. Demand is set by the blending and fuel-switching choices in the consuming sectors and by renewable-fuel policy, rather than by an exogenous production target.

Adjacent activities are handled elsewhere: biodiesel has its own [Biodiesel](#) chapter; gasoline is produced in [Petroleum Refining](#); the combustion of the ethanol blend happens in the transport sectors; and feedstock cultivation is represented here only through an **Agricultural Input** service rather than the full [Agriculture](#) sub-model.

### 13.2 Real-world context

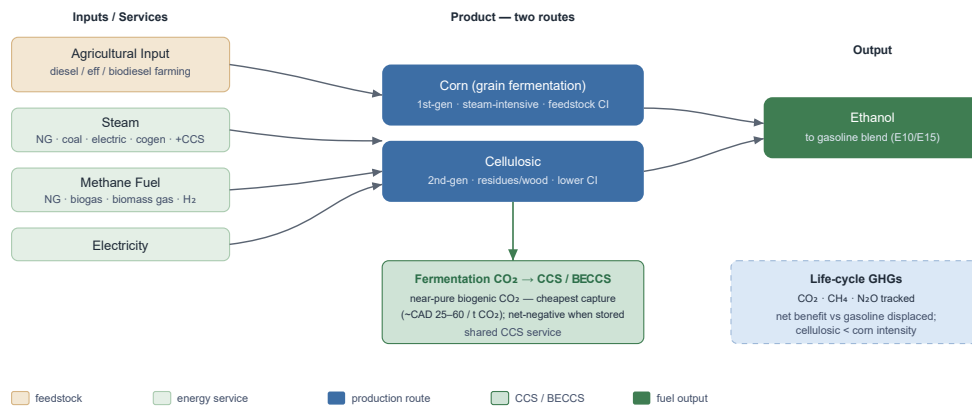
Canadian ethanol is predominantly first-generation: corn in Eastern Canada (Ontario in particular) and wheat in the West, fermented to ethanol and blended into gasoline under the federal Clean Fuel Regulations and provincial renewable-content mandates (Natural Resources Canada [2024d](#)). Cellulosic ethanol — made from crop residues, wood, and other lignocellulosic feedstock — is a second-generation technology that is much less developed but carries a

substantially lower carbon intensity, because it is not burdened by the cultivation emissions (fertiliser  $\text{N}_2\text{O}$ , farm diesel, land use) of a dedicated grain crop and uses material that would otherwise decay or be burned (Tullo 2023).

The sector has one striking decarbonisation feature. The fermentation step that turns sugars into ethanol releases roughly an equal mass of  $\text{CO}_2$ , and that  $\text{CO}_2$  is biogenic and almost pure — it needs little separation before it can be compressed and stored. This makes ethanol plants among the lowest-cost carbon-capture opportunities anywhere, and capturing biogenic fermentation  $\text{CO}_2$  delivers net-negative emissions (BECCS). The CIMS structure reflects both points: it separates the corn and cellulosic routes, carries life-cycle  $\text{CO}_2$ ,  $\text{CH}_4$ , and  $\text{N}_2\text{O}$  accounting through the feedstock and process services, and provides a CCS service the routes can draw on.

### 13.3 Model structure

The sub-model divides into a **Product** sub-tree — the two production routes — and a **Service** sub-tree of the inputs they draw on. Figure 13.1 shows the structure. Each node carries a competition rule (see Chapter 4); the production chains are fixed-ratio sequences and the technology choices occur at the service leaves.



**Figure 13.1:** Energy-service flow model of the ethanol sub-model. Two production routes — Corn (grain fermentation) and Cellulosic — convert feedstock into ethanol, drawing on an Agricultural Input service, process Steam, a gaseous Methane Fuel supply, electricity, and a shared CCS service well suited to the pure biogenic  $\text{CO}_2$  from fermentation; life-cycle  $\text{CO}_2$ ,  $\text{CH}_4$ , and  $\text{N}_2\text{O}$  are tracked.

### 13.3.1 Products: corn and cellulosic routes

- **Corn** — first-generation starch fermentation. It draws on the **Agricultural Input** service (the grain feedstock and its cultivation energy) and on **Steam** for cooking, distillation, and drying, and is the more steam-intensive of the two routes.
- **Cellulosic** — second-generation conversion of lignocellulosic feedstock (crop residues, wood). In the model it draws less on the agricultural-cultivation input and instead consumes some industrial diesel for feedstock handling, plus **Steam**; its lower cultivation burden gives it a lower life-cycle carbon intensity.

### 13.3.2 Services: feedstock, heat, fuel, and capture

- **Agricultural Input** — feedstock cultivation, competing diesel-powered, efficient-diesel, and biodiesel-powered farming, so the energy and emissions of growing the grain are represented (including the option to displace farm diesel with biodiesel).
- **Steam** — process heat, competing natural-gas, coal, and electric boilers, cogeneration, and CCS-equipped variants.
- **Methane Fuel** — the gaseous-fuel supply, competing natural gas against biogas, biomass gasification, and hydrogen.
- **CCS** — the shared carbon-capture service, especially valuable here because of the pure fermentation CO<sub>2</sub>.

TODO: The current regional input files predate the full Product/Service split (the routes sit under a `Production` node rather than directly under the master tree's `Product`). Migrate them to the region-agnostic master tree and the shared General Services chapters.

## 13.4 Decarbonisation technologies

Ethanol decarbonisation means lowering life-cycle carbon intensity — shifting toward cellulosic feedstock, cleaning up process energy, and, distinctively, capturing the pure fermentation CO<sub>2</sub> for net-negative emissions. Table 13.1 summarises maturity and Table 13.2 gives indicative costs.

**Table 13.1:** Indicative performance and maturity of key ethanol decarbonisation options

| Technology                                   | Indicative benefit                                         | Maturity                            |
|----------------------------------------------|------------------------------------------------------------|-------------------------------------|
| Fermentation CO <sub>2</sub> capture (BECCS) | Captures near-pure biogenic CO <sub>2</sub> ; net-negative | Early commercial (low-cost capture) |
| Cellulosic route (vs corn)                   | Much lower life-cycle CI; avoids cultivation burden        | Pilot / early commercial            |

| Technology                                   | Indicative benefit                          | Maturity              |
|----------------------------------------------|---------------------------------------------|-----------------------|
| Biodiesel-powered / efficient farming        | Lowers feedstock-cultivation emissions      | Commercial            |
| Electric process steam / boilers             | Removes fossil process-heat CO <sub>2</sub> | Commercial            |
| Biogas / biomass-gas / hydrogen process fuel | Decarbonises gaseous process fuel           | Commercial / emerging |
| CCS on process steam                         | Captures process-heat CO <sub>2</sub>       | Demonstration         |

The cost estimates in Table 13.2 are order-of-magnitude figures from public literature (Natural Resources Canada 2024d; Tullo 2023), expressed in 2025 Canadian dollars (US-dollar sources converted at roughly 1.37 CAD/USD and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 13.2:** Indicative techno-economic characteristics of ethanol decarbonisation options, 2025 CAD

| Decarbonisation option                       | Capex                               | Operating cost / cost of CO <sub>2</sub>                     | Energy / CI effect                          |
|----------------------------------------------|-------------------------------------|--------------------------------------------------------------|---------------------------------------------|
| Fermentation CO <sub>2</sub> capture (BECCS) | low (compression, dehydration only) | ~CAD 25–60 / t CO <sub>2</sub> (cheapest industrial capture) | net-negative; near-pure CO <sub>2</sub>     |
| Cellulosic conversion (vs corn)              | higher than corn plant              | feedstock-dependent                                          | much lower life-cycle CI                    |
| Electric process boiler                      | ~150–400 CAD/kWth                   | electricity-priced heat                                      | removes process-heat fossil CO <sub>2</sub> |
| Biomass / biogas process fuel                | ~800–1,400 CAD/kWth (biomass)       | low-carbon fuel                                              | decarbonises process heat                   |
| CCS on process steam                         | +30–50% on boiler plant             | ~CAD 100–200 / t CO <sub>2</sub>                             | captures process CO <sub>2</sub>            |

#### 13.4.1 Capturing fermentation CO<sub>2</sub> (the cheapest BECCS)

The standout option is capturing the fermentation CO<sub>2</sub>. Unlike a flue gas, it emerges nearly pure, so only compression and dehydration are needed before storage — capture costs are a small fraction of those for dilute industrial streams. Because the carbon is biogenic, storing it yields net-negative emissions. The model represents this through the CCS service; given the cost advantage, it is the single most valuable lever in the sub-model.

TODO: Make the fermentation-CO<sub>2</sub> capture pathway explicit and distinct from process-heat CCS, with its own (much lower) capture cost, so the model captures the unusually cheap BECCS opportunity at ethanol plants rather than treating all capture at a single cost.

#### 13.4.2 Feedstock and process energy

Shifting from corn to cellulosic feedstock lowers the cultivation burden, and within the corn route the **Agricultural Input** options (efficient or biodiesel-powered farming) reduce field emissions. On the plant side, switching **Steam** to electric, biomass, or biogas, or adding CCS to gas-fired steam, cuts the conversion footprint — the same low-carbon heat options described in the [Low-Temperature Industrial Heat](#) chapter.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

### 13.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output basis, service requests for the inputs it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Ethanol technologies are discounted at rates of roughly 10–40% in the current data. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the payback horizons and adoption barriers that govern real investment decisions; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 10–25 across the competitions.

As with biodiesel, the emissions accounting is **life-cycle**: the sub-model carries CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O factors so that agricultural N<sub>2</sub>O, process CH<sub>4</sub>, and CO<sub>2</sub> are all represented, and the net benefit of ethanol emerges from the difference between these emissions and the gasoline it displaces. The CCS service applies an emissions-removal factor to the routes that request it — and because the captured fermentation CO<sub>2</sub> is biogenic, that removal can drive the route's net emissions below zero. The carbon price therefore rewards the cellulosic route, clean process energy, and above all fermentation-CO<sub>2</sub> capture.

## 13.6 Regional coverage and differences

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry both the corn and cellulosic routes. In practice corn ethanol is concentrated in Ontario and the East, wheat-based grain ethanol in the Prairies, and cellulosic feedstock potential follows crop-residue and forestry availability. Regional differences show up in feedstock mix and in the carbon intensity of the grid supplying process electricity.

TODO: Calibrate the regional split between corn/wheat and cellulosic routes, and the associated carbon intensities, against Clean Fuel Regulations pathway data and Canadian crop-residue supply estimates.

## 13.7 Limitations and planned extensions

The ethanol sub-model is compact and supply-side. Its main limitations: it treats feedstock through a single Agricultural Input service rather than linking to the agriculture sub-model; it applies a single CCS cost rather than recognising the unusually cheap fermentation- $\text{CO}_2$  capture; and its corn/cellulosic split and intensities are inherited from older calibrations. The TODOs above record these priorities, of which making the cheap fermentation- $\text{CO}_2$  BECCS pathway explicit is the most consequential for emissions outcomes.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled ethanol output and carbon intensity against Natural Resources Canada and Clean Fuel Regulations data.

## 14 Biodiesel

Biodiesel production is one of CIMS's energy-supply sub-models rather than an end-use sector: it produces a low-carbon liquid fuel that other sectors — chiefly freight and passenger transport, and industrial and agricultural diesel use — consume in place of, or blended with, fossil diesel. The emphasis is on the structure of the energy-service flow model: the production node, the technologies that compete at each service, and the engineering ratios that link them. The tree is region-agnostic. The defining feature of biodiesel is that most of its life-cycle greenhouse-gas footprint sits upstream of the plant, in the feedstock, so the sub-model carries explicit agricultural inputs and life-cycle CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O accounting rather than treating the fuel as automatically carbon-neutral.

### 14.1 Scope

CIMS represents the production of biodiesel (fatty-acid methyl ester, FAME) by transesterification of vegetable oils and animal fats. The sub-model covers the conversion plant and its main inputs — the agricultural feedstock, process steam, electricity, and a gaseous process fuel — and produces biodiesel as an output that flows into the fuel supply. Demand is set by the blending and fuel-switching choices made in the consuming sectors and by policy (renewable-fuel mandates), rather than by an exogenous production target.

Adjacent activities are handled elsewhere: ethanol is a separate fuel with its own [Ethanol](#) chapter; fossil diesel is produced in [Petroleum Refining](#); and the combustion of biodiesel happens in the transport and other end-use sectors, where the blended fuel is burned. The cultivation of the feedstock crop is represented here only through an **Agricultural Input** service that carries its energy and emissions, not through the full [Agriculture](#) sub-model.

### 14.2 Real-world context

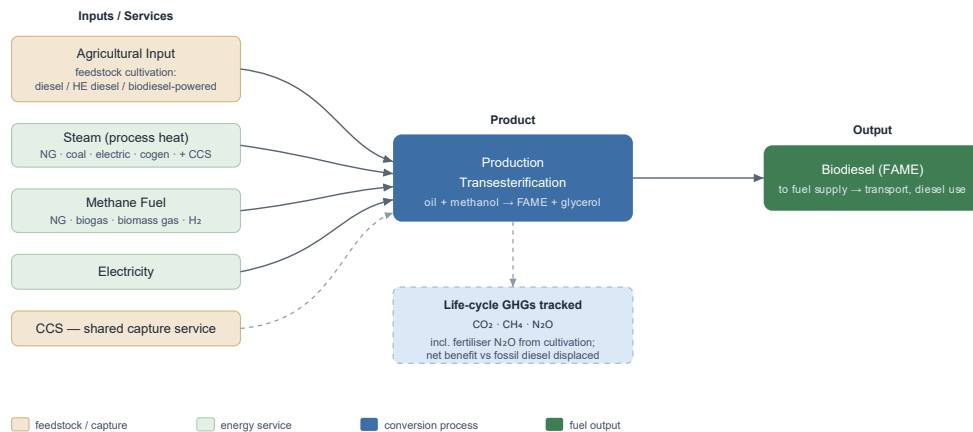
Biodiesel is a mature, drop-in-capable diesel substitute, and in Canada its production is tied to agriculture and to fuel policy. The country has a handful of biodiesel plants (and a growing number of renewable diesel plants, a different, hydrotreating route), and the dominant feedstock is canola — around 90% of domestic biomass-based diesel is made from Canadian canola — supplemented by tallow and used cooking oil (Natural Resources Canada [2024a](#);

Canada Energy Regulator 2023). Demand is driven by the federal Clean Fuel Regulations and provincial renewable-content mandates, which reward fuels by their measured carbon intensity.

That last point is central. Biodiesel is not automatically low-carbon: its life-cycle intensity depends heavily on the feedstock and on the energy used to grow and process it. Crop feedstocks carry the emissions of cultivation — diesel for farm equipment, and  $\text{N}_2\text{O}$  from nitrogen fertiliser, a potent greenhouse gas — whereas waste and residue feedstocks such as tallow and used cooking oil avoid most of that burden and can achieve 79–86% lower life-cycle emissions than fossil diesel (Canada Energy Regulator 2023). The CIMS structure mirrors this by separating the conversion step from an explicit **Agricultural Input** service and by carrying life-cycle  $\text{CO}_2$ ,  $\text{CH}_4$ , and  $\text{N}_2\text{O}$  factors, so the modelled benefit of biodiesel reflects how it is made rather than assuming neutrality.

### 14.3 Model structure

The sub-model divides into a **Product** sub-tree — the conversion process — and a **Service** sub-tree of the inputs it draws on. Figure 14.1 shows the structure. Each node carries a competition rule (see Chapter 4); the production chain is a fixed-ratio sequence and the technology choices occur at the tech-compete leaves of the services.



**Figure 14.1:** Energy-service flow model of the biodiesel sub-model. A transesterification Production node converts agricultural feedstock into biodiesel, drawing on an Agricultural Input service (feedstock cultivation, diesel- or biodiesel-powered), process Steam, a gaseous Methane Fuel supply, electricity, and a shared CCS service; life-cycle  $\text{CO}_2$ ,  $\text{CH}_4$ , and  $\text{N}_2\text{O}$  are tracked.

### 14.3.1 Product: transesterification

The **Production** node is biodiesel production by transesterification — reacting the feedstock oil with methanol to yield FAME and glycerol. In the model it consumes three things per unit of biodiesel: electricity, an **Agricultural Input** (the feedstock), and **Steam** (process heat). The master tree names this product node Transesterification.

### 14.3.2 Services: feedstock, heat, fuel, and capture

- **Agricultural Input** — the feedstock-cultivation service, competing diesel-powered, high-efficiency diesel, and biodiesel-powered farming. This is how the model captures the energy (and the diesel-versus-biodiesel choice) of growing the crop; the associated N<sub>2</sub>O and other field emissions enter the life-cycle accounting here.
- **Steam** — process heat for the reaction and separation, competing natural-gas, coal, and electric boilers, cogeneration units, and CCS-equipped boiler variants.
- **Methane Fuel** — the gaseous-fuel supply, competing natural gas against biogas, biomass gasification, and hydrogen.
- **CCS** — the shared carbon-capture service that the CCS-equipped steam routes request.

### 14.3.3 Renewable diesel — a gap

The model represents only the transesterification (FAME) route. The fast-growing renewable diesel (hydrotreated vegetable oil, HVO/HDRD) route — chemically a true drop-in diesel made by hydrogenation rather than transesterification — is not represented, even though new Canadian capacity is overwhelmingly of this type.

TODO: Add a renewable diesel (HVO/HDRD) production pathway (hydrotreating, with its hydrogen demand drawn from the [Hydrogen](#) sub-model), since most new Canadian biomass-based diesel capacity is renewable diesel rather than FAME biodiesel.

TODO: The current regional input files predate the full Product/Service split (the product node is named `Production` rather than the master tree's `Transesterification`). Migrate them to the region-agnostic master tree and the shared General Services chapters.

## 14.4 Decarbonisation technologies

Because biodiesel is already a low-carbon fuel relative to fossil diesel, “decarbonising biodiesel” means lowering its life-cycle carbon intensity — choosing better feedstocks and cleaning up the energy used to grow and process it — so that it earns more credit under

carbon-intensity-based policy. Table 14.1 summarises maturity and Table 14.2 gives indicative costs.

**Table 14.1:** Indicative performance and maturity of key biodiesel decarbonisation options

| Technology                                           | Indicative benefit                                         | Maturity                             |
|------------------------------------------------------|------------------------------------------------------------|--------------------------------------|
| Waste / residue feedstock (tallow, used cooking oil) | 79–86% lower life-cycle CO <sub>2</sub> than fossil diesel | Commercial                           |
| Biodiesel-powered / efficient farming inputs         | Lowers feedstock-cultivation emissions                     | Commercial                           |
| Electric process steam / boilers                     | Removes fossil process-heat CO <sub>2</sub>                | Commercial                           |
| Biogas / biomass-gas / hydrogen process fuel         | Decarbonises gaseous process fuel                          | Commercial / emerging                |
| CCS on process steam                                 | Captures process-heat CO <sub>2</sub>                      | Demonstration                        |
| Renewable diesel (HVO) route                         | True drop-in; very low CI with waste oils                  | Commercial — <i>not yet in model</i> |

The cost estimates in Table 14.2 are order-of-magnitude figures from public literature (Natural Resources Canada 2024a; Canada Energy Regulator 2023), expressed in 2025 Canadian dollars (US-dollar sources converted at roughly 1.37 CAD/USD and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 14.2:** Indicative techno-economic characteristics of biodiesel decarbonisation options, 2025 CAD

| Decarbonisation option                   | Capex                                             | Operating cost                                          | Energy / CI effect                          |
|------------------------------------------|---------------------------------------------------|---------------------------------------------------------|---------------------------------------------|
| Waste/residue feedstock switch           | none (feed handling)                              | feedstock-cost dependent; often <i>higher</i> feed cost | 79–86% lower life-cycle CO <sub>2</sub>     |
| Electric process boiler                  | ~150–400 CAD/kWth                                 | electricity-priced heat                                 | removes process-heat fossil CO <sub>2</sub> |
| Biomass / biogas process fuel            | ~800–1,400 CAD/kWth (biomass)                     | low-carbon fuel                                         | decarbonises process heat                   |
| CCS on process steam                     | +30–50% on boiler plant                           | ~CAD 100–200 / t CO <sub>2</sub>                        | captures process CO <sub>2</sub>            |
| Renewable diesel (HVO) plant (reference) | higher than FAME (hydrotreater + H <sub>2</sub> ) | hydrogen-cost dependent                                 | true drop-in, very low CI                   |

#### 14.4.1 Feedstock is the main lever

The largest determinant of biodiesel's carbon intensity is the feedstock. Switching from crop oils (canola) to waste and residue oils (tallow, used cooking oil) removes most of the cultivation emissions — including the fertiliser-driven  $\text{N}_2\text{O}$  — and can cut life-cycle emissions to 79–86% below fossil diesel (Canada Energy Regulator 2023). The model captures the cultivation side through the **Agricultural Input** service, including the option to run farm equipment on biodiesel rather than fossil diesel, which closes part of the loop. Feedstock supply is, however, constrained: waste oils are limited, so feedstock competition with renewable diesel and other uses is a real-world limit the model should eventually reflect.

#### 14.4.2 Cleaning up process energy

The plant's own emissions come from process heat and electricity. Switching steam to electric boilers (clean where the grid is clean), to biomass or biogas, or adding CCS to gas-fired steam reduces the conversion step's footprint; the **Methane Fuel** node lets natural gas be displaced by biogas, biomass gas, or hydrogen. These are the same low-carbon heat options described in the [Low-Temperature Industrial Heat](#) chapter.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files, and add an explicit feedstock-supply constraint so waste-oil availability limits the low-CI pathways.

### 14.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output basis, service requests for the inputs it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Biodiesel technologies are discounted at rates of roughly 10–40% in the current data. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the payback horizons and adoption barriers that govern real investment decisions; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 10–25 across the competitions.

The emissions accounting is **life-cycle** in spirit, which distinguishes biodiesel from the end-use sectors. The sub-model carries  $\text{CO}_2$ ,  $\text{CH}_4$ , and  $\text{N}_2\text{O}$  factors, so the agricultural  $\text{N}_2\text{O}$  from

fertiliser and the  $\text{CH}_4$  and  $\text{CO}_2$  from process energy are all represented, and the net benefit of biodiesel emerges from the difference between these life-cycle emissions and the fossil diesel it displaces — rather than from an assumption that biofuel combustion is automatically zero-carbon. The CCS service applies an emissions-removal factor to the process-heat routes that request it. The carbon price therefore rewards low-intensity feedstocks and clean process energy, which is the mechanism by which carbon-intensity policy is represented.

## 14.6 Regional coverage and differences

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the biodiesel structure. In practice the Western provinces (Saskatchewan, Alberta, Manitoba) are the natural home of canola-based production, while waste-oil and tallow feedstocks are more widely distributed; the regional differences show up in feedstock availability and in the carbon intensity of the grid that supplies process electricity.

TODO: Calibrate regional feedstock mixes (canola vs tallow vs used cooking oil) and the resulting carbon intensities against Clean Fuel Regulations pathway data, since the credit value — and therefore the modelled production — depends sensitively on feedstock.

## 14.7 Limitations and planned extensions

The biodiesel sub-model is compact and supply-side, and its main limitations follow from that: it represents only the transesterification route (omitting renewable diesel), it treats feedstock through a single Agricultural Input service rather than linking to the full agriculture sub-model, and it lacks an explicit feedstock-supply constraint. The TODOs above record these priorities, of which adding the renewable diesel (HVO) pathway is the most important, since it now dominates new Canadian capacity.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled biodiesel output and carbon intensity against Natural Resources Canada and Clean Fuel Regulations data.

## 15 Hydrogen

Hydrogen production is one of CIMS's energy-supply sub-models: it produces a versatile energy carrier and chemical feedstock that other sectors consume — in oil refining, ammonia and methanol synthesis, potentially in low-carbon steelmaking (direct reduction), and in transport. The emphasis is on the structure of the energy-service flow model: the production routes that compete, the services they draw on, and the engineering ratios that link them. The tree is region-agnostic. The defining feature of hydrogen is that the molecule is identical however it is made, but its emissions are entirely a function of the production route — the familiar “colours” of hydrogen — so the whole sub-model is essentially one competition among production technologies that differ enormously in carbon intensity.

### 15.1 Scope

CIMS represents the production of hydrogen by all the major routes — natural-gas reforming, coal and biomass gasification, water electrolysis, and ethanol reforming — at both centralised and distributed scale, together with the delivery infrastructure and a shared CCS service. The sub-model produces hydrogen as an output that flows to the consuming sectors; its demand is set by those sectors' fuel-switching and feedstock choices and by policy, rather than by an exogenous production target.

Adjacent activities are handled elsewhere: the end uses of hydrogen — as a refinery input ([Petroleum Refining](#), [Petroleum Crude](#)), an ammonia/methanol feedstock ([Chemical Products](#)), a reductant in steel ([Iron and Steel](#)), or a transport fuel — are represented in those sectors. This chapter covers only how the hydrogen is made.

### 15.2 Real-world context

Almost all hydrogen made in Canada today is “grey” — produced by steam methane reforming (SMR) of natural gas without capture, mostly in Alberta to serve oil-sands upgrading, refining, and ammonia production. Grey hydrogen is the cheapest, at roughly CAD 1.5–2.5/kg, but emits on the order of 9–10 kg of CO<sub>2</sub> per kg of hydrogen (Natural Resources Canada [2024e](#)). Canada's Hydrogen Strategy and a new investment tax credit (15–40% of capital, the

top rate for the cleanest hydrogen) aim to shift production toward low-carbon routes (Natural Resources Canada [2024e](#)).

Two such routes are advancing. “Blue” hydrogen — SMR or gasification with carbon capture (CCS) — is the basis of large Alberta projects such as the Air Products net-zero hydrogen complex (1,500 t/day with ~3 Mt CO<sub>2</sub>/yr captured) and Linde’s clean-hydrogen facility; it costs roughly CAD 2–3.5/kg (Air Products [2024](#); Natural Resources Canada [2024e](#)). “Green” hydrogen — electrolysis powered by low-carbon electricity — is the lowest-emission route and is attractive where power is clean and cheap (Québec and BC hydro, Atlantic wind), but at roughly CAD 4.8–6.3/kg today it remains the most expensive (Natural Resources Canada [2024e](#)). The CIMS structure represents this whole ladder as a single production competition, with explicit life-cycle emissions and a CCS service, so the model can shift the production mix from grey toward blue and green as costs and the carbon price evolve.

### 15.3 Model structure

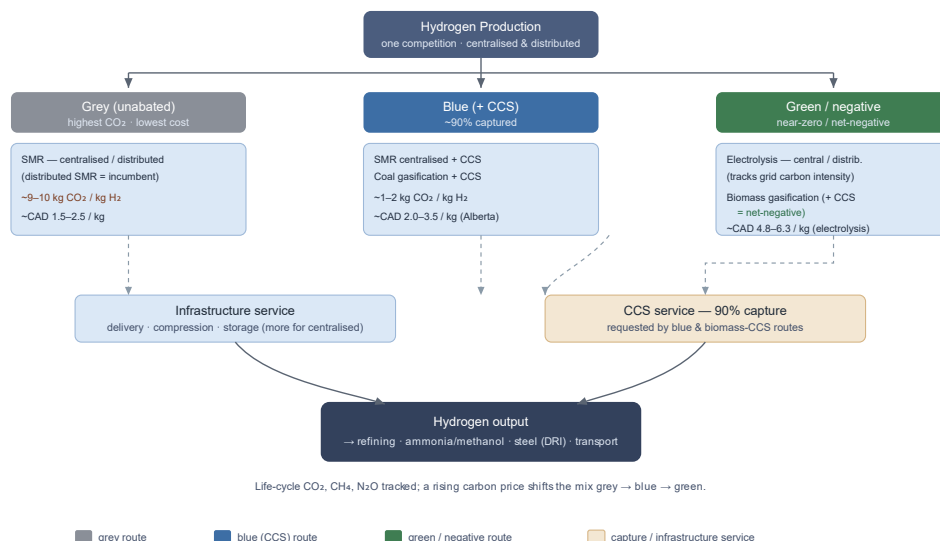
The sub-model divides into a **Product** sub-tree — the hydrogen-production competition — and a **Service** sub-tree holding the delivery **Infrastructure** and the **CCS** service. Figure [15.1](#) shows the structure. The production node is a single tech-compete node (see Chapter [4](#)); the incumbent in the current data is distributed SMR.

#### 15.3.1 Product: the production competition

The **Production** node competes the full set of routes, each consuming a characteristic mix of fuel, feedstock, and electricity and (for the capture routes) requesting the CCS service:

- **Steam methane reforming (SMR)** — centralised and distributed, and a centralised + CCS variant. NG-based; the distributed route is today’s incumbent. (Grey, and blue with CCS.)
- **Coal gasification + CCS** — centralised; coal-based blue hydrogen.
- **Biomass gasification** — centralised, and a + CCS variant. Renewable feedstock; with CCS it is net-negative (the captured CO<sub>2</sub> is biogenic).
- **Electrolysis** — centralised and distributed; splits water using electricity, so its emissions track the grid (green where the grid is clean). Electricity-dominated, at roughly 0.2 GJ electricity per unit of hydrogen energy.
- **Ethanol reforming** — distributed; reforms ethanol from the [Ethanol](#) sub-model.

The centralised versus distributed distinction matters because it trades production economies of scale against the **Infrastructure** (delivery, compression) cost: distributed production is made near the point of use and needs less delivery infrastructure, while centralised production is cheaper per unit but draws more heavily on the Infrastructure service.



**Figure 15.1:** Energy-service flow model of the hydrogen sub-model. A single Production competition spans the “colours” of hydrogen — grey (SMR, coal/biomass gasification), blue (the same routes with CCS), and green (electrolysis; biomass-with-CCS is net-negative) — at centralised and distributed scale, drawing on a delivery Infrastructure service and a shared 90%-capture CCS service; life-cycle CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O are tracked.

### 15.3.2 Services: infrastructure and capture

- **Infrastructure** — the delivery, compression, and storage needed to get hydrogen to users; every production route requests it (distributed routes less so).
- **CCS** — the shared carbon-capture service, removing 90% of the CO<sub>2</sub> presented to it, requested by the SMR-, coal-, and biomass-with-CCS routes.

TODO: The hydrogen consumed elsewhere in CIMS is in several places drawn through generic fuel nodes (for example the Methane Blend service in industry, and the green-ammonia and HDRI-steel routes) rather than explicitly sourced from this sub-model. Link those hydrogen demands to the hydrogen Production competition so that the cost and carbon intensity of the hydrogen actually used is consistent across the model.

## 15.4 Decarbonisation technologies

Decarbonising hydrogen means shifting the production mix away from unabated SMR toward capture-equipped and electrolytic routes. Because the product is identical, this is a pure

cost-and-carbon competition; Table 15.1 summarises maturity and Table 15.2 gives indicative costs.

**Table 15.1:** Indicative performance and maturity of the hydrogen production routes

| Route (colour)                     | Indicative benefit                              | Maturity                      |
|------------------------------------|-------------------------------------------------|-------------------------------|
| SMR + CCS (blue)                   | Captures ~90% of SMR CO <sub>2</sub>            | Commercial (Alberta projects) |
| Coal gasification + CCS (blue)     | Captures most gasification CO <sub>2</sub>      | Commercial                    |
| Electrolysis (green)               | Near-zero direct CO <sub>2</sub> on clean power | Commercial (scaling)          |
| Biomass gasification (green)       | Renewable feedstock; low CO <sub>2</sub>        | Demonstration                 |
| Biomass gasification + CCS (BECCS) | Net-negative emissions                          | Demonstration                 |
| Ethanol reforming (distributed)    | Low-CI where ethanol is low-CI                  | Niche / distributed           |
| SMR, unabated (grey)               | Incumbent; lowest cost, highest CO <sub>2</sub> | Commercial (to be displaced)  |

The cost estimates in Table 15.2 are order-of-magnitude figures from public literature (Natural Resources Canada 2024e; International Energy Agency 2019), expressed in 2025 Canadian dollars (US- and euro-denominated sources converted at roughly 1.37 CAD/USD and 1.48 CAD/EUR and adjusted to 2025). Levelised costs are per kilogram of hydrogen; electrolyser capital is per kilowatt of electrolyser. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 15.2:** Indicative techno-economic characteristics of hydrogen production routes, 2025 CAD

| Production route             | Levelised cost (2025 CAD/kg H <sub>2</sub> ) | Capex note                        | Energy / emission                                          |
|------------------------------|----------------------------------------------|-----------------------------------|------------------------------------------------------------|
| Grey SMR (incumbent)         | ~1.5–2.5                                     | mature, low capex                 | ~9–10 kg CO <sub>2</sub> /kg H <sub>2</sub>                |
| Blue SMR + CCS               | ~2.0–3.5                                     | SMR + capture (+30–50% equipment) | ~1–2 kg CO <sub>2</sub> /kg H <sub>2</sub> (~90% captured) |
| Blue coal gasification + CCS | ~2.5–4.0                                     | gasifier + capture                | mostly captured                                            |
| Green electrolysis           | ~4.8–6.3                                     | electrolyser ~1,400–2,700 CAD/kW  | ~0 direct; ~50–55 kWh/kg                                   |

| Production route               | Levelised cost (2025<br>CAD/kg H <sub>2</sub> ) | Capex note           | Energy / emission              |
|--------------------------------|-------------------------------------------------|----------------------|--------------------------------|
| Biomass gasification<br>(+CCS) | ~3–6                                            | gasifier (+ capture) | low / net-negative<br>with CCS |

#### 15.4.1 Blue hydrogen: SMR and gasification with CCS

The nearest-term low-carbon route keeps the established reforming or gasification process but adds carbon capture, cutting CO<sub>2</sub> by around 90%. It is the basis of Canada’s largest announced clean-hydrogen capacity, in Alberta, where low-cost natural gas and viable CO<sub>2</sub> storage favour it (Air Products 2024). In the model this is the SMR centralised + CCS, coal gasification + CCS, and biomass gasification + CCS routes, each requesting the shared CCS service.

#### 15.4.2 Green and net-negative hydrogen

Electrolysis produces hydrogen with no direct CO<sub>2</sub>; its footprint is the carbon intensity of the electricity, so it is genuinely “green” on Canada’s hydro and wind power but only as clean as the grid elsewhere. It is the most expensive route today, dominated by electricity cost and electrolyser capital, both of which are falling. Biomass gasification with CCS is distinctive: because the carbon is biogenic, capturing it yields net-negative hydrogen, one of the few production routes anywhere that removes CO<sub>2</sub> from the atmosphere.

TODO: Add a turquoise hydrogen route (methane pyrolysis, producing solid carbon instead of CO<sub>2</sub>) and represent the federal clean-hydrogen investment tax credit, both of which materially change the relative economics of the low-carbon routes.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

### 15.5 Economics

Technology choice at the production node follows the lifecycle-cost competition of Chapter 4. Each route carries a capital cost, fixed operating and maintenance cost, an output basis, service requests for the fuel, feedstock, electricity, infrastructure, and capture it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Hydrogen technologies are discounted at roughly 25% in the current data. As elsewhere in CIMS this is a **behavioural** (revealed or empirical) discount rate rather than a true financial

cost of capital, capturing the payback horizons and adoption barriers that govern real investment; CIMS applies it throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 25.

The emissions accounting is **life-cycle**, carrying CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O factors, so the upstream methane and the combustion and process CO<sub>2</sub> of each route are represented; the CCS service applies a 90% removal factor to the routes that request it, which can drive biomass-gasification hydrogen net-negative. Because the routes share an identical product, the carbon price acts directly on the production mix: as it rises, it raises the effective cost of grey SMR (through its large CO<sub>2</sub> factor) relative to the blue and green routes, shifting market share toward them. This makes hydrogen one of the clearest illustrations in CIMS of a carbon-price-driven technology transition.

## 15.6 Regional coverage and differences

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the hydrogen structure, but the economic route differs sharply by region. Alberta and Saskatchewan, with cheap natural gas and CO<sub>2</sub> storage, favour blue hydrogen; Québec, British Columbia, and Manitoba, with abundant low-carbon hydro electricity, favour green electrolytic hydrogen; and the Atlantic region's wind resource supports green production. The same competition therefore resolves to different winners in different regions, driven by local fuel and electricity prices.

TODO: Confirm the regional electricity and natural-gas prices and CO<sub>2</sub>-storage availability that drive the blue-versus-green split, since the regional production mix is highly sensitive to them.

## 15.7 Limitations and planned extensions

The hydrogen sub-model is compact but captures the essential structure of the production-route competition. Its main limitations are the loose coupling to the sectors that consume hydrogen, the absence of a turquoise (pyrolysis) route and of the investment tax credit, and reliance on inherited cost assumptions. The most important is the first: until the hydrogen demands in refining, ammonia, steel, and transport are explicitly sourced from this competition, the model can understate the system value of building clean-hydrogen supply. The TODOs above record these priorities.

TODO: Replace the illustrative figures and example discount rate in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled hydro-

gen production cost and carbon intensity against the Hydrogen Strategy for Canada and IEA data.

## 16 Mining

The mining sector is represented in CIMS by an energy-service flow model. The emphasis is on the structure of that model — the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them — rather than on the geology or economics of individual mines. The model tree is region-agnostic: a single master structure is instantiated for every province and territory, and regional differences are expressed through which branches carry production and through the data attached to each node, not through separate per-region structures.

### 16.1 Scope

CIMS represents both metal mining and non-metal mining. Metal mining is split into open-pit and underground operations, and within open-pit into iron and non-iron ores. Non-metal mining is dominated, in energy terms, by potash, which is modelled in full; other non-metal commodities are lower-energy but are within the scope of the sector and are noted below where they are not yet built out. The driving variable for the sub-model is the physical throughput of ore, expressed in tonnes; a forecast of ore throughput drives the demand for every downstream process and energy service, and is set exogenously in the same way as the output of the other industrial branches.

Several activities that are colloquially called “mining” are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- Coal mining is covered in the [Coal Mining](#) chapter and the energy-supply component.
- Oil and gas extraction and oil sands mining are covered under [Petroleum Crude](#).
- Cement, lime, glass, brick, and other non-metallic *mineral products* are covered under [Industrial Minerals](#) — these are downstream manufacturing branches, distinct from the extraction of the non-metal ores treated here.

The smelting and refining of mined concentrates is likewise excluded and handled in the [Metal Smelting and Refining](#) chapter; the mining sub-model stops at the mine gate, with a saleable concentrate or product as its final output.

## 16.2 Real-world context

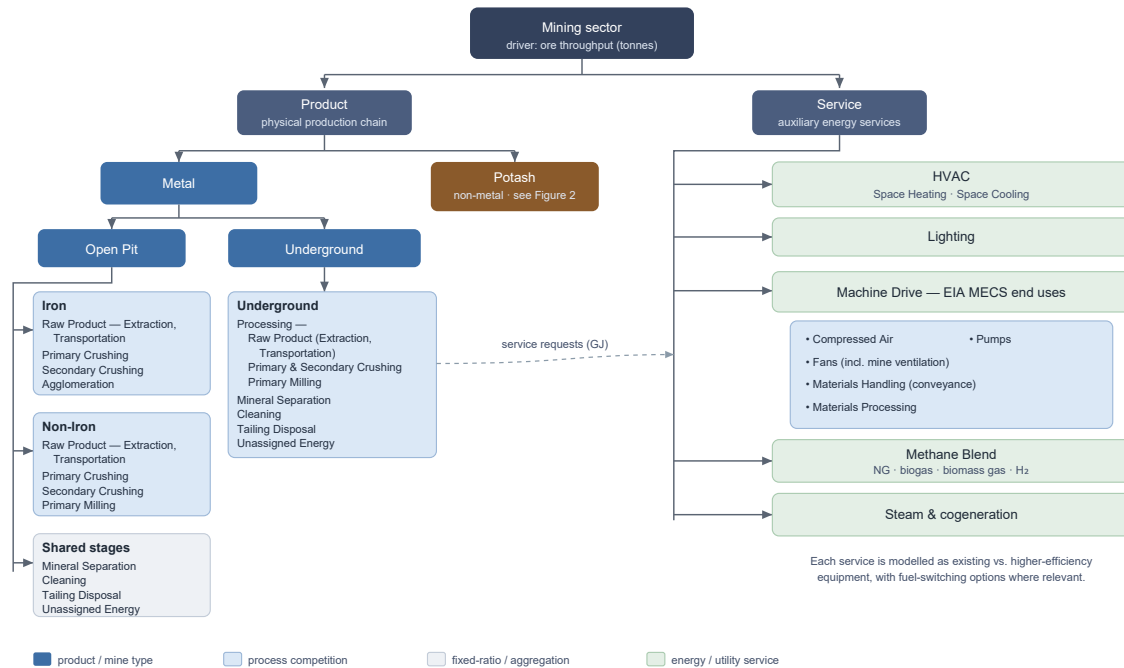
Canada is one of the world’s leading mining nations, and the industry is highly regionalised. Metal mines produce base and precious metals — copper, nickel, zinc, gold, iron ore, and others — from both open-pit and underground operations. Saskatchewan is the world’s largest potash-producing region, accounting on the order of 30% of global potash production (Natural Resources Canada 2025b), and also supplies roughly a quarter of the world’s uranium from the high-grade deposits of the Athabasca Basin (World Nuclear Association 2025). The Northwest Territories host Canada’s diamond industry, and Nunavut has growing gold and iron-ore production. Mining is capital-intensive and strongly export-oriented, so activity levels are driven by world commodity prices and ore grades rather than by domestic demand — which is why CIMS drives the sector with an exogenous throughput forecast rather than an endogenous demand response.

Mining is energy-intensive, and the energy is concentrated in a few activities. Comminution — the crushing, grinding, and milling that reduce run-of-mine rock to particles fine enough for mineral separation — is typically the single largest energy consumer at a metal mine; published estimates put it at roughly a quarter of final energy at an average mine site and a majority of mine *electricity* use (Coalition for Eco-Efficient Comminution 2021). Diesel dominates mobile equipment, particularly haulage, and is the largest single fuel at many sites; many remote northern mines are off-grid and rely heavily on diesel generation for all loads. Ventilation is a large and distinctively underground electricity load, since deep mines must move enormous volumes of air for air quality and temperature control. The CIMS structure is built around exactly these distinctions: it separates open-pit from underground mining, breaks the ore-processing chain into its comminution stages, and represents ventilation, fans, pumping, and the other utility loads through the shared machine-drive service (see the [Industrial Machine Drive Systems](#) chapter).

## 16.3 Model structure

Following the standard CIMS pattern, the sector divides into two sub-trees: a **Product** sub-tree, which is the physical production chain that turns ore into a saleable product, and a **Service** sub-tree, which holds the auxiliary energy and utility services that the production technologies draw on. Figure 16.1 shows the overall structure.

As described in the [Sector Overview](#), each node carries a competition rule. **Fixed-ratio** nodes impose the unavoidable engineering sequence of process steps, while **tech-compete** nodes host competitions among alternative technologies on the basis of lifecycle cost (see Chapter 4). In the mining tree, the disaggregation of products and process stages is built from fixed-ratio nodes, and the technology choices — which extraction method, which crusher, which boiler, which class of motor — occur at the tech-compete leaves.



**Figure 16.1:** Energy-service flow model of the mining sector (region-agnostic master tree). The sector divides into a Product sub-tree — metal (open-pit iron and non-iron, and underground) and potash — and a Service sub-tree of auxiliary energy services structured on the EIA MECS end uses.

### 16.3.1 Products: metal mining

Metal mining divides first into Open Pit and Underground, reflecting the substantial difference in their energy profiles (most importantly the ventilation load that underground operations carry). Open-pit mining is then split into an Iron path and a Non-Iron path, because iron ore and base/precious-metal ores follow different processing routes:

- The Iron path runs ore through a Raw Product stage (Extraction and Transportation), Primary and Secondary Crushing, and an Agglomeration stage that binds fine iron concentrate into a form suitable for downstream ironmaking.
- The Non-Iron path runs ore through Raw Product (Extraction, Transportation), Primary and Secondary Crushing, and Primary Milling (fine grinding), but no agglomeration.

Both open-pit paths feed a set of shared stages — Mineral Separation, Cleaning (concentrate drying), Tailing Disposal, and an Unassigned Energy node. Underground mining carries the same processing chain (grouped under a Processing node: Raw Product, Primary and Sec-

ondary Crushing, Primary Milling) together with its own Mineral Separation, Cleaning, Tailings Disposal, and Unassigned Energy nodes.

The Unassigned Energy node is a reconciliation device: it carries metered sector energy that has not yet been allocated to a specific process stage, so that the modelled total matches reported energy use while the more detailed allocation is still being developed.

TODO: Progressively reassign the Unassigned Energy at each mine type to specific process stages and technologies, and reduce it toward zero as the bottom-up representation improves.

### 16.3.2 Products: non-metal mining

Non-metal mining is represented in the same Product sub-tree as a sibling of Metal. The dominant non-metal commodity by energy use is potash, whose structure is shown in Figure 16.2. Potash production is met from two distinct mining methods that have markedly different energy profiles, and the model represents both:

- Conventional mining — underground extraction (at depths around 1,000 m), with its own Extraction, Transportation, Crush-and-Screen, and Dissolution stages.
- Solution mining — used for deeper deposits (beyond roughly 1,500 m), where the ore is dissolved in situ and pumped to surface, with a simpler Extraction and Transportation chain.

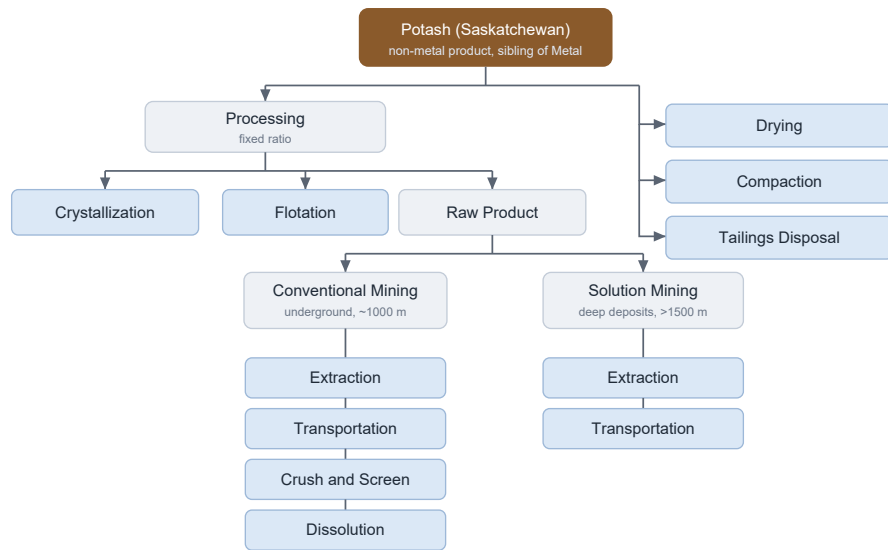
Both methods feed a shared surface-processing and finishing train — Crystallization and Flotation (under Processing), followed by Drying, Compaction, and Tailings Disposal — and, like the metal mines, both draw on the cross-cutting energy and utility services.

Canada's other non-metal mining is smaller in energy terms but is within the scope of this sector and is included here for completeness. The most significant omissions are uranium and diamonds, both of which are material parts of the Canadian mining industry even though their energy intensity is lower than that of metal milling.

TODO: Add uranium mining (Saskatchewan, Athabasca Basin — e.g. McArthur River and Cigar Lake). Saskatchewan supplies roughly a quarter of world uranium; it is currently absent from the sector.

TODO: Add diamond mining (Northwest Territories) and other northern non-metal mining, once the corresponding regional data are in place.

TODO: Confirm whether lower-energy non-metal extraction such as salt, gypsum, and aggregates should be represented explicitly or remain aggregated, given their small share of mining energy.



Conventional and solution mining feed shared processing (crystallization, flotation) and finishing (drying, compaction, tailings disposal); both draw on the cross-cutting energy and utility services shown in Figure 1.

**Figure 16.2:** Potash flow model. Potash is a non-metal product sibling of Metal; conventional (underground) and solution mining feed a shared processing and finishing train.

### 16.3.3 Auxiliary energy and utility services

Mining’s production technologies draw on the shared **Service** sub-tree rather than specifying their own utility equipment. These services are common to the industrial sectors and are documented in their own chapters in the General Services part; their general structure — and the “existing versus more efficient equipment, with fuel-switching where relevant” convention they follow — is set out in the [Sector Overview](#). This chapter does not repeat that detail. The services the mining sector draws on are:

- **Machine Drive** — see the [Industrial Machine Drive Systems](#) chapter. Mining’s largest electrical loads sit here: comminution under Materials Processing, mine ventilation under Fans, and haulage and conveyance under Materials Handling, alongside Pumps and Compressed Air.
- **Steam** — see [Low-Temperature Industrial Heat](#), which covers boiler-raised steam and its fuel options.
- **Cogeneration** — see [Cogeneration](#) for combined heat-and-power.
- **Methane Blend** — the gaseous-fuel supply node, where natural gas competes with bio-gas, biomass gasification, and hydrogen.
- **HVAC** (space heating and cooling) and **Lighting** for mine facilities.

TODO: The current regional input files still implement an earlier, more detailed auxiliary structure (per-horsepower machine-drive classes; separate pumping, compression, ventilation, and conveyance sub-trees) and predate the Product/Service split. Migrate these files to the region-agnostic master tree and the shared General Services chapters referenced above.

## 16.4 Decarbonisation technologies

Mining's emissions are dominated by diesel in mobile equipment and by the electricity and combustion that drive comminution, process heat, and ventilation. The main decarbonisation levers therefore sit at the Extraction and Transportation nodes (mobile equipment), the comminution stages (Crushing, Milling), Cleaning (concentrate drying), and underground ventilation, alongside the shared services documented in the General Services chapters. The technologies below are admitted to the relevant tech-compete nodes; representative performance and maturity are summarised in Table 16.1. The figures quoted are indicative ranges from industry and techno-economic reviews (Mungyeko Bisulandu et al. 2026), not the calibrated values used in the model.

**Table 16.1:** Indicative performance and maturity of key mining decarbonisation technologies

| Technology                                     | Indicative benefit                                                                      | Maturity                      |
|------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------|
| Trolley-assist diesel-electric haulage         | 40–60% energy saving on loaded climbs; ~28–36% CO <sub>2</sub> cut on a low-carbon grid | Commercial                    |
| Battery-electric haul truck                    | Eliminates tailpipe diesel; range / charging constrained                                | Early commercial              |
| Hydrogen fuel-cell haul truck                  | Zero tailpipe emissions; H <sub>2</sub> supply constrained                              | Demonstration                 |
| Biodiesel / renewable diesel                   | Drop-in lifecycle CO <sub>2</sub> reduction, no new equipment                           | Commercial                    |
| High-pressure grinding rolls (HPGR)            | ~20–30% lower downstream grinding energy                                                | Commercial                    |
| Vertical / stirred mills                       | ~40% energy saving vs ball mills; integral drying                                       | Commercial                    |
| Ore sorting / pre-concentration                | Rejects barren rock; lowers energy per unit metal                                       | Commercial (deposit-specific) |
| Ventilation on demand (VOD)                    | Up to ~50% ventilation energy saving                                                    | Commercial                    |
| Electric / heat-pump drying and boilers        | Displaces gas- and oil-fired process heat                                               | See Low-Temp Heat             |
| High-efficiency motors + variable-speed drives | Lower auxiliary drive energy                                                            | See Machine Drive             |

#### 16.4.1 Mobile equipment and haulage

Diesel haulage is the largest single diesel load at most surface mines. Trolley-assist systems feed the truck's electric drive from an overhead catenary on the energy-intensive loaded uphill segments, cutting energy use on those segments by roughly 40–60% and lifecycle CO<sub>2</sub> by about 28–36% where the connected grid is low-carbon; the technology is commercial and deployed at several open-pit operations (Mungyeko Bisulandu et al. 2026). Battery-electric haul trucks, now offered by the major equipment makers, remove tailpipe diesel entirely but are constrained by range and charging time and remain early-commercial; hydrogen fuel-cell haul trucks are at the demonstration stage. Biodiesel and renewable diesel are drop-in fuels that cut lifecycle CO<sub>2</sub> without new equipment and are already represented in the tree (the “Extraction Biodiesel” and “Biodiesel truck” technologies). In CIMS these options compete at the Extraction and Transportation nodes.

#### 16.4.2 Comminution efficiency

Comminution is the single largest electricity consumer at a metal mine, so efficiency gains here have an outsized effect. High-pressure grinding rolls (HPGR) pre-fracture ore between counter-rotating rolls and reduce downstream grinding energy by about 20–30%. Vertical roller and stirred mills can deliver on the order of 40% energy savings relative to conventional ball mills and combine grinding with drying in a single unit. Ore sorting and pre-concentration reject barren rock before milling, lowering the energy spent per unit of contained metal, though their applicability is deposit-specific (Mungyeko Bisulandu et al. 2026). These compete at the Crushing and Milling nodes, while ore sorting acts as a service-side reduction in the feed sent to milling.

#### 16.4.3 Electrified and low-carbon process heat

Concentrate drying (Cleaning) and process steam are met largely by combustion today. The low- and zero-emission alternatives — electric and heat-pump drying, electrode and electric-resistance boilers, biomass and hydrogen boilers, and mechanical vapour recompression — are described in the [Low-Temperature Industrial Heat](#) chapter and reached through the **Steam** service; the **Methane Blend** node additionally substitutes biogas, biomass gas, and hydrogen for natural gas in any remaining gas-fired duty.

#### 16.4.4 Ventilation and auxiliary drive (underground)

Ventilation can account for roughly 40–50% of an underground mine's energy use. Ventilation on demand (VOD) matches airflow to where vehicles and crews are actually working and can cut ventilation energy by up to about 50% (ABB 2024); high-efficiency fans and motors with

variable-speed drives further reduce auxiliary drive energy. These are represented through the Fans and other machine-drive end uses described in the [Industrial Machine Drive Systems](#) chapter.

#### 16.4.5 Potash-specific options

Conventional potash mining already relies on electric continuous miners and electric hoisting, so its decarbonisation depends mainly on grid emissions intensity and on electrifying surface haulage; VOD applies in the underground workings. Solution mining and the surface Drying and Compaction stages are heat-intensive and decarbonise through the same low-carbon heat options as above — in particular mechanical vapour recompression for crystallization and evaporation duties, and electric or biomass boilers for drying.

#### 16.4.6 Indicative capital and operating costs and energy use

Table 16.2 gathers order-of-magnitude capital and operating costs and energy characteristics for the main comminution and mobile-equipment technologies. The estimates are drawn from public techno-economic literature (Coalition for Eco-Efficient Comminution 2021; Mungyeke Bisulandu et al. 2026; SRK Consulting 2024) and expressed in 2025 Canadian dollars: US-dollar sources were converted at the mid-2025 rate of roughly 1.37 CAD/USD and adjusted to 2025 where the source year differed. Because mining capital is dominated by site-specific concentrators and mobile fleets, the figures are given per piece of equipment or relative to a baseline rather than per tonne of capacity where that is more meaningful; energy intensity is per tonne of ore. Fixed O&M follows the heavy-industry convention of roughly 4–6% of capital cost per year. The values are for relative comparison only, not project budgeting, and are separate from the plant-specific values the model carries internally.

**Table 16.2:** Indicative techno-economic characteristics of mining technologies, 2025 CAD

| Technology                                | Overnight capex<br>(2025 CAD)                          | Fixed O&M | Energy intensity                                      | Main energy<br>carrier |
|-------------------------------------------|--------------------------------------------------------|-----------|-------------------------------------------------------|------------------------|
| SAG / ball-mill<br>comminution            | site-specific<br>concentrator<br>capital               | ~4–6%/yr  | ~20–60 kWh/t<br>ore ( $\approx 0.07$ – $0.2$<br>GJ/t) | electricity            |
| High-pressure<br>grinding rolls<br>(HPGR) | $\approx$ SAG circuit<br>( $\pm$ ~10 M per<br>circuit) | ~4%/yr    | ~10–35% less<br>than SAG                              | electricity            |
| Vertical / stirred<br>mills               | moderate                                               | ~4%/yr    | ~40% less than<br>ball mills                          | electricity            |

| Technology                                          | Overnight capex<br>(2025 CAD)                 | Fixed O&M                          | Energy intensity                            | Main energy<br>carrier |
|-----------------------------------------------------|-----------------------------------------------|------------------------------------|---------------------------------------------|------------------------|
| Diesel haul truck<br>( $\approx 150\text{--}290$ t) | $\sim 2\text{--}3.4$ M per truck              | high fuel cost                     | diesel                                      | diesel                 |
| Battery-electric haul truck                         | $\sim 4\text{--}7$ M per truck<br>(+100–150%) | $\sim 65\%$ lower cost per t moved | electricity                                 | electricity            |
| Hydrogen fuel-cell haul truck                       | higher than battery-electric (demonstration)  | —                                  | hydrogen                                    | hydrogen               |
| Trolley-assist (diesel-electric)                    | overhead catenary infrastructure              | —                                  | 40–60% energy saving on loaded climbs       | grid electricity       |
| Ventilation on demand (VOD)                         | retrofit controls (low)                       | —                                  | up to $\sim 50\%$ ventilation energy saving | electricity            |

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

## 16.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, a service request for the energy or upstream service it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

A few sector-wide conventions are worth noting. Mining technologies are discounted at a high rate (35% in the current data). These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital: they sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions, and CIMS applies this behavioural rate throughout the lifecycle-cost calculation (see the discount-rate discussion in the [Theory](#) chapter and Chapter 4). **Heterogeneity (variance) parameters** at the process nodes are set so that competitions respond to cost differences without collapsing to winner-take-all. Fuel prices entering the service-cost calculation are scaled by a sector-level

block of price multipliers, allowing the mining sector to face prices that differ from the generic delivered price of each fuel.

Greenhouse-gas costs enter through the fuels: combustion emissions follow from each technology's fuel service requests and the prevailing carbon price, so fuel-switching — from diesel haulage to biodiesel or electric, or from natural gas to hydrogen or biogas in the methane blend — is the model's main channel for emissions reduction.

TODO: Mining has no carbon-capture (CCS) service in the current tree (see the [Carbon Capture, Utilization, and Storage](#) chapter), on the assumption that its emissions are predominantly combustion-related and dispersed. Confirm this is appropriate, and add process-emissions representation if any mined commodity warrants it.

## 16.6 Regional coverage and differences

Because the tree is region-agnostic, every province and territory is built from the same master structure, and a region simply carries no production on the branches that do not apply to it. Regional differences therefore show up in *where* production sits rather than in the shape of the tree:

- Potash production is concentrated in Saskatchewan, which carries essentially all of the potash branch.
- Iron ore is concentrated in the Atlantic region (the Labrador Trough, on the Québec-Newfoundland border) and Québec, which carry most of the open-pit iron path.
- Base- and precious-metal mining is spread across British Columbia, Ontario, Manitoba, Québec, and the Atlantic region (for example nickel in Manitoba and Ontario, and copper and gold in British Columbia), populating the non-iron and underground branches.
- Territorial mining — diamonds in the Northwest Territories and gold and iron ore in Nunavut — populates the metal and (future) non-metal branches in those regions.

TODO: Several regions are not yet populated in the current input files. Build out mining throughput and technology data for Alberta and the three territories (the territories are not yet represented anywhere in CIMS — see the Architecture chapter), so that the region-agnostic tree is exercised everywhere it should be.

TODO: The Atlantic, Manitoba, Ontario, and Québec metal mines are currently represented generically rather than by commodity. Consider whether the Iron / Non-Iron split, plus any commodity-specific energy intensities, adequately capture the real mix in each region, and refine where the averaging is too coarse.

## 16.7 Limitations and planned extensions

The mining sub-model is a deliberately simplified representation. The TODOs above record the main gaps between this design and both the current input files and the real Canadian mining sector: the auxiliary-service migration to the MECS structure; the reassignment of Unassigned Energy; the addition of uranium, diamonds, and the smaller non-metal commodities; full provincial and territorial coverage including Alberta and the territories; and the question of commodity-level detail within metal mining. Together these define the path from the current state of the model to the region-agnostic, fully populated design described in this chapter.

TODO: Replace the illustrative real-world figures and the example discount-rate value in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled mining energy use against CIEEDAC / Natural Resources Canada data.

## 17 Industrial Minerals

The industrial minerals sector — the non-metallic mineral products industry — is represented in CIMS by an energy-service flow model. The emphasis is on the structure of that model: the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them, rather than the chemistry or economics of individual plants. The model tree is region-agnostic: a single master structure is instantiated for every province, and regional differences are expressed through which branches carry production and through the data attached to each node. The sector is defined by high-temperature thermal processing — calcining, clinkering, and glass melting — and, like chemicals, a large share of its emissions are process emissions released by the chemistry of calcination rather than by burning fuel. That single fact shapes both the sub-model and its decarbonisation options.

### 17.1 Scope

CIMS represents the four energy-intensive branches of the non-metallic mineral products industry: cement, lime, glass, and bricks. Cement and lime dominate the sector's energy and emissions because both depend on calcining carbonate rock at high temperature. The driving variable is the physical output of product in tonnes; a forecast of demand — closely tied to construction activity — drives every upstream process and energy service and is set exogenously, as for the other industrial branches.

Several adjacent activities are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- Quarrying and extraction of the raw minerals (limestone, silica, clay, aggregates) are part of the [Mining](#) sector and the upstream supply; the industrial-minerals sub-model begins with raw-material preparation inside the plant gate.
- Concrete batching and downstream fabrication (ready-mix concrete, precast products, finished glazing) are downstream manufacturing and are not part of this sub-model, which ends with cement, lime, glass, and brick as saleable products.
- Fuel and electricity supply enter through the energy-supply sub-models and the shared fuel nodes.

## 17.2 Real-world context

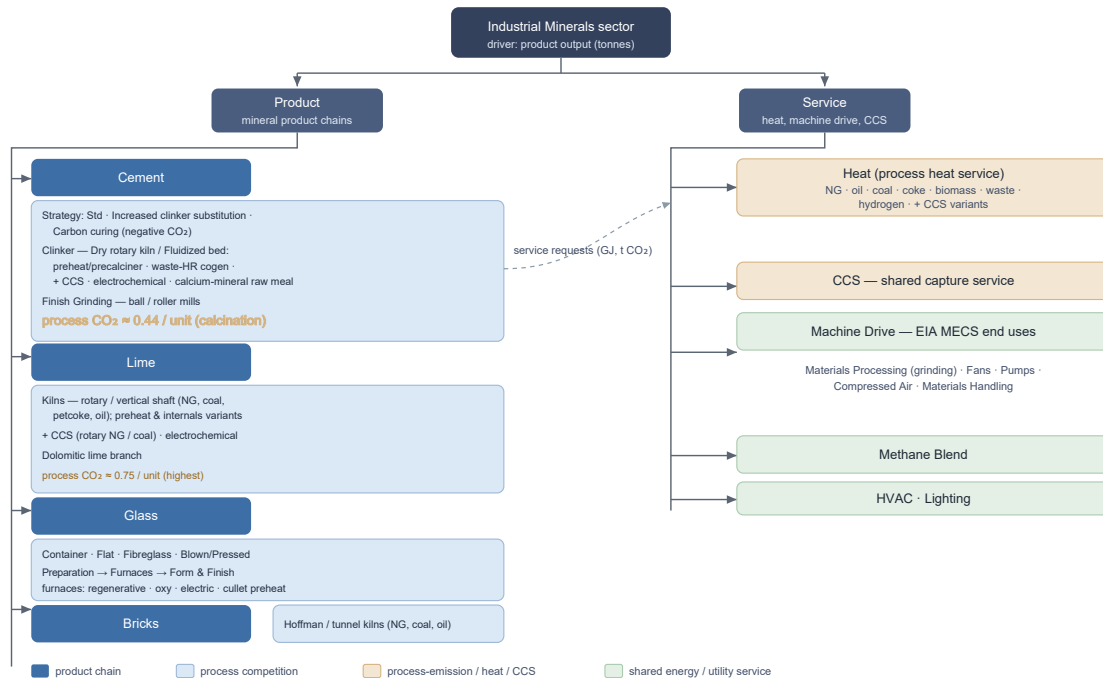
The industrial minerals sector is small in tonnage relative to its emissions, because calcination releases CO<sub>2</sub> as an unavoidable part of the chemistry. Canadian cement manufacturing alone emitted about 11.2 Mt CO<sub>2</sub> in 2019, roughly 1.5% of national emissions, and calcination accounts for some 60–65% of the CO<sub>2</sub> from clinker production — released when limestone (CaCO<sub>3</sub>) is broken down into lime (CaO) and CO<sub>2</sub> — with the rest coming from the fuel burned to reach clinkering temperature (Cement Association of Canada and Innovation, Science and Economic Development Canada 2022; International Energy Agency 2024b). Lime production is chemically the same calcination step without the clinkering, and carries an even higher process-CO<sub>2</sub> intensity per tonne of product. Glass is melted rather than calcined, so its emissions are dominated by furnace fuel and by the carbonate batch materials, and brick is fired clay.

Production is concentrated in the populous provinces and tracks construction demand, so unlike most heavy industry it is growing in the model. Ontario is the largest producer, followed by Québec and a fast-growing Alberta, with British Columbia and the Atlantic region smaller. Because so much of the sector's CO<sub>2</sub> is intrinsic to the chemistry, deep decarbonisation depends on options that most other sectors do not need — carbon capture on the kiln, substitution of clinker with other materials, alternative raw meals that avoid carbonate, and CO<sub>2</sub> mineralisation (carbon curing) — alongside the fuel-switching and efficiency levers common to all industry. The CIMS structure reflects this: it carries explicit process-emission factors on the clinker and lime nodes, a shared CCS service, and a family of clinker-substitution, alternative-fuel, electrified-kiln, and carbon-curing options.

## 17.3 Model structure

Following the standard CIMS pattern, the sector divides into a **Product** sub-tree, holding the cement, lime, glass, and brick production chains, and a **Service** sub-tree, holding the auxiliary energy and utility services including a sector-level **Heat** service and the shared **CCS** service. Figure 17.1 shows the overall structure.

As described in the [Sector Overview](#), each node carries a competition rule. **Fixed-ratio** nodes impose the unavoidable engineering sequence of process steps, while **tech-compete** nodes host competitions among alternative technologies on the basis of lifecycle cost (see Chapter 4). In the industrial-minerals tree, the disaggregation into products and process stages is built from fixed-ratio nodes, while the technology choices — which kiln, which furnace, which fuel, which clinker route — occur at the tech-compete leaves.



**Figure 17.1:** Energy-service flow model of the industrial minerals sector (region-agnostic master tree). The Product sub-tree holds cement, lime, glass, and brick chains; the Service sub-tree holds a sector-level Heat service, the MECS-aligned Machine Drive, and the shared CCS service that the calcination routes draw on.

### 17.3.1 Products: cement

Cement is the most important chain. At the top, a Cement competition selects the overall production strategy — conventional Std process, Increased clinker substitution (replacing clinker with supplementary cementitious materials), and Carbon curing (curing concrete by mineralising CO<sub>2</sub>, which the model treats as a net-negative emission). Beneath this, Portland cement resolves into:

- **Clinker** — the energy-and-emissions core, produced in a Dry process rotary kiln or a Fluidized bed. The rotary-kiln node competes a wide range of routes: conventional and efficiency-retrofitted kilns, new preheater and precalciner designs, waste-heat-recovery cogeneration, a new preheat Eff with CCS route, an electrochemical synthesis (electrified clinker) route, and a Std process with calcium minerals route that uses a non-carbonate raw meal to avoid process CO<sub>2</sub> altogether. The fluidized-bed node competes a range of fuels — natural gas, coal, petcoke, distillate oil, and waste-derived and hazardous-waste fuels.

- **Finish Grinding** — ball mills and roller mills with high-efficiency separators that grind clinker into cement.
- **Dry Ground Cement** — raw-meal preparation.

The clinker routes carry an explicit process-CO<sub>2</sub> factor (about 0.44 t per unit in the data) reflecting calcination; the calcium-mineral and carbon-curing options are the model's way of breaking that link.

### 17.3.2 Products: lime, glass, and bricks

- **Lime** is produced in Kilns (and a Dolomitic branch) competing rotary and vertical-shaft kilns on a range of fuels (NG, coal, petcoke, residual oil), with preheater and internals variants for efficiency, CCS-equipped rotary kilns (NG and coal), and an electrochemical synthesis route. Lime carries the sector's highest process-CO<sub>2</sub> factor (about 0.75 t per unit).
- **Glass** divides into Container, Flat, Fibreglass, and Blown/Pressed glass, each a chain of Preparation, Furnaces, and Form and Finish (flat glass adds Temper; blown/pressed adds Fire Polish). The furnace competitions are where the energy and emissions are decided, and they include regenerative and oxy-boosted melters, fluidized-bed cullet/batch preheaters, advanced glass melters, and all-electric melters (including fluidized-bed electric melters).
- **Bricks** are fired in Hoffman or tunnel kilns on natural gas, coal, or residual oil.

### 17.3.3 Services, including process heat and CCS

The services follow the standard pattern and are documented in their own chapters in the General Services part; their structure is set out in the [Sector Overview](#). Two are worth noting here:

- **Heat** — a sector-level process-heat service whose technology set is unusually broad, competing natural-gas (standard and efficient), oil, coal, coke, biomass, waste-fuel, hydrogen, and CCS-equipped (NG and coal) heat supply. This is a key fuel-switching lever for the non-calcination heat demand.
- **CCS** — the shared carbon-capture service that the CCS-equipped clinker, lime, and heat routes request; its broader treatment is in the [Carbon Capture, Utilization, and Storage](#) chapter.

The remaining services — **Machine Drive** (see [Industrial Machine Drive Systems](#)), **Methane Blend**, **HVAC**, and **Lighting** — are the cross-cutting utilities common to all sectors.

TODO: The current regional input files still implement an earlier, more detailed auxiliary structure (six per-horsepower machine-drive size classes; separate Compression, Conveyance,

Pumping, Ventilation, and Direct Drive sub-trees) and predate the Product/Service split and the MECS-aligned Machine Drive grouping. They also name the product branch `Products` rather than the master tree's `Product`. Migrate these files to the region-agnostic master tree and the shared General Services chapters.

## 17.4 Decarbonisation technologies

Because most of the sector's CO<sub>2</sub> comes from calcining carbonate rock, its decarbonisation problem is unusual: efficiency and fuel-switching can only reach the combustion share, so deep cuts require capturing or avoiding the process CO<sub>2</sub> as well. Roughly the first 30–40% of cement emissions can be abated with clinker substitution, alternative fuels, and efficiency, while the remaining 60–70% — the calcination CO<sub>2</sub> — needs CCUS or alternative production chemistry (U.S. Department of Energy 2023; Cement Association of Canada and Innovation, Science and Economic Development Canada 2022). The model represents the full ladder. Table 17.1 summarises maturity and Table 17.2 gives indicative costs.

**Table 17.1:** Indicative performance and maturity of key industrial minerals decarbonisation technologies

| Technology                                     | Indicative benefit                                                        | Maturity                           |
|------------------------------------------------|---------------------------------------------------------------------------|------------------------------------|
| Clinker substitution / SCMs (incl. LC3)        | ~20–40% less clinker CO <sub>2</sub> per tonne cement                     | Commercial                         |
| Alternative & waste-derived fuels              | Displaces fossil kiln fuel; up to ~30% combustion CO <sub>2</sub>         | Commercial                         |
| Preheater / precalciner & efficiency retrofits | Lower kiln fuel use                                                       | Commercial                         |
| Waste-heat-recovery cogeneration               | On-site power from kiln waste heat                                        | Commercial                         |
| Calcium-mineral (non-carbonate) raw meal       | Eliminates the calcination process CO <sub>2</sub>                        | Pilot / demonstration              |
| Carbon curing / CO <sub>2</sub> mineralisation | Stores CO <sub>2</sub> in concrete (net-negative in model)                | Early commercial                   |
| Cement / lime kiln + CCS (CCUS)                | Captures ~60% of total; most of process CO <sub>2</sub>                   | Demonstration                      |
| Electrochemical / electrified clinker & lime   | Removes combustion CO <sub>2</sub> ; enables pure-CO <sub>2</sub> capture | Pilot                              |
| Electric & oxy-fuel glass melters              | Electrifies / concentrates glass-furnace CO <sub>2</sub>                  | Commercial (electric, small scale) |
| Cullet / batch preheating (glass)              | Lower furnace energy                                                      | Commercial                         |

| Technology                      | Indicative benefit              | Maturity          |
|---------------------------------|---------------------------------|-------------------|
| Biomass / hydrogen process heat | Low-carbon non-calcination heat | See Low-Temp Heat |

The cost estimates in Table 17.2 are order-of-magnitude figures from public techno-economic literature (U.S. Department of Energy 2023; International Energy Agency 2024b; Cement Association of Canada and Innovation, Science and Economic Development Canada 2022), expressed in 2025 Canadian dollars (US- and euro-denominated sources converted at roughly 1.37 CAD/USD and 1.48 CAD/EUR and adjusted to 2025). Capture and avoidance costs are given per tonne of CO<sub>2</sub>; incremental capital is relative to a conventional plant. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 17.2:** Indicative techno-economic characteristics of industrial minerals decarbonisation technologies, 2025 CAD

| Decarbonisation technology         | Incremental capex                                 | Operating cost / cost of CO <sub>2</sub>                     | Energy / input                       |
|------------------------------------|---------------------------------------------------|--------------------------------------------------------------|--------------------------------------|
| Clinker substitution / SCMs (LC3)  | low (grinding, blending)                          | often cost- <i>neutral</i> or saving                         | calcined clay / limestone / pozzolan |
| Alternative & waste-derived fuels  | low–moderate (feed handling)                      | often a fuel-cost <i>saving</i>                              | waste, biomass, RDF                  |
| Preheater / precalciner efficiency | moderate                                          | fuel saving                                                  | —                                    |
| Carbon curing / mineralisation     | moderate                                          | ~CAD 40–110 / t CO <sub>2</sub> stored                       | CO <sub>2</sub> + concrete           |
| Cement kiln + CCS                  | +30–50% plant equipment                           | ~CAD 100–200 / t CO <sub>2</sub> avoided                     | steam / electricity for capture      |
| Electrified / electrochemical kiln | high (new process)                                | ~CAD 130 / t CO <sub>2</sub> avoided (≈87 €/t clinker added) | low-carbon electricity               |
| Lime kiln + CCS                    | +30–50% plant equipment                           | ~CAD 110–200 / t CO <sub>2</sub> avoided                     | steam / electricity                  |
| Electric glass melter              | lower capex than regenerative; higher energy cost | electricity-priced melt                                      | electricity                          |
| Oxy-fuel glass melter              | moderate (oxygen plant)                           | fuel saving; CO <sub>2</sub> concentration                   | oxygen + gas                         |

#### 17.4.1 Reducing and substituting clinker

The cheapest cement reductions come from making less clinker per tonne of cement. Supplementary cementitious materials (SCMs) — slag, fly ash, calcined clay, limestone, and natural pozzolans — replace a share of clinker; limestone calcined clay cement (LC3) can cut cement CO<sub>2</sub> by around 40% and is commercial (U.S. Department of Energy 2023). The model represents this as the Increased clinker substitution route at the Cement node. A complementary lever is the Std process with calcium minerals clinker route, which uses a non-carbonate raw meal so that calcination releases no CO<sub>2</sub> at all.

#### 17.4.2 Alternative fuels, efficiency, and waste-heat recovery

The combustion share of kiln emissions is addressed by alternative and waste-derived fuels (biomass, refuse-derived and hazardous-waste fuels in the fluidized-bed routes and the Heat service), by preheater/precaliner designs and efficiency retrofits that cut fuel use, and by waste-heat-recovery cogeneration that turns kiln exhaust heat into on-site power. These are mature and often cost-saving, but they cannot touch the calcination CO<sub>2</sub>.

#### 17.4.3 Carbon capture, electrification, and mineralisation

Deep cuts require attacking the process CO<sub>2</sub>. The model offers three routes: CCS on the kiln (cement and lime), which captures most of the CO<sub>2</sub> but adds 30–50% more plant equipment and a capture energy penalty; electrochemical/electrified clinker and lime, which both remove the combustion CO<sub>2</sub> and produce a purer CO<sub>2</sub> stream that is cheaper to capture; and carbon curing, which mineralises CO<sub>2</sub> into concrete and is the one route the model scores as net-negative. On the glass side, electric and oxy-fuel melters and cullet preheating electrify or concentrate furnace emissions. Low-carbon process heat — biomass, hydrogen, and CCS-equipped boilers in the Heat service — addresses the non-calcination thermal demand, drawing on the [Low-Temperature Industrial Heat](#) chapter.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

TODO: The model represents clinker substitution as a single aggregate route. Consider distinguishing the main SCM pathways (slag, fly ash, calcined clay/LC3, limestone) since their availability and CO<sub>2</sub> benefit differ markedly and fly-ash/slag supply is constrained as other heavy industries decarbonise.

## 17.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, service requests for the energy and upstream services it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Industrial minerals technologies are discounted at a high rate — 35% at most process nodes in the current data, with some auxiliary and capital-intensive nodes at 40–50%. These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital. They sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions; CIMS applies this behavioural rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** at the process nodes are mostly set around 10 so that competitions respond to cost differences without collapsing to winner-take-all.

The sector's emissions accounting is distinctive. Combustion emissions follow from each technology's fuel service requests and the carbon price, as in the other sectors. In addition, the clinker and lime technologies carry explicit process-emission factors (about 0.44 and 0.75 t CO<sub>2</sub> per unit) representing calcination, applied independently of fuel use; the carbon-curing route carries a net-negative emission factor; and the **CCS** service applies an emissions-removal factor to the technologies that request it. Switching among conventional, substituted, alternative-fuel, captured, and electrified production is therefore the model's main channel for emissions reduction, driven by the interaction of fuel and material costs with a carbon price that applies to both the combustion and the process CO<sub>2</sub>.

## 17.6 Regional coverage and differences

Because the tree is region-agnostic, each modelled province is built from the same master structure and carries no production on branches that do not apply. Five regions are represented — Alberta, the Atlantic region, British Columbia, Ontario, and Québec — and differences show up in scale and growth rather than in the shape of the tree:

- Ontario is the largest producer and grows over the projection.
- Québec and a fast-growing Alberta follow, reflecting construction-driven demand.
- British Columbia and the Atlantic region are smaller.

Unlike most heavy industry, modelled output rises in every region, because cement, lime, and glass demand tracks population and construction rather than declining commodity markets.

TODO: Only five provinces are represented (Saskatchewan and the territories are absent). Confirm whether cement, lime, or glass capacity elsewhere should be added, and build it out on the region-agnostic tree.

## 17.7 Limitations and planned extensions

The industrial minerals sub-model is detailed on the product side but carries the same structural debts as the other older sector files. The TODOs above record the main gaps: the migration of the auxiliary services to the MECS-aligned Machine Drive and the Product/Service split; the disaggregation of clinker substitution into specific SCM pathways; and broader provincial coverage. The sector's growth trajectory and its large process-emission share make it one of the most important to keep current, since it is one of the few industrial branches whose absolute emissions are projected to rise without intervention.

TODO: Replace the illustrative real-world figures and the example discount-rate values in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled cement, lime, and glass energy use and process emissions against CIEEDAC, Environment and Climate Change Canada (NIR), and the Cement Association of Canada's net-zero roadmap.

## 18 Chemical Products

The chemical products sector is represented in CIMS by an energy-service flow model. The emphasis is on the structure of that model: the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them, rather than the chemistry or economics of individual plants. The model tree is region-agnostic: a single master structure is instantiated for every province, and regional differences are expressed through which branches carry production and through the data attached to each node. Unlike most other industrial branches, chemicals is not dominated by a single energy-intensive product; it is a collection of distinct production chains, and a large share of its emissions are **process emissions** — CO<sub>2</sub> released by the reforming and cracking of hydrocarbon feedstock, and N<sub>2</sub>O from acid production — rather than combustion of fuel. This shapes both the structure of the sub-model and the decarbonisation options it admits.

### 18.1 Scope

CIMS represents the energy-intensive chemical products that dominate the sector's energy and emissions, organised as separate production chains: ammonia and methanol; the chlor-alkali group (chlorine, caustic soda) together with sodium chlorate and hydrogen peroxide; adipic acid; and other petrochemicals — the olefins and aromatics made by steam cracking and reforming, and the polymers made from them. The driving variable is the physical output of chemical product in tonnes; a forecast of demand drives every upstream process and energy service and is set exogenously, as for the other industrial branches.

Several adjacent activities are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- Petroleum refining — the production of fuels and refinery streams from crude oil — is covered in the [Petroleum Refining](#) chapter; the chemicals sub-model takes refinery and gas-plant streams (ethane, naphtha, natural gas) as feedstock rather than producing them.
- Merchant hydrogen production is covered in the [Hydrogen](#) chapter; where the chemicals sub-model uses hydrogen as a reactant (for example in electrolytic ammonia) it draws on that representation rather than rebuilding it.

- Fuel and feedstock supply (natural gas, oil, ethane) enters through the energy-supply sub-models and the shared fuel nodes; the chemicals sub-model consumes these as priced inputs.

The sub-model stops at saleable primary and intermediate chemicals and polymers; downstream compounding and plastics fabrication belong to the manufacturing branches.

## 18.2 Real-world context

Canada's chemical industry is large, export-oriented, and strongly regionalised, and it sits on the country's cheap natural gas. Alberta hosts the largest concentration of chemical and petrochemical plants in Canada, centred on the Industrial Heartland northeast of Edmonton: its four ethane crackers have a combined capacity of about 4.1 Mt/yr of ethylene, roughly 80% of national capacity, and the province is also the centre of ammonia, urea, and methanol production, which locate near low-cost gas feedstock (Invest Alberta 2025; Canada Energy Regulator 2024). Ontario has the most diverse slate, including the country's adipic-acid production, alongside petrochemicals and chlor-alkali; Québec and Saskatchewan add ammonia/fertilizer and petrochemical capacity; and British Columbia is dominated by the pulp-bleaching chemicals — chlor-alkali, sodium chlorate, and hydrogen peroxide — rather than petrochemicals.

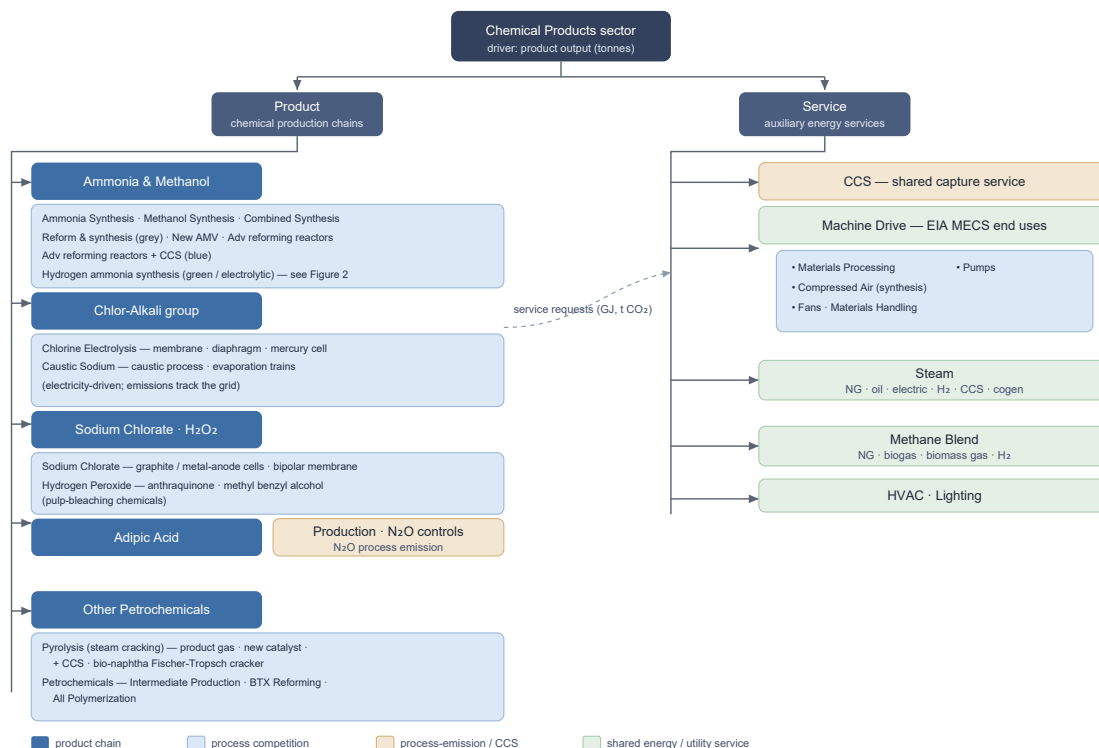
What distinguishes chemicals from the other industrial sectors is the central role of feedstock and process chemistry in its emissions. In ammonia and methanol, natural gas is reformed to make synthesis gas, and the carbon in that gas is released as CO<sub>2</sub> as an unavoidable part of the chemistry; grey ammonia carries roughly 1.8–2.4 t CO<sub>2</sub> per tonne on this basis (RMI 2024b). In steam cracking, high-temperature furnaces break naphtha or ethane into olefins, releasing both combustion and process CO<sub>2</sub>. In adipic and nitric acid production, nitrous oxide (N<sub>2</sub>O) — a greenhouse gas with roughly 270–300 times the warming potential of CO<sub>2</sub> — is the dominant emission. The CIMS structure is built around exactly these distinctions: it carries explicit process-emission factors (CO<sub>2</sub> and N<sub>2</sub>O) on the relevant technologies in addition to their fuel use, and it admits the abatement options — carbon capture, electrolytic hydrogen, N<sub>2</sub>O destruction, and bio-feedstock — that target each.

## 18.3 Model structure

Following the standard CIMS pattern, the sector divides into a **Product** sub-tree, which holds the distinct chemical production chains, and a **Service** sub-tree, which holds the auxiliary energy and utility services the production technologies draw on. Figure 18.1 shows the overall structure.

As described in the [Sector Overview](#), each node carries a competition rule. **Fixed-ratio** nodes impose the unavoidable engineering sequence of process steps, while **tech-compete** nodes

host competitions among alternative technologies on the basis of lifecycle cost (see Chapter 4). In the chemicals tree, the disaggregation into products and process stages is built from fixed-ratio nodes, while the technology choices — which synthesis route, which electrolysis cell, which cracker — occur at the tech-compete leaves.



**Figure 18.1:** Energy-service flow model of the chemical products sector (region-agnostic master tree). The sector divides into a Product sub-tree of distinct chemical chains — ammonia and methanol, chlor-alkali, sodium chlorate, hydrogen peroxide, adipic acid, and other petrochemicals — and a Service sub-tree of auxiliary energy services structured on the EIA MECS end uses, plus the shared CCS service.

### 18.3.1 Products: ammonia and methanol

The Ammonia Methanol chain is the most emissions-significant in the sector, because reforming natural gas to synthesis gas releases process CO<sub>2</sub> before any fuel is burned. The model carries three production nodes — Ammonia Synthesis, Methanol Synthesis, and a Combined Ammonia Methanol Synthesis node for co-production plants — each a tech-compete node among:

- Reform and synthesis — the conventional steam-methane-reforming route (grey ammonia/methanol), carrying a process-CO<sub>2</sub> factor of about 0.9 t per unit in the data.

- New AMV process and Advanced reforming reactors — more efficient reforming routes.
- Advanced reforming reactors CCS — the same route with carbon capture (blue ammonia/methanol).
- Hydrogen ammonia synthesis — ammonia made from electrolytic (“green”) hydrogen, which displaces reforming entirely.

This is exactly the grey → blue → green progression along which the real industry is moving, and the model represents it as a cost-and-carbon-price-driven competition.

### 18.3.2 Products: chlor-alkali, sodium chlorate, and hydrogen peroxide

These electrochemical and bleaching chemicals are grouped together and are largely electricity-driven, so their emissions track the grid:

- **Chlor-Alkali** splits into Chlorine Electrolysis — competing membrane, diaphragm, and the legacy mercury cells — and Caustic Sodium, with a Caustic Process (electrolytic or non-electrolytic) and an energy-intensive Chlor-Alkali Evaporation stage offering progressively more efficient evaporator trains.
- **Sodium Chlorate** competes graphite-electrode and metal-anode electrolysis cells and a bipolar membrane route that yields caustic as a by-product.
- **Hydrogen Peroxide** competes the dominant anthraquinone route against a methylbenzyl-alcohol route.

### 18.3.3 Products: adipic acid and other petrochemicals

**Adipic Acid** (a nylon precursor) is a tech-compete node whose significance is its  $\text{N}_2\text{O}$  process emission; the competition is between conventional Production and Production with  $\text{N}_2\text{O}$  controls, the latter destroying most of the nitrous oxide.

**Other Petrochemicals** is the olefins-and-aromatics chain. A Pyrolysis (steam-cracking) node makes the primary product gas and competes:

- Product gas — conventional steam cracking (process- $\text{CO}_2$  factor  $\sim 0.85$ );
- Product gas new catalyst — a higher-efficiency variant ( $\sim 0.68$ );
- Product gas CCS — cracking with carbon capture ( $\sim 0.24$ );
- Ligno Fischer-Tropsch naphtha steam cracker — a bio-based feedstock route.

Downstream, Petrochemicals covers Intermediate Production, BTX Reforming (benzene-toluene-xylene aromatics), and All Polymerization (conversion of monomers to polymers).

### 18.3.4 Auxiliary energy and utility services

The remaining services are common to the industrial sectors and are documented in their own chapters in the General Services part; their general structure — and the “existing versus more efficient equipment, with fuel-switching where relevant” convention they follow — is set out in the [Sector Overview](#). This chapter does not repeat that detail. The chemicals sector draws on:

- **Machine Drive** — see the [Industrial Machine Drive Systems](#) chapter. The sector’s large motive loads sit here: compressors (a major load in synthesis and refrigeration), pumps, fans, and materials handling and processing.
- **Steam** — see [Low-Temperature Industrial Heat](#), covering boiler- and turbine-raised steam and its fuel options, including electric, hydrogen, and CCS-equipped variants. Steam is a large load in chemicals.
- **Methane Blend** — the gaseous-fuel supply node, where natural gas competes with bio-gas, biomass gasification, and hydrogen.
- **HVAC** (space heating and cooling) and **Lighting** for plant facilities.
- **CCS** — the shared carbon-capture service that the reforming, cracking, and other equipped routes request.

TODO: The current regional input files still implement an earlier, more detailed auxiliary structure (six per-horsepower machine-drive size classes; separate Compression, Conveyance, Pumping, Ventilation, and Direct Drive sub-trees) and predate the Product/Service split and the MECS-aligned Machine Drive grouping. Migrate these files to the region-agnostic master tree and the shared General Services chapters.

## 18.4 Decarbonisation technologies

Because so much of the sector’s emissions are tied to feedstock chemistry rather than to energy use, its decarbonisation levers are unusual: the decisive choices are the synthesis and cracking routes (and whether their process CO<sub>2</sub> is captured or avoided), and the destruction of N<sub>2</sub>O in acid production, alongside the electrification of heat and the decarbonisation of the grid that already powers the electrochemical products. The technologies below are admitted to the relevant tech-compete nodes; representative performance and maturity are summarised in Table 18.1. The figures quoted are indicative ranges from techno-economic reviews (RMI 2024a; RMI 2024b), not the calibrated values used in the model.

**Table 18.1:** Indicative performance and maturity of key chemicals decarbonisation technologies

| Technology                                                     | Indicative benefit                                                                   | Maturity                         |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| Blue ammonia/methanol (reforming + CCS)                        | Captures most reforming process CO <sub>2</sub>                                      | Commercial / early commercial    |
| Green ammonia (electrolytic hydrogen)                          | ~95–98% CO <sub>2</sub> cut vs grey ammonia                                          | Demonstration / early commercial |
| Adipic-acid N <sub>2</sub> O abatement (catalytic destruction) | Removes most N <sub>2</sub> O ( $\approx 270\text{--}300\times$ CO <sub>2</sub> GWP) | Commercial                       |
| Steam cracking + CCS                                           | Process-CO <sub>2</sub> factor falls to ~0.24 in the model                           | Demonstration                    |
| Higher-efficiency cracking catalysts                           | Process-CO <sub>2</sub> factor ~0.85 $\rightarrow$ ~0.68 in the model                | Commercial                       |
| Bio-naphtha (Fischer-Tropsch) steam cracker                    | Bio-based feedstock displaces fossil carbon                                          | Pilot / demonstration            |
| Electrified steam cracking                                     | Up to ~90% CO <sub>2</sub> cut on cracking heat; ~95% heat to process                | Demonstration                    |
| Membrane chlor-alkali / metal-anode cells                      | Lower electrolysis energy than diaphragm/mercury                                     | Commercial                       |
| Electric / hydrogen process steam                              | Displaces gas-fired steam                                                            | See Low-Temp Heat                |
| High-efficiency motors + variable-speed drives                 | Lower compressor and pump energy                                                     | See Machine Drive                |

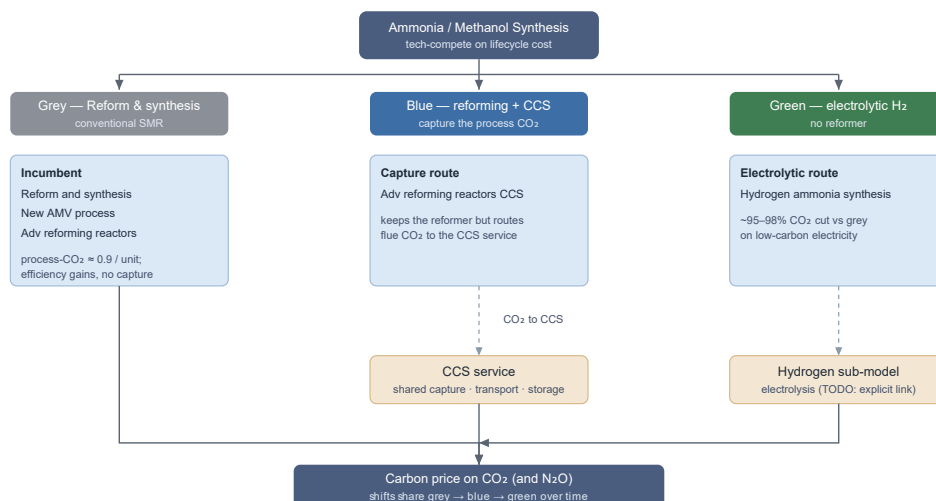
The cost estimates in Table 18.2 gather order-of-magnitude capital and operating costs and energy characteristics for the main chemical products and routes, drawn from public techno-economic literature (International Renewable Energy Agency and Ammonia Energy Association 2022; RMI 2024b; RMI 2024a) and expressed in 2025 Canadian dollars: US-dollar sources were converted at the mid-2025 rate of roughly 1.37 CAD/USD and adjusted to 2025 where the source year differed. Overnight capital costs are normalised per tonne of annual product capacity (the specific product is noted where it is not the headline product), fixed O&M follows the heavy-industry convention of roughly 3–6% of capital cost per year, and energy intensity is per tonne of product and includes feedstock energy where the feedstock is also the energy source (as in reforming and cracking). The values are for relative comparison only, not project budgeting, and are separate from the plant-specific values the model carries internally.

**Table 18.2:** Indicative techno-economic characteristics of chemical products technologies, 2025 CAD

| Technology                                      | Overnight capex<br>(2025 CAD per<br>t/yr capacity) | Fixed O&M | Energy intensity<br>(per t product)                  | Main energy<br>carrier /<br>feedstock                  |
|-------------------------------------------------|----------------------------------------------------|-----------|------------------------------------------------------|--------------------------------------------------------|
| Grey ammonia<br>(SMR)                           | ~900–2,400                                         | ~3–4%/yr  | ~28–38 GJ                                            | natural gas<br>(feedstock +<br>fuel)                   |
| Blue ammonia<br>(SMR + CCS)                     | ~1,400–2,800                                       | ~4%/yr    | ~30–40 GJ                                            | natural gas +<br>capture energy                        |
| Green ammonia<br>(electrolytic)                 | ~1,800–2,700<br>(excl.<br>renewables)              | ~2–3%/yr  | ~36–43 GJ<br>(≈10–12 MWh)                            | electricity (+<br>water)                               |
| Methanol<br>(SMR)                               | ~800–1,400                                         | ~4%/yr    | ~28–33 GJ                                            | natural gas                                            |
| Ethane steam<br>cracker                         | ~1,500 (per t/yr<br>ethylene)                      | ~3–5%/yr  | ~15–25 GJ                                            | ethane / gas<br>(feedstock +<br>fuel)                  |
| Naphtha steam<br>cracker                        | ~2,200 (per t/yr<br>ethylene)                      | ~3–5%/yr  | ~20–30 GJ                                            | naphtha + fuel<br>gas                                  |
| Electric steam<br>cracker (not yet<br>in model) | cracker capex +<br>electrification<br>premium      | —         | furnace heat<br>electrified                          | electricity                                            |
| Chlor-alkali<br>(membrane<br>cell)              | ~700–1,200                                         | ~3%/yr    | ~8–10 GJ<br>(≈2,400–2,700<br>kWh/t Cl <sub>2</sub> ) | electricity                                            |
| Sodium chlorate                                 | ~700–1,300                                         | ~3%/yr    | ~18–22 GJ<br>(≈5,000–6,000<br>kWh/t)                 | electricity                                            |
| Hydrogen<br>peroxide (an-<br>thraquinone)       | ~600–1,200                                         | ~3%/yr    | hydrogen +<br>moderate<br>electricity                | hydrogen +<br>electricity                              |
| Adipic acid +<br>N <sub>2</sub> O abatement     | abatement<br>add-on <5% of<br>plant capex          | —         | thermal +<br>electric                                | (abatement<br>destroys N <sub>2</sub> O,<br>~300 kg/t) |

#### 18.4.1 Feedstock and synthesis routes: blue and green ammonia and methanol

Reforming natural gas is the single largest source of process CO<sub>2</sub> in the sector, and Figure 18.2 shows how the model represents the alternatives. Two routes cut it. Blue production keeps



**Figure 18.2:** Ammonia and methanol synthesis routes. The conventional reform-and-synthesis route carries a large process- $\text{CO}_2$  emission; the model competes it against efficiency improvements, a CCS variant (blue), and an electrolytic-hydrogen route (green), with CCS-equipped routes drawing on the shared CCS service.

the reformer but captures its  $\text{CO}_2$  — represented as Advanced reforming reactors CCS, drawing on the shared **CCS** service. Green production replaces the reformer with electrolytic hydrogen — represented as Hydrogen ammonia synthesis — cutting emissions by roughly 95–98% relative to grey ammonia when the electricity is low-carbon (RMI 2024b). Both compete against the conventional and efficiency-improved reforming routes as costs and the carbon price evolve.

#### 18.4.2 Electrified and low-carbon cracking

Steam cracking is both energy- and emissions-intensive. The model represents three decarbonisation paths at the Pyrolysis node: carbon capture on the cracker (cutting the process- $\text{CO}_2$  factor from  $\sim 0.85$  to  $\sim 0.24$  in the data), a higher-efficiency catalyst (to  $\sim 0.68$ ), and a bio-naphtha feedstock from lignocellulosic Fischer-Tropsch synthesis, which substitutes renewable carbon for fossil. Beyond what the current tree contains, the most prominent emerging option is electrified steam cracking, demonstrated at pilot scale by a BASF–SABIC–Linde plant, which can transfer about 95% of its heat to the process and cut cracking-furnace  $\text{CO}_2$  by up to  $\sim 90\%$  when run on low-carbon electricity (BASF 2024; RMI 2024a).

TODO: Add an electrified steam cracking technology to the Pyrolysis competition (electricity-driven furnace, near-zero direct combustion emissions), so the model can represent the electrification pathway now being demonstrated commercially.

### 18.4.3 N<sub>2</sub>O abatement in acid production

Adipic and nitric acid production emit N<sub>2</sub>O, which dominates their greenhouse-gas footprint because of its very high global warming potential. Catalytic N<sub>2</sub>O destruction is a mature, high-impact abatement option, represented in the model as Production with N<sub>2</sub>O controls at the Adipic Acid node. Because the abatement targets a non-CO<sub>2</sub> gas, its value in the model depends on the carbon price being applied to N<sub>2</sub>O on a CO<sub>2</sub>-equivalent basis.

### 18.4.4 Carbon capture

Carbon capture is represented as a shared **CCS** service that any equipped route requests. It is especially well suited to chemicals because several of the sector's CO<sub>2</sub> streams — from reforming and from cracking — are relatively concentrated and therefore cheaper to capture than dilute flue gas. The CCS-equipped variants (Advanced reforming reactors CCS, Product gas CCS) route their CO<sub>2</sub> to this service; its broader treatment, including transport and storage and the associated costs, is in the [Carbon Capture, Utilization, and Storage](#) chapter.

### 18.4.5 Electrified heat and efficient drives

Process steam and lower-temperature heat can shift from gas to electricity and hydrogen through the **Steam** and **Methane Blend** services described in the [Low-Temperature Industrial Heat](#) chapter, and the large compressor and pump loads can move to high-efficiency motors and variable-speed drives in the **Machine Drive** service. The electrochemical products (chlor-alkali, sodium chlorate) are already electricity-driven, so their decarbonisation depends mainly on grid intensity and on choosing the more efficient cell and evaporator technologies the model already competes.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

TODO: Electrolytic (“green”) ammonia is represented as a single technology drawing hydrogen at a blended price. Consider linking it explicitly to the [Hydrogen](#) sub-model and an electrolysis electricity load, so the route's economics reflect hydrogen cost and availability rather than an assumed price.

## 18.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, service requests for the energy and upstream services it consumes, and

availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

A few sector-wide conventions are worth noting. Chemicals technologies are discounted at a high rate — 35% at most process nodes in the current data, with some auxiliary and capital-intensive nodes at 40–50%. These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital. They sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions; CIMS applies this behavioural rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** at the process nodes are set around 10 so that competitions respond to cost differences without collapsing to winner-take-all.

Greenhouse-gas costs enter through three channels, which is what makes chemicals distinctive. Combustion emissions follow from each technology's fuel service requests and the carbon price, as in the other sectors. In addition, many chemical technologies carry an explicit process-emission factor — CO<sub>2</sub> from reforming (~0.9 per unit) and cracking (~0.85, falling with efficient catalysts and capture), and N<sub>2</sub>O from adipic acid and intermediate production — applied independently of fuel use. Finally, the CCS service applies an emissions-removal factor to the technologies that request it. Route-switching among grey, blue, green, and bio-based production is therefore the model's main channel for emissions reduction, and it is driven by the interaction of feedstock prices, capital costs, and the carbon price applied to both CO<sub>2</sub> and N<sub>2</sub>O.

## 18.6 Regional coverage and differences

Because the tree is region-agnostic, each modelled province is built from the same master structure and simply carries no production on the branches that do not apply to it. Five regions are represented — Alberta, British Columbia, Ontario, Québec, and Saskatchewan — and regional differences show up in which products a region makes and at what scale:

- Alberta is by far the largest, and is dominated by ammonia/methanol and petrochemicals, consistent with its gas-based Industrial Heartland; its modelled output grows over the projection.
- Ontario carries the most diverse slate and is the only region with adipic acid, alongside ammonia/methanol, chlor-alkali, hydrogen peroxide, petrochemicals, and sodium chlorate.
- Québec and Saskatchewan carry ammonia/methanol, the chlor-alkali group, and petrochemicals at smaller scale.
- British Columbia carries the pulp-bleaching chemicals — chlor-alkali, sodium chlorate, and hydrogen peroxide — and ammonia/methanol, but no petrochemicals branch.

TODO: Only five provinces are represented. Confirm whether any other region hosts chemical production that should be included (and whether the absence of a petrochemicals branch in British Columbia is still accurate), and build out the region-agnostic tree accordingly.

TODO: The exogenous product-demand trajectories and regional product mixes are inherited from older calibrations. Revisit them against announced Alberta petrochemical and low-carbon ammonia/hydrogen projects, which will materially change the sector's scale and emissions profile within the projection period.

## 18.7 Limitations and planned extensions

The chemicals sub-model is a deliberately simplified representation of an unusually heterogeneous sector. The TODOs above record the main gaps between this design and both the current input files and the real Canadian sector: the migration of the auxiliary services to the MECS-aligned Machine Drive and the Product/Service split; the addition of electrified steam cracking; an explicit link from green ammonia to the hydrogen sub-model; and the refresh of the exogenous demand trajectories. A further structural simplification is that Other Petrochemicals is represented as an aggregated olefins-aromatics-polymers chain rather than by individual product (ethylene, propylene, BTX, specific polymers); this keeps the tree tractable but limits product-level resolution.

TODO: Consider disaggregating Other Petrochemicals into the major products (ethylene, propylene, aromatics, key polymers) if product-level energy intensities and demand projections warrant the additional resolution.

TODO: Replace the illustrative real-world figures and the example discount-rate values in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled chemicals energy use and emissions (including process CO<sub>2</sub> and N<sub>2</sub>O) against CIEEDAC, Environment and Climate Change Canada (NIR), and Natural Resources Canada data.

## 19 Pulp and Paper

As with the other industrial sub-models, the emphasis in the pulp and paper sector is on the structure of the energy-service flow model — the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them — rather than on the chemistry or economics of individual mills. Pulp and paper is distinctive among the industrial branches in two ways. First, it is already largely powered by biomass — the spent pulping liquor and wood residues the mills produce themselves — so its on-site fossil emissions are concentrated in a few stages rather than spread across the whole process. Second, much of the traditional mill products have been declining for many years in Canada, as the long fall in newsprint demand works through the fleet. Both shape the sub-model and the decarbonisation options it admits.

### 19.1 Scope

CIMS represents the manufacture of **pulp, paper and paperboard** — the energy-intensive core of the sector — organised as a pulp-making chain feeding a set of paper-grade chains. The driving variable is the physical output of pulp and paper in tonnes; a forecast of demand drives every upstream process and energy service and is set exogenously, as for the other industrial branches.

Several adjacent activities are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- **Forest harvesting and the supply of roundwood and residues** are covered in the [Forestry](#) chapter; the pulp and paper sub-model takes logs, chips and recovered paper as inputs.
- **Converted paper products** — boxes, bags, asphalt roofing, and other converting industries — are downstream manufacturing and are handled in the [Light Industrial](#) branch, not here.
- **Fuel and electricity supply** enter through the energy-supply sub-models and the shared fuel nodes; the sub-model consumes these as priced inputs and, importantly, can also *generate* electricity and steam on site (see below).

The sub-model stops at saleable pulp and paper grades.

## 19.2 Real-world context

Canadian pulp and paper is large, export-oriented, regionally concentrated, and in long-run decline (unless significant changes occur within the sector). The industry is largest in **Québec, Ontario and British Columbia**, with significant capacity in the **Atlantic** region and a smaller presence in **Alberta**. Its economic output peaked around 2004 and fell by roughly 45% to 2018 as newsprint demand collapsed (Canada Energy Regulator 2019); the model's exogenous demand trajectories reflect this, with output declining over the projection in every region except Alberta.

What most distinguishes the sector energetically is its reliance on **self-produced biomass energy**. Kraft (chemical) mills concentrate and burn the **black liquor** — the spent cooking chemicals and dissolved wood lignin left after pulping — in a recovery boiler that both regenerates the pulping chemicals and raises high-pressure steam; mills also burn **hog fuel** (bark and wood residues). Together these supply the majority of mill energy, and a growing number of mills run **biomass cogeneration**, meeting their own steam and power needs and selling surplus electricity to the grid: self-generation supplied on the order of a quarter of the sector's electricity in recent years, across dozens of cogeneration units (Canada Energy Regulator 2019; Canadian Climate Institute 2023). Because this energy is biogenic, the sector's *fossil* emissions are comparatively small and concentrated — most prominently in the **lime kiln**, which calcines lime for the kraft chemical cycle and has traditionally burned natural gas or oil, and in any remaining fossil-fired steam and drying (Pulp and Paper Canada 2024). The CIMS structure mirrors this: it carries an explicit **Black Liquor** service representing the recovery-and-energy cycle, it offers biomass and cogeneration options throughout the Steam service, and it concentrates the fossil-fuel choices at the lime kiln and the dryers.

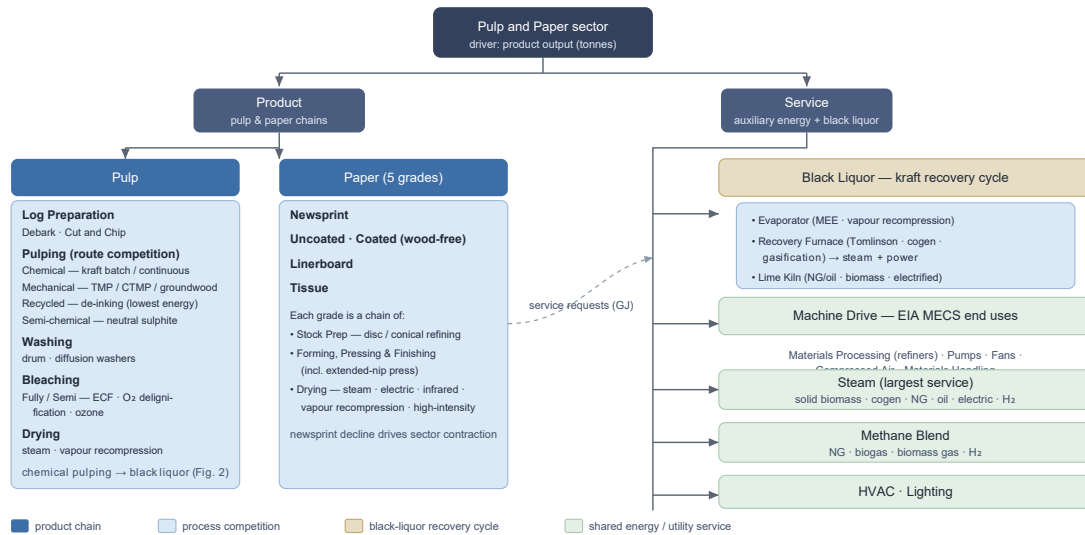
## 19.3 Model structure

Following the standard CIMS pattern, the sector divides into two sub-trees: a **Product** sub-tree, which holds the pulp-making and paper-making chains, and a **Service** sub-tree, which holds the auxiliary energy and utility services — including the sector-specific black-liquor recovery cycle. Figure 19.1 shows the overall structure.

In the pulp and paper tree, the disaggregation into products and process stages is built from fixed-ratio nodes, while the technology choices — which pulping process, which bleaching sequence, which dryer, which recovery furnace — occur at the tech-compete leaves.

### 19.3.1 Products: pulp

The **Pulp** chain turns wood (or recovered paper) into prepared pulp through a sequence of fixed-ratio stages, with technology competitions at the end of each:



**Figure 19.1:** Energy-service flow model of the pulp and paper sector (region-agnostic master tree). The sector divides into a Product sub-tree — a pulp-making chain (log preparation, pulping, washing, bleaching, drying) and five paper-grade chains — and a Service sub-tree that holds the auxiliary energy services plus the sector-specific Black Liquor recovery-and-energy cycle.

- **Log Preparation** — debark (ring, Rosser-head or biodegradable mechanical debarkers) and cut and chip.
- **Pulping** — the key fork, where four distinct routes compete and, as in the real industry, generally serve different products rather than competing head-to-head:
  - **Chemical (kraft)** — batch and continuous digesters (including superbatches and blow-heat-recovery variants), the dominant route for strong, bleachable pulp; it is the route that produces black liquor.
  - **Mechanical** — stone groundwood, refiner, thermo-mechanical (TMP) and chemi-thermo-mechanical (CTMP) pulping, which use wood fibre efficiently but are very electricity-intensive.
  - **Semi-chemical** — neutral-sulphite digesters, used for specific grades.
  - **Recycled** — de-inking of recovered paper by washing, flotation or explosion; the lowest-energy route, limited by feedstock supply.
- **Washing** — drum and diffusion washers that recover pulping chemicals.

- **Bleaching** — competing fully and semi bleaching sequences built around elemental-chlorine-free (ECF) chemistry, oxygen delignification, ozone bleaching and chlorine-dioxide substitution.
- **Drying** — pulp dryers (steam, and steam with electric vapour recompression) for market pulp.

### 19.3.2 Products: paper

The **Paper** chain branches into the five grades the sector reports, each a fixed-ratio sequence of **Stock Preparation** (disc or conical refining and screening), **Forming, Pressing and Finishing**, and **Drying**:

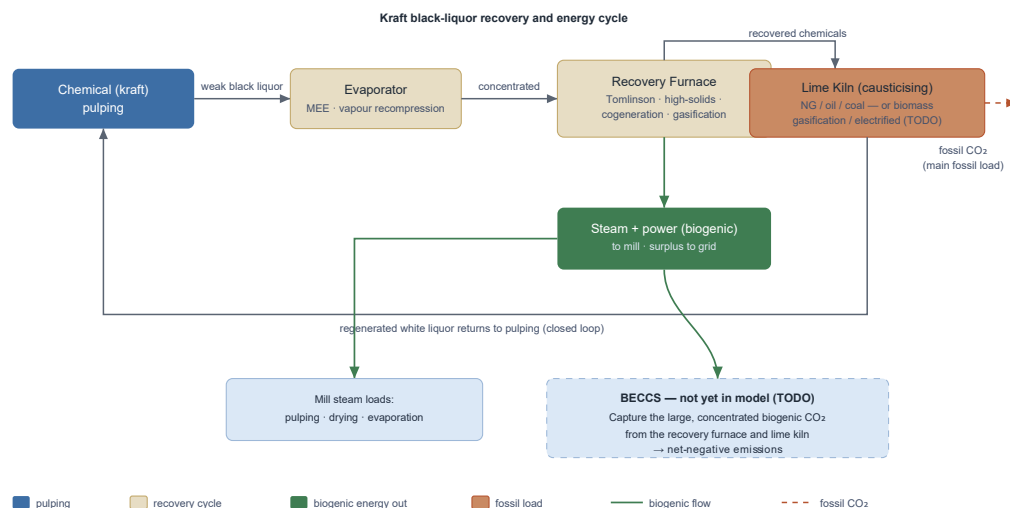
- **Newsprint** — historically the largest grade, and the one whose decline drives the sector's contraction.
- **Uncoated** and **Coated** wood-free (fine) papers.
- **Linerboard** — packaging board.
- **Tissue**.

The Forming, Pressing and Finishing competitions include energy-saving extended-nip presses, which remove more water mechanically before the energy-intensive drying step. The Drying competitions are where much of the grade's energy is decided, offering steam, electric, infrared, vapour-recompression, high-humidity-hood and high-intensity dryers.

### 19.3.3 Sector-specific service: the black-liquor recovery and energy cycle

The **Black Liquor** service is the energy and chemical heart of a kraft mill, and the feature that most sets this sector apart. It represents the closed loop, shown in Figure 19.2, that recovers pulping chemicals while generating most of the mill's energy:

- The **Evaporator** concentrates weak black liquor, competing multiple-effect evaporators (with computer control and integrated concentrators) and **vapour-recompression** evaporators.
- The **Recovery Furnace** burns the concentrated liquor, regenerating cooking chemicals and raising steam. Technologies range from the conventional **Tomlinson** boiler through low-odour, high-solids-firing variants to **cogeneration** configurations (e.g. high-solids firing with 900–1200 psig cogeneration) and **gasification**, which raise the energy recovered from the liquor.
- The **Lime Kiln** calcines lime to causticise the recovered chemicals. It is the sector's main fossil-fuel choke point, and the model competes conventional natural-gas, oil and coal kilns against **wood-waste gasification** and **lignin-precipitation** retrofits that displace fossil fuel.



**Figure 19.2:** The kraft black-liquor recovery and energy cycle as represented in CIMS. Chemical pulping produces weak black liquor, which is concentrated in the Evaporator and burned in the Recovery Furnace; the furnace regenerates pulping chemicals (via the Lime Kiln) and raises steam and power for the mill, with surplus electricity exported. Biomass and black liquor are biogenic, so this energy carries no fossil CO<sub>2</sub>; the kiln is the main fossil load.

### 19.3.4 Auxiliary energy and utility services

The remaining services are common to the industrial sectors and are documented in their own chapters in the General Services part; their general structure — and the “existing versus more efficient equipment, with fuel-switching where relevant” convention they follow — is set out in the [Sector Overview](#). This chapter does not repeat that detail. The Pulp and Paper sector draws on:

- **Machine Drive** — see the [Industrial Machine Drive Systems](#) chapter. Pulp and Paper’s very large motor loads sit here, especially the refiners of mechanical pulping (materials processing), along with pumps, fans, compressed air and materials handling.
- **Steam** — see [Low-Temperature Industrial Heat](#). This is the sector’s largest energy service, and its technology set is unusually rich: alongside natural-gas, oil, coal, electric and hydrogen boilers it includes **solid-biomass** boilers, **biomass steam turbines** and grate firing, and gas- and biomass-fired **cogeneration**, reflecting the mills’ reliance on hog fuel and on combined heat and power.
- **Methane Blend** — the gaseous-fuel supply node, where natural gas competes with bio-gas, biomass gasification and hydrogen.
- **HVAC and Lighting** for mill facilities.

## 19.4 Decarbonisation technologies

Because the sector is already largely bioenergy-powered, its decarbonisation problem is unusual: it is less about replacing a single dominant fossil process than about (i) squeezing the remaining fossil fuel out of the **lime kiln** and any fossil-fired **steam and drying**, (ii) raising the efficiency of the energy-intensive **drying, pressing and mechanical-pulping** steps, and (iii) capturing the large, concentrated, **biogenic** CO<sub>2</sub> streams to deliver *negative* emissions. The technologies below are admitted to the relevant tech-compete nodes; representative performance and maturity are summarised in Table 19.1. The figures quoted are indicative ranges from techno-economic reviews (Canadian Climate Institute 2023; Pulp and Paper Canada 2024), not the calibrated values used in the model.

**Table 19.1:** Indicative performance and maturity of key pulp and paper decarbonisation technologies

| Technology                                      | Indicative benefit                                           | Maturity         |
|-------------------------------------------------|--------------------------------------------------------------|------------------|
| Recycled / de-inked pulping                     | Lowest-energy pulp route; limited by feedstock               | Commercial       |
| Mechanical pulping with vapour recompression    | Recovers refiner heat as usable steam                        | Commercial       |
| Extended-nip pressing                           | More water removed before drying; less drying energy         | Commercial       |
| Electric / infrared / high-intensity drying     | Displaces fossil-fired drying                                | Commercial       |
| ECF, oxygen-delignification, ozone bleaching    | Lower energy and chemical use than legacy bleaching          | Commercial       |
| Black-liquor recovery cogeneration              | Raises on-site power from spent liquor                       | Commercial       |
| Black-liquor gasification                       | Recovery efficiency ~65% → ~75–80%; syngas for power/biofuel | Demonstration    |
| Lime-kiln biomass gasification / lignin firing  | Displaces fossil fuel in the kiln                            | Early commercial |
| Solid-biomass boilers and steam turbines        | Bio-steam and power in place of fossil boilers               | Commercial       |
| Electrified lime kiln (e.g. plasma calcination) | Removes the main fossil load                                 | Pilot            |
| BECCS on recovery boiler / kiln                 | Captures biogenic CO <sub>2</sub> for net-negative emissions | Demonstration    |

#### 19.4.1 Efficient pulping, pressing and drying

The lowest-cost reductions come from using less energy per tonne. **Recycled (de-inked) pulping** is the least energy-intensive route and is admitted where recovered-paper feedstock is available; **mechanical pulping with vapour recompression** recovers the refiners' large heat output as usable steam. On the paper machine, **extended-nip presses** remove more water mechanically so that less must be evaporated, and the **Drying** competitions admit **electric, infrared, high-intensity and vapour-recompression** dryers that either cut drying energy or shift it from fossil fuel to electricity. In the model's bleaching competitions, the **ECF, oxygen-delignification and ozone** sequences reduce both energy and chemical use relative to legacy chlorine bleaching.

#### 19.4.2 Black-liquor recovery, cogeneration and gasification

The recovery furnace is both a chemical reactor and the mill's main power plant. The model competes increasingly capable configurations — high-solids firing, low-odour designs, and **cogeneration** at 900–1200 psig — that raise the steam and electricity recovered from each tonne of liquor. **Black-liquor gasification** goes further, lifting energy-recovery efficiency from roughly 65% for a conventional boiler to about 75–80% and producing a syngas that can drive a gas turbine or be converted to biofuels (IEA Bioenergy 2007). These options let a kraft mill meet more of its own demand and export more surplus power.

#### 19.4.3 Decarbonising the lime kiln

The lime kiln is the single most stubborn fossil load in a kraft mill, because calcination needs high-temperature heat that has traditionally come from natural gas or oil (Pulp and Paper Canada 2024). The model competes wood-waste gasification and lignin-precipitation retrofits that substitute renewable fuel for fossil in the kiln. The emerging frontier option — **electrified calcination** (for example electric plasma kilns) — would remove the load entirely.

TODO: Add an **electrified lime kiln** (e.g. electric plasma calcination) and **high-temperature heat-pump** options to the relevant competitions, so the model can represent full electrification of the kraft mill's remaining fossil heat.

#### 19.4.4 Carbon capture and negative emissions (BECCS)

The pulp and paper sector is widely regarded as one of the most promising hosts for **bioenergy with carbon capture and storage (BECCS)**, because the recovery boiler and lime kiln emit large, relatively concentrated streams of *biogenic* CO<sub>2</sub>; capturing them yields **net-negative** emissions at comparatively low cost.

TODO: Add a **CCS / BECCS** service for pulp and paper, attached to the Recovery Furnace and Lime Kiln, so the model can represent capture of biogenic CO<sub>2</sub> and the resulting negative emissions.

#### 19.4.5 Indicative capital and operating costs and energy use

Table 19.2 gathers order-of-magnitude capital and operating costs and energy characteristics for the main pulping and papermaking steps and the black-liquor recovery cycle. The estimates are drawn from public techno-economic literature (Canadian Climate Institute 2023; Pulp and Paper Canada 2024; IEA Bioenergy 2007; Tran and Vakkilainen 2016) and expressed in 2025 Canadian dollars. Overnight capital costs are normalised per tonne of annual product capacity (except where another basis is noted), fixed O&M follows the heavy-industry convention of roughly 3–6% of capital cost per year, and energy intensity is per tonne of product. A defining feature of the sector is that much of the heat and power below is **self-supplied** from biomass (black liquor and hog fuel), so the relevant comparison is often net purchased energy rather than gross use.

**Table 19.2:** Indicative techno-economic characteristics of pulp and paper technologies, 2025 CAD

| Technology                                  | Overnight capex<br>(per t/yr<br>capacity) | Fixed O&M (%<br>of capex) | Energy intensity<br>(per t product)                           | Main energy<br>carrier                     |
|---------------------------------------------|-------------------------------------------|---------------------------|---------------------------------------------------------------|--------------------------------------------|
| Kraft (chemical)<br>pulping                 | ~2,000–3,500<br>(integrated<br>mill)      | ~4–5%/yr                  | heat ~10–14 GJ<br>+ ~0.6–0.8<br>MWh; largely<br>self-supplied | biomass (black<br>liquor) +<br>electricity |
| Mechanical<br>pulping (TMP /<br>CTMP)       | ~1,000–1,800                              | ~4%/yr                    | ~4–15 GJ<br>electricity<br>(1,100–4,300<br>kWh/t)             | electricity                                |
| Recycled<br>(de-inked)<br>pulping           | ~600–1,200                                | ~3–4%/yr                  | ~1–3 GJ                                                       | electricity (+<br>some heat)               |
| Bleaching (ECF<br>/ O <sub>3</sub> / ozone) | moderate<br>(site-specific)               | ~4%/yr                    | steam +<br>bleaching<br>chemicals                             | steam /<br>electricity                     |
| Paper machine<br>+ drying                   | ~1,400–2,700                              | ~4%/yr                    | drying ~4–8 GJ                                                | steam (biomass<br>/ gas) ±<br>electricity  |

| Technology                                    | Overnight capex<br>(per t/yr<br>capacity)                | Fixed O&M (%<br>of capex) | Energy intensity<br>(per t product)          | Main energy<br>carrier            |
|-----------------------------------------------|----------------------------------------------------------|---------------------------|----------------------------------------------|-----------------------------------|
| Recovery boiler<br>(cogeneration)             | ≈35% of mill<br>capital; 25–35<br>MWe per 1,000<br>t/day | ~4%/yr                    | net energy<br>exporter                       | black liquor<br>(biogenic)        |
| Black-liquor<br>gasification                  | higher than<br>Tomlinson<br>boiler                       | —                         | recovery<br>efficiency<br>~75–80% vs<br>~65% | black liquor →<br>syngas          |
| Lime kiln                                     | moderate                                                 | —                         | ~5.4–8.5 GJ/t<br>lime                        | natural gas / oil<br>(or biomass) |
| Biomass<br>cogeneration<br>(steam +<br>power) | ~5,500–9,600<br>CAD/kWe                                  | ~3–4%/yr                  | bio-steam +<br>power                         | solid biomass /<br>hog fuel       |

## 19.5 Economics

A few sector-wide conventions are worth noting. Pulp and paper technologies are discounted at a high rate — 35% at most process nodes, with some auxiliary and capital-intensive nodes at 40–50%. These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital. They sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions; CIMS applies this behavioural rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity** (variance) parameters at the process nodes are set around 10 so that competitions respond to cost differences without collapsing to winner-take-all. Fuel prices entering the service-cost calculation are scaled by a sector-level block of price multipliers.

The sector's emissions accounting is distinctive. Unlike iron and steel or chemicals, the pulp and paper tree carries no explicit process-emission factors: essentially all of its emissions arise from fuel combustion, and they enter the model through each technology's fuel service requests and the prevailing carbon price. Crucially, the large biomass flows — black liquor and hog fuel — are treated as biogenic and therefore carbon-neutral in the combustion accounting, so the carbon price bites mainly on the fossil fuels that remain, above all in the lime kiln and in any fossil-fired steam and drying. Fuel-switching from these fossil fuels to biomass, electricity, and hydrogen is therefore the model's main channel for emissions reduction, reinforced by the mills' ability to self-generate steam and power and export surplus electricity.

## 19.6 Regional coverage and differences

Five regions are represented — **Alberta, Atlantic, British Columbia, Ontario, and Québec** — and regional differences show up in scale, grade mix, and trajectory:

- **Québec** and **Ontario** are the largest, and both decline substantially over the projection as newsprint demand falls.
- The **Atlantic** region carries significant capacity with a particularly steep modelled decline.
- **British Columbia** is a major kraft- and mechanical-pulp region with biomass cogeneration, declining more gently.
- **Alberta** is the smallest and the only region whose modelled output grows.

TODO: Only five regions are represented, and Saskatchewan and the territories are absent. Confirm whether any other region hosts pulp and paper capacity that should be included.

TODO: The exogenous demand trajectories are inherited from older calibrations of the newsprint decline. Revisit them against current production and the shift toward packaging (linerboard), tissue and market pulp, which are not declining in the way newsprint is.

## 20 Metal Smelting and Refining

The metal smelting and refining sector — the production of finished non-ferrous metals — is represented in CIMS by an energy-service flow model. The emphasis is on the structure of that model: the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them, rather than the metallurgy of individual smelters. The model tree is region-agnostic: a single master structure is instantiated for every province, and regional differences are expressed through which branches carry production and through the data attached to each node. The sector is unusual in two respects. It is dominated by a single metal, aluminium, whose production is overwhelmingly electricity-driven and located in Québec on cheap hydropower; and its hardest-to-cut emissions are process emissions — CO<sub>2</sub> from the consumable carbon anode in aluminium electrolysis, and SF<sub>6</sub> from magnesium melt protection — rather than ordinary combustion.

### 20.1 Scope

CIMS represents the smelting and refining of the major non-ferrous metals: aluminium, copper, nickel, zinc, and magnesium, with lead and titanium appearing in some regions. The driving variable is the physical output of refined metal in tonnes; a forecast of demand drives every upstream process and energy service and is set exogenously, as for the other industrial branches.

Several adjacent activities are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- Iron and steel are explicitly excluded and covered in their own [Iron and Steel](#) chapter.
- Mining and concentration of the ores are part of the [Mining](#) sector; the smelting sub-model takes concentrates (and, for aluminium, alumina) as inputs.
- Downstream fabrication — rolling, extrusion, casting of finished goods — is downstream manufacturing and is not part of this sub-model, which ends with refined and cast metal.

Minor and precious metals (gold, silver, platinum-group, cadmium) are not represented explicitly because they are recovered in conjunction with the base metals or are produced in quantities too small to require their own representation.

## 20.2 Real-world context

Canada is one of the world's leading non-ferrous metal producers, and the sector's structure is dominated by aluminium. Almost all Canadian primary aluminium is smelted in Québec, where abundant low-cost hydroelectricity makes the energy-intensive Hall-Héroult electrolysis economic; Canadian smelters are among the lowest-carbon in the world for this reason (International Energy Agency 2024a). Nickel and copper smelting and refining are concentrated in Ontario (Sudbury) and Manitoba (Thompson), with zinc and other base metals across several provinces.

The sector's emissions profile is distinctive. Hall-Héroult electrolysis consumes roughly 14 MWh of electricity per tonne of aluminium, but even on perfectly clean electricity it emits CO<sub>2</sub> because the carbon anode is consumed in the reaction — about 1.6–1.7 t CO<sub>2</sub> per tonne of aluminium — and intermittent anode effects release perfluorocarbons (PFCs), extremely potent greenhouse gases (International Energy Agency 2024a). Magnesium production and casting use SF<sub>6</sub> as a melt-protection cover gas; SF<sub>6</sub> has roughly 22,000 times the warming potential of CO<sub>2</sub>, so even small quantities matter. Pyrometallurgical copper, nickel, and zinc smelting burns fuel for roasting, smelting, and converting. The CIMS structure reflects all of this: it carries explicit process-emission factors for the aluminium carbon anode and for magnesium SF<sub>6</sub>, and it admits the technologies — inert anodes, recycling, oxygen-enriched and flash smelting, and hydrometallurgy — that target each.

## 20.3 Model structure

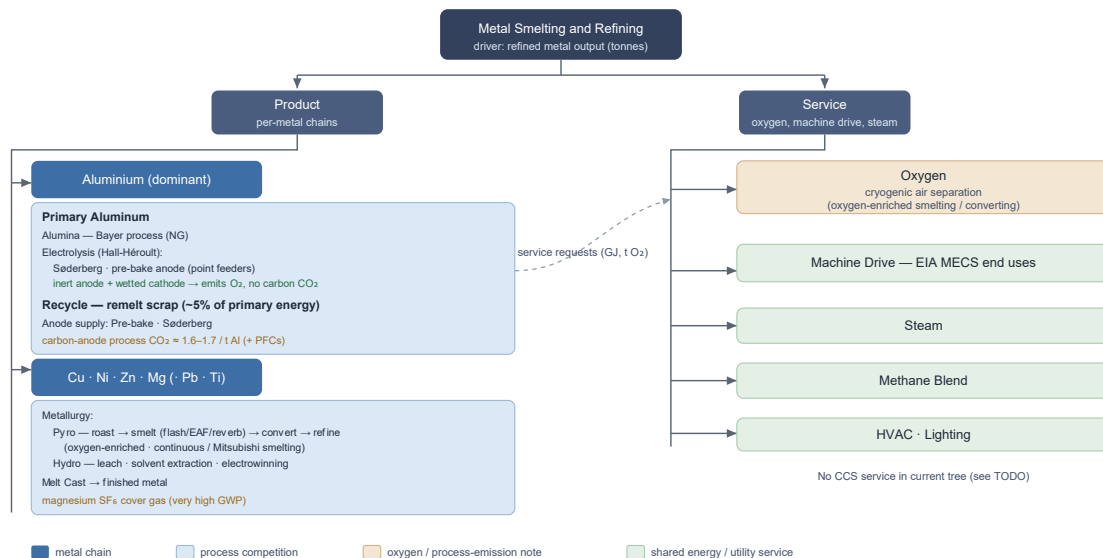
Following the standard CIMS pattern, the sector divides into a **Product** sub-tree, holding the per-metal production chains, and a **Service** sub-tree of auxiliary energy services. Figure 20.1 shows the overall structure.

As described in the [Sector Overview](#), each node carries a competition rule. **Fixed-ratio** nodes impose the unavoidable engineering sequence of process steps, while **tech-compete** nodes host competitions among alternative technologies on the basis of lifecycle cost (see Chapter 4). In the metal-smelting tree, the disaggregation into metals and process stages is built from fixed-ratio nodes, while the technology choices — which anode, which smelting furnace, which metallurgical route — occur at the tech-compete leaves.

### 20.3.1 Products: aluminium

Aluminium is the largest chain and has its own structure. Primary Aluminum runs through:

- **Alumina** — refining bauxite to alumina by the Bayer process (natural-gas fired, with an efficiency retrofit). (Canada imports alumina, but the step is represented.)



**Figure 20.1:** Energy-service flow model of the metal smelting and refining sector (region-agnostic master tree). The Product sub-tree holds per-metal chains — aluminium (alumina, electrolysis, recycle) and the pyro-/hydro-metallurgical base metals — and the Service sub-tree holds the auxiliary energy services and oxygen supply.

- **Electrolysis** — the Hall-Héroult cell, the energy and emissions core, competing Søderberg and several pre-bake anode designs (side- and centre-point feeders, with automation) against the inert anode with wetted cathode route that eliminates the carbon anode entirely.

A Recycle branch remelts scrap aluminium at a small fraction of primary energy, and dedicated Pre-bake Anode and Søderberg Anode nodes supply the consumable anodes for the conventional routes.

The conventional electrolysis routes carry a process- $\text{CO}_2$  factor of about 1.6–1.7 t per tonne, representing carbon-anode consumption; the inert-anode route is the model’s way of breaking that link.

### 20.3.2 Products: copper, nickel, zinc, and magnesium

The base metals share a common structure under Primary Metals, each resolving into a Metallurgy route and a Melt Cast finishing step:

- **Pyro Metallurgy** — the conventional high-temperature route, broken into a Smelting Process (conventional roast-smelt-convert, or continuous smelting such as Mitsubishi) and a Refine step (electrolytic, anode-furnace, or fire refining). The smelting technologies compete fuels and furnace types — flash furnaces, reverberatory and electric-arc furnaces, fluidized-bed and hearth roasting, Pierce-Smith and top-blown rotary converters — with oxygen enrichment options.
- **Hydro Metallurgy** — the aqueous route (pressure or percolation leaching, solvent extraction, electrolytic precipitation/electrowinning), which is more electricity-intensive but avoids high-temperature smelting.

Magnesium carries an explicit SF<sub>6</sub> process-emission factor on its melt-cast and refining steps, reflecting cover-gas use.

### 20.3.3 Services

The services follow the standard pattern and are documented in their own chapters in the General Services part. The sector draws on **Oxygen** (cryogenic air separation, feeding oxygen-enriched smelting and converting), **Machine Drive** (see [Industrial Machine Drive Systems](#)), **Steam** (see [Low-Temperature Industrial Heat](#)), **Methane Blend**, **HVAC**, and **Lighting**. Notably, the sector has no CCS service in the current tree, which is a significant omission for the pyrometallurgical routes whose off-gases are relatively concentrated.

TODO: The current regional input files still implement an earlier, more detailed auxiliary structure (six per-horsepower machine-drive size classes; separate Compression, Conveyance, Pumping, Ventilation, and Direct Drive sub-trees) and predate the Product/Service split and the MECS-aligned Machine Drive grouping. Migrate these files to the region-agnostic master tree (which places the sector under `Metal . Smelting` with the Product/Service split) and the shared General Services chapters.

## 20.4 Decarbonisation technologies

Because Canadian smelting already runs largely on clean electricity, its decarbonisation problem is less about the energy supply than about the process chemistry — the carbon anode, the SF<sub>6</sub> cover gas, and the fuel burned in pyrometallurgical smelting — and about using less primary metal through recycling. The model represents most of these levers; Table [20.1](#) summarises maturity and Table [20.2](#) gives indicative costs.

**Table 20.1:** Indicative performance and maturity of key metal smelting decarbonisation technologies

| Technology                                                          | Indicative benefit                                                     | Maturity                                |
|---------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------|
| Inert anode (Hall-Héroult)                                          | Eliminates carbon-anode CO <sub>2</sub> and PFCs; emits O <sub>2</sub> | Pilot / pre-commercial (ELYSIS)         |
| Low-carbon electricity (hydro)                                      | Removes most indirect CO <sub>2</sub> from electrolysis                | Commercial (in place in Québec)         |
| Secondary (recycled) aluminium                                      | ~95% less energy than primary                                          | Commercial                              |
| Oxygen-enriched / flash / continuous smelting                       | Lower fuel use per tonne (base metals)                                 | Commercial                              |
| Hydrometallurgy (vs pyrometallurgy)                                 | Avoids high-temperature smelting fuel                                  | Commercial (metal-specific)             |
| Electrified smelting / process heat                                 | Displaces fossil smelting fuel                                         | Emerging                                |
| SF <sub>6</sub> -free magnesium cover gas (SO <sub>2</sub> , novac) | Removes a very high-GWP emission                                       | Commercial                              |
| Smelter off-gas CCS                                                 | Captures pyrometallurgical CO <sub>2</sub>                             | Demonstration — <i>not yet in model</i> |

The cost estimates in Table 20.2 are order-of-magnitude figures from public techno-economic literature (International Energy Agency 2024a; ELYSIS 2024; Intergovernmental Panel on Climate Change 2000), expressed in 2025 Canadian dollars (US- and euro-denominated sources converted at roughly 1.37 CAD/USD and 1.48 CAD/EUR and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 20.2:** Indicative techno-economic characteristics of metal smelting decarbonisation technologies, 2025 CAD

| Decarbonisation technology             | Capex                               | Operating cost                                                              | Energy / emission effect                       |
|----------------------------------------|-------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------|
| Inert anode (new cells)                | comparable to new pre-bake cells    | conversion cost ~CAD 100/t Al, falling below carbon-anode cost in new cells | eliminates ~1.6 t CO <sub>2</sub> /t Al + PFCs |
| Greenfield primary smelter (reference) | ~CAD 7,400–10,400 per t/yr capacity | electricity-dominated (~14 MWh/t)                                           | indirect CO <sub>2</sub> tracks grid           |

| Decarbonisation technology                | Capex                       | Operating cost                   | Energy / emission effect                   |
|-------------------------------------------|-----------------------------|----------------------------------|--------------------------------------------|
| Secondary (recycled) aluminium            | far lower than primary      | ~95% less energy                 | large energy/CO <sub>2</sub> saving        |
| Oxygen-enriched / flash smelting          | moderate (oxygen plant)     | fuel saving                      | lower smelting fuel CO <sub>2</sub>        |
| Hydrometallurgy                           | metal-specific              | more electricity, less fuel      | shifts energy to electricity               |
| SF <sub>6</sub> -free magnesium cover gas | low (gas handling retrofit) | similar or lower gas cost        | removes ~22,000× GWP emission              |
| Smelter off-gas CCS (not in model)        | +30–50% plant equipment     | ~CAD 100–200 / t CO <sub>2</sub> | captures pyrometallurgical CO <sub>2</sub> |

#### 20.4.1 Inert anodes and clean-power aluminium

Because Québec’s aluminium already runs on hydroelectricity, its remaining direct emissions come almost entirely from the consumable carbon anode. The inert anode route — represented in the model as *Hall-Héroult inert anode with wetted cathode* — replaces the carbon anode with a non-consumable material that releases oxygen instead of CO<sub>2</sub>, eliminating both the anode CO<sub>2</sub> and the PFCs from anode effects. This is the single largest lever for the sector. It is being commercialised by the ELYSIS joint venture in Québec, which produced primary aluminium at industrial scale with inert anodes in 2021 and targets broader commercial availability later this decade (ELYSIS 2024). Conversion cost is expected to fall below that of carbon anodes in new cells.

#### 20.4.2 Recycling and base-metal process efficiency

Secondary (recycled) aluminium uses on the order of 5% of the energy of primary production and is represented as the Recycle branch; expanding it is a major lever as scrap availability grows. For the base metals, oxygen-enriched, flash, and continuous smelting cut fuel use per tonne, and hydrometallurgy avoids high-temperature smelting altogether for metals where it is applicable — both are in the model. Further electrification of smelting and process heat, and CCS on the concentrated off-gas streams of pyrometallurgical smelters, are the main options not yet fully represented.

### 20.4.3 Magnesium SF<sub>6</sub> replacement

Magnesium melt protection traditionally uses SF<sub>6</sub>, one of the most potent greenhouse gases known. Replacing it with alternative cover gases (sulphur dioxide, or fluorinated ketones such as Novec 612) removes almost all of this emission at low cost and is mature; the model carries the SF<sub>6</sub> emission explicitly, so an SF<sub>6</sub>-free option is a natural addition.

TODO: Add a CCS service to the metal-smelting tree, attached to the pyrometallurgical smelting and converting nodes whose off-gases (especially in copper and nickel flash smelting) are relatively concentrated and capture-amenable.

TODO: Represent PFC emissions from aluminium anode effects explicitly (currently only the carbon-anode CO<sub>2</sub> is carried), and add an SF<sub>6</sub>-free magnesium cover gas technology so the high-GWP magnesium emission can be abated in-model.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

## 20.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, service requests for the energy and upstream services it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Metal smelting technologies are discounted at a high rate — 35% at most process nodes in the current data, with some auxiliary and capital-intensive nodes at 40–50%. These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital. They sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions; CIMS applies this behavioural rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** at the process nodes are set around 10 so that competitions respond to cost differences without collapsing to winner-take-all.

The sector's emissions accounting is distinctive. Combustion emissions follow from each technology's fuel service requests and the carbon price, as in the other sectors. In addition, the aluminium electrolysis technologies carry an explicit process-CO<sub>2</sub> factor (about 1.6–1.7 t per tonne) for carbon-anode consumption, and the magnesium melt-cast and refining technologies carry an SF<sub>6</sub> factor — both applied independently of fuel use, with SF<sub>6</sub> entering on a CO<sub>2</sub>-equivalent basis at its very high warming potential. Because there is no CCS service, the model's main channels for emissions reduction are switching to the inert anode, expanding

recycling, switching to lower-fuel smelting routes, and (indirectly) the carbon intensity of the electricity the sector draws.

## 20.6 Regional coverage and differences

Because the tree is region-agnostic, each modelled region is built from the same master structure and carries no production on branches that do not apply. Five regions are represented — the Atlantic region, British Columbia, Manitoba, Ontario, and Québec — and the differences are stark:

- Québec dominates the sector by a wide margin, carrying essentially all primary aluminium on its hydro power, and grows over the projection.
- Ontario (Sudbury) carries nickel and copper smelting and refining.
- Manitoba (Thompson) carries nickel; British Columbia and the Atlantic region carry smaller base-metal operations.

TODO: Only five provinces are represented (Alberta and Saskatchewan are absent, consistent with the absence of major non-ferrous smelters there). Confirm the regional metal mix — including the lead and titanium branches that appear only in some regions — against current operations.

## 20.7 Limitations and planned extensions

The metal-smelting sub-model is detailed on the metallurgical side but carries the structural debts common to the older sector files, plus two sector-specific gaps: the absence of a CCS service and the incomplete representation of the non-CO<sub>2</sub> greenhouse gases (PFCs and the SF<sub>6</sub>-abatement option). The TODOs above record these along with the auxiliary-service migration. Because the sector is already largely clean-powered, its remaining emissions are concentrated and high-leverage — the carbon anode, magnesium SF<sub>6</sub>, and pyrometallurgical fuel — so closing these representational gaps matters more here than the sector's modest energy total might suggest.

TODO: Replace the illustrative real-world figures and the example discount-rate values in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled non-ferrous energy use and process emissions against CIEEDAC and Environment and Climate Change Canada (NIR) data.

## 21 Iron and Steel

The iron and steel sector is represented in CIMS by an energy-service flow model. The emphasis is on the structure of that model — the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them — rather than on the metallurgy or economics of individual mills. The model tree is region-agnostic: a single master structure is instantiated for every province, and regional differences are expressed through which branches carry production and through the data attached to each node, not through separate per-region structures. Iron and steel is one of the most energy- and emissions-intensive branches of Canadian industry, and almost all of its emissions are concentrated in a single step — the reduction of iron ore to metallic iron — which is also where the sector’s main decarbonisation choices sit.

### 21.1 Scope

CIMS represents the integrated process chain of primary and secondary steelmaking, from the preparation of iron-bearing feed through to finished steel products. The sub-model begins with the on-site agglomeration of iron ore and the production of oxygen and process gases, takes in scrap, runs the metallic charge through the ironmaking and steelmaking step that produces molten steel, and ends with casting and finishing into the three basic forms — slabs, blooms, and billets — and their many sub-forms. The driving variable is the physical output of steel in tonnes; a forecast of crude-steel demand drives every upstream process and energy service and is set exogenously, as for the other industrial branches.

Several adjacent activities are deliberately handled elsewhere in CIMS to avoid double-counting energy and emissions:

- Iron-ore mining and ore beneficiation are covered in the [Mining](#) chapter; the iron and steel sub-model takes ore (or pellets) as an input and stops short of the mine gate on the upstream side.
- Smelting and refining of non-ferrous metals — aluminium, nickel, copper, zinc, and the rest — are covered in the [Metal Smelting and Refining](#) chapter, which explicitly excludes iron and steel.
- Downstream fabrication of steel into final goods (structural fabrication, stamping, machining) belongs to the manufacturing branches and is not part of this sub-model, which ends with saleable cast-and-rolled steel forms.

Coke-making and the on-site reducing gases are treated within the sector because they are integral to the ironmaking step and cannot be cleanly separated from it.

## 21.2 Real-world context

Canada produces on the order of 12 Mt of crude steel a year, placing it around sixteenth among steelmaking nations, against world output of roughly 1,900 Mt (World Steel Association 2024). Production is concentrated overwhelmingly in Ontario, which hosts the large integrated and converting plants at Hamilton (ArcelorMittal Dofasco and Stelco) and Sault Ste. Marie (Algoma Steel); Québec, with electric-arc operations at Contrecoeur and Longueuil; and Saskatchewan, where the Regina works runs a scrap-based electric-arc mini-mill of about 1 Mt/yr (World Steel Association 2024). This geography maps directly onto the two production routes the model represents.

Two routes dominate world and Canadian steelmaking, and they differ enormously in energy and emissions:

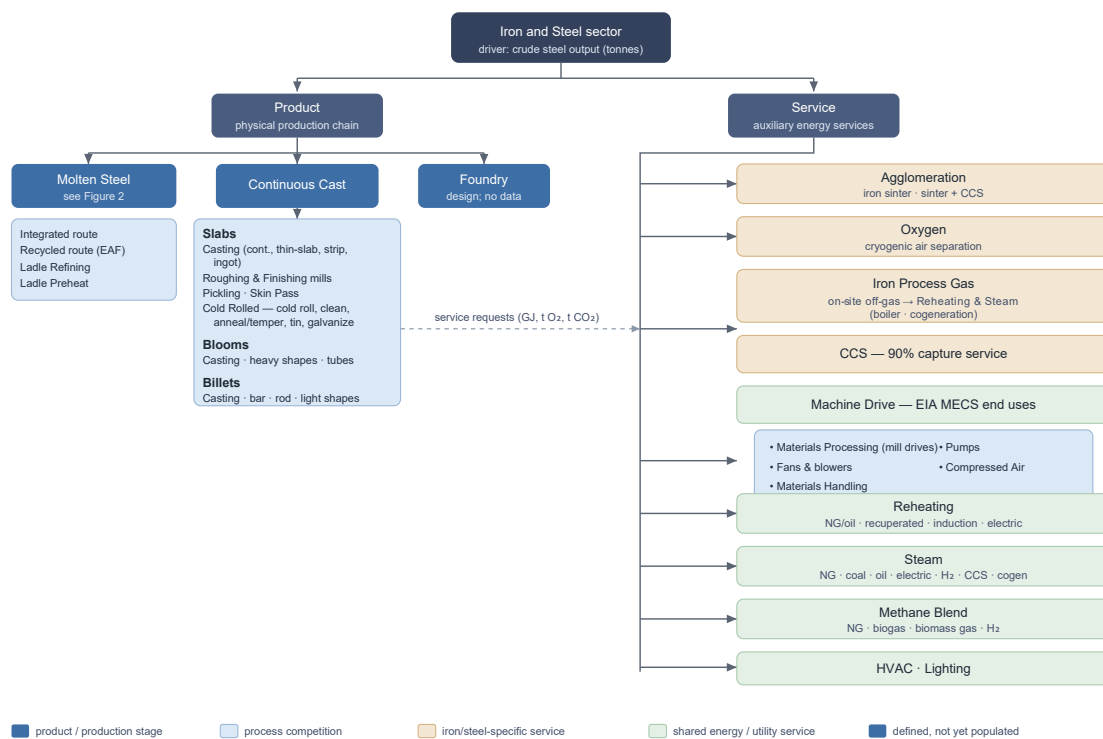
- The integrated, or primary, route reduces iron ore to pig iron in a blast furnace using coke (itself made on site from coal), then refines the liquid iron together with some scrap in a basic oxygen furnace (BOF) by injecting high-purity oxygen. The blast-furnace/BOF (BF-BOF) route is the most energy-intensive, at roughly 18–21 GJ and around 1.8–2.3 t CO<sub>2</sub> per tonne of crude steel, with most of the carbon entering as the coke that chemically reduces the ore (International Energy Agency 2020; Fan and Friedmann 2021).
- The secondary, or recycled, route melts 100% scrap in an electric arc furnace (EAF). Because it skips ore reduction entirely, it uses on the order of 2–4 GJ of electricity per tonne and a small fraction of the emissions of the integrated route; its footprint is dominated by the carbon intensity of the grid that supplies it (Fan and Friedmann 2021).

The Canadian industry is in the middle of a historic shift from the first route to the second. Algoma Steel and ArcelorMittal Dofasco are both replacing blast furnaces with electric arc furnaces, conversions expected to cut each plant's CO<sub>2</sub> by around 70% (CBC News 2025). Globally the EAF share of crude-steel output passed 29% in 2024 and is rising (World Steel Association 2024). The CIMS structure is built around exactly this distinction: it separates an Integrated ironmaking-and-steelmaking node from a Recycled (scrap-EAF) node, and it admits to the integrated node a family of emerging low-carbon primary routes — direct-reduced iron, hydrogen direct reduction, smelting reduction, and carbon capture — so that the model can represent the route shift endogenously as costs and carbon prices change.

## 21.3 Model structure

Following the standard CIMS pattern, the sector divides into two sub-trees: a **Product** sub-tree, which is the physical chain that turns ore and scrap into saleable steel, and a **Service** sub-tree, which holds the auxiliary energy and utility services the production technologies draw on. Figure 21.1 shows the overall structure.

As described in the [Sector Overview](#), each node carries a competition rule. **Fixed-ratio** nodes impose the unavoidable engineering sequence of process steps, while **tech-compete** nodes host competitions among alternative technologies on the basis of lifecycle cost (see Chapter 4). In the iron and steel tree, the disaggregation of products and process stages is built from fixed-ratio nodes, while the technology choices — which steelmaking route, which casting method, which rehear furnace, which class of motor — occur at the tech-compete leaves.



**Figure 21.1:** Energy-service flow model of the iron and steel sector (region-agnostic master tree). The sector divides into a Product sub-tree — molten-steel production (Integrated and Recycled routes), continuous casting and finishing, and foundry — and a Service sub-tree of auxiliary energy services structured on the EIA MECS end uses, plus the sector-specific Agglomeration, Oxygen, Iron Process Gas, and CCS services.

### 21.3.1 Products: molten steel

The heart of the sub-model is the **Molten Steel** node, where the metallic charge becomes liquid steel. The exogenous split between primary and secondary production is set here as a service request: a region directs part of its steel demand to the Integrated route and the remainder to the Recycled route. In the current data Ontario runs roughly two-thirds integrated and one-third recycled, drifting toward more EAF over the horizon; Québec starts similarly mixed and shifts to a majority-EAF split by 2050; and Saskatchewan is modelled as 100% recycled, reflecting its scrap-based mini-mill. The molten steel then passes through Ladle Refining and Ladle Preheat (electric or natural-gas) before casting. Figure 21.2 details the route competition.

The Integrated node is a tech-compete node whose incumbent is the conventional BF-BOF, but to which a full set of lower-carbon primary routes is admitted so they can win share as their costs and the carbon price evolve:

- Direct-reduced iron with EAF (DRI + EAF) and its hydrogen variant (HDRI + EAF) reduce ore in the solid state with reformed natural gas or hydrogen and melt the DRI in an electric furnace, avoiding the blast furnace and coke entirely.
- Smelting-reduction routes — Corex smelter, HIsarna, Smelter + EAF — reduce ore without coke ovens or sinter plants.
- Top gas recycling strips CO<sub>2</sub> from blast-furnace gas and reinjects it to cut coke use.
- Each of these (and BF-BOF itself, via the shared CCS service) has a + CCS variant that routes its flue CO<sub>2</sub> to the capture service.

The Recycled node is a tech-compete node among electric arc furnaces of increasing intensity — High power EAF, High power EAF + O<sub>2</sub>, Ultra power EAF + O<sub>2</sub> — with an Ultra power EAF + O<sub>2</sub> + CCS option that captures the small process CO<sub>2</sub> from electrode and charge carbon. The EAF carries a small direct process-emission factor in addition to its electricity draw.

TODO: The Integrated/Recycled split is currently an exogenous service-request ratio per region. Consider making it (or the scrap-supply constraint behind it) responsive to scrap availability and price, so that the secondary route's growth is limited by domestic and imported scrap rather than set by assumption.

### 21.3.2 Products: casting, finishing, and foundry

Cast and finished products are organised under a Continuous Cast node that branches into the three basic forms and their sub-forms, mirroring the way the industry reports product mix:

- Slabs run through Casting (continuous, thin-slab, strip, or the older ingot route), Slab Roughing and Finishing mills, Pickling, Skin Pass, and a Cold Rolled chain (cold reduction, electrolytic cleaning, annealing and tempering, electric/tin-plate and galvanizing lines).

- Blooms run through Bloom Casting to heavy structural shapes, tubes, and other heavy products.
- Billets run through Billet Casting to bar, rod, and light structural shapes.

Casting and rolling technologies generally compete as natural-gas versus fuel-oil fired variants, and the casting node also carries the energy-saving near-net-shape options (thin-slab and strip casting) that reduce or eliminate downstream reheating.

The master tree also defines a Foundry product node alongside Continuous Cast, for cast-iron and cast-steel products that bypass the rolling chain. It is part of the intended design but is not yet populated in the regional input files.

TODO: Build out the Foundry product branch (technologies, energy intensities, and product split) in the regional input files; it currently exists in the master tree but carries no data.

### 21.3.3 Sector-specific services

Three services are specific to ironmaking and sit in the Service sub-tree alongside the cross-cutting utilities:

- **Agglomeration** binds fine iron ore into sinter suitable for the blast furnace; it competes a conventional sinter technology against a sinter + CCS variant.
- **Oxygen** is supplied by cryogenic air separation (compression and distillation) and feeds the BOF and oxygen-enriched EAF routes; oxygen is itself an energy-intensive input.
- **Iron Process Gas** captures the energy value of blast-furnace, coke-oven, and BOF off-gases and uses them on site for **Reheating** and for **Steam** (process-gas boilers and co-generators), displacing purchased fuel.

### 21.3.4 Auxiliary energy and utility services

The remaining services are common to the industrial sectors and are documented in their own chapters in the General Services part; their general structure — and the “existing versus more efficient equipment, with fuel-switching where relevant” convention they follow — is set out in the [Sector Overview](#). This chapter does not repeat that detail. The iron and steel sector draws on:

- **Machine Drive** — see the [Industrial Machine Drive Systems](#) chapter. The sector’s large motive loads sit here: mill drives and materials processing, fans and blowers, materials handling, pumps, and compressed air.
- **Reheating** — reheat furnaces (natural-gas or oil; unrecuperated, recuperated, induction, or direct-rolling) that bring semi-finished steel back to rolling temperature.
- **Steam** — see [Low-Temperature Industrial Heat](#), covering boiler- and turbine-raised steam and its fuel options, including electric, hydrogen, and CCS-equipped variants.

- **Methane Blend** — the gaseous-fuel supply node, where natural gas competes with bio-gas, biomass gasification, and hydrogen.
- **HVAC** (space heating and cooling) and **Lighting** for mill facilities.

TODO: The current regional input files still implement an earlier, more detailed auxiliary structure (six per-horsepower machine-drive size classes; separate Compression, Conveyance, Pumping, Ventilation, and Direct Drive sub-trees) and predate the Product/Service split and the MECS-aligned Machine Drive grouping. Migrate these files to the region-agnostic master tree and the shared General Services chapters.

## 21.4 Decarbonisation technologies

The sector's emissions are dominated by the chemical reduction of iron ore in the integrated route — the carbon in coke is the reductant, not merely a fuel — so the decisive decarbonisation lever is the choice of primary route at the Molten Steel Integrated node, backed by carbon capture and by the shift of melting and heating loads to low-carbon electricity and fuels. The technologies below are admitted to the relevant tech-compete nodes; representative performance and maturity are summarised in Table 21.1. The figures quoted are indicative ranges from technology roadmaps and techno-economic reviews (International Energy Agency 2020; Fan and Friedmann 2021), not the calibrated values used in the model.

**Table 21.1:** Indicative performance and maturity of key iron and steel decarbonisation technologies

| Technology                                        | Indicative benefit                                                        | Maturity                         |
|---------------------------------------------------|---------------------------------------------------------------------------|----------------------------------|
| Scrap-based EAF (recycled route)                  | ~2–4 GJ/t and a small fraction of BF-BOF CO <sub>2</sub> ; grid-dependent | Commercial                       |
| Natural-gas DRI + EAF                             | ~1.4–1.7 t CO <sub>2</sub> /t vs ~1.8–2.3 for BF-BOF                      | Commercial                       |
| Hydrogen DRI + EAF (HDRI)                         | Up to ~95–98% CO <sub>2</sub> cut with green hydrogen                     | Demonstration / early commercial |
| Smelting reduction (HIsarna, Corex, Smelter+EAF)  | ~20% energy and CO <sub>2</sub> cut alone; up to ~80% with CCS            | Pilot / demonstration            |
| Top gas recycling blast furnace                   | ~25% CO <sub>2</sub> avoided by recycling decarbonised top gas            | Pilot                            |
| Carbon capture (BF-BOF, DRI, smelting, EAF) + CCS | ~90% capture of treated flue CO <sub>2</sub> in the model                 | Demonstration                    |
| Oxygen-enriched / ultra-high-power EAF            | Lower melt energy and tap-to-tap time                                     | Commercial                       |
| Near-net-shape casting (thin-slab, strip)         | Reduces or eliminates reheating                                           | Commercial                       |

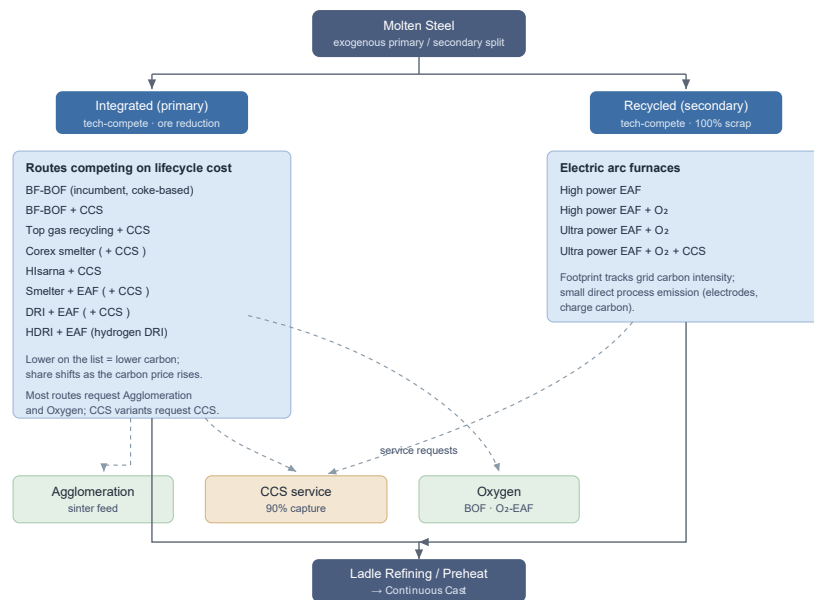
| Technology                                     | Indicative benefit                | Maturity          |
|------------------------------------------------|-----------------------------------|-------------------|
| Electric / hydrogen reheat and process steam   | Displaces gas- and oil-fired heat | See Low-Temp Heat |
| High-efficiency motors + variable-speed drives | Lower auxiliary drive energy      | See Machine Drive |

The cost estimates in Table 21.2 are order-of-magnitude figures from public techno-economic literature (International Energy Agency 2020; Fan and Friedmann 2021; SteelOnTheNet 2025) and expressed in 2025 Canadian dollars: US-dollar sources were converted at the mid-2025 rate of roughly 1.37 CAD/USD and adjusted to 2025 where the source year differed. Overnight capital costs are normalised per tonne of annual crude-steel capacity (except where another basis is noted), fixed operating and maintenance (O&M) costs follow the heavy-industry convention of roughly 3–6% of capital cost per year, and energy intensity is per tonne of crude steel. The values are for relative comparison only, not project budgeting, and are separate from the plant-specific capital and O&M values the model carries internally for each technology.

**Table 21.2:** Indicative techno-economic characteristics of iron and steel technologies, 2025 CAD

| Technology                                   | Overnight capex<br>(2025 CAD per<br>t/yr capacity)                          | Fixed O&M | Energy intensity<br>(per t crude<br>steel) | Main energy<br>carrier /<br>reductant        |
|----------------------------------------------|-----------------------------------------------------------------------------|-----------|--------------------------------------------|----------------------------------------------|
| BF–BOF<br>(integrated,<br>incumbent)         | ~2,000                                                                      | ~5%/yr    | ~18–21 GJ                                  | metallurgical<br>coal / coke (+<br>some gas) |
| Scrap EAF<br>(high-power)                    | ~450–700                                                                    | ~4%/yr    | ~2–4 GJ<br>(≈0.4–0.6<br>MWh)               | electricity (+<br>scrap)                     |
| Natural-gas<br>DRI–EAF                       | ~850–1,100                                                                  | ~5%/yr    | ~16–18 GJ                                  | natural gas +<br>electricity                 |
| Hydrogen<br>DRI–EAF<br>(HDRI)                | ~1,000–1,300                                                                | ~5%/yr    | ~11–13 GJ H <sub>2</sub> +<br>≈0.6 MWh     | hydrogen +<br>electricity                    |
| Smelting<br>reduction<br>(HIsarna,<br>Corex) | ~1,000–1,400                                                                | ~5%/yr    | ~15–17 GJ                                  | coal (no coke<br>ovens / sinter)             |
| Carbon capture<br>(±CCS add-on)              | +10–40% on<br>plant capex<br>(capture<br>≈70–140 CAD/t<br>CO <sub>2</sub> ) | —         | +2–4 GJ per t<br>CO <sub>2</sub> captured  | steam /<br>electricity                       |

| Technology                              | Overnight capex<br>(2025 CAD per<br>t/yr capacity) | Fixed O&M | Energy intensity<br>(per t crude<br>steel)        | Main energy<br>carrier /<br>reductant |
|-----------------------------------------|----------------------------------------------------|-----------|---------------------------------------------------|---------------------------------------|
| Sinter<br>agglomeration                 | ~70–140 (per<br>t/yr sinter)                       | ~5%/yr    | ~1.5–2.3 GJ/t<br>sinter                           | coke breeze /<br>gas                  |
| Cryogenic<br>oxygen (air<br>separation) | ~200–400 (per<br>t/yr O <sub>2</sub> )             | ~5%/yr    | ~0.7–0.9 GJ/t O <sub>2</sub><br>(≈200–250<br>kWh) | electricity                           |



**Figure 21.2:** Molten Steel route competition. The exogenous primary/secondary split feeds an Integrated node (BF-BOF and emerging low-carbon primary routes, each with a CCS variant) and a Recycled node (scrap EAFs of increasing power, with an oxygen-and-CCS option). Most routes draw on the shared Agglomeration, Oxygen, and CCS services.

#### 21.4.1 Primary-route shift: DRI, hydrogen, and smelting reduction

The largest single emissions reduction available to the sector is to displace the coke-based blast furnace. Direct reduced iron (DRI) reduces ore in the solid state with a reducing gas and melts the product in an EAF; with reformed natural gas it already cuts CO<sub>2</sub> to roughly 1.4–1.7 t/t from the 1.8–2.3 t/t of BF-BOF, and when the reducing gas is green hydrogen the reduction step’s emissions essentially disappear, for an overall cut of up to about 95–98% relative to the blast furnace (Fan and Friedmann 2021). The model represents both as DRI + EAF and

HDRI + EAF at the Integrated node. Smelting-reduction processes — HIsarna, Corex, and the generic Smelter + EAF — eliminate the coke ovens and sinter plant and cut energy and CO<sub>2</sub> by around a fifth on their own, rising toward 80% when paired with CCS (International Energy Agency 2020). These compete directly against BF-BOF as the carbon price rises.

#### 21.4.2 Scrap-based electric steelmaking

Where scrap is available, the Recycled route is already the lowest-emission way to make steel, because it bypasses ore reduction altogether; its footprint then tracks the carbon intensity of the electricity grid, which is low in hydro-rich Québec and increasingly so in Ontario. The model represents progressively more capable furnaces — High power and Ultra power EAFs, with oxygen enrichment to speed melting — and an Ultra power EAF + O<sub>2</sub> + CCS option to capture the residual process CO<sub>2</sub>. The practical limit on this route is scrap supply and quality, which is why the primary/secondary split is currently set exogenously rather than allowed to go fully to scrap.

#### 21.4.3 Carbon capture

Carbon capture is represented as a shared CCS service that any equipped route requests; in the model it removes 90% of the CO<sub>2</sub> presented to it. It is the only way to deeply decarbonise the routes that still reduce ore with carbon — hence the BF-BOF + CCS, Corex + CCS, HIsarna CCS, Top gas recycling CCS, Smelter + EAF + CCS, and DRI + EAF + CCS variants. The broader treatment of capture, transport, and storage, and its costs, is in the [Carbon Capture, Utilization, and Storage](#) chapter.

#### 21.4.4 Electrified and low-carbon heat, and efficient casting

Beyond the reduction step, reheating and process steam are met largely by combustion today. The low- and zero-emission alternatives — electric and hydrogen reheat furnaces, electrode and electric-resistance boilers, and process-gas cogeneration — are reached through the **Re-heating, Steam, and Iron Process Gas** services and are described in the [Low-Temperature Industrial Heat](#) chapter; the **Methane Blend** node substitutes biogas, biomass gas, and hydrogen for natural gas in any remaining gas-fired duty. On the product side, near-net-shape casting (thin-slab and strip casting) reduces or removes the reheating step entirely, and oxygen-enriched, ultra-high-power EAFs cut melting energy. High-efficiency motors and variable-speed drives in the [Machine Drive](#) service reduce the large auxiliary load.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

TODO: Hydrogen direct reduction is represented as a single HDRI + EAF technology drawing hydrogen through the Methane Blend node. Consider an explicit green-hydrogen reducing-gas supply (electrolysis, storage, and the associated electricity load) so the route's economics reflect hydrogen cost and availability rather than a blended gas price.

#### 21.4.5 Indicative capital and operating costs and energy use

Table 21.2 (above) gathers order-of-magnitude capital and operating costs and energy characteristics for the main steelmaking routes and the sector-specific services. The values are for relative comparison only, not project budgeting, and they are separate from the plant-specific capital and O&M values the model carries internally for each technology.

### 21.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, service requests for the energy and upstream services it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

A few sector-wide conventions are worth noting. Iron and steel technologies are discounted at a high rate — 35% at most process nodes in the current data, with some auxiliary and capital-intensive nodes at 40–50%. These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital. They sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions; CIMS applies this behavioural rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** at the process nodes are set around 10 so that competitions respond to cost differences without collapsing to winner-take-all. Fuel prices entering the service-cost calculation are scaled by a sector-level block of price multipliers, allowing the sector to face prices that differ from the generic delivered price of each fuel.

Greenhouse-gas costs enter through two channels. Most emissions follow from each technology's fuel service requests and the prevailing carbon price, so route-switching — from coke-based ironmaking to gas- or hydrogen-based DRI, or from natural gas to hydrogen and biogas in the methane blend — is the model's main channel for emissions reduction. In addition, the EAF routes carry a small direct process-emission factor (electrode and charge carbon), and the **CCS** service applies an emissions-removal factor (90% of the CO<sub>2</sub> routed to it) to the technologies that request it.

## 21.6 Regional coverage and differences

Because the tree is region-agnostic, each modelled province is built from the same master structure and simply carries no production on the branches that do not apply. Regional differences show up in where production sits and in the primary/secondary split rather than in the shape of the tree:

- Ontario carries most of Canada's steel and runs a mixed integrated and recycled fleet (about two-thirds integrated today), consistent with its large Hamilton and Sault Ste. Marie plants — and with the BF-to-EAF conversions now under way there.
- Québec is modelled as mixed but shifting strongly to EAF, reaching a majority-recycled split by 2050.
- Saskatchewan is modelled as 100% recycled, reflecting its single scrap-based mini-mill.

TODO: Only Ontario, Québec, and Saskatchewan are present in the current input files. Confirm whether any other province (for example Alberta or Manitoba) hosts steelmaking that should be represented, and build it out on the region-agnostic tree; otherwise document explicitly that national coverage is limited to these three regions.

TODO: The exogenous primary/secondary trajectories and the regional product mixes are inherited from older calibrations. Revisit them against the announced BF-to-EAF conversions at Algoma and ArcelorMittal Dofasco, which will materially raise Ontario's EAF share within the projection period.

## 21.7 Limitations and planned extensions

The iron and steel sub-model is a deliberately simplified representation of a complex, fast-changing industry. The TODOs above record the main gaps between this design and both the current input files and the real Canadian sector: the migration of the auxiliary services to the MECS-aligned Machine Drive and the Product/Service split; the population of the Foundry branch; an explicit scrap-supply constraint behind the primary/secondary split; an explicit green-hydrogen reducing-gas supply for the HDRI route; broader provincial coverage; and the refresh of the exogenous route trajectories in light of the conversions now under construction. Together these define the path from the current state of the model to the region-agnostic, fully populated design described in this chapter.

TODO: Replace the illustrative real-world figures and the example discount-rate values in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled iron and steel energy use and emissions against CIEEDAC, Environment and Climate Change Canada (NIR), and Natural Resources Canada data.

## 22 Light Industrial

The light industrial sector — the “other manufacturing” industries — is represented in CIMS by an energy-service flow model. The emphasis is on the structure of that model: the hierarchy of nodes, the technologies that compete at each node, and the engineering ratios that link them. The model tree is region-agnostic: a single master structure is instantiated for every province, and regional differences are expressed through which branches carry activity and through the data attached to each node. Light industrial is the aggregate of the many manufacturing industries that are individually too small to merit their own sub-model but collectively consume a large share of regional industrial energy. Because it has no single physical product, it is the one industrial sector CIMS drives with a monetary output measure rather than a tonnage, and it is modelled as a set of generic energy services rather than specific chemical or metallurgical processes.

### 22.1 Scope

CIMS groups under light industrial the manufacturing industries not represented elsewhere: food, beverage, and tobacco; rubber and plastics; leather and textiles; wood products; furniture, printing, and machinery; transportation equipment; and electronics and other manufacturing. The driving variable is the real domestic product (RDP) of these industries — their economic output in constant dollars — rather than a physical tonnage, because there is no common physical unit across such a diverse group. A forecast of RDP drives the demand for each energy service and is set exogenously.

Several adjacent activities are deliberately handled elsewhere to avoid double-counting. The energy-intensive industries that do have their own sub-models — iron and steel, chemicals, pulp and paper, industrial minerals, metal smelting, mining, and the rest — are excluded here and covered in their own chapters; light industrial is, by construction, “everything else” in manufacturing. Fuel and electricity supply enter through the energy-supply sub-models and the shared fuel nodes.

## 22.2 Real-world context

The “other manufacturing” group is heterogeneous but shares an energy signature: its demand is dominated by low- and medium-temperature heat, steam, motor-driven systems, and building services (space conditioning, water heating, lighting), rather than by high-temperature process chemistry. Food and beverage processing is typically the largest energy user in the group, and it is overwhelmingly low-temperature: on the order of 97% of its thermal demand is below about 130 °C — squarely in the range that modern industrial heat pumps can serve at a coefficient of performance of 3.5–4 (American Council for an Energy-Efficient Economy 2024; Arpagaus et al. 2018). Wood products, plastics, textiles, and the rest add their own low- and medium-temperature loads.

This matters for decarbonisation. Because the sector’s energy is general-purpose rather than locked into a specific reaction, it is among the most readily electrified parts of industry: heat pumps, electric boilers, and resistance heating can replace fossil combustion for much of the heat, high-efficiency motors and drives cut the large motive-power load, and the remaining high-temperature or hard-to-electrify duty can switch to biomass or hydrogen. The CIMS structure reflects this by representing the sector almost entirely through the shared energy-service competitions — the same Heat, Steam, Machine Drive, and building-service structures used across industry — applied to an aggregate manufacturing output.

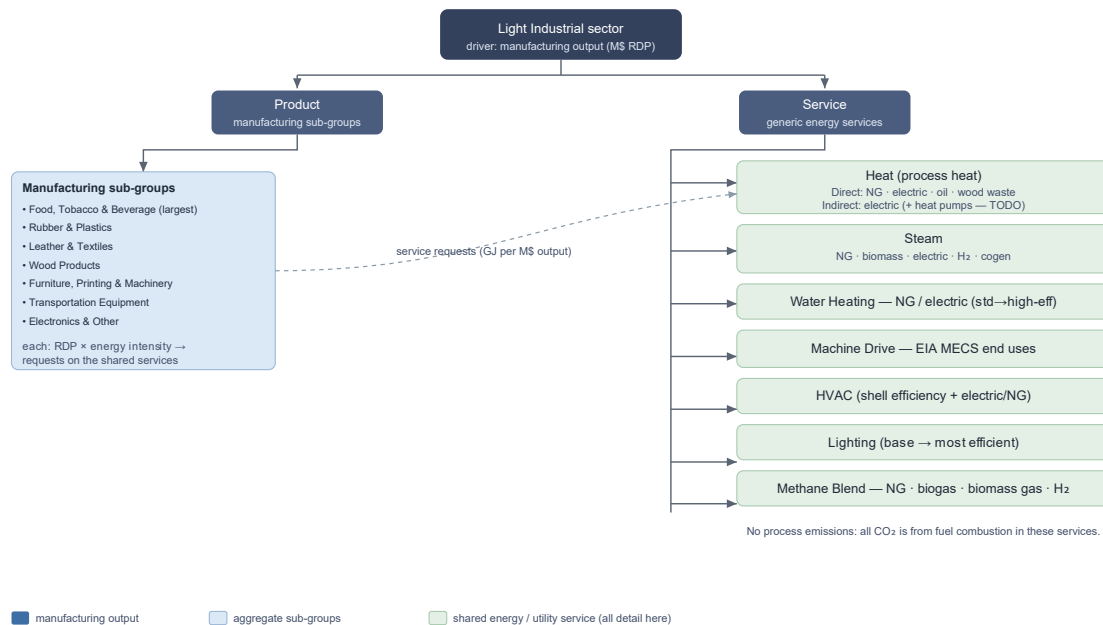
## 22.3 Model structure

The sector divides into a **Product** sub-tree — the manufacturing sub-groups whose aggregate output drives demand — and a **Service** sub-tree holding the energy services that supply them. Figure 22.1 shows the structure. Unlike the process-heavy sectors, almost all of the modelling detail is in the Service sub-tree.

As described in the [Sector Overview](#), each node carries a competition rule. The manufacturing sub-groups are fixed-ratio output nodes that translate RDP into energy-service demand; the technology choices occur at the tech-compete leaves of the services (see Chapter 4).

### 22.3.1 Products: the manufacturing sub-groups

The **Product** sub-tree divides manufacturing into seven sub-groups: Food, Tobacco, and Beverage; Rubber and Plastics; Leather and Textiles; Wood Products; Furniture, Printing, and Machinery; Transportation Equipment; and Electronics and Other. Each is an output node whose RDP, multiplied by an energy-intensity ratio, generates requests on the shared energy services. The sub-groups differ in how much heat versus motive power versus building service they draw, but they share the same service technologies.



**Figure 22.1:** Energy-service flow model of the light industrial sector (region-agnostic master tree). The Product sub-tree holds the seven manufacturing sub-groups whose aggregate output (RDP) drives demand; the Service sub-tree holds the generic energy services — process heat, steam, water heating, HVAC, lighting, and the MECS-aligned machine drive — through which essentially all of the sector’s energy is modelled.

### 22.3.2 Services: where the energy is decided

Essentially all of the sector’s energy use — and all of its emissions — is decided in the **Service** sub-tree:

- **Heat** — process heat, split into Direct (standard and efficient natural gas, electric, oil/HFO, and wood-waste) and Indirect (electric) firing. This is the central fuel-switching competition.
- **Steam** — boiler- and turbine-raised steam competing natural gas, oil, coal, electric, hydrogen, solid-biomass, and cogeneration units.
- **Water Heating** — standard, efficient, and high-efficiency natural-gas and electric units.
- **HVAC** — space conditioning, competing building-shell efficiency levels with electric and gas systems.
- **Lighting** — base, more-efficient, and most-efficient lighting.
- **Machine Drive** — the motor-driven loads (pumps, fans, compressed air, materials handling, and processing), structured on the EIA MECS end uses; see the [Industrial Machine Drive Systems](#) chapter. (The current files label this Motive Power with high- and

low-efficiency options.)

- **Methane Blend** — the gaseous-fuel supply node, where natural gas competes with bio-gas, biomass gas, and hydrogen.

The detailed structure and conventions of these shared services are set out in the [Sector Overview](#) and the General Services chapters and are not repeated here.

TODO: The current regional input files predate the full Product/Service split and the MECS-aligned Machine Drive grouping (they carry a single Motive Power node with high-/low-efficiency options rather than the Compressed Air / Fans / Materials Handling / Materials Processing / Pumps breakdown). Migrate these files to the region-agnostic master tree and the shared General Services chapters, so light industrial draws on the same machine-drive representation as the other sectors.

## 22.4 Decarbonisation technologies

Because light industrial energy is general-purpose and low-to-medium temperature, its decarbonisation is unusually tractable: it does not require new process chemistry, only electrification, fuel-switching, and efficiency applied to heat, steam, and motors. Table 22.1 summarises maturity and Table 22.2 gives indicative costs.

**Table 22.1:** Indicative performance and maturity of key light industrial decarbonisation technologies

| Technology                                     | Indicative benefit                               | Maturity   |
|------------------------------------------------|--------------------------------------------------|------------|
| Industrial heat pumps (low-temp heat)          | COP ~3.5–4; serves heat up to ~120 °C            | Commercial |
| Electric boilers / resistance heating          | Eliminates combustion where heat pumps don't fit | Commercial |
| Biomass / wood-waste heat and steam            | Low-carbon heat from residues                    | Commercial |
| Hydrogen / biogas in the methane blend         | Decarbonises remaining gas-fired duty            | Emerging   |
| Cogeneration (steam + power)                   | Higher fuel utilisation                          | Commercial |
| High-efficiency motors + variable-speed drives | Lower motive-power energy                        | Commercial |
| Efficient lighting (LED)                       | Lower lighting energy                            | Commercial |
| Building-shell & HVAC efficiency               | Lower space-conditioning energy                  | Commercial |

The cost estimates in Table 22.2 are order-of-magnitude figures from public techno-economic literature (American Council for an Energy-Efficient Economy 2024; Arpagaus et al. 2018; Madeddu et al. 2020), expressed in 2025 Canadian dollars (US- and euro-denominated sources converted at roughly 1.37 CAD/USD and 1.48 CAD/EUR and adjusted to 2025). Capital costs are per kilowatt of delivered service capacity. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 22.2:** Indicative techno-economic characteristics of light industrial decarbonisation technologies, 2025 CAD

| Decarbonisation technology                        | Capex (2025 CAD)                  | Operating characteristics                 | Energy / input        |
|---------------------------------------------------|-----------------------------------|-------------------------------------------|-----------------------|
| Industrial heat pump ( $\leq 120^\circ\text{C}$ ) | ~1,100–2,000 / kWth               | COP 3.5–4; high electricity use, low fuel | electricity           |
| Electric boiler / resistance                      | ~150–400 / kWth                   | ~99% efficient; electricity-priced heat   | electricity           |
| Biomass / wood-waste boiler                       | ~800–1,400 / kWth                 | fuel handling; low-carbon if sustainable  | solid biomass         |
| Hydrogen burner / blend                           | retrofit of existing burners      | tracks hydrogen cost                      | hydrogen / blend      |
| Cogeneration (gas, ~6–40 MW)                      | ~1,500–2,700 / kWe                | sells surplus power; higher fuel use      | natural gas / biomass |
| High-efficiency motor + VSD                       | small premium over standard motor | ~2–8% drive-energy saving                 | electricity           |
| LED lighting                                      | low; fast payback                 | large lighting-energy saving              | electricity           |

#### 22.4.1 Electrified heat: heat pumps and electric boilers

The largest lever is electrifying the sector’s abundant low- and medium-temperature heat. Industrial heat pumps deliver heat up to about  $120^\circ\text{C}$  at a coefficient of performance of 3.5–4, so each unit of electricity yields three to four units of heat — a strong economic and emissions case wherever the grid is clean, as in much of Canada (American Council for an Energy-Efficient Economy 2024; Arpagaus et al. 2018). Where temperatures or duty cycles do not suit a heat pump, electric boilers and resistance heating electrify the remainder at high efficiency but higher running cost. These appear in the model as the electric options at the **Heat**, **Steam**, and **Water Heating** services, drawing on the [Low-Temperature Industrial Heat](#) chapter.

#### 22.4.2 Biomass, hydrogen, and cogeneration

For heat that is harder to electrify, biomass and wood-waste firing (already a Heat and Steam option, and a natural fit for wood-products manufacturers with on-site residues) and hydrogen and biogas through the **Methane Blend** node decarbonise the remaining combustion. Cogeneration raises the fuel utilisation of any remaining gas use and can export surplus power.

#### 22.4.3 Efficiency in motors, lighting, and buildings

The motive-power load is reduced by high-efficiency motors and variable-speed drives in the **Machine Drive** service (see the [Industrial Machine Drive Systems](#) chapter), lighting by LED upgrades, and space-conditioning by building-shell and HVAC efficiency — all mature, low-cost, and already represented as the efficient options in their respective competitions.

TODO: Add explicit industrial heat-pump technologies to the Heat and Water Heating competitions. The current files represent electric heat as resistance/standard-electric only; heat pumps' COP of 3.5–4 makes them materially cheaper to run than resistance heating and is the most important missing low-temperature electrification option for this sector.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

### 22.5 Economics

Technology choice at every tech-compete node follows the lifecycle-cost competition of Chapter 4. Each technology carries a capital cost, fixed operating and maintenance cost, an output (node-unit) basis, service requests for the energy it consumes, and availability and lifetime years; these are annualised, combined with service costs into a lifecycle cost, and converted to market shares by the inverse-power choice rule.

Light industrial technologies are discounted at a high rate — 35% at most nodes in the current data, with some at 40–50%. These values are **behavioural** (revealed or empirical) discount rates, not true financial costs of capital. They sit well above the single-digit rates a firm's cost of capital would imply because they summarise the short payback horizons, risk aversion, and other non-financial barriers that govern real technology-adoption decisions; CIMS applies this behavioural rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 10 so that competitions respond to cost differences without collapsing to winner-take-all.

The sector's emissions accounting is the simplest kind: there are no process emissions, so all CO<sub>2</sub> comes from fuel combustion in the Heat, Steam, and Water Heating services, entering through each technology's fuel service requests and the carbon price. Electrification (heat pumps), fuel-switching, and energy efficiency are therefore the only channels for emissions reduction, and their effect depends on the carbon intensity of the regional grid. Because output is measured in real dollars, energy intensities are expressed per unit of RDP, and the sector's projected energy use is driven by the interaction of economic growth with these efficiency and fuel-switching choices.

## 22.6 Regional coverage and differences

Light industrial has the most complete regional coverage of any industrial sector in the model: all seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the full structure, because every region has a substantial “other manufacturing” base. Differences are matters of scale and growth: Ontario and Québec are the largest, with Québec and Alberta growing strongly over the projection, while Ontario's modelled output is roughly flat to declining. Because the sector is driven by economic output rather than a commodity, its energy use tends to rise with regional GDP unless offset by efficiency and electrification.

TODO: The relative energy intensities and sub-group mix are inherited from older calibrations. Revisit them against current CIEEDAC manufacturing energy-use data, since the composition of “other manufacturing” (and the share of food and beverage in particular) varies considerably by region.

## 22.7 Limitations and planned extensions

The light industrial sub-model is deliberately a generic-services representation: it trades process detail for breadth, capturing a large and diverse slice of manufacturing through a compact set of energy-service competitions. Its main limitations follow from that design — it cannot represent process-specific options, and it depends entirely on the shared service technologies being kept current. The TODOs above record the priorities: migrating to the MECS-aligned Machine Drive, adding industrial heat pumps to the heat competitions, and refreshing the intensities and sub-group mix against current data.

TODO: Replace the illustrative real-world figures and the example discount-rate values in this chapter with quantitative values drawn directly from the model input files, and add a benchmark of modelled light industrial energy use against CIEEDAC and Natural Resources Canada data, by sub-group where possible.

## 23 Construction

The industry's on-site greenhouse-gas footprint comes almost entirely from the diesel burned by off-road equipment — excavators, loaders, dozers, cranes, and the like working on building and civil-engineering sites — so the sector is represented as a single mobile-equipment energy service in which diesel competes against electric. The tree is region-agnostic, and the sector's activity is measured directly in tonnes of CO<sub>2</sub>-equivalent, signalling that emissions, not output, are its organising variable. The structure is deliberately coarse; the limitations below set out what a fuller representation would add.

### 23.1 Scope

CIMS represents the **fuel use of construction's mobile equipment** — the off-road machinery that excavates, lifts, grades, and moves material on site — as one aggregate energy service. The driving variable is an exogenous forecast of the sector's emissions in tonnes of CO<sub>2</sub>-equivalent. Importantly, this is the operational fuel of the construction process only. The far larger embodied-carbon question — the emissions of making the cement, steel, and other materials that go into a building — is accounted in the industries that produce those materials, such as [Industrial Minerals](#) (cement) and [Iron and Steel](#), not here. The energy the finished building later consumes belongs to [Residential](#) and [Commercial](#).

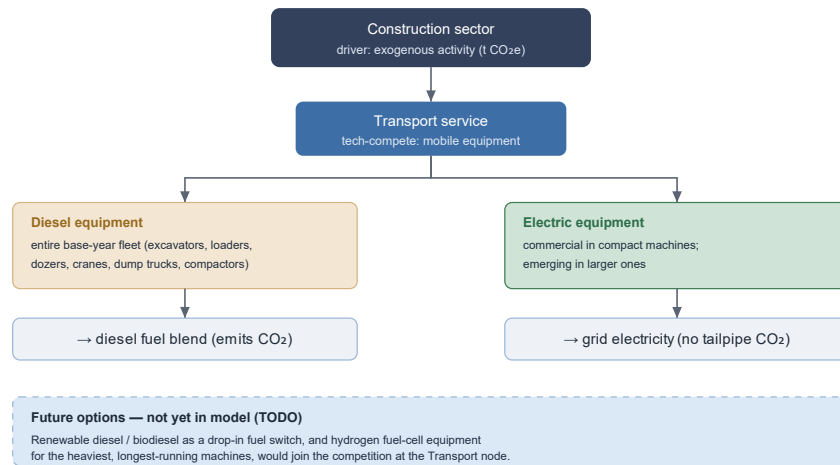
### 23.2 Real-world context

Construction's direct emissions are an off-road diesel problem. Sites run a wide mix of heavy machines — excavators, wheel loaders, dozers, graders, cranes, dump trucks, and compactors — most of them diesel, often idling and working in stop-start cycles. The decarbonisation options are the familiar three for off-road fleets: drop-in **renewable diesel** or biodiesel that cuts lifecycle carbon with no equipment change, battery-electric machines, and hydrogen fuel-cell machines for the heaviest duty. Electrification has moved fastest here of any off-road sector, because many construction machines are compact, work fixed-hour shifts, and operate in cities where noise and local air quality matter: major manufacturers now ship battery-electric excavators and loaders, and electric machines increasingly match diesel on power and cycle time, at a purchase premium on the order of 50–100% that is partly offset by lower fuel and

maintenance costs (IDTechEx 2024). The CIMS structure mirrors the core choice directly: a single competition between diesel and electric equipment.

### 23.3 Model structure

The sector is a single tech-compete node. Figure 23.1 shows the structure: the exogenous activity drives one **Transport** service, at which two technologies compete to provide the same mobile-equipment work.



**Figure 23.1:** Energy-service flow model of the construction sector (region-agnostic master tree). An exogenous activity forecast in tonnes of CO<sub>2</sub>-equivalent drives a single **Transport** service node, a tech-compete between diesel equipment (which draws on the diesel fuel blend and emits CO<sub>2</sub>) and electric equipment (which draws grid electricity). Lower-carbon drop-in fuels and hydrogen are noted as future options not yet in the model.

At the **Transport** node, diesel equipment (the entire base-year fleet) requests the diesel transportation fuel blend, while electric equipment requests grid electricity; both carry a twenty-year lifetime and the same energy requirement per unit of activity. Because the diesel option draws on the blended fuel, any renewable-diesel content in that blend lowers its lifecycle carbon automatically (see [Biodiesel](#)). Fuel prices at the node are currently borrowed from [Agriculture](#) as a proxy and flagged for review.

## 23.4 Decarbonisation technologies

Construction decarbonisation means getting the carbon out of the off-road fleet's fuel: cleaner liquid fuel now, electrification where it fits, and hydrogen for the hardest duty cycles. Table 23.1 summarises maturity and Table 23.2 gives indicative costs.

**Table 23.1:** Indicative performance and maturity of key construction decarbonisation options

| Technology / practice                  | Indicative benefit                                      | Maturity                                                                  |
|----------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------|
| Renewable diesel / biodiesel (drop-in) | Cuts lifecycle CO <sub>2</sub> with no equipment change | Commercial                                                                |
| Battery-electric equipment             | Removes tailpipe CO <sub>2</sub> ; shifts load to grid  | Commercial (compact) / emerging (large) — <i>electric option in model</i> |
| Hybrid diesel-electric machines        | Lowers fuel use on stop-start cycles                    | Commercial — <i>not yet in model</i>                                      |
| Hydrogen fuel-cell equipment           | Zero-tailpipe for heavy, long-run machines              | Early / demonstration — <i>not yet in model</i>                           |
| Idle reduction & efficient machines    | Lower fuel per unit of work                             | Commercial — <i>not explicit in model</i>                                 |

The estimates in Table 23.2 are order-of-magnitude figures from public literature (IDTechEx 2024; Canada Energy Regulator 2023; SRK Consulting 2024), expressed in 2025 Canadian dollars. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 23.2:** Indicative techno-economic characteristics of construction decarbonisation options, 2025 CAD

| Option                       | Capital vs diesel          | Operating effect                        | Energy / emission                              |
|------------------------------|----------------------------|-----------------------------------------|------------------------------------------------|
| Renewable diesel / biodiesel | none (drop-in fuel)        | fuel-cost dependent                     | lifecycle CO <sub>2</sub> cut, no fleet change |
| Battery-electric machine     | ~+50–100% on machine price | ~50% lower maintenance; lower fuel cost | grid electricity; no tailpipe CO <sub>2</sub>  |
| Hydrogen fuel-cell machine   | high (early-stage)         | fuel-cost dependent                     | zero-tailpipe; needs H <sub>2</sub> supply     |

The cheapest near-term lever is the fuel: switching the diesel blend toward **renewable diesel** or biodiesel cuts lifecycle carbon across the whole fleet with no change to the machines. Battery-electric equipment — the model’s electric option — removes tailpipe emissions and cuts maintenance and fuel cost; it is already commercial in compact machines and emerging in larger ones, helped by the urban, fixed-shift nature of much construction work. Hydrogen fuel cells remain early-stage and depend on on-site hydrogen supply.

TODO: Add the renewable-diesel/biodiesel fuel-switching and hydrogen fuel-cell pathways as explicit competing options at the Transport node, and populate the technologies with capital, operating, and emission data rather than the current Agriculture-proxied fuel prices.

## 23.5 Economics

Technology choice at the **Transport** node follows the lifecycle-cost competition of Chapter 4. As elsewhere in CIMS, the intended discount rate is **behavioural** (revealed or empirical) rather than a true financial cost of capital, capturing the short payback horizons and adoption barriers typical of the trade (see the [Theory](#) chapter). In the current input files, however, the two technologies carry only a lifetime and their energy requests — capital cost, operating cost, the discount rate, and the heterogeneity (variance) parameter are not yet populated — so the diesel-versus-electric competition cannot yet resolve on cost and the fleet holds near its base-year shares. Emissions enter upstream through the fuels the technologies request: the diesel option carries the carbon of the diesel blend, while the electric option moves the load to the grid, where it is counted in [Electricity](#).

## 23.6 Regional coverage

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the construction structure, scaled by an exogenous emissions forecast that tracks the scale of building activity: Ontario and Québec are the largest, with Alberta and British Columbia next, and the smaller provinces well below. The structure itself is identical across regions.

## 23.7 Limitations and planned extensions

The construction sub-model is a placeholder: it captures the sector’s dominant direct emission source — off-road diesel equipment — and the central diesel-to-electric choice, but little else. The off-road fleet is collapsed into a single aggregate **Transport** service rather than resolved by machine type or duty cycle; the competing technologies are not yet costed; the fuel prices are proxied from agriculture; and lower-carbon liquid fuels and hydrogen are not yet explicit

options. The sub-model also covers only operational site fuel, not the embodied carbon of construction materials, which lives in the materials-producing sectors. The TODOs above record the priorities; populating the technology costs is the minimum needed for the diesel-to-electric transition to respond to policy at all.

TODO: Replace the illustrative figures and the Agriculture-proxied prices in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled construction fuel use against Statistics Canada and Environment and Climate Change Canada inventory data.

## 24 Residential

The residential buildings sector is represented in CIMS as a demand-side sub-model, like the commercial sector: it consumes energy to deliver household **energy services** — space and water heating, cooling, lighting, appliances, and cooking — across the full range of dwellings. The emphasis is on the structure of the energy-service flow model: the dwelling types and vintages that drive demand, the end-use services, and the technologies that compete to supply them. The tree is region-agnostic but climate-sensitive, so the housing stock is resolved by building type, construction vintage, and climate zone. The sector has **no process emissions**; all of its CO<sub>2</sub> comes from the fuels burned for space and water heating, so — as in commercial buildings — decarbonisation is fundamentally a story of electrification and efficiency.

### 24.1 Scope

CIMS represents the energy use of all residential dwellings — single-family detached houses, attached/row housing, low-rise walk-ups, and large apartment buildings (the last included here, rather than in commercial, because their consumption patterns fit the residential model). The driving variable is the number of households, projected exogenously from population growth; it sets the demand for every service except water heating and furnace-fan demand, which are determined by the heating and hot-water technologies themselves.

The residential sector accounts for roughly 17% of all energy consumed in Canada. Space heating is about 60% of that, water heating about 21%, and the remainder is lighting, appliances, and other uses (Natural Resources Canada [2024b](#)). The combustion of the delivered fuels is accounted here; the upstream supply of those fuels and of electricity is in the energy-supply sub-models.

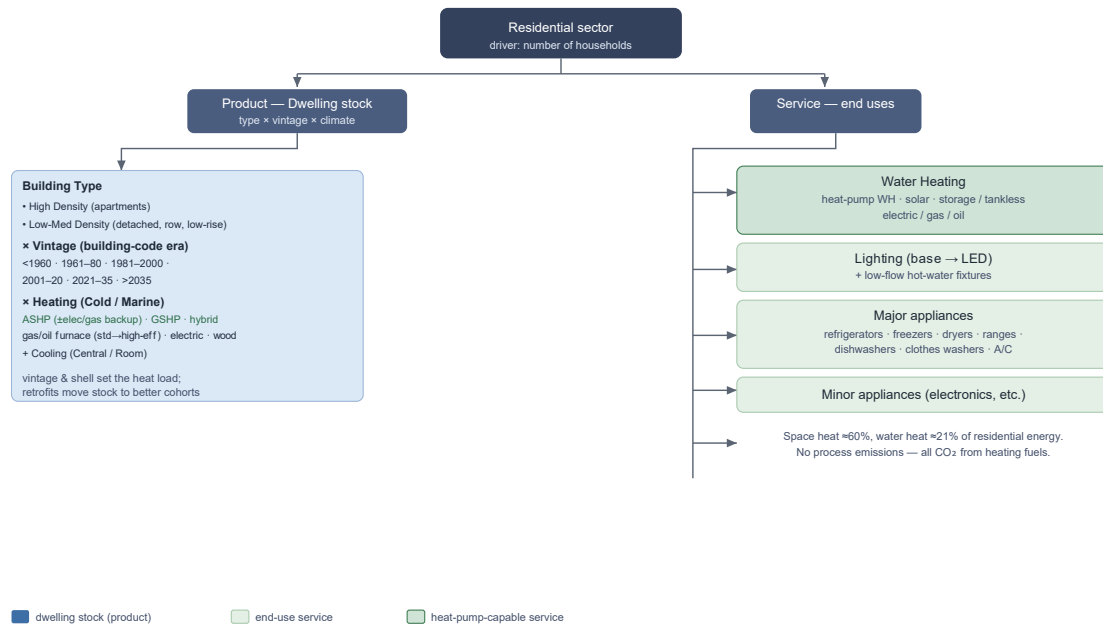
### 24.2 Real-world context

Household energy is dominated by space and water heating, and that is where the direct emissions sit, because lighting and most appliances are already electric. Heating demand depends on the dwelling — its size, envelope, vintage, and climate — and on the heating system, which in Canada ranges from oil and natural-gas furnaces through electric baseboards and wood stoves to heat pumps.

The decarbonisation path mirrors the commercial sector: because household heat is low-temperature, the stock is highly electrifiable. Air- and ground-source heat pumps can deliver space and water heat at a coefficient of performance well above one, cold-climate and hybrid (heat-pump-with-fuel-backup) systems extend that into Canadian winters, and building-envelope improvements (tracked through construction-vintage cohorts and shell upgrades) cut the heat load at source. Efficient appliances, lighting (LED), and water fixtures reduce the rest. The CIMS structure captures all of this and, like the commercial model, links the building shell to the heating demand it creates.

## 24.3 Model structure

The sector divides into a **Product** sub-tree — the dwelling stock, resolved by building type, vintage, and climate — and a **Service** sub-tree of the household end-uses. Figure 24.1 shows the structure. Each node carries a competition rule (see Chapter 4).



**Figure 24.1:** Energy-service flow model of the residential sector (region-agnostic master tree). The Product sub-tree is the dwelling stock — split by building type (High Density / Low-Med Density), construction vintage (building-code era), and climate zone (Cold / Marine), with cooling — and the Service sub-tree holds the household end-uses: water heating, lighting, and the major and minor appliances.

### 24.3.1 Products: the dwelling stock

The **Building Type** node resolves the housing stock along three dimensions:

- **Building type** — High Density (apartments) and Low-Med Density (single-family detached, attached/row, low-rise), reflecting their very different heat loads per dwelling.
- **Vintage** — construction-code cohorts (<1960, 1961–1980, 1981–2000, 2001–2020, 2021–2035, >2035), so that the model carries the thermal performance of each era of the stock and can represent both new construction to current codes and retrofits of older stock.
- **Climate zone** — each vintage carries Heating (Cold) and Heating (Marine) competitions, since heating demand and heat-pump performance differ sharply between Canada’s cold interior and milder coastal regions.
- **Cooling** — Central and Room air-conditioning, where applicable.

Within each dwelling/vintage/climate combination, the Heating node competes the systems: electric standard and baseboard, natural-gas and oil furnaces (existing, standard, and high-efficiency), wood, air-source heat pumps (with electric or gas backup), ground-source heat pumps, and gas-ASHP hybrids. The vintage-and-shell structure is how the model represents the interaction between envelope quality and heating demand: a better shell (newer vintage or a retrofit) lowers the load the heating system must meet.

### 24.3.2 Services: water heating, lighting, and appliances

- **Water Heating** — storage and tankless units on electricity, gas, and oil; heat-pump water heaters (Electric- and NG-ASHP); and solar pre-heat with electric or gas backup. Water-heating demand is itself driven by the hot-water-using services (showers/faucets, dishwashing, clothes washing).
- **Lighting** — efficiency levels from base stock to LED.
- **Major appliances** — refrigerators, freezers, dryers (air-dry or machine), ranges, dishwashers (machine and non-machine), clothes washers, and air conditioners, each modelled by unit and market penetration with a range of efficiency levels.
- **Minor appliances** — a diverse, growing category (electronics, block heaters, etc.) whose efficiency gains are often outweighed by rising ownership.

TODO: The current regional input files have moved to a vintage-and-climate structure that is close to, but not identical with, the master tree’s Product/Service split, and some appliance services remain aggregated. Complete the migration to the region-agnostic master tree and the shared General Services chapters.

## 24.4 Decarbonisation technologies

Residential decarbonisation is electrification plus efficiency: replace fossil furnaces with heat pumps, electrify water heating, tighten the envelope, and improve appliances and lighting. Table 24.1 summarises maturity and Table 24.2 gives indicative costs.

**Table 24.1:** Indicative performance and maturity of key residential decarbonisation technologies

| Technology                          | Indicative benefit                       | Maturity   |
|-------------------------------------|------------------------------------------|------------|
| Air-source heat pump (space heat)   | COP 2–3.5; displaces furnace fuel        | Commercial |
| Cold-climate / hybrid heat pump     | Heat pump in winter, fuel backup on peak | Commercial |
| Ground-source heat pump             | Higher COP; higher capital               | Commercial |
| Heat-pump water heater              | Efficient electric hot water             | Commercial |
| Solar water heating                 | Pre-heats hot water                      | Commercial |
| Envelope retrofit / new-code shell  | Lowers heat load at source               | Commercial |
| Efficient appliances + LED lighting | Lower electricity use                    | Commercial |
| Low-flow hot-water fixtures         | Lowers hot-water (and heating) demand    | Commercial |

The estimates in Table 24.2 are order-of-magnitude figures from public literature (Arpagaus et al. 2018; American Council for an Energy-Efficient Economy 2024; Natural Resources Canada 2024b), in 2025 Canadian dollars (US sources converted at roughly 1.37 CAD/USD and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 24.2:** Indicative techno-economic characteristics of residential decarbonisation technologies, 2025 CAD

| Technology                  | Capex (2025 CAD)      | Operating characteristics            | Energy / input         |
|-----------------------------|-----------------------|--------------------------------------|------------------------|
| Air-source heat pump (home) | ~6,000–18,000 / home  | COP 2–3.5; fuel backup on cold days  | electricity (+ backup) |
| Cold-climate heat pump      | ~10,000–22,000 / home | maintains output to low temperatures | electricity            |
| Ground-source heat pump     | ~25,000–55,000 / home | COP 3.5–5; ground loop cost          | electricity            |

| Technology                 | Capex (2025 CAD)      | Operating characteristics   | Energy / input |
|----------------------------|-----------------------|-----------------------------|----------------|
| Heat-pump water heater     | ~3,000–6,000 / unit   | COP 2.5–3.5                 | electricity    |
| Envelope retrofit          | ~10,000–40,000 / home | reduces heat & cooling load | — (passive)    |
| Efficient appliances / LED | low incremental       | lower electricity use       | electricity    |

#### 24.4.1 Heat pumps and the cold-climate question

As in commercial buildings, the central lever is the heat pump. The model competes air-source and ground-source heat pumps and, importantly for Canada, hybrid systems pairing a heat pump with an electric or gas backup for the coldest hours — which is why heating is split into Cold and Marine competitions. Cold-climate heat pumps that maintain output at low temperatures are the key enabling technology for electrifying heat across most of the country. Heat-pump water heaters and solar water heating electrify domestic hot water. These draw on the [Low-Temperature Industrial Heat](#) chapter’s broader heat-pump treatment.

#### 24.4.2 Envelope, appliances, and behaviour

The vintage-and-shell structure makes envelope improvement a first-class lever: new dwellings built to tighter codes (the later vintage cohorts) and retrofits of older stock both reduce the heat load before any system runs. Efficient appliances, LED lighting, and low-flow hot-water fixtures cut the non-heating loads — the last also reducing water-heating demand.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files, and ensure cold-climate heat-pump performance curves are reflected in the Cold-zone heating competition.

### 24.5 Economics

Technology choice at every node follows the lifecycle-cost competition of Chapter 4, with extensive use of retrofit competitions: older-vintage dwellings can retrofit to better-insulated archetypes, and existing heating and water-heating stock can be upgraded, so the model represents improvements to the existing stock and not only new-build choices.

Residential technologies are discounted at high rates — around 25–30% in the current data. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than financial costs of capital: households consistently behave as though they discount future energy savings far more steeply than a financial calculation would, and the behavioural rate encodes that wedge — risk aversion, up-front-cost sensitivity, information, and transaction costs. CIMS applies it throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** vary across the end-uses.

The emissions accounting carries **no process emissions**: all CO<sub>2</sub> comes from fuel combustion in space and water heating (and gas cooking), entering through each technology’s fuel service requests and the carbon price. Electrification (heat pumps), fuel-switching, and efficiency are the only channels for reduction, and their emissions benefit depends on the carbon intensity of the regional grid — large where the grid is clean (Québec, BC, Manitoba), smaller where it is fossil-heavy (Alberta, Saskatchewan). Because the high behavioural discount rates make households reluctant to pay up-front for efficient equipment, the model reproduces the well-documented gap between cost-effective and actually-adopted efficiency — which is itself a target of policy.

## 24.6 Regional coverage and differences

All seven modelled regions carry the residential structure, and regional differences are pronounced because climate, fuel availability, and grid carbon intensity all vary:

- The Cold / Marine climate split distinguishes most of Canada (Cold) from milder coastal British Columbia and parts of the Atlantic (Marine); coastal and interior British Columbia are modelled as distinct climates for single-family homes.
- Fuel mix varies: natural gas dominates heating in the West and Ontario, electricity in Québec and parts of the Atlantic, and oil and wood remain significant in the Atlantic region and rural areas.
- Air conditioning is a large load in hot-summer regions such as Ontario and much smaller elsewhere.

TODO: Confirm the regional dwelling-type, vintage, and fuel-mix shares against current housing-stock and Comprehensive Energy Use Database data, since heating demand and heat-pump economics depend heavily on all three.

## 24.7 Limitations and planned extensions

The residential sub-model is a detailed, vintage- and climate-resolved buildings representation with explicit retrofit competitions and a full set of heat-pump and efficiency options. Its

limitations are the usual ones for a stock model: reliance on archetype generalisations, inherited calibration of dwelling and appliance intensities, and incomplete representation of behavioural and rebound effects (e.g. minor-appliance growth outpacing efficiency). The TODOs above record the priorities, of which cold-climate heat-pump performance and refreshed stock data matter most for current analysis.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled residential energy use by end-use and dwelling type against Natural Resources Canada's Comprehensive Energy Use Database.

## 25 Commercial and Institutional

The commercial and institutional buildings sector is represented in CIMS as a demand-side sub-model: it does not make a physical product but consumes energy to deliver building **energy services** — heating, cooling, ventilation, lighting, refrigeration, cooking, hot water, and plug loads — across offices, stores, schools, hospitals, hotels, and the rest of the non-residential building stock. The emphasis, as elsewhere, is on the structure of the energy-service flow model: the building segments that drive demand, the end-use services, and the technologies that compete to supply them. The tree is region-agnostic, but building energy use is climate-sensitive, so the model splits the stock by climate zone. The sector has **no process emissions**; all of its CO<sub>2</sub> comes from the fuels burned for heating, so its decarbonisation is a story of electrification and efficiency.

### 25.1 Scope

CIMS represents the energy services of commercial and institutional buildings, driven by floor space. It deliberately excludes light industrial facilities (in [Light Industrial](#)) and high-rise residential apartments (in [Residential](#)). The commercial building sector accounts for roughly 20% of all energy consumed in Canada, of which lighting and HVAC together are about 60% (Natural Resources Canada [2024b](#)). The model includes the end-uses significant enough to matter — space heating and cooling, ventilation, lighting, refrigeration, cooking, hot water, and plug load — and drives each through engineering ratios that link it to floor space; floor space itself is projected exogenously from population and economic growth. Exterior lighting (street, parking, architectural) is not represented for lack of data.

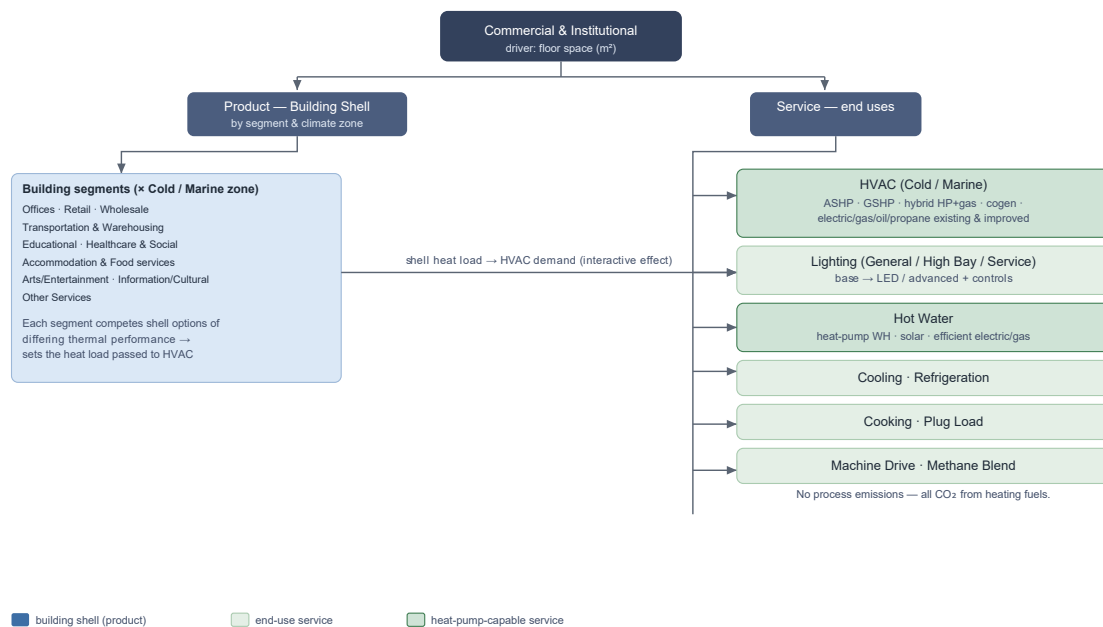
### 25.2 Real-world context

Commercial buildings are diverse — a warehouse, a hospital, and a small office have very different energy profiles — but their energy use shares a common structure dominated by space conditioning and lighting. The most common fuels are, in order, electricity, natural gas, and light fuel oil; electricity is the sole source for lighting, refrigeration, plug load, cooling, fans, and pumps, while space and water heating use all three. Heating is where most of the sector's direct emissions arise, since lighting and plug loads are already electric.

The decarbonisation path is correspondingly clear. Because the loads are general-purpose and mostly low-temperature, the sector is highly electrifiable: heat pumps (air- and ground-source) can serve space and water heating at a coefficient of performance well above one; LED lighting and efficient HVAC, controls, and refrigeration cut electricity use; and a tighter building envelope reduces the heating and cooling load at source. Larger facilities with coincident heat and power demand can use cogeneration. The CIMS structure captures all of these, and — distinctively for a buildings model — it represents the interaction between the building shell and the HVAC system, since a better envelope reduces the heat load the HVAC must meet.

## 25.3 Model structure

The sector divides into a **Product** sub-tree — the building shell, disaggregated by building segment and climate zone — and a **Service** sub-tree of the energy end-uses. Figure 25.1 shows the structure. The building shell is the organising “product” because the choice of envelope determines the heating and cooling demand placed on the services. Each node carries a competition rule (see Chapter 4).



**Figure 25.1:** Energy-service flow model of the commercial sector (region-agnostic master tree). The Product sub-tree is the building shell, split by building segment (offices, retail, healthcare, education, accommodation, etc.) and by climate zone (Cold / Marine); the Service sub-tree holds the end-use services — HVAC, lighting, cooling, hot water, cooking, refrigeration, plug load, and machine drive — that the shell’s demand draws on.

### 25.3.1 Products: the building shell, by segment and climate

The **Building Shell** node disaggregates the stock into about a dozen building segments — Offices, Retail, Wholesale, Transportation and Warehousing, Educational, Healthcare and Social Assistance, Accommodation and Food Services, Arts/Entertainment/Recreation, Information and Cultural, and Other Services — each split by climate zone (Cold / Marine). Each segment competes shell options of differing thermal performance, which set the heat load passed to the HVAC service. This shell-to-HVAC link is the model's way of capturing the main interactive effect in commercial buildings: investing in a better envelope reduces the energy the heating and cooling systems must deliver.

### 25.3.2 Services: the end-uses

The **Service** sub-tree holds the end-uses, each linked to floor space (or to the shell's heat load) by historical engineering ratios:

- **HVAC** (split Cold / Marine) — the dominant service, modelled as integrated single systems that compete electric and gas existing/improved units, air-source heat pumps (ASHP) and ground-source heat pumps (GSHP), hybrid heat-pump-with-gas-backup systems, oil and propane, and natural-gas cogeneration.
- **Lighting** — three sub-categories (General Area, High Bay, Service), competing efficiency levels from base stock to advanced/LED.
- **Hot Water** — efficient electric, gas, and oil units, heat-pump water heaters (Electric-ASHP), solar pre-heat, and service-load-reduction options.
- **Cooling, Refrigeration, Cooking, Plug Load** — the remaining electric (and, for cooking, gas) end-uses, modelled at average intensities that evolve with equipment efficiency.
- **Machine Drive and Methane Blend** — the shared motor and gaseous-fuel services.

TODO: The current regional input files predate parts of the Product/Service split and label some services differently (e.g. Motive Power rather than the MECS-aligned Machine Drive). Migrate them fully to the region-agnostic master tree and the shared General Services chapters.

## 25.4 Decarbonisation technologies

Commercial decarbonisation is electrification plus efficiency: replace fossil heating with heat pumps, cut electricity demand with efficient lighting, HVAC, and refrigeration, and reduce loads at source with a better envelope. Table 25.1 summarises maturity and Table 25.2 gives indicative costs.

**Table 25.1:** Indicative performance and maturity of key commercial decarbonisation technologies

| Technology                        | Indicative benefit                      | Maturity   |
|-----------------------------------|-----------------------------------------|------------|
| Air-source heat pump (space heat) | COP 2–3.5; displaces gas/oil heat       | Commercial |
| Ground-source heat pump           | Higher COP; higher capital              | Commercial |
| Hybrid heat pump + gas backup     | Heat pump for most hours, gas on peak   | Commercial |
| Heat-pump water heater            | Efficient electric hot water            | Commercial |
| Building-envelope / shell upgrade | Lowers heating & cooling load at source | Commercial |
| LED lighting + controls           | Large lighting-electricity cut          | Commercial |
| Efficient HVAC + smart controls   | Lower HVAC energy                       | Commercial |
| Efficient / low-GWP refrigeration | Lower energy and refrigerant emissions  | Commercial |
| Cogeneration (large facilities)   | Combined heat and power; surplus export | Commercial |

The estimates in Table 25.2 are order-of-magnitude figures from public literature (Arpagaus et al. 2018; American Council for an Energy-Efficient Economy 2024; Natural Resources Canada 2024b), in 2025 Canadian dollars (US sources converted at roughly 1.37 CAD/USD and adjusted to 2025). Capital costs are per kW of delivered capacity unless noted. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 25.2:** Indicative techno-economic characteristics of commercial decarbonisation technologies, 2025 CAD

| Technology               | Capex (2025 CAD)    | Operating characteristics      | Energy / input             |
|--------------------------|---------------------|--------------------------------|----------------------------|
| Air-source heat pump     | ~1,400–2,800 / kWth | COP 2–3.5; backup on cold days | electricity (+ gas backup) |
| Ground-source heat pump  | ~3,000–6,000 / kWth | COP 3.5–5; ground loop cost    | electricity                |
| Heat-pump water heater   | ~2,000–4,000 / unit | COP 2.5–3.5                    | electricity                |
| Envelope / shell upgrade | building-specific   | reduces heat & cool load       | — (passive)                |
| LED lighting + controls  | low; fast payback   | large electricity saving       | electricity                |

| Technology                    | Capex (2025 CAD)   | Operating characteristics           | Energy / input |
|-------------------------------|--------------------|-------------------------------------|----------------|
| Gas cogeneration (large bldg) | ~1,800–3,000 / kWe | sized to heat load; exports surplus | natural gas    |

#### 25.4.1 Heat pumps: the central lever

The largest direct-emissions source in commercial buildings is fossil space and water heating, and the central decarbonisation lever is the heat pump. The model compares air-source and ground-source heat pumps, and hybrid systems that pair a heat pump with a gas backup for the coldest hours — a configuration well suited to Canada’s cold climate, which is why the model splits HVAC into Cold and Marine zones. Heat-pump water heaters electrify domestic hot water. These options draw on the broader treatment in the [Low-Temperature Industrial Heat](#) and [Cogeneration](#) chapters.

#### 25.4.2 Efficiency: envelope, lighting, HVAC, and refrigeration

Reducing demand is as important as switching fuel. A tighter building envelope lowers the heating and cooling load before any equipment runs — the shell-to-HVAC link in the model captures exactly this interaction. LED lighting with controls sharply cuts the single largest electricity end-use, efficient HVAC with smart controls trims the next largest, and efficient, low-GWP refrigeration addresses both energy and refrigerant emissions.

#### 25.4.3 Cogeneration in large facilities

Large institutions with steady, coincident heat and power demand — hospitals, universities, large offices, hotels — can use natural-gas cogeneration, which the model admits to those segments (limited exogenously to 30–50% of the relevant demand) and sizes to the facility’s heat load, selling any surplus electricity back to the grid.

TODO: Replace the indicative literature ranges above with the capital costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files, and add cold-climate (low-temperature) heat-pump performance curves so the Cold-zone HVAC competition reflects real winter coefficients of performance.

## 25.5 Economics

Technology choice at every node follows the lifecycle-cost competition of Chapter 4, applied here to building equipment and shells. The sector also makes notable use of retrofit technologies: once existing stock retires, CIMS sets up a competition between leaving the “remembered” stock as-is (operating and maintenance cost only) and attaching a retrofit (adding annualised retrofit capital), so that efficiency upgrades to the existing stock are represented, not just new-build choices.

Commercial technologies are discounted at a high rate — generally 40% in the current data — reflecting the rapid-payback requirements, risk aversion, and capital constraints typical of commercial energy decisions. Two exceptions are encoded: cogeneration and the hospital and school shell/HVAC technologies are discounted at 30%, reflecting their longer, owner-operated planning horizons. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than financial costs of capital — the chapter’s own data notes they are “ex post measures ... meant to encompass many more factors than simple rate-of-return criterion” — and CIMS applies them throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4.

The emissions accounting is the simplest kind: there are **no process emissions**, so all CO<sub>2</sub> comes from fuel combustion in the heating, hot-water, and cooking services, entering through each technology’s fuel service requests and the carbon price. Electrification (heat pumps), fuel-switching, and efficiency are therefore the only channels for emissions reduction, and their effect depends on the carbon intensity of the regional grid. A known simplification is that energy prices are average per-unit charges, so the demand-charge savings of efficiency measures are not captured; the model also captures only the main shell–HVAC interactive effect, not the full set of inter-end-use interactions.

## 25.6 Regional coverage and differences

All seven modelled regions carry the commercial structure, and the Cold / Marine climate split is the key regional dimension: most of Canada is in the Cold zone, while coastal British Columbia and parts of the Atlantic are Marine, with milder heating demand and different heat-pump performance. Beyond climate, regional differences show up in the fuel mix — the model assumes essentially no new oil heating after 1990 except in the Atlantic region, where natural gas is less available and oil heating remains significant — and in grid carbon intensity, which determines how much electrification actually reduces emissions.

TODO: Confirm the regional building-segment floor-space mix and the Cold/Marine zone assignments against current commercial building-stock data, since both the heating demand and the heat-pump economics depend on them.

## 25.7 Limitations and planned extensions

The commercial sub-model is a mature, detailed buildings representation — segment- and climate-resolved shells, an explicit shell–HVAC interaction, retrofit competitions, and a full set of heat-pump and efficiency options. Its acknowledged limitations are the partial treatment of inter-end-use interactive effects (only the shell–HVAC link is quantified), the average-price treatment of energy (missing demand-charge savings), and reliance on inherited floor-space and intensity calibrations. The TODOs above record the priorities, of which cold-climate heat-pump performance and refreshed floor-space data are the most important for present-day analysis.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled commercial energy use by end-use against Natural Resources Canada’s Comprehensive Energy Use Database.

## 26 Transportation (Passenger)

Passenger transportation is the personal-mobility module of CIMS-Transportation (CIMS-T), a demand-side sub-model: it meets a demand for personal mobility (people moved, in passenger-kilometres) by competing transport modes and, within the dominant road mode, competing vehicle powertrains. The emphasis is on the structure of the service flow model: the mobility-demand nodes, the modes and vehicles that supply them, and the costs — financial and intangible — that drive the competition. The tree is region-agnostic. Passenger transport is distinctive in two ways: it is among Canada’s largest emitting sectors, and its technology choices are unusually shaped by consumer preferences (range, refuelling convenience, vehicle size, and power), which CIMS represents through explicit intangible costs alongside financial cost.

### 26.1 Scope

CIMS represents passenger mobility — urban and inter-city travel of people — driven by passenger-kilometre demand. It covers personal vehicles (cars and light trucks), urban transit (public bus and rapid transit), inter-city bus and rail, and inter-city air. The driving demand is projected exogenously and can itself respond to cost (the model can shift demand for mobility).

Adjacent activities are handled elsewhere: freight movement is in [Transportation \(Freight\)](#); off-road vehicles are handled separately; the fuels (gasoline, diesel, electricity, hydrogen, bio-fuels) are supplied by the energy-supply sub-models, including [Ethanol](#), [Biodiesel](#), [Hydrogen](#), and [Electricity](#); and vehicle manufacturing is in [Light Industrial](#). This chapter covers the use of vehicles to move people.

### 26.2 Real-world context

Transportation is responsible for roughly a quarter of Canada’s greenhouse-gas emissions, and light-duty passenger vehicles are the single largest part of it. The emissions come overwhelmingly from burning gasoline and diesel, with associated health costs estimated in the billions of dollars annually (Environment and Climate Change Canada [2023](#)). Policy is now driving a rapid transition: Canada’s zero-emission-vehicle (ZEV) mandate requires 100% of

new light-duty vehicle sales to be zero-emission by 2035, with interim targets of 20% by 2026 and 60% by 2030, where a ZEV is a battery-electric, plug-in-hybrid, or fuel-cell vehicle (Environment and Climate Change Canada 2023).

Two forces shape the transition in the model. First, vehicle electrification: battery-electric vehicles now cost roughly CAD 50,000–70,000 (entry models from ~\$40,000) but have 66–77% lower fuel-and-maintenance costs than gasoline vehicles, and the gap is closing (Natural Resources Canada 2026). Second, mode shift and transit electrification: moving trips from single-occupant vehicles to higher-occupancy vehicles, transit, and active modes cuts energy per passenger-km, and electrifying transit buses cuts their emissions by around 93% (“National Assessment of Transit Electrification in Canada: Infrastructure Costs, Energy Demand, and Greenhouse Gas Reduction Potential” 2025). But consumer choice is not driven by cost alone — range anxiety, refuelling availability, and preferences for larger, more powerful vehicles all matter, which is why CIMS-T carries intangible costs to reproduce observed behaviour. The structure captures all three levers: powertrain competition, mode competition, and the intangible-cost preferences that temper both.

## 26.3 Model structure

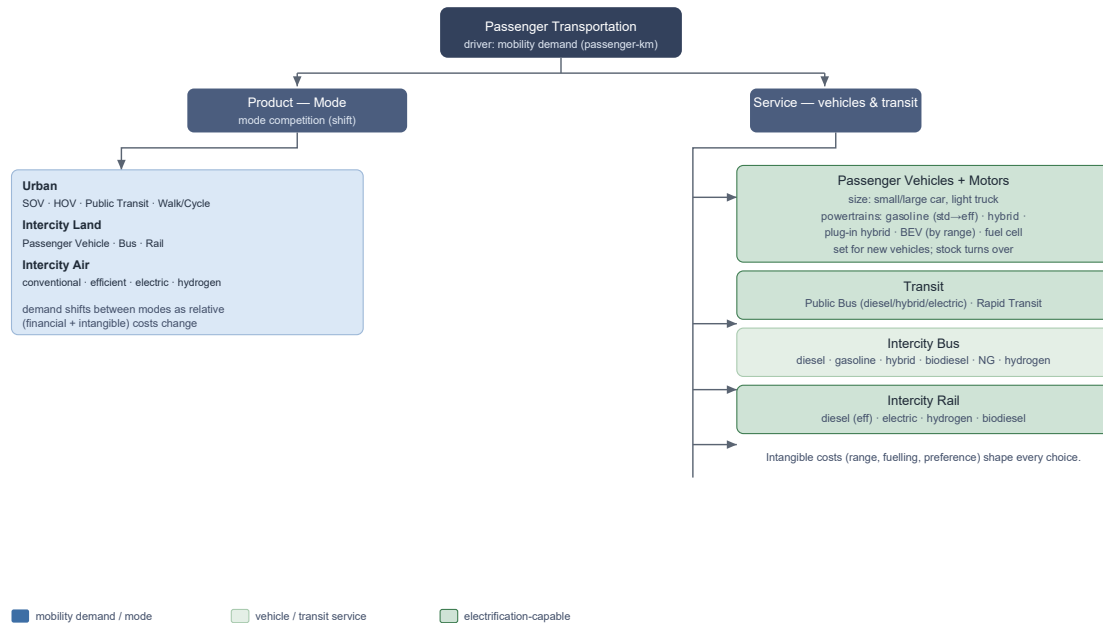
The sector divides into a **Product** sub-tree — the mobility demand by mode — and a **Service** sub-tree of the vehicles and transit systems that supply it. Figure 26.1 shows the structure. Each node carries a competition rule (see Chapter 4); mode and powertrain shares are determined by lifecycle cost plus intangible cost.

### 26.3.1 Products: mobility demand by mode

The **Mode** node splits personal mobility into:

- **Urban** — single-occupant vehicle (SOV), high-occupant vehicle (HOV), public transit, and walk/cycle. The competition among these is the model’s representation of urban mode shift.
- **Intercity Land** — passenger vehicle, bus, and rail.
- **Intercity Air** — competing conventional, efficient, electric, and hydrogen aircraft.

Demand can shift between modes (SOV↔HOV↔transit↔active; vehicle↔bus↔rail) as relative costs change, one of the endogenous actions CIMS-T simulates.



**Figure 26.1:** Service flow model of passenger transportation (region-agnostic master tree). The Product sub-tree is mobility demand by mode — Urban (single- and high-occupancy vehicles, public transit, walk/cycle), Intercity Land (vehicle, bus, rail), and Intercity Air — and the Service sub-tree supplies it through passenger vehicles (size classes and competing powertrains), transit (public bus, rapid transit), and intercity bus and rail.

### 26.3.2 Services: vehicles and transit

- **Passenger Vehicles** — the size classes (small/large cars, small/large light trucks) — and **Passenger Vehicle Motors**, where the powertrains compete: gasoline (existing/standard/efficient), hybrid, plug-in hybrid, battery-electric (by range, e.g. 500 and 800 km), and hydrogen fuel-cell. Powertrain and efficiency are specified for *new* vehicles; older vintages are tracked and retired by a stock-turnover function.
- **Transit** — Public Bus (diesel, hybrid, electric) and Rapid Transit.
- **Intercity Bus** — diesel, gasoline, hybrid, biodiesel-hybrid, natural-gas, hydrogen.
- **Intercity Rail** — diesel (and efficient diesel), electric, hydrogen, biodiesel-hybrid.

TODO: Add sustainable aviation fuel (SAF) and e-fuels to the Intercity Air competition (carried over from the chapter's original to-do). These are the principal decarbonisation routes for aviation, which is otherwise hard to electrify beyond short-haul, and they are currently absent.

## 26.4 Decarbonisation technologies

Passenger-transport decarbonisation works through three channels: electrifying vehicles (and transit and rail), shifting modes toward higher-occupancy and active travel, and low-carbon fuels where electrification is hard (aviation, some intercity). Table 26.1 summarises maturity and Table 26.2 gives indicative costs.

**Table 26.1:** Indicative performance and maturity of key passenger-transport decarbonisation options

| Technology / action                 | Indicative benefit                          | Maturity                           |
|-------------------------------------|---------------------------------------------|------------------------------------|
| Battery-electric vehicle (BEV)      | Zero tailpipe; 66–77% lower running cost    | Commercial                         |
| Plug-in hybrid (PHEV)               | Large tailpipe cut; gasoline backup         | Commercial                         |
| Hydrogen fuel-cell vehicle (FCEV)   | Zero tailpipe; fast refuel; sparse fuelling | Early commercial                   |
| Hybrid-electric vehicle             | ~20–30% fuel cut, no plug                   | Commercial                         |
| Electric / hydrogen transit bus     | ~90%+ emissions cut per bus                 | Commercial                         |
| Electric / hydrogen intercity rail  | Displaces diesel                            | Commercial (electric)              |
| Mode shift (SOV→HOV→transit→active) | Lower energy per passenger-km               | Policy / behavioural               |
| SAF / e-fuels / electric aircraft   | Aviation decarbonisation                    | Emerging — <i>not yet in model</i> |

The estimates in Table 26.2 are order-of-magnitude figures from public Canadian sources (Environment and Climate Change Canada 2023; Natural Resources Canada 2026; “[National Assessment of Transit Electrification in Canada: Infrastructure Costs, Energy Demand, and Greenhouse Gas Reduction Potential](#)” 2025), in 2025 Canadian dollars. Vehicle costs are per vehicle; running-cost effects are relative to a comparable gasoline vehicle. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 26.2:** Indicative techno-economic characteristics of passenger-transport decarbonisation options, 2025 CAD

| Technology               | Capex (2025 CAD)               | Running cost                    | Energy / input |
|--------------------------|--------------------------------|---------------------------------|----------------|
| Gasoline car (reference) | ~35,000–45,000                 | baseline                        | gasoline       |
| Battery-electric vehicle | ~45,000–70,000 (entry ~40,000) | 66–77% lower fuel + maintenance | electricity    |

| Technology                   | Capex (2025 CAD)                    | Running cost                     | Energy / input         |
|------------------------------|-------------------------------------|----------------------------------|------------------------|
| Plug-in hybrid               | ~40,000–55,000                      | lower; gasoline on long trips    | electricity + gasoline |
| Hydrogen fuel-cell vehicle   | ~60,000–90,000 (early)              | hydrogen-cost dependent          | hydrogen               |
| Battery-electric transit bus | ~1.0–1.5 M / bus (+ depot charging) | lower energy; ~93% emissions cut | electricity            |
| Hydrogen fuel-cell bus       | higher than BEV bus                 | hydrogen-cost dependent          | hydrogen               |

#### 26.4.1 Electrifying the light-duty fleet

The dominant lever is electrifying personal vehicles. The model competes battery-electric (at different ranges), plug-in hybrid, hybrid, and fuel-cell powertrains against gasoline, with the ZEV transition emerging from their falling capital costs and much lower running costs — tempered by the intangible costs that capture range anxiety, charging/refuelling availability, and the consumer preference for larger vehicles. Because powertrain is set for *new* vehicles and the fleet turns over through retirement, the modelled transition is gradual even under a sales mandate.

#### 26.4.2 Mode shift and transit/rail electrification

Beyond the vehicle, shifting trips from single-occupant cars to high-occupancy vehicles, public transit, and walking and cycling lowers energy per passenger-km — one of CIMS-T’s endogenous actions, governed by the intangible cost differences between modes. Electrifying transit buses (and, where lines justify it, rail) cuts those modes’ emissions by around 90% (“[National Assessment of Transit Electrification in Canada: Infrastructure Costs, Energy Demand, and Greenhouse Gas Reduction Potential](#)” 2025), and the model competes electric and hydrogen buses and electric and hydrogen rail directly.

#### 26.4.3 Decarbonising aviation

Inter-city air is the hardest to abate. The model already competes efficient, electric, and hydrogen aircraft (the last two viable mainly for short-haul), but the principal near-term route — sustainable aviation fuel and e-fuels — is not yet represented (see the TODO above).

TODO: Replace the indicative literature ranges above with the capital costs, energy intensities, availability years, and intangible-cost parameters actually used at these nodes in the model input files.

## 26.5 Economics

Technology and mode choice follow the lifecycle-cost competition of Chapter 4, but passenger transport relies more than any other sector on intangible costs — the non-financial component of the choice. CIMS-T’s intangible costs encode consumers’ revealed willingness to pay for vehicle size and power, their preference for the range and refuelling convenience of gasoline over alternative fuels, and their preferences among urban modes. These are calibrated from market data and stated-preference surveys, and they are what allow the model to reproduce observed adoption rather than the cost-minimising choice a purely financial calculation would predict.

Financial costs are discounted at behavioural rates — around 8% for some capital-intensive, long-lived assets (rail, transit infrastructure) and around 25% for personal-vehicle decisions in the current data. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than pure financial costs of capital; together with the intangible costs they govern technology choice, as set out in the discount-rate and intangible-cost discussion of the [Theory](#) chapter and in Chapter 4.

Emissions arise from fuel combustion in the vehicles, buses, trains, and aircraft, entering through each technology’s fuel service requests and the carbon price; battery-electric options have zero tailpipe emissions and inherit the carbon intensity of the grid. The emissions benefit of electrification therefore depends on regional grid intensity, and the benefit of biofuels on their life-cycle intensity from the [Ethanol](#) and [Biodiesel](#) sub-models. A note of caution carried from the original documentation: many of CIMS-T’s intangible-cost parameters are under active review against stated-preference survey data, so the modelled pace of the ZEV transition is sensitive to them.

## 26.6 Regional coverage and differences

All seven modelled regions carry the passenger-transport structure. Regional differences show up in urban form — denser regions (the large metropolitan areas) support more transit and active-mode share, while dispersed regions are more car-dependent — and in grid carbon intensity, which determines how much electrification actually reduces emissions (large in hydro-rich Québec, British Columbia, and Manitoba; smaller in Alberta and Saskatchewan until their grids decarbonise). Fuel availability (charging and hydrogen networks) also varies and enters through the intangible costs.

TODO: Calibrate the regional mode-share baselines and the intangible costs of transit and active modes against current travel-survey data, since urban mode shift is sensitive to both and varies widely by region.

## 26.7 Limitations and planned extensions

The passenger-transport sub-model captures the essential structure — mobility demand, mode competition, powertrain competition, and a rich intangible-cost representation of consumer preference. Its main limitations are the absence of sustainable aviation fuels and e-fuels, the reliance on intangible-cost parameters that are still being recalibrated, the exogenous treatment of some transit-technology choices, and stock-turnover dynamics that make the modelled transition gradual. The TODOs above record the priorities, of which adding SAF/e-fuels and refreshing the intangible costs are the most important.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled passenger-transport energy use and mode shares against Transport Canada and Natural Resources Canada data.

## 27 Freight Transportation

Freight transportation is the goods-movement module of CIMS-Transportation (CIMS-T), a demand-side sub-model: it meets a demand for freight movement (goods moved, in tonne-kilometres) by competing transport modes — road, rail, marine, and air — and, within each, competing vehicle powertrains and fuels. The emphasis is on the structure of the service flow model: the freight-demand nodes, the modes and vehicles that supply them, and the refuelling and charging infrastructure they need. The tree is region-agnostic. The general CIMS-T framework — service flow model, intangible costs, and stock-turnover dynamics — is described in the [Transportation \(Passenger\)](#) chapter and is not repeated here.

### 27.1 Scope

CIMS represents the movement of goods by road (heavy trucks and light/medium commercial vehicles), rail, marine, and air, plus an off-road category (construction, agricultural, and other off-highway equipment). The driving variable is freight demand in tonne-kilometres, projected exogenously and able to shift between modes. The model also carries explicit infrastructure services — charging, overhead catenary, and hydrogen refuelling — recognising that zero-emission freight depends as much on infrastructure as on the vehicles.

Adjacent activities are handled elsewhere: passenger mobility is in [Transportation \(Passenger\)](#); the fuels (diesel, electricity, hydrogen, biofuels, natural gas) are supplied by the energy-supply sub-models; and vehicle manufacturing is in [Light Industrial](#). This chapter covers the use of vehicles to move goods.

### 27.2 Real-world context

Freight is a large and stubborn share of transport emissions, and it is dominated by on-road heavy trucks. In 2019, on-road freight produced about 64.8 Mt of Canada’s transport emissions, against 7.8 Mt from rail and 4.5 Mt from marine (International Council on Clean Transportation 2023). The disparity reflects efficiency as much as volume: rail freight is more than five times as emissions-efficient per tonne-kilometre as road, so shifting goods from truck to

rail or intermodal can cut emissions by up to 30% (International Council on Clean Transportation 2023). Heavy trucks are also among the hardest road vehicles to decarbonise, because their weight, range, and duty cycles strain battery technology.

Three zero-emission pathways are competing for heavy freight, and the model represents all of them. Battery-electric trucks are favoured for lighter and shorter-range duty; hydrogen fuel-cell trucks are emerging for heavy long-haul, where range and refuelling time matter and where — as techno-economic studies including intangible costs find — hydrogen tends to win over batteries (International Council on Clean Transportation 2023; Pembina Institute 2023); and overhead catenary (eHighway) systems power trucks directly from the road on major corridors. By the 2030s, catenary and battery-electric trucks are projected to cost roughly 30–35% less to own and run than diesel, and hydrogen 15–25% less (International Council on Clean Transportation 2023). For rail, electrification, hydrogen, and biofuels apply; for marine, hydrogen, ammonia, and biofuels; and for aviation, sustainable aviation fuels and e-fuels. The CIMS structure captures these technology competitions and, distinctively, the infrastructure they require.

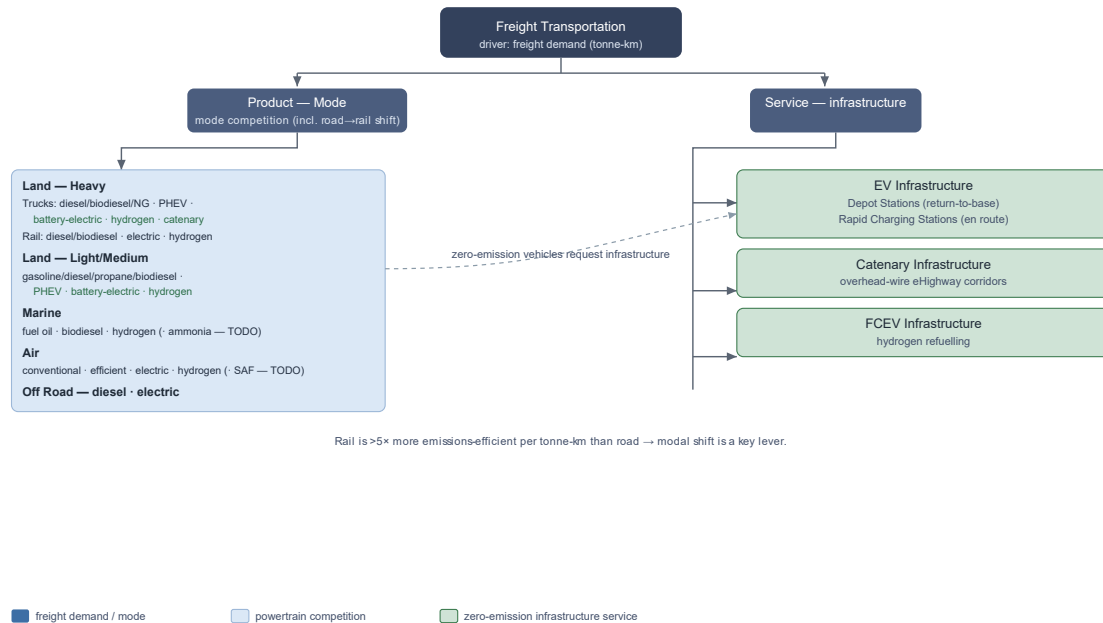
## 27.3 Model structure

The sector divides into a **Product** sub-tree — freight demand by mode — and a **Service** sub-tree of the refuelling and charging infrastructure. Figure 27.1 shows the structure. Each node carries a competition rule (see Chapter 4); mode and powertrain shares are determined by lifecycle cost plus intangible cost.

### 27.3.1 Products: freight demand by mode

- **Land — Heavy** — split into Trucks and Rail. Heavy Trucks compete diesel (existing/standard/efficient), biodiesel, natural gas, plug-in hybrid, battery-electric, hydrogen fuel-cell, and catenary (eHighway) powertrains. Heavy Rail competes diesel (and efficient diesel), biodiesel, electric, and hydrogen.
- **Land — Light/Medium** — commercial vans and medium trucks, competing gasoline and diesel (existing/standard/efficient), propane, biodiesel, plug-in hybrid, battery-electric, and hydrogen.
- **Marine** — competing fuel oil (existing/standard/efficient), biodiesel, and hydrogen.
- **Air** — competing conventional, efficient, electric, and hydrogen aircraft.
- **Off Road** — construction, agricultural, and other off-highway equipment, competing diesel (standard/medium/high-efficiency) and electric.

Freight demand can shift between modes — most importantly road to rail — as relative costs change.



**Figure 27.1:** Service flow model of freight transportation (region-agnostic master tree). The Product sub-tree is freight demand by mode — Land (heavy trucks and rail; light/medium vehicles), Marine, Air, and Off-Road — each competing diesel/biofuel against electric, hydrogen, and (for trucks) catenary powertrains; the Service sub-tree provides the EV charging, overhead-catenary, and hydrogen-refuelling infrastructure that zero-emission freight depends on.

### 27.3.2 Services: zero-emission infrastructure

A distinctive feature of the freight sub-model is that it carries the infrastructure zero-emission vehicles depend on as explicit services:

- **EV Infrastructure** — Depot Stations (overnight/return-to-base charging) and Rapid Charging Stations (en-route fast charging).
- **Catenary Infrastructure** — the overhead-wire (eHighway) system for catenary trucks.
- **FCEV Infrastructure** — hydrogen refuelling.

Representing these explicitly means the model can capture the chicken-and-egg coupling between zero-emission trucks and the networks they need, rather than assuming the infrastructure is free.

## 27.4 Decarbonisation technologies

Freight decarbonisation works through powertrain switching (electric, hydrogen, catenary, biofuels), modal shift (road to rail/marine), and low-carbon fuels for the hard-to-electrify modes (marine, aviation). Table 27.1 summarises maturity and Table 27.2 gives indicative costs.

**Table 27.1:** Indicative performance and maturity of key freight decarbonisation options

| Technology / action                                   | Indicative benefit                                  | Maturity                         |
|-------------------------------------------------------|-----------------------------------------------------|----------------------------------|
| Battery-electric truck (light/medium, regional heavy) | Zero tailpipe; ~30–35% lower lifecycle cost by 2030 | Early commercial                 |
| Hydrogen fuel-cell truck (heavy long-haul)            | Zero tailpipe; range/refuel suit heavy duty         | Demonstration / early commercial |
| Catenary (eHighway) truck                             | Direct road power; ~30–35% lower cost on corridors  | Demonstration                    |
| Electric / hydrogen rail                              | Displaces diesel; rail already efficient            | Commercial (electric)            |
| Hydrogen / ammonia / biofuel marine                   | Decarbonises shipping fuel                          | Emerging                         |
| SAF / e-fuels / electric / hydrogen aviation          | Aviation decarbonisation                            | Emerging (SAF)                   |
| Modal shift road → rail / intermodal                  | Up to ~30% emissions cut                            | Policy / logistics               |
| Electric off-road equipment                           | Zero tailpipe off-highway                           | Emerging                         |

The estimates in Table 27.2 are order-of-magnitude figures from public sources (International Council on Clean Transportation 2023; Pembina Institute 2023; SRK Consulting 2024), in 2025 Canadian dollars (US sources converted at roughly 1.37 CAD/USD and adjusted to 2025). They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 27.2:** Indicative techno-economic characteristics of freight decarbonisation options, 2025 CAD

| Technology                     | Capex / cost             | Operating effect                     | Energy / input           |
|--------------------------------|--------------------------|--------------------------------------|--------------------------|
| Diesel heavy truck (reference) | baseline                 | baseline                             | diesel                   |
| Battery-electric heavy truck   | ~50–100% higher purchase | ~30–35% lower lifecycle cost by 2030 | electricity (+ charging) |

| Technology                | Capex / cost                     | Operating effect                         | Energy / input          |
|---------------------------|----------------------------------|------------------------------------------|-------------------------|
| Hydrogen fuel-cell truck  | higher than BEV truck            | ~15–25% lower lifecycle cost by 2030     | hydrogen (+ refuelling) |
| Catenary (eHighway) truck | truck premium + road catenary    | ~30–35% lower cost on equipped corridors | grid electricity        |
| Electric rail             | electrification capex (corridor) | low energy cost; rail already efficient  | electricity             |
| Hydrogen / biofuel marine | engine/fuel-system + fuel        | fuel-cost dependent                      | hydrogen / biofuel      |

#### 27.4.1 Zero-emission trucks and their infrastructure

Heavy trucks are the centre of gravity for freight emissions, and the model competes the three contenders — battery-electric, hydrogen fuel-cell, and catenary — against diesel and biodiesel. Which wins depends on duty cycle and on the intangible costs of range, refuelling time, and infrastructure availability: studies that include these costs find battery-electric dominating lighter and regional duty while hydrogen emerges for heavy long-haul (International Council on Clean Transportation [2023](#)). Crucially, all three depend on infrastructure — depot and rapid charging, overhead catenary, and hydrogen refuelling — which the model carries as explicit services, so the cost of building out that infrastructure is part of the competition.

#### 27.4.2 Modal shift and the hard-to-electrify modes

Because rail is over five times as efficient per tonne-km as road, shifting freight from truck to rail and intermodal is a powerful lever, and the model allows freight demand to move between modes. Rail itself decarbonises through electrification, hydrogen, and biofuels. Marine and aviation are the hardest: marine switches to hydrogen, ammonia, and biofuels, and aviation to sustainable aviation fuels and e-fuels (the model currently competes efficient, electric, and hydrogen aircraft; SAF/e-fuels remain to be added, consistent with the passenger-aviation TODO).

TODO: Add sustainable aviation fuel (SAF) and e-fuels to the freight Air competition, and consider ammonia as a marine fuel option, since these are the principal decarbonisation routes for the modes that cannot electrify.

TODO: Replace the indicative literature ranges above with the capital costs, energy intensities, availability years, and intangible-cost parameters actually used at these nodes in the model input files.

## 27.5 Economics

Technology and mode choice follow the lifecycle-cost competition of Chapter 4, combined — as throughout CIMS-T — with intangible costs that capture the non-financial barriers to alternative-fuel freight: vehicle range, refuelling time and availability, payload penalties, and operator familiarity. These are decisive for heavy trucks, where they tip the competition between battery-electric and hydrogen.

Financial costs are discounted at behavioural rates — around 12.5% for some capital-intensive, long-lived assets (rail) and around 25–30% for road-vehicle decisions in the current data. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than pure financial costs of capital; together with the intangible costs they govern technology choice, as set out in the discount-rate and intangible-cost discussion of the [Theory](#) chapter and in Chapter 4.

Emissions arise from fuel combustion in the trucks, trains, ships, and aircraft, entering through each technology’s fuel service requests and the carbon price; battery-electric and catenary options have zero tailpipe emissions and inherit the carbon intensity of the grid, while hydrogen inherits the intensity of the [Hydrogen](#) production mix and biofuels their life-cycle intensity from the [Biodiesel](#) sub-model. The emissions benefit of each option therefore depends on the upstream energy system, and the infrastructure services mean the model accounts for the capital needed to enable the switch, not just the vehicles.

## 27.6 Regional coverage and differences

All seven modelled regions carry the freight structure. Regional differences show up in mode mix — long inland corridors (the Prairies, Ontario–Québec) favour rail and long-haul trucking, coastal and Great Lakes regions add marine — in grid carbon intensity (which sets the benefit of electric and catenary trucks and electric rail), and in the hydrogen supply available regionally (Alberta’s blue hydrogen, hydro-based electrolytic hydrogen elsewhere). Infrastructure build-out — charging, catenary corridors, hydrogen refuelling — is inherently regional and enters through the infrastructure services and their intangible costs.

TODO: Calibrate the regional freight mode-share baselines (road/rail/marine/air) and the corridor-level feasibility of catenary and hydrogen refuelling, since both modal shift and the heavy-truck powertrain competition depend on them.

## 27.7 Limitations and planned extensions

The freight sub-model captures the essential structure — freight demand, mode competition, a full set of zero-emission truck powertrains, and explicit refuelling/charging infrastructure

— which makes it well suited to analysing the heavy-duty transition. Its main limitations are the absence of sustainable aviation fuels, e-fuels, and marine ammonia; reliance on intangible-cost parameters still being recalibrated; the exogenous treatment of some freight-technology shares (noted in the CIMS-T overview); and stock-turnover dynamics that make the transition gradual. The TODOs above record the priorities.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled freight energy use and mode shares against Transport Canada and Natural Resources Canada data.

## 28 Waste

The waste sector is unlike most others CIMS represents: its greenhouse-gas footprint is overwhelmingly a single non-CO<sub>2</sub> gas — methane generated as organic waste decomposes in landfills — rather than energy use. The sub-model is therefore built around the management of that landfill methane: whether it is vented, flared, or captured for energy. The emphasis, as elsewhere, is on the structure of the energy-service flow model — the nodes, the technologies that compete at each, and the engineering ratios that link them. The tree is region-agnostic. Tellingly, the sector's activity is measured directly in tonnes of CO<sub>2</sub>-equivalent, signalling that emissions, not energy or output, are its organising variable.

### 28.1 Scope

CIMS represents the **management of landfill gas** at municipal solid waste disposal sites — the dominant source of greenhouse gases in the waste sector. Landfills are grouped by size (large, medium, and small), because the economics of gas collection and use depend strongly on how much gas a site produces. The driving variable is the methane-generation potential of the waste in place, expressed in tonnes of CO<sub>2</sub>-equivalent; a forecast of that potential — which declines over the projection as waste diversion reduces the organic material sent to landfill — drives the sub-model.

Adjacent activities are handled elsewhere or fall outside the current model. Waste collection and haulage are part of the [Transportation \(Freight\)](#) sector; the energy services of waste-management facilities are small and not separately represented; and waste used as a fuel (waste-derived fuel in cement kilns, biomass boilers) is accounted in the consuming sectors. The current sub-model focuses on landfill gas; wastewater treatment methane and waste incineration are not yet represented (see the limitations below).

### 28.2 Real-world context

The waste sector is a small but stubborn share of Canada's greenhouse-gas emissions, and it is dominated by **landfill methane**. As the organic fraction of municipal solid waste — food, paper, wood, yard trimmings — decomposes anaerobically in a landfill, it generates landfill

gas that is roughly half methane and half CO<sub>2</sub>. Because methane has a global-warming potential many times that of CO<sub>2</sub> (on the order of 28–30 over 100 years, and far higher over 20 years), uncontrolled landfill gas is the sector’s defining problem and its largest abatement opportunity.

The levers are well established. Capturing the gas with a collection system and **flaring** it converts the methane to CO<sub>2</sub>, cutting the warming impact of that gas by roughly a factor of 25 at modest cost; capturing it and using it for **energy** — electricity generation, or upgrading to pipeline-quality renewable natural gas (RNG) — does the same while displacing fossil energy and earning revenue. Upstream, diverting organics from landfill through composting and anaerobic digestion prevents the methane from being generated in the first place. Canada has committed under the Global Methane Pledge to cut methane sharply by 2030, and federal regulations targeting methane from large municipal landfills (gas capture and control requirements) are a central part of the waste-sector effort (Environment and Climate Change Canada [2024b](#)). The CIMS structure mirrors the end-of-pipe choice directly: each landfill chooses among venting, flaring, and energy recovery, with the size split capturing the very different economics of large versus small sites.

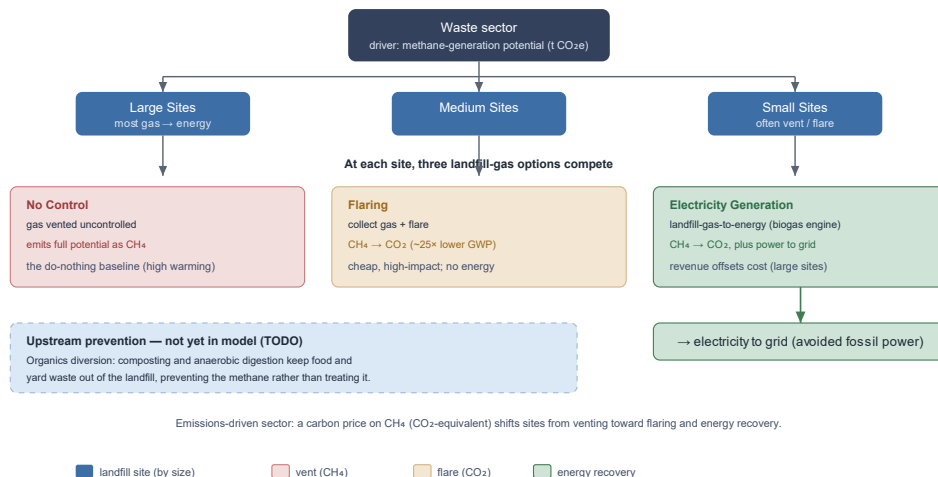
## 28.3 Model structure

The sector divides into a **Product** sub-tree — the landfill sites, grouped by size — with the gas-management technologies competing at each. Figure [28.1](#) shows the structure. Each node carries a competition rule (see Chapter [4](#)); the landfill-gas options compete on lifecycle cost, with the carbon price (applied to methane on a CO<sub>2</sub>-equivalent basis) as the main driver.

### 28.3.1 Products: landfill sites and their gas-management options

The **Sites** node splits the landfill stock into **Large Sites**, **Medium Sites**, and **Small Sites**. At each, three technologies compete to manage the landfill gas:

- **No Control** — the gas is vented uncontrolled, so the site emits its full methane potential as CH<sub>4</sub>. This is the do-nothing baseline.
- **Flaring** — a gas-collection system captures the landfill gas and flares it, combusting the methane. The emission becomes CO<sub>2</sub> rather than CH<sub>4</sub>, cutting the warming impact of the captured gas roughly twenty-five-fold while producing no useful energy.
- **Electricity Generation** — landfill-gas-to-energy: the captured gas (modelled as a biogas fuel) is burned in an engine to generate electricity, again emitting CO<sub>2</sub> rather than CH<sub>4</sub> and, in addition, displacing grid electricity (the technology returns power to the grid).



**Figure 28.1:** Energy-service flow model of the waste sector (region-agnostic master tree). Methane-generation potential (in tonnes of CO<sub>2</sub>-equivalent) is allocated across landfill **Sites** by size (Large, Medium, Small); at each, three landfill-gas options compete — No Control (vents CH<sub>4</sub>), Flaring (captures and burns the gas, converting CH<sub>4</sub> to CO<sub>2</sub>), and Electricity Generation (landfill-gas-to-energy, which also produces electricity). Larger sites more readily justify capture and energy recovery.

The site-size grouping is what lets the model represent the central real-world fact that large landfills generate enough gas to justify a collection system and an engine, while small sites often cannot and are more likely to vent or, at most, flare.

**TODO:** The model represents only the end-of-pipe management of landfill gas. Add upstream **organics diversion** (composting and anaerobic digestion) as a competing way to reduce the methane-generation potential itself, since diverting food and yard waste prevents the gas rather than treating it — and is the lever federal and municipal policy increasingly emphasises.

## 28.4 Decarbonisation technologies

Waste-sector decarbonisation means **capturing and using landfill methane** and, further upstream, **preventing its generation**. Table 28.1 summarises maturity and Table 28.2 gives indicative costs (here as cost of abatement per tonne of CO<sub>2</sub>-equivalent, since the options are gas-management systems rather than process plant).

**Table 28.1:** Indicative performance and maturity of key waste decarbonisation options

| Technology / practice                        | Indicative benefit                                                    | Maturity                             |
|----------------------------------------------|-----------------------------------------------------------------------|--------------------------------------|
| Gas collection + flaring                     | Converts captured CH <sub>4</sub> to CO <sub>2</sub> (~25× lower GWP) | Commercial                           |
| Landfill-gas-to-electricity                  | As flaring, plus displaces grid power                                 | Commercial                           |
| Landfill-gas-to-RNG (pipeline upgrading)     | As above, plus displaces fossil gas                                   | Commercial — <i>not yet in model</i> |
| Improved collection efficiency / final cover | Captures a larger share of generated gas                              | Commercial                           |
| Organics diversion (composting)              | Prevents methane generation                                           | Commercial — <i>not yet in model</i> |
| Anaerobic digestion of organics              | Prevents landfill methane; yields biogas                              | Commercial — <i>not yet in model</i> |
| Source reduction / waste minimisation        | Less organic waste landfilled                                         | Policy / behavioural                 |

The estimates in Table 28.2 are order-of-magnitude figures from public literature (Environment and Climate Change Canada 2024b; U.S. Environmental Protection Agency 2024), expressed in 2025 Canadian dollars. Because the options are gas-management systems whose benefit is avoided methane, costs are given as a cost (or net saving) per tonne of CO<sub>2</sub>-equivalent abated. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 28.2:** Indicative cost of waste-sector methane abatement, 2025 CAD

| Option                          | Cost of abatement (2025 CAD/t CO <sub>2</sub> e) | Capital                  | Notes                            |
|---------------------------------|--------------------------------------------------|--------------------------|----------------------------------|
| Gas collection + flaring        | ~5–30                                            | collection wells + flare | cheap, high-impact; no revenue   |
| Landfill-gas-to-electricity     | ~20–60 (net of power revenue)                    | collection + genset      | revenue offsets cost             |
| Landfill-gas-to-RNG             | ~30–100 (net of gas revenue)                     | collection + upgrading   | best at large sites              |
| Organics diversion (composting) | ~20–80                                           | facility + collection    | prevents generation; co-benefits |
| Anaerobic digestion of organics | ~40–120                                          | digester                 | biogas revenue offsets           |

### 28.4.1 Capturing and using landfill gas

The largest and cheapest lever is to stop venting. A **gas-collection system** with **flaring** converts captured methane to  $\text{CO}_2$  at low cost and is the workhorse of landfill methane control; it is the Flaring technology in the model. Where a site generates enough gas, **electricity generation** does the same while earning power revenue — the model's Electricity Generation technology, which returns electricity to the grid — and **upgrading to renewable natural gas** (not yet a distinct option in the model) displaces fossil gas instead. Larger sites favour energy recovery; smaller sites favour flaring or remain uncontrolled, which is exactly why the model resolves the stock by size.

### 28.4.2 Preventing methane: organics diversion

The deeper lever is to keep the organic material out of the landfill in the first place. **Composting** and **anaerobic digestion** of food and yard waste prevent the anaerobic decomposition that generates landfill methane, and anaerobic digestion additionally yields biogas. These upstream options reduce the methane-generation potential itself rather than managing the gas after the fact, and are the focus of much current policy — but the model does not yet represent them (see the TODO above).

TODO: Replace the indicative literature ranges above with the abatement costs, gas-capture efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

## 28.5 Economics

Technology choice at each site follows the lifecycle-cost competition of Chapter 4. Each landfill-gas option carries a capital cost (collection system, flare or engine), an operating cost, the methane-generation potential it serves, any electricity it returns to the grid, and availability and lifetime years; these are combined into a lifecycle cost and converted to market shares by the inverse-power choice rule.

Waste technologies are discounted at rates of roughly 20–35% in the current data. As elsewhere in CIMS these are **behavioural** (revealed or empirical) discount rates rather than true financial costs of capital, capturing the payback horizons and adoption barriers that govern real investment; CIMS applies this rate throughout the lifecycle-cost calculation, as set out in the discount-rate discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** govern how sharply the gas-management competitions respond to cost.

The emissions accounting is the heart of this sub-model and is almost entirely **non- $\text{CO}_2$** . The No Control technology emits the site's full potential as  $\text{CH}_4$ , while Flaring and Electricity Generation emit the same quantity of carbon as  $\text{CO}_2$  — so the difference between them, on

a CO<sub>2</sub>-equivalent basis, is the large methane-versus-CO<sub>2</sub> warming gap. Methane enters at its high global-warming potential, which is what makes capture so valuable per dollar. The Electricity Generation route additionally returns power to the grid, an avoided-emissions and revenue benefit that improves its lifecycle cost. Because the activity variable is itself in CO<sub>2</sub>-equivalent and the abatement options are inexpensive relative to their methane benefit, even a modest carbon price (or a landfill methane regulation) drives a large shift from venting toward flaring and energy recovery.

## 28.6 Regional coverage and differences

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the waste structure, each with its large, medium, and small landfill sites. The methane-generation potential differs by region with population and waste arisings, and it declines over the projection as diversion programmes reduce the organic fraction sent to landfill. Regional differences also show up in the value of the recovered electricity, which depends on the local grid and power price, and in the share of large sites (which can justify energy recovery) versus small ones.

TODO: Calibrate the regional split of waste across large, medium, and small sites and the baseline control levels (vent / flare / energy) against Environment and Climate Change Canada's National Inventory Report and provincial landfill data, since both the emissions and the abatement potential depend heavily on the site-size distribution.

## 28.7 Limitations and planned extensions

The waste sub-model captures the sector's dominant emission — landfill methane — and the core end-of-pipe management choice between venting, flaring, and energy recovery, resolved by site size. But it is deliberately narrow, and several real-world levers and sources are missing. The TODOs above record the priorities: adding upstream **organics diversion** (composting and anaerobic digestion) that prevents methane generation, adding a **renewable natural gas** upgrading pathway alongside electricity generation, and representing **gas-capture efficiency** explicitly rather than as a binary control choice. Two further sources — **wastewater treatment** methane and **waste incineration** — are not represented at all and would round out the sector. Because waste emissions are largely non-CO<sub>2</sub> and cheap to abate, a faithful representation of these levers is valuable out of proportion to the sector's small energy footprint.

TODO: Add wastewater-treatment methane and waste incineration to complete the sector, and represent the methane-generation potential's dependence on waste composition and diversion policy rather than as a single exogenous CO<sub>2</sub>-equivalent forecast.

TODO: Replace the illustrative figures and example discount rates in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled landfill methane against Environment and Climate Change Canada's National Inventory Report.

## 29 Agriculture

Agriculture is unlike the other sectors CIMS represents: most of its greenhouse-gas emissions are not from burning fuel but from biological and chemical processes — methane ( $\text{CH}_4$ ) from livestock digestion and manure, and nitrous oxide ( $\text{N}_2\text{O}$ ) from soils — so the sub-model is built around those process emissions and the options that abate them, rather than around energy services. The emphasis, as elsewhere, is on the structure of the energy-service flow model — the process and service nodes, the technologies that compete at each, and the engineering ratios that link them. The tree is region-agnostic. Tellingly, the sector's activity is measured directly in tonnes of  $\text{CO}_2$ -equivalent, signalling that emissions, not energy or output, are its organising variable.

### 29.1 Scope

CIMS represents the greenhouse-gas-emitting activities of agriculture and the on-farm energy that supports them: livestock enteric fermentation and manure management (the  $\text{CH}_4$  sources), agricultural soils (the  $\text{N}_2\text{O}$  source, together with the land-management practices that affect it), farm tractors (mobile diesel equipment), and farm process and building heat. The driving variable is the level of agricultural activity, expressed in  $\text{CO}_2$ -equivalent terms.

Adjacent activities are handled elsewhere to avoid double-counting: the manufacture of nitrogen fertiliser is in [Chemical Products](#) (ammonia synthesis); food and beverage processing is in [Light Industrial](#); the biofuels that use agricultural feedstock are in [Biodiesel](#) and [Ethanol](#) (which carry their own Agricultural Input services); and the fuels burned on-farm are supplied by the energy-supply sub-models. This chapter covers the emissions and energy use of farming itself.

### 29.2 Real-world context

Agriculture is responsible for roughly 8% of Canada's greenhouse-gas emissions, and the composition is distinctive: it is overwhelmingly non- $\text{CO}_2$ . Enteric fermentation — the methane cattle and other ruminants produce as they digest forage — is the single largest source, around 46% of agricultural emissions; manure management ( $\text{CH}_4$  and  $\text{N}_2\text{O}$ ) adds about 27%; and

agricultural soils emit  $\text{N}_2\text{O}$ , chiefly from nitrogen fertiliser and crop residues (National Farmers Union 2022). On-farm fuel use (tractors, grain drying, heating) is comparatively small and is mostly diesel and natural gas.

This composition shapes the decarbonisation problem. Because the emissions are biological, the levers are agronomic and dietary rather than purely technological: feed additives and rumen modification to cut enteric methane, manure management changes, nitrogen and tillage management to cut soil  $\text{N}_2\text{O}$ , and land-use practices (agroforestry, wetland restoration, reduced tillage) that store carbon in soils and biomass. The feed additive 3-nitrooxypropanol (3-NOP), for example, consistently cuts enteric methane by around 30% on average (Kebreab et al. 2021). The CIMS structure mirrors this: it carries explicit  $\text{CH}_4$  and  $\text{N}_2\text{O}$  process-emission factors on the livestock and soils nodes and competes the abatement practices against conventional ones, while representing the on-farm energy through a small set of tractor and heat technologies.

## 29.3 Model structure

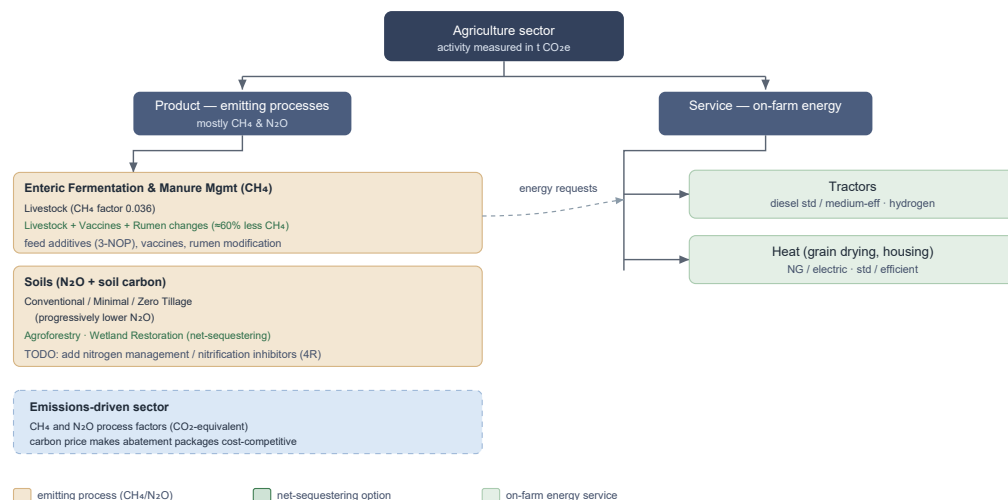
The sector divides into a **Product** sub-tree — the emitting agricultural processes — and a **Service** sub-tree of the on-farm energy services. Figure 29.1 shows the structure. Each node carries a competition rule (see Chapter 4); the abatement options compete against conventional practice on a lifecycle-cost basis, with the carbon price (applied to  $\text{CH}_4$  and  $\text{N}_2\text{O}$  on a  $\text{CO}_2$ -equivalent basis) as the main driver.

### 29.3.1 Products: the emitting processes

- **Enteric Fermentation and Manure Management** — the livestock methane source. It competes conventional Livestock against Livestock + Vaccines + Rumen changes, a package of dietary and biological interventions (feed additives such as 3-NOP, methane vaccines, rumen modification) that cuts the enteric- $\text{CH}_4$  factor by roughly 60% in the data.
- **Soils** — the cropland  $\text{N}_2\text{O}$  source, competing land-management practices that differ in their soil- $\text{N}_2\text{O}$  intensity and carbon storage: Conventional Tillage, Minimal Tillage, and Zero Tillage (progressively lower  $\text{N}_2\text{O}$ ), and Agroforestry and Wetland Restoration, which store carbon and can be net-sequestering.

### 29.3.2 Services: tractors and heat

- **Tractors** — on-farm mobile equipment, competing standard and medium-efficiency diesel against a hydrogen option.
- **Heat** — farm process and building heat (grain drying, livestock housing), competing standard and efficient natural-gas and electric units.



**Figure 29.1:** Energy-service flow model of the agriculture sector (region-agnostic master tree). The Product sub-tree holds the emitting processes — Enteric Fermentation and Manure Management (CH<sub>4</sub>) and Soils (N<sub>2</sub>O, with tillage and land-use options including net-sequestering practices) — and the Service sub-tree holds on-farm Tractors and Heat.

TODO: The model lumps manure management together with enteric fermentation in a single node. Separate manure management so that its distinct abatement options — anaerobic digestion (biogas capture), covered storage, and solid–liquid separation — can be represented, since manure is about a quarter of agricultural emissions and has different, often cost-effective, mitigation.

## 29.4 Decarbonisation technologies

Agricultural decarbonisation means abating CH<sub>4</sub> and N<sub>2</sub>O and storing carbon in soils and biomass, with on-farm energy a secondary concern. Table 29.1 summarises maturity and Table 29.2 gives indicative costs (here as cost of abatement per tonne of CO<sub>2</sub>-equivalent, since most options are practice changes rather than capital equipment).

**Table 29.1:** Indicative performance and maturity of key agricultural decarbonisation practices

| Technology / practice                            | Indicative benefit                                            | Maturity         |
|--------------------------------------------------|---------------------------------------------------------------|------------------|
| Feed additives / rumen modification (e.g. 3-NOP) | ~30% (avg) enteric CH <sub>4</sub> cut; up to ~60% in package | Early commercial |

| Technology / practice                               | Indicative benefit                                 | Maturity                                       |
|-----------------------------------------------------|----------------------------------------------------|------------------------------------------------|
| Anaerobic manure digestion (biogas)                 | Captures manure CH <sub>4</sub> ; yields biogas    | Commercial — <i>not yet separated in model</i> |
| Zero / minimal tillage                              | Lower soil N <sub>2</sub> O; soil carbon retention | Commercial                                     |
| Agroforestry                                        | Stores carbon in biomass and soil                  | Commercial                                     |
| Wetland restoration                                 | Avoids/stores emissions; co-benefits               | Commercial                                     |
| Nitrogen management / nitrification inhibitors (4R) | Lower soil N <sub>2</sub> O                        | Commercial — <i>not yet in model</i>           |
| Hydrogen / efficient tractors                       | Lower on-farm diesel CO <sub>2</sub>               | Emerging (hydrogen)                            |
| Electric / efficient farm heat                      | Lower heating fuel CO <sub>2</sub>                 | Commercial                                     |

The estimates in Table 29.2 are order-of-magnitude figures from public literature (National Farmers Union 2022; Kebreab et al. 2021), expressed in 2025 Canadian dollars. Because most options are practices rather than plant, costs are given as a cost (or saving) per tonne of CO<sub>2</sub>-equivalent abated. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 29.2:** Indicative cost of agricultural abatement practices, 2025 CAD

| Practice                  | Cost of abatement (2025 CAD/t CO <sub>2</sub> e) | Capital / input             | Notes                                             |
|---------------------------|--------------------------------------------------|-----------------------------|---------------------------------------------------|
| Feed additives (3-NOP)    | ~20–80                                           | additive cost per head      | ~30% enteric CH <sub>4</sub> cut                  |
| Anaerobic manure digester | ~30–120                                          | digester capex (farm-scale) | captures CH <sub>4</sub> ; biogas revenue offsets |
| Zero / minimal tillage    | often <i>cost-saving</i>                         | none (equipment/practice)   | fuel and labour savings                           |
| Agroforestry              | ~10–60                                           | land opportunity cost       | carbon storage, co-benefits                       |
| Wetland restoration       | ~10–50                                           | land opportunity cost       | sequestration + co-benefits                       |
| Hydrogen tractor          | high (early)                                     | hydrogen + new equipment    | small share of ag emissions                       |

#### 29.4.1 Cutting enteric and manure methane

The largest single source — enteric methane — is addressed by dietary and biological means: feed additives (notably 3-NOP), methane-suppressing vaccines, and rumen modification, represented in the model as the Livestock + Vaccines + Rumen changes package. Manure methane is best cut by anaerobic digestion, which captures the gas for energy, and by covered storage — high-value options the model does not yet separate from enteric emissions (see the TODO above).

#### 29.4.2 Cutting soil N<sub>2</sub>O and storing carbon

Soil N<sub>2</sub>O falls with reduced and zero tillage and with better nitrogen management (right rate, source, timing, and placement — “4R” stewardship — and nitrification inhibitors). Agroforestry and wetland restoration go further by storing carbon in biomass and soils, and are among the few agricultural options that can be net-sequestering. The model competes the tillage and land-use practices directly at the Soils node.

#### 29.4.3 On-farm energy

On-farm energy is a smaller lever: efficient and hydrogen tractors cut mobile-diesel CO<sub>2</sub>, and electric and efficient farm heat cut heating emissions, drawing on the same low-carbon options as the other sectors.

TODO: Add explicit nitrogen-management / nitrification-inhibitor (4R) options at the Soils node, since fertiliser-driven N<sub>2</sub>O is a major and cost-effectively abatable source not currently represented as a distinct technology.

TODO: Replace the indicative literature ranges above with the abatement costs, efficiencies, availability years, and emission factors actually used at these nodes in the model input files.

### 29.5 Economics

Technology choice at every node follows the lifecycle-cost competition of Chapter 4. Each practice carries a capital cost (often small), operating cost, an activity basis, any service requests, and availability and lifetime years; these are combined into a lifecycle cost and converted to market shares by the inverse-power choice rule.

Agricultural practices are discounted at about 30% in the current data. As elsewhere in CIMS this is a **behavioural** (revealed or empirical) discount rate rather than a true financial cost of capital, capturing the payback horizons and adoption barriers that govern real farm decisions; CIMS applies it throughout the lifecycle-cost calculation, as set out in the discount-rate

discussion of the [Theory](#) chapter and in Chapter 4. **Heterogeneity (variance) parameters** are set around 10–15.

The emissions accounting is the heart of this sub-model and is almost entirely non-CO<sub>2</sub>: it carries explicit CH<sub>4</sub> factors on the enteric/manure node and N<sub>2</sub>O factors on the soils node, applied per unit of activity and entering on a CO<sub>2</sub>-equivalent basis at their high global-warming potentials. Some land-use practices (agroforestry, wetland restoration) can carry net-negative factors, representing carbon storage. Because the activity variable is itself in CO<sub>2</sub>-equivalent, the carbon price acts very directly here: it makes the abatement packages — feed additives, reduced tillage, sequestering land use — cost-competitive against conventional practice, which is the model's channel for agricultural mitigation.

## 29.6 Regional coverage and differences

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the agriculture structure. The mix of sources differs by region: the Prairie provinces (Alberta, Saskatchewan, Manitoba) dominate both cattle (enteric and manure CH<sub>4</sub>) and cropland (soil N<sub>2</sub>O from extensive grain and oilseed production); Ontario and Québec have mixed livestock and cropping; and the Atlantic and British Columbia have smaller, more specialised agricultural sectors.

TODO: Calibrate the regional balance of livestock versus cropland emissions and the baseline tillage mix, since the effectiveness and cost of the abatement practices depend heavily on the regional activity mix.

## 29.7 Limitations and planned extensions

The agriculture sub-model captures the sector's dominant emissions — enteric and manure methane, and soil N<sub>2</sub>O — and the main practice-based abatement levers, but it is deliberately coarse. Its principal limitations are the lumping of manure with enteric fermentation (hiding the cost-effective digester and storage options), the absence of explicit nitrogen-management options, and the simplified treatment of soil-carbon sequestration, whose permanence and measurement are genuinely difficult. The TODOs above record these priorities. Because agricultural emissions are largely non-CO<sub>2</sub> and biological, they are among the hardest in the whole model to abate deeply, which makes a faithful representation of the available practices especially valuable.

TODO: Replace the illustrative figures and example discount rate in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled agricultural CH<sub>4</sub> and N<sub>2</sub>O against Environment and Climate Change Canada's National Inventory Report and Agriculture and Agri-Food Canada data.

## 30 Forestry

The logging industry's on-site greenhouse-gas footprint comes almost entirely from the diesel burned by mobile harvesting and handling equipment — feller-bunchers, skidders, harvesters, forwarders, and loaders working in the bush — so the sector is represented as a single mobile-equipment energy service in which diesel competes against electric. The tree is region-agnostic, and the sector's activity is measured directly in tonnes of CO<sub>2</sub>-equivalent, signalling that emissions, not output, are its organising variable. The structure is deliberately coarse; the limitations below set out what a fuller representation would add.

### 30.1 Scope

CIMS represents the **fuel use of forestry's mobile equipment** — the off-road machinery that fells, gathers, and moves timber at the harvest site — as one aggregate energy service. The driving variable is an exogenous forecast of the sector's emissions in tonnes of CO<sub>2</sub>-equivalent. Everything downstream of the roadside is handled elsewhere: log haulage on public roads belongs to [Transportation \(Freight\)](#), and the milling of logs into lumber and pulp is in [Pulp and Paper](#) and the wood-products part of [Light Industrial](#). The biological carbon of the forest itself — growth, harvest, and regrowth — is the domain of [LULUCF](#), not this sector.

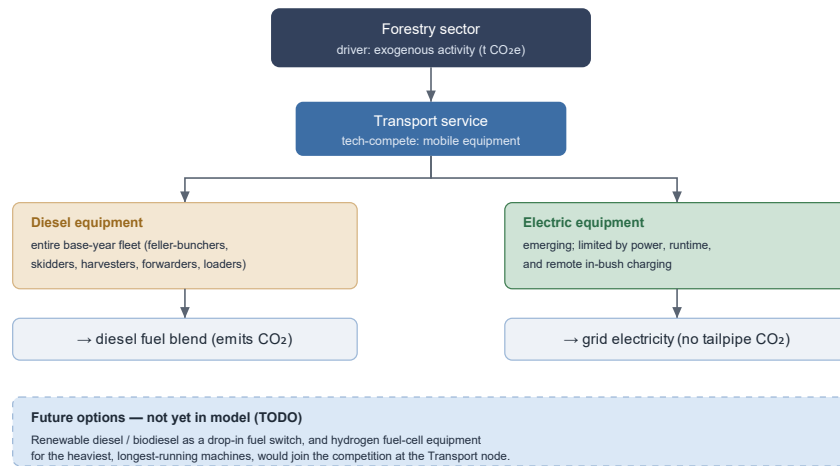
### 30.2 Real-world context

Logging is energy-light compared with heavy industry, but what energy it uses is overwhelmingly diesel in rugged off-road machines that run long duty cycles far from grid power. That combination — high power demand, remote sites, and no charging infrastructure — is exactly what makes the harvest fleet hard to electrify, and it is why diesel still does essentially all the work today. The decarbonisation options are the same three that face any off-road diesel fleet: drop-in **renewable diesel** or biodiesel that cuts lifecycle carbon with no equipment change, battery-electric machines where duty cycles and charging allow, and hydrogen fuel-cell machines for the heaviest, longest-running equipment. Canadian work on zero- and low-emission options for the forest sector — much of it through FPInnovations — has so far concentrated on the haulage fleet, where battery-electric Class 8 trucks have demonstrated more than 60% lower energy use and at least 80% lower greenhouse-gas emissions than diesel,

with the in-bush machines a harder and later problem (FPInnovations 2024). The CIMS structure mirrors the core choice directly: a single competition between diesel and electric equipment.

### 30.3 Model structure

The sector is a single tech-compete node. Figure 30.1 shows the structure: the exogenous activity drives one **Transport** service, at which two technologies compete to provide the same mobile-equipment work.



**Figure 30.1:** Energy-service flow model of the forestry sector (region-agnostic master tree). An exogenous activity forecast in tonnes of CO<sub>2</sub>-equivalent drives a single **Transport** service node, a tech-compete between diesel equipment (which draws on the diesel fuel blend and emits CO<sub>2</sub>) and electric equipment (which draws grid electricity). Lower-carbon drop-in fuels and hydrogen are noted as future options not yet in the model.

At the **Transport** node, diesel equipment (the entire base-year fleet) requests the diesel transportation fuel blend, while electric equipment requests grid electricity; both carry a twenty-year lifetime and the same energy requirement per unit of activity. Because the diesel option draws on the blended fuel, any renewable-diesel content in that blend lowers its lifecycle carbon automatically (see [Biodiesel](#)). Fuel prices at the node are currently borrowed from [Agriculture](#) as a proxy and flagged for review.

## 30.4 Decarbonisation technologies

Forestry decarbonisation means getting the carbon out of the off-road fleet's fuel: cleaner liquid fuel now, electrification where it fits, and hydrogen for the hardest duty cycles. Table 30.1 summarises maturity and Table 30.2 gives indicative costs.

**Table 30.1:** Indicative performance and maturity of key forestry decarbonisation options

| Technology / practice                  | Indicative benefit                                      | Maturity                                        |
|----------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| Renewable diesel / biodiesel (drop-in) | Cuts lifecycle CO <sub>2</sub> with no equipment change | Commercial                                      |
| Battery-electric equipment             | Removes tailpipe CO <sub>2</sub> ; shifts load to grid  | Emerging — <i>electric option in model</i>      |
| Hybrid diesel-electric machines        | Lowers fuel use on variable duty cycles                 | Emerging — <i>not yet in model</i>              |
| Hydrogen fuel-cell equipment           | Zero-tailpipe for heavy, long-run machines              | Early / demonstration — <i>not yet in model</i> |
| Efficient machines & idle reduction    | Lower fuel per unit harvested                           | Commercial — <i>not explicit in model</i>       |

The estimates in Table 30.2 are order-of-magnitude figures from public literature (FPIInnovations 2024; Canada Energy Regulator 2023; SRK Consulting 2024), expressed in 2025 Canadian dollars. They are indicative, for relative comparison rather than budgeting, and are separate from the values the model carries internally.

**Table 30.2:** Indicative techno-economic characteristics of forestry decarbonisation options, 2025 CAD

| Option                       | Capital vs diesel          | Operating effect                  | Energy / emission                              |
|------------------------------|----------------------------|-----------------------------------|------------------------------------------------|
| Renewable diesel / biodiesel | none (drop-in fuel)        | fuel-cost dependent               | lifecycle CO <sub>2</sub> cut, no fleet change |
| Battery-electric machine     | ~+50–100% on machine price | lower energy and maintenance cost | grid electricity; no tailpipe CO <sub>2</sub>  |
| Hydrogen fuel-cell machine   | high (early-stage)         | fuel-cost dependent               | zero-tailpipe; needs H <sub>2</sub> supply     |

The cheapest near-term lever is the fuel itself: switching the diesel blend toward **renewable diesel** or biodiesel cuts lifecycle carbon across the whole fleet with no change to the machines. Battery-electric equipment — the model's electric option — removes tailpipe emissions and lowers energy and maintenance cost, but is held back by the power, runtime, and remote-charging demands of in-bush work, so it is emerging fastest in lighter machines and at the

roadside. Hydrogen fuel cells remain early-stage and depend on hydrogen supply reaching remote sites.

TODO: Add the renewable-diesel/biodiesel fuel-switching and hydrogen fuel-cell pathways as explicit competing options at the Transport node, and populate the technologies with capital, operating, and emission data rather than the current Agriculture-proxied fuel prices.

### 30.5 Economics

Technology choice at the **Transport** node follows the lifecycle-cost competition of Chapter 4. As elsewhere in CIMS, the intended discount rate is **behavioural** (revealed or empirical) rather than a true financial cost of capital, capturing the short payback horizons typical of resource industries (see the [Theory](#) chapter). In the current input files, however, the two technologies carry only a lifetime and their energy requests — capital cost, operating cost, the discount rate, and the heterogeneity (variance) parameter are not yet populated — so the diesel-versus-electric competition cannot yet resolve on cost and the fleet holds near its base-year shares. Emissions enter upstream through the fuels the technologies request: the diesel option carries the carbon of the diesel blend, while the electric option moves the load to the grid, where it is counted in [Electricity](#).

### 30.6 Regional coverage

All seven modelled regions — Alberta, the Atlantic region, British Columbia, Manitoba, Ontario, Québec, and Saskatchewan — carry the forestry structure, scaled by an exogenous emissions forecast that differs sharply with the size of each region's logging industry: British Columbia is by far the largest, with Québec and Ontario next, and the Prairie provinces much smaller. The structure itself is identical across regions.

### 30.7 Limitations and planned extensions

The forestry sub-model is a placeholder: it captures the sector's dominant emission source — off-road diesel equipment — and the central diesel-to-electric choice, but little else. The off-road fleet is collapsed into a single aggregate **Transport** service rather than resolved by machine type or duty cycle; the competing technologies are not yet costed; the fuel prices are proxied from agriculture; and lower-carbon liquid fuels and hydrogen are not yet explicit options. The TODOs above record the priorities. Because forestry's footprint is small, this coarse treatment is a reasonable interim, but populating the technology costs is the minimum needed for the diesel-to-electric transition to respond to policy at all.

TODO: Replace the illustrative figures and the Agriculture-proxied prices in this chapter with quantitative values drawn directly from the model input files, and benchmark modelled forestry fuel use against Natural Resources Canada and FPInnovations data.

## 31 Carbon Dioxide Removal / Direct Air Capture

- Run year as normal with carbon price
- Re-run year with shadow carbon price, iterate until between:
  - total emissions difference between runs is equal to maximum DAC quantity
    - \* have maximum annual growth rate for DAC?
    - \* Expert opinion?
  - minimum shadow price equals DAC cost
- Emissions difference between runs is DAC request by tech
  - Only change shadow price in sectors that are allowed to purchase DAC
- Total emissions

## 32 LULUCF

Need to include emissions for accounting. Want to have total emissions from CIMS equal the NIR for easy comparison.

Also include credits in GHG accounting, again for easy comparison. Or should this be done

## **Part IV**

# **General Services**

## 33 Industrial Machine Drive Systems

TODO: Describe how the model represents industrial machine drive systems and their contribution to industrial electricity demand across manufacturing and resource-processing sectors. The section should explain the treatment of end uses such as motors, pumps, compressors, fans, conveyors, and other mechanically driven equipment, including how energy consumption is allocated across industries consistent with sources such as the U.S. Energy Information Administration Manufacturing Energy Consumption Survey (MECS). It should document assumptions related to motor efficiency improvements, variable speed drive adoption, equipment turnover, process electrification, and production-output linkages. The section should also explain how machine drive demand responds to changes in industrial activity, efficiency policies, and technology adoption, as well as any interactions with waste heat recovery, onsite generation, or demand-side flexibility measures.

Nodes shaded in green represent unique industry processes, while nodes shaded in other colors represent auxiliary systems. These are generic energy services that are supplied to the major process technologies and are shared by all CIMS industry sub-models. The auxiliary systems fall into four general categories: steam generation (boilers and cogenerators), lighting, heating, ventilation, and air conditioning (HVAC), and electric motor systems including pumps, fans, compressors, and conveyors. Figure 9 shows the energy flow diagrams for auxiliary systems not included in Figure 8. In some cases, the energy service meets the direct need for steam, pumping, or compression, while in other cases, it serves only to provide suitable conditions for production to continue, as in the case of lighting and HVAC systems.

Figure 9. Auxiliary Flow Model Diagram

The sections that follow describe how key auxiliary systems and process specific components are captured by the CIMS industrial model. The section on auxiliary systems covers steam generation and electric auxiliary services. Lighting and HVAC are not described in detail because the energy demanded by these services is small. The section on process specific systems covers each of the CIMS industrial sub-models.

### 33.0.1 Auxilliary Systems

#### 33.0.1.1 Steam Generation

A number of industry process technologies generate steam as a by-product. CIMS accounts for steam generated by these technologies in the overall steam demand in each industry. Any

shortage of steam must be met by boilers and cogenerators.

Boiler efficiency can vary greatly depending on boiler design, age, and fuel used. For modern oil and gas boilers, thermal (first law) efficiencies may be 85% or higher. Boiler efficiencies can be increased by introducing non-condensing and condensing heat recovery systems and by installing regenerative burners with computerized fuel/air mixtures to maximize fuel efficiency.

Cogeneration is the sequential production of electricity and useful thermal energy, usually as steam or shaft drive power. In cogeneration a boiler is used in conjunction with a turbine system to generate electricity, with the “waste” steam going to meet process steam requirements. In comparison to conventional electricity generation, cogeneration facilities are capable of conversion efficiencies of up to 85% (including both the steam and electricity produced) as compared to utility condensing turbine systems which have an efficiency in the 30% to 40% range.

#### **33.0.1.2 Electric Auxiliary Systems**

The vast majority of electricity consumed by industry is used by motor systems. A motor is the core component of a much broader system of electrical and mechanical equipment that provides services, including hydraulic power, compressed air, motive power, and air flow. Opportunities for efficiency improvement exist in both the motor itself, and in the latter systems – pumping, air displacement, compression, conveyance as well as other types of machine drive that are unique to a given production process (direct drive).

Figure 3.3 above is a generic energy flow model representing motor and steam auxiliary systems as they appear in the various CIMS industrial sub-models. As the figure indicates, CIMS simulates six auxiliary groups related to process drive (in red). The first five demand shaft or mechanical drive provided by the sixth group, motors (machine drive). Each group simulates technology evolution of a specific system.

Increased disaggregation in some of the groups reflects differing end-uses and levels of efficiency. For example, compressor systems vary in electrical efficiency by size. Therefore, competition between different technologies must be separated by size. The energy flow model reflects this by displaying two sizes of compressors, which are available for use by the model.

Pumping, air displacement, compressing, and conveyance systems have been represented as packages within CIMS. The components of each of these systems are associated with the key end-use. For example, a pipe, throttle, and speed control device is attached to a pump, and the entire package is called a pump system. In the model, there are at least two types of systems:

1. The least efficient system comprised of the least efficient components
2. The most efficient system comprised of the most efficient components

The system efficiency is then a function of the efficiency of all the components within that system. In some cases intermediate efficiencies have also been listed. New technologies may be added to reflect the intermediate range.

Electronic variable speed drives (VSDs) provide speed control by varying the rotating speed of the induction motor to match variable speed requirements of driven equipment. Pumps and fans, for example, are often required to deliver varying qualities of fluid flow. When this control procedure is used the energy consumption of the fan or pump is dramatically reduced during part load operation, as compared to the conventional practice of using a valve or d

## 34 Low-Temperature Industrial Heat

### 34.1 Scope and Definition

Low-temperature industrial process heat in CIMS is modeled as the steam, hot-water, and direct-heating services supplied at end-use temperatures below approximately 200 °C. Process heat is partitioned into three temperature bands — low (below 200 °C), medium (200–400 °C), and high (above 400 °C). The bands are defined by end-use temperature rather than by energy or equipment type, because the technology set admitted to the relevant service nodes differs substantially across them. The 200 °C boundary follows the convention adopted by the U.S. Department of Energy, Natural Resources Canada, and applied in the recent peer-reviewed techno-economic literature (e.g., Schoeneberger et al. [2020](#)). Industrial heat globally accounts for approximately two-thirds of final industrial energy use and roughly one-fifth of energy-related CO<sub>2</sub> emissions; analyses across IEA member countries place 30–45 % of industrial heat demand within the low-temperature band (International Energy Agency [2022](#); International Energy Agency [2024c](#); Madeddu et al. [2020](#)).

The allocation of base-year low-temperature heat demand across sub-models on a provincial basis is the result of a calibration exercise bringing together information from Statistics Canada and Natural Resources Canada’s Comprehensive Energy Use Database (CEUD), Environment and Climate Change Canada’s National Inventory Report (NIR), and U.S. Energy Information Administration Manufacturing Energy Consumption Survey (MECS) data used for analogous subsector benchmarks. The CEUD does not publish process heat disaggregated by temperature band; end-use accounting nevertheless indicates that conventional steam systems, boilers, and process heating account for the majority of fossil-fuel consumption in Canadian manufacturing (Natural Resources Canada [2024b](#)).

### 34.2 Industrial Subsectors and End-Uses

Low-temperature heat demand is concentrated in a defined set of CIMS industrial sub-models — Pulp and Paper, Chemicals, Mining, Industrial Minerals, Metal Smelting, Iron and Steel, and Light Industrial (which aggregates food and beverage, textiles, wood, and prefabricated building products, and general light manufacturing). Each sub-model is calibrated independently for base-year energy consumption, and service demand is assumed to scale with industrial

sectoral output projections in future years. The relative share of low-, medium-, and high-temperature heat within each subsector is assumed to remain constant in projection years. Subsectors and representative end-use temperature ranges relevant to the Canadian industrial mix are summarized in Table 34.1; note that food and beverage processing, textiles and laundry services, wood products, and light manufacturing and assembly are not standalone CIMS sub-models but are represented within the Light Industrial sub-model.

**Table 34.1:** Representative low-temperature end-uses and temperature ranges by industrial subsector

| Subsector                                         | Representative end-uses                                                                           | Range (°C) |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------|------------|
| Pulp and paper                                    | Black-liquor evaporation, paper drying on steam-heated cylinders, digester heating                | 100–180    |
| Chemicals and pharmaceuticals                     | Distillation reboilers for low-boiling solvents, batch reactor heating, crystallization           | 80–180     |
| Mining and oil and gas                            | Produced-water heating, glycol regeneration, remote-site building heat, mid-stream gas processing | 60–200     |
| Food and beverage processing                      | Pasteurization, sterilization, spray and tray drying, clean-in-place wash water                   | 60–200     |
| Textiles, leather, and laundry services           | Dyeing baths, drying, finishing                                                                   | 60–160     |
| Wood products and prefabricated building products | Lumber kiln drying, panel pressing                                                                | 60–200     |
| Light manufacturing and assembly                  | Paint cure, parts washing, fabrication-hall space conditioning                                    | 50–180     |

### 34.3 Technologies for Low- and Zero-Emission Heat Supply

This section describes the low- and zero-emission supply technologies admitted to the low-temperature service nodes in the CIMS industrial sub-models, together with their assumed performance parameters and data sources. Technology efficiencies and coefficients of performance (COPs) used to parameterize the technology competitions are summarized in Table 34.2. Combustion boiler efficiencies are estimated based on data from MECS; heat pump COPs are based on inputs from IEA Heat Pumping Technologies Programme Annex 58 (U.S. Energy Information Administration 2021; IEA Heat Pumping Technologies Programme 2023).

**Table 34.2:** Assumed efficiencies and coefficients of performance for low-temperature heat supply technologies

| Technology                                 | Efficiency / COP | Maximum temperature        |
|--------------------------------------------|------------------|----------------------------|
| Natural gas boiler                         | 80–85 %          | 250 °C                     |
| Liquid fuel boiler                         | 80–85 %          | 250 °C                     |
| Coal boiler                                | 85–90 %          | 250 °C                     |
| Biomass boiler                             | 75 %             | 250 °C                     |
| Electric resistance / electrode boiler     | 99 %             | 250 °C                     |
| Heat pump                                  | COP 4.0–6.0      | ≤ 100 °C                   |
| Heat pump                                  | COP 2.5–4.0      | 100–150 °C                 |
| Heat pump                                  | COP 2.0–3.0      | 150–200 °C (demonstration) |
| Mechanical vapour recompression            | COP 5–10         | Evaporator duty only       |
| Solar thermal, flat-plate / evacuated-tube | n/a              | 100 °C                     |
| Solar thermal, linear concentrating        | n/a              | 200 °C                     |

Combustion boilers and electric resistance boilers are assumed to have a lifetime of 25 years; industrial heat pumps and mechanical vapour recompression units a lifetime of 15 years; and solar thermal collectors a lifetime of 20 years.

### 34.3.1 Industrial Heat Pumps

Closed-cycle vapour-compression heat pumps upgrade ambient or recovered low-grade heat to a useful end-use temperature. Commercially available products reach temperatures of approximately 165 °C; units operating at 200 °C with novel working fluids are at the demonstration stage (Arpagaus et al. 2018; Zühlsdorf et al. 2019). Mechanical vapour recompression, an open-cycle heat pump configuration applied to evaporators in dairy, sugar, and pulp operations, is the most mature variant and is represented as a distinct technology in the food and beverage and pulp and paper sub-models.

### 34.3.2 Direct Electrification

Electric resistance and electrode boilers deliver low-pressure steam at near-100% end-use efficiency at lower capital cost than heat pumps, and are represented as the simplest like-for-like replacement for packaged gas-fired boilers. Application is favoured in retrofit configurations where capital constraints favour low first-cost equipment (Environment and Climate Change Canada 2024a). Direct-contact electric processes (induction, infrared, microwave, and dielectric heating) are commercially deployed for specific drying, curing, and surface-heating duties in the corresponding service nodes (Madeddu et al. 2020).

### 34.3.3 Solar Thermal

Flat-plate and evacuated-tube collectors economically supply heat up to approximately 100 °C; linear concentrating systems (parabolic-trough and linear-Fresnel) extend the addressable range to 200 °C and beyond. Schoeneberger et al. (2020) estimate that approximately half of global industrial heat demand falls within the temperature window addressable by current commercial solar thermal technologies. Within Canada, solar resources are most favourable in the Prairie region and the southern interior of British Columbia; representation in CIMS uses capacity factors calibrated to regional direct normal irradiance (International Energy Agency 2024c).

### 34.3.4 Bioenergy

Biomass-fired boilers, anaerobic-digestion biogas, and renewable natural gas from landfill and agricultural sources are admitted as drop-in low-carbon supply technologies in facilities configured around combustion equipment. Biomass is the dominant on-site thermal energy source in the Canadian pulp and paper sector and is also relevant in food processing operations co-located with feedstock supply (Natural Resources Canada, CanmetENERGY 2026). Outside the pulp and paper sector, deployed bioheat in Canada is concentrated at the 50 kW–5 MW thermal capacity range tracked by CanmetENERGY’s Canadian Bioheat Database; the 2023 survey update reports that more than 90 % of catalogued commercial, institutional, and small-industrial systems use either wood chips (approximately 45 %, dominant in institutional, agricultural, and light industrial installations) or wood pellets (approximately 46 %, dominant in institutional and commercial applications), with hog fuel and agricultural residues uncommon at this scale and concentrated in pulp and paper and adjacent forest-products operations (Natural Resources Canada, CanmetENERGY 2023). Regional fuel-type preferences are pronounced: wood chips dominate in Quebec, Ontario, British Columbia, and Nova Scotia, while pellets account for nearly all use in New Brunswick and the Northwest Territories, with lesser amounts in Quebec and BC. These distributional facts inform the technologies admitted to the bioenergy supply node feeding the Light Industrial sub-model, and the regional weighting applied to wood-chip versus pellet capital and operating cost assumptions. Sustainable supply is the binding constraint at the system level and is represented in CIMS through energy supply curves rather than at the device level.

### 34.3.5 Geoechange and Recovered Waste Heat

Shallow enhanced geoechange coupled with a heat pump is deployable across all provinces for facility heat. Independent of geoechange extraction, every industrial site is itself a source of low-grade waste heat — compressors, condensers, flue gases, and effluents — that is conventionally rejected to the environment. National-laboratory analyses estimate that 20–50 % of industrial process heat could in principle be supplied via recovered waste heat (Thekdi

and Nimbalkar [2014](#)). Process integration and heat-pump-assisted recovery are represented in CIMS as a service-side reduction in the demand directed to other supply technologies, parameterized by sub-model based on recoverable-heat fractions reported in MECS and CanmetENERGY studies, rather than as a competing supply technology at the service node.

### 34.3.6 Thermal Energy Storage

Pressurized hot water, molten salt, and solid-media storage shift electrified heat loads to off-peak hours and extend solar thermal coverage to continuous-process duty cycles. Storage is treated in CIMS as a distinct technology that competes at the same service node as the heat-supply technologies it enables, with round-trip efficiency, capacity, and discharge-duration parameters drawn from IEA reference values (International Energy Agency [2022](#)).

### 34.3.7 Hydrogen

Direct combustion of hydrogen below 200 °C is not competitive with electrification under most modelled conditions: round-trip energy efficiency from electricity to hydrogen to heat is substantially lower than direct electrification or heat-pumps, and the system value of hydrogen is concentrated in genuinely high-temperature or feedstock applications. Hydrogen is therefore not included in the default low-temperature technology set; it remains available in the broader CIMS technology library for the medium- and high-temperature sub-models (Thiel and Stark [2021](#); Rissman et al. [2020](#)).

## 34.4 Implications for the CIMS Industrial Sub-Models

The technology competitions in the food and beverage, pulp, and paper, chemicals, mining, and light manufacturing sub-models admit, at the relevant service nodes, conventional natural gas and (where regionally relevant) heavy fuel oil boilers, used for calibration to CEUD and NIR baseline data; electric resistance and electrode boilers, parameterized by provincial industrial electricity rates; industrial heat pumps disaggregated by sink-temperature class ( $\leq 100$  °C, 100–150 °C, 150–200 °C) with explicit COP and source-temperature assumptions drawn from IEA HPT Annex 58; mechanical vapour recompression where evaporation duties exist; biomass and biogas boilers, with energy availability governed by the bioenergy supply sub-model; and solar thermal in provincial sub-models, parameterized to regional irradiance (Natural Resources Canada [2024b](#); Environment and Climate Change Canada [2024a](#); IEA Heat Pumping Technologies Programme [2023](#); Natural Resources Canada, CanmetENERGY [2026](#); Schoeneberger et al. [2020](#)).

Existing boiler stock, distinguished by sub-model and temperature class, is estimated based on CEUD and NIR base-year data. All existing boiler stock is assumed to stay online for modeled lifetimes past the base year, after which replacement is required. Policy or other drivers may drive earlier retirement and replacement of some of these existing units. Capital costs, retrofit feasibility (as defined in [the technology chapter](#)), and electricity-price sensitivity vary substantially across these technologies.

## 35 Cogeneration

### 35.1 Scope and Definition

Cogeneration (also referred to as combined heat and power) is the sequential or simultaneous generation of multiple forms of useful energy (usually mechanical and thermal) in a single, integrated system. Cogeneration systems consist of a number of individual components—prime mover (heat engine), generator, heat recovery, and electrical interconnection—configured into an integrated whole. The type of equipment that drives the overall system (i.e., the prime mover) typically identifies the cogeneration system. Prime movers for cogeneration systems include steam turbines, gas turbines (also called combustion turbines), spark ignition engines, diesel engines, microturbines, and fuel cells. These prime movers are capable of burning a variety of fuels, including biomass/biogas, natural gas, or coal to produce shaft power or mechanical energy. Additional technologies are also used in configuring a complete cogeneration system, including boilers, absorption chillers, desiccants, engine-driven chillers, and gasifiers. Boilers and gasifiers are discussed in Chapter 5 of this document. (U.S. Environmental Protection Agency (EPA) and Partnership (2007))

Although mechanical energy from the prime mover is most often used to drive a generator to produce electricity, it can also be used to drive rotating equipment such as compressors, pumps, and fans. Thermal energy from the system can be used in direct process applications or indirectly to produce steam, hot water, hot air for drying, or chilled water for process cooling.

The industrial sector currently produces both thermal output and electricity from biomass in CHP facilities in the paper, chemical, wood products, and food processing industries. These industries are major users of biomass fuels—utilizing the heat and steam in their processes can improve energy efficiencies by more than 35 percent. In these applications, the typical CHP system configuration consists of a biomass-fired boiler whose steam is used to propel a steam turbine in addition to the extraction of steam or heat for process use.

## 36 Carbon Capture, Utilization, and Storage

TODO: Describe how the model represents carbon capture, utilization, and storage across sectors, including capture technologies, capture rates, energy penalties, transport and storage infrastructure, and associated costs. The section should explain which emissions sources are eligible for capture, how captured CO<sub>2</sub> flows are tracked, and whether utilization pathways or permanent sequestration are distinguished. It should also document assumptions related to technology availability, deployment constraints, storage capacity, policy incentives, and interactions with hydrogen production, industrial processes, and power generation.

## 37 Energy Storage

TODO: Describe how the model represents energy storage technologies and their role in balancing supply and demand across the energy system. The section should document the types of storage included (e.g., batteries, pumped hydro, thermal storage, hydrogen storage), storage duration assumptions, charging and discharging efficiencies, capacity constraints, and operational behavior. It should also explain how storage interacts with variable renewable generation, grid reliability requirements, peak demand management, and transmission infrastructure, along with assumptions for costs, learning rates, and deployment limits.

## **Part V**

# **Data and Calibration**

## 38 Data Inputs

TODO: A reader-oriented description of the `data/` directory. Cover: (1) `data/model_inputs/` — what each region/sector CSV represents, the meaning of the Branch / Type / Parameter / Service / Technology columns, how rows are interpreted as nodes in the CIMS tree, what `service_provide` / `service_request` / `competition` / etc. parameter values mean. (2) `data/mapping_conversions/` — `region_primary`, `sector_map`, `energy_map`, the region aggregation files, and how they're used during processing and rendering (the dynamic sector-by-region table in the architecture chapter is one example). (3) `data/raw_data/` and `data/processed_data/` — source datasets and how they flow into model inputs. (4) `data/calibration/` — what lives there. Cross-reference the file-format appendix for column-by-column specifications.

May want to include information here about including geographic levels other than provinces and territories, like energy markets, other countries (for trade dynamics), and cities (for building and transportation policies).

## 39 Calibration

TODO: Expand the outline below into a full chapter. Each bullet is a section heading or paragraph topic in its own right — the calibration period and counterfactual data sources, the FIC-optimization workflow, the manual-adjustment fallback, how the forecast-period FIC is derived from the historical estimates, and the long-term FIC trend-to-zero convention all deserve explicit treatment.

- Optimization by year to counterfactual data
- Change FIC values to match objective criteria (counterfactual)
- Can use market share for tech competitions at a node, or energy or emissions at same or adjacent node
- Also have option for manual adjustment of FIC values for more disconnected objective and adjustment locations in the cims tree
- Optimize FIC year by year over historical counterfactual period
- Use median or filtered mean (remove top/bottom 10% before mean) to estimate FIC for forecast period
- Trend all FIC values to zero over very long-term (next 50-70 years?)

Calibration of endogenous supply sectors:

1. Set Financial lifecycle cost to 1
2. Set Price Multiplier to expected price (including sector markup)
3. Calibrate demand sector
4. Calibrate energy supply sector to achieve expected production cost at node one level below top level
5. Remove Financial lifecycle cost
6. Reset Price Multiplier to only include sector markup

## 40 Validation

TODO: Document how the calibrated model is checked against independent data. Common validation comparisons for CIMS: historical fit against the NRCan Comprehensive Energy Use Database (CEUD), comparison of emissions against the ECCC National Inventory Report (NIR), comparison of new-stock investment against Statistics Canada figures, sensitivity tests for key behavioural parameters (variance  $v$ , discount rates). Include a discussion of known limitations and where validation is weak (e.g. provinces with sparse historical data).

## **Part VI**

# **Running the Model**

## 41 Setup the model

Welcome to the CIMS repository! This guide will help you set up and run the CIMS Python package for energy-economy modelling.

### 41.1 Prerequisites

Before you begin, ensure you have the following installed:

- **uv:** A Python package and environment manager. Install it with one command — no separate Python installation required.

- Mac/Linux:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

- Windows (PowerShell):

```
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/i
```

- **Git:** Used to clone the repository. Follow the [installation instructions](#) for your operating system.
- **Terminal or Command Prompt:**
  - Windows: Git Bash or PowerShell
  - macOS/Linux: Terminal

### 41.2 Setup

#### 41.2.1 1. Clone the Repository

Open your terminal and navigate to the directory where you want to work:

```
cd ~/Documents/projects/
```

Clone the repository:

```
git clone https://github.com/EMRG-SFU/cims.git
cd cims
```

### 41.2.2 2. Install Dependencies

From the `cims` directory, run:

```
uv sync --extra notebooks
```

This will automatically select a compatible Python version, create a virtual environment, and install all dependencies. You only need to do this once, or again when dependencies are updated.

### 41.2.3 3. Launch Marimo

```
uv run marimo edit
```

Marimo will open in your default browser. From there you can open, run, and edit notebooks.

## 41.3 Common Workflows

| I want to...                    | Command                                              |
|---------------------------------|------------------------------------------------------|
| Set up or update my environment | <code>uv sync --extra notebooks</code>               |
| Launch Marimo                   | <code>uv run marimo edit</code>                      |
| Run a Python script             | <code>uv run python my_script.py</code>              |
| Open a Python shell             | <code>uv run python</code>                           |
| Use a specific Python version   | <code>uv sync --python 3.12 --extra notebooks</code> |

## 41.4 Subsequent Runs

To re-launch Marimo in future sessions, simply run from the `cims` directory:

```
uv run marimo edit
```

`uv` will use your existing environment and skip reinstalling dependencies unless something has changed.

## 41.5 Troubleshooting

- **uv not found after install:** Restart your terminal session so the `uv` command is available on your `PATH`.
- **Dependency conflicts:** Run `uv sync --extra notebooks` again to bring your environment up to date with the latest `pyproject.toml`.

## 41.6 Additional Resources

- **Submit Issues:** If you encounter any problems, please submit an issue on our [GitHub Issues](#) page.

## 42 Getting started

how to run a scenario

Discuss model operational structure (inputs, marimo notebook, results)

## 43 Policy Costing & Outputs

### 43.0.1 Policy costing methods associated with ISTUM and early CIMS

Issues associated with deriving costs from its outputs guided much of CIMS' development. The cost methodologies in the following sections are described in brief and outline some of the issues in estimating costs.

The ISTUM model was initially designed for energy policy analysis, mainly for assessing how fuel consumption and emissions would change with a given energy policy. The emphasis was on marginal costs, not total or absolute expenditures. The first consistent costing concept was the techno-economic cost, or TEC.

#### **Techno-economic Cost (TEC)**

TEC is the difference in estimated (ex-ante) financial expenditure on capital, labour, and energy between the Business-as-Usual and Policy cases. It does not usually (unless otherwise specified) contain carbon taxes as these are considered a transfer internal to society. It is possible to get negative TEC costs, or benefits, especially in transportation where increased use of transit and walking and cycling reduces financial costs, implying that some actions induced by policy may be profitable. There are significant complications as well – do those who switch to transit and walking and cycling continue to own and pay for cars? This reflects the challenging question about how vehicle acquisition will be affected by policies to reduce vehicle use. TEC for all consuming sectors are reported with and without electricity price increases, which are sub-regional transfers that balance out to zero at the regional level once the financial effects on electricity producers' revenues are included.

The ISTUM sub-models were integrated under the energy demand and price feedback system in the mid-1999 to 2000 period, thus creating the first version of CIMS. During this period, the model was used for a series of large GHG abatement analyses for Canada. The thrust of these analyses was to integrate and assess a series of emissions reduction actions proposed by panels of interested parties, the Issue Tables of Canada's Climate Change National Assessment Process. The most effective actions were determined by minimizing their welfare costs, as opposed to their pure financial costs. This led to the development of the Perceived Private Cost methodology (Bataille et al. 2002, Laurin et al. 2003, MKJA/EMRG 2000).

#### **Perceived Private Costs (PPC)**

PPC are calculated as the GHG tax multiplied by the marginal amount of GHG reductions that occurred at each GHG tax level. The area under the resulting GHG abatement cost curve

represents the perceived cost of the actions by the GHG charge. The PPC methodology estimates welfare losses that represent in part the unwillingness of all consumers to switch to technologies that reduce emissions. This fostered development of the Expected Resource Cost methodology, which attempts to more realistically depict the costs and risks associated with a given GHG reduction policy.

### **Expected Resource Costs (ERC)**

ERC is PPC that has been adjusted for risk, option value, and general inefficiency. It is equal to  $TEC + (PPC - TEC) \times 0.75$ . The difference between TEC and PPC that is allowed as ERC is a “rule-of-thumb” estimate for what percentage of the difference can be considered real equipment failure risk, as opposed to inefficient resistance to price signals. It represents that perceived cost, which could turn into real financial cost. Negative ERC is possible, especially at lower GHG charges, but not likely (Bataille et al. 2002, Laurin et al, 2003, MKJA/EMRG, 2000).

Table 10 summarizes the three early costing methods used for CIMS.

Table 8. Types of Costs Used in Early Incarnation of CIMS

### **Distribution of Costs Under Electricity and Refining Demand Changes**

Electricity demand changes in all sectors in response to increased electricity prices, leading to an issue of how to distribute GHG reductions as well as TEC and PPC charges. GHG reductions from reduced electricity consumption were accorded to the direct consumer (based on the policy GHG intensity of electricity), who faced increased electricity prices, while electricity producers were accorded the remaining GHG intensity reductions, the TEC costs generated by their intensity reductions plus their changes in revenue. These net transfers, while positive at the sub-regional level, disappear at the regional level.

### **The Relationship Between TEC, ERC, Option Value and Welfare Costs**

The cost difference between ERC and TEC estimated during early analyses was described as risk, or “failure risk”, which is related to the concept of option-value (Pindyck 1991, Dixit 1992, Dixit and Pindyck 1994 ). Option value is the expected gain from delaying or avoiding an investment while waiting for new information that might lead to a better decision. Thus, ERC are anticipated ex-post costs, and it is calculated as the sum of expected financial costs (TEC), lost option value, consumers’ surplus losses internal to residential and transportation (internal shifts away from business-as-usual (BAU) preferences within the sub-models) and finally, when demand feedbacks are included, consumers’ surplus losses related to demand adjustments for domestic consumers of domestic goods. It may therefore be considered a measure of personal welfare costs, not including negative environmental externalities (e.g. poorer quality air) or positive public goods (e.g. education, defence, or public parks).

### 43.0.2 Policy Costs Associated With Changes in Demand

The inclusion of final demand feedbacks required the addition of five new policy-costing concepts. In all these concepts, financial output is defined as a sector's sales revenue, which includes both value-added and purchased inputs. Value-added is net profits plus the cost of all expenditures on land (unprocessed "gifts of nature"), labour and capital (Lipsey et al. 1988).

- The cost of production effect on sector financial output, which is the increased financial expenditures required to adapt to the policy, or TEC.
- The demand effect on sector financial output, which is the reduction in production expenditures associated with lower output.
- The absolute change in sector financial output, which is the sum of the cost of production and demand effects.
- Consumers' surplus losses associated with demand changes in transportation and residences.
- Changes in value-added associated with cost of production and demand changes in manufacturing.

#### 43.0.2.1 The Cost of Production and Demand Effects on Sector Financial Output

The cost of production effect captures the internal capital investment and fuel choice adjustments of the model, or the TEC costs endogenous to the model. It is the anticipated financial cost of technical adjustment to the policy. The demand effect captures the effects of the demand feedbacks system, or reduction in output, as measured by expenditure compared to the business-as-usual case (BAU). The third is the sum of the first two. These three concepts capture a portion of the anticipated financial cost of a policy. They do not include the cost of buying emission permits from foreign sources or the cost of imports that replace domestically produced goods for domestic consumption. The fourth item, consumers' surplus losses, represents lost consumer welfare associated with reduced residential expenditure and mobility, while the fifth represents the lost value-added to manufacturing firms due to policy.

Most GHG policies will increase the cost of production for a given service, increase its price, and thus reduce its demand. This increase in the cost of production will also include the value of carbon charges that are imposed on GHG emissions. These charges are transfers and as such must be subtracted from regional and national costs. Additionally, if demand is inelastic (unresponsive to price) for a given service, the dollar value of the sector, as valued by cost of production, may increase faster than demand reductions, leading to an overall increase in sector value to the economy. If demand is elastic, the opposite may occur.

The standard outputs from the sub-models include BAU and policy values for new physical capital investment associated with consuming energy, labour associated with the latter capital, and energy costs, as well as changes in physical output. It also produces BAU and policy costs

per unit output, including GHG charges. These GHG charges are isolated by calculating policy emissions multiplied by the GHG price. When the micro and macro economic dynamics are turned on, part of the difference between the BAU and policy cases may be physical output changes while the rest will be cost of production differences due to internal stock adjustments. CIMS only captures a portion of the price that the consumer sees; both the BAU and Policy cost per unit are adjusted accordingly. The cost of production and demand effects on sector financial output (SFO) (Equations 23 & 24), and the absolute change in financial expenditure are calculated as follows (Equation 25).

(Equation 28)

(Equation 29)

(Equation 30)

Where:

$Q_{GHG}$  = GHG emissions (Tonnes CO<sub>2</sub>e)

$P_{GHG}$  = GHG price (\$/Tonne CO<sub>2</sub>e)

$Q_{policy}$  = Policy physical output of good or service X

$Q_{BAU}$  = BAU physical output of good or service X

$P_{policy}$  = Policy cost per unit of good or service X

$P_{BAU}$  = BAU cost per unit of good or service X

The absolute change in sector financial output could be positive or negative depending on the elasticity of demand – if the cost of production rises faster than demand falls, then the absolute change in financial output could be positive. This result is common with less aggressive policies. The demand effect on sector financial output will always be negative for normal goods, whose demand fall as their price rises. The cost of production effect on sector financial output is usually positive, except for the case where there are decreasing returns to scale.

#### **43.0.2.2 Consumers' Surplus Adjustments in Residential and Transportation**

Reductions in residential activity and mobility reflect consumers' surplus losses. Consumers' surplus is the area between the demand curve, which symbolizes the marginal willingness to pay for a service or good, and the market-clearing price of the good or service (Figure 43.1).

The consumers' surplus loss between BAU and Policy is then calculated as BAU minus the policy. When calculating the direct welfare costs of a policy, consumers' surplus loss is added to the ERC value already calculated. Table 11 summarizes the financial and welfare costs adjustments made to the various sectors when their output changes.

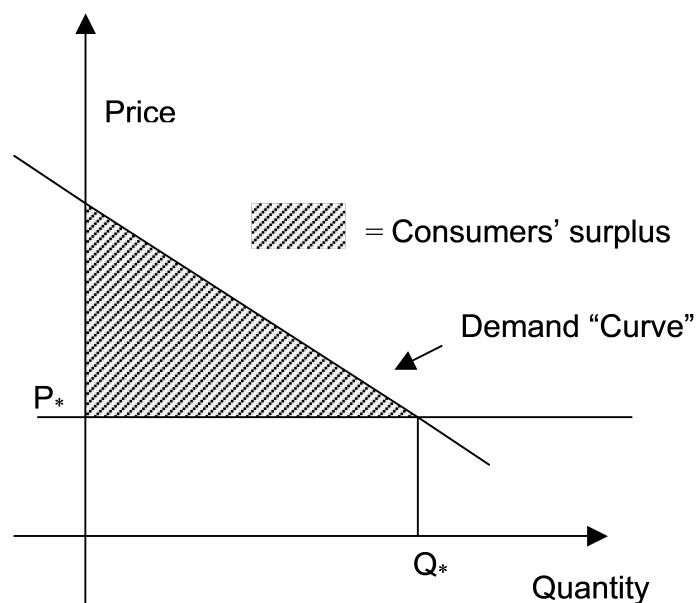


Figure 43.1: Consumer Surplus

Table 43.1: Financial, welfare cost, and value-added adjustments induced by changes in output

|                                                | Output adjustment to financial costs                                 | Output adjustment to welfare costs / value-added                                                                                               |
|------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Transportation</b>                 | Demand effect on sector financial value, as measured by expenditure  | Difference in <b>consumers' surplus</b> under BAU and policy calculated using transportation <i>pkt</i> elasticity                             |
| <b>Industry (&amp; Freight Transportation)</b> | Demand effect on sector financial output, as measured by expenditure | Difference in <b>value-added</b> between BAU and Policy                                                                                        |
| <b>Residential</b>                             | Demand effect on sector financial value, as measured by expenditure  | Difference in <b>consumers' surplus</b> under BAU and policy calculated using combination of savings/consumption and leisure/work elasticities |
| <b>Commercial</b>                              | Demand effect on sector financial value, as measured by expenditure  | None                                                                                                                                           |

The lack of an output adjustment for welfare costs or value-added for the commercial sector

is partly due to the nature of the Commercial and Institutional sub-model. It is composed mainly of building shells, HVAC, miscellaneous electricity, and air-conditioning systems for government and office buildings, retail space, hospitals, schools, etc. CIMS does not model what these spaces are used for, and thus cannot account for how financial output or value-added may have changed.

### 43.0.3 Outputs from CIMS

Table 12 provides the outputs specific to the sub-models (i.e., for when they are run by themselves without feedbacks), while Table 13 provides the extra outputs necessary for working with the CIMS' micro and macro economic systems of dynamics (also referred to as feedback systems).

#### Table 10. Sub-model Outputs Existing Prior to This Research

Sub-models Annualized lifecycle costs by sector by year by technology based on behavioural discount rates (20-80%) (See Table 5).

Fuel use and emissions by sector by year by technology.

Service and auxiliary use by sector by year by technology.

Total stock use by sector by year by technology.

New stock use by sector by year by technology.

Retrofits by sector by year by technology.

Technology cost changes due to declining capital cost function

#### Table 11. Supply and Demand Feedback Outputs Added as a Result of This Research

Supply and Demand feedbacks Annualized lifecycle cost per unit produced for all sectors by year based on financial cost of capital for both the BAU and Policy case; the model operator sets the cost of capital (e.g. 7.5 or 10%)

BAU and Policy final demand for each sector's product or service by year.

BAU and Policy disaggregated energy demand for all sectors by year.

Changes in disaggregated capital, operations and maintenance, and energy costs by sector by year.

Changes in GHGs and selected criteria air contaminants (CACs), energy and service use by technology, sorted by technology competition, for BAU and Policy by sector by year.

Changes in net exports of key energy commodities from BAU to Policy

Changes in energy prices by year

Changes in the financial cost of production split into the cost of production and demand effects for industry and manufacturing

Changes consumers' surplus for personal transportation and residences

Changes in value-added for industry and

## 44 Building the Model Tree

TODO: Write up best-practices for designing nodes in the CIMS tree. Cover: when to request services directly at a node vs. through technology competition, how to lay out service nodes for a new sector, naming and Branch-path conventions, where intangible costs are typically anchored, and how to expose policy levers cleanly. Worked examples (one residential heating, one industrial process-heat) would communicate the patterns better than abstract prose.

### Note

If setting market shares after the base year, need to use lifetime = 1, otherwise stock is not calculated correctly.

## **Part VII**

# **Outputs and Analysis**

## 45 Outputs

Three main classes of output are created by the model:

- 1) Sub-sector specific reports
  - a) fuel and emissions for a given sector (FUELA) for BAU and Policy, (B or C-Region-FUELA, e.g. BONFUELA.
  - b) technology levelized costs for a given sector (LEVEL) – same pattern as FUELA.
  - c) new and total technology market share reports for a given sector (NEWMA and TOTAL) – same pattern as FUELA.
  - d) technology node competition reports for a given sector (REGION-NC-YEAR, e.g. “ABNC2005”
- 2) Semi-aggregate reports
  - a) changes in physical output, sorted by sector and by region (CCOMP)
  - b) changes in sector capital, fuel, emissions, and labour costs, sorted by sector and by region (CIMCOSTA & CIMCOSTB)
  - c) changes in sector energy use (by market class Elec., NG, coal and RPP) and GHG emissions, sorted by sector and by region (DCOMP)
  - d) changes in sector cost per unit (inclusive of emissions charges), sorted by sector and by region (LCOMP)
- 3) Aggregate report
  - a) CIMS Outputs (As of March 2007)

### 45.1 Sub-sector Reports

Four report types are available. In each type of report, the analyst opens a spreadsheet where data can be manipulated using spreadsheet functions. The report can then be reformatted, printed, or saved under another name.

#### 45.1.1 Fuel and Emissions Report

Data concerning fuel consumption and emission generation are saved for each year of the simulation. The spreadsheet entitled B(BAU) or C(Policy) Region (e.g. BC) FUEL, e.g. CBC-FUELA, receives these data. The model's report generator produces a table containing: (1) a list of all fuels and emissions, (3) consumption/production of those fuels and emissions for each year, and (4) a summation of fuels consumed and emissions output. Note that, if fuel/emission units are not uniform, this summation will be erroneous. Figure 3.1 displays the layout of a typical aggregate fuel report.

Figure 3.1 ISTUM's Fuel Report

Typically, technologies are grouped into Aggregate, Major processes, Service categories and Product type categories.

Aggregate reports include the cumulative energy consumption of all the technologies in the industrial branch under analysis and are generated for every industry. Note that this is the sum of consumed and produced energy. To determine the amount of cogenerated or self-generated energy, the analyst should refer to the NC files which show all energy inputs and outputs by technology competition. No technology can contribute to the cumulative consumption/production of more than one group in each category.

Figure 3.1 ISTUM's Fuel Report

#### 45.1.2 Total Stock Report

The table generated in the TOTAL spreadsheets contains data on the sum of operating old and new stock inventories for that year. The stock is listed in output units for that technology. As with the other report types, the table's header contains specific information on the scenario and economic criteria applied to produce that output. To the right of the total stock table, one finds a table of the share of total stock for competing technologies (in percent) that each technology has obtained. Figure 3.23 displays the layout of a typical Total Stock Report.

The lower portion of this table provides information concerning changes in stock consequent to retrofitting. Each set of lead and retrofit technologies show the lead technology first followed by the technology(ies) that can be retrofit to it. The data indicate the number of units not retrofit (the value associated with parent technologies) and those that were retrofit (the values associated with the retrofit technologies).

Figure 3.2 ISTUM's Total Stock Report

### 45.1.3 New Stock Report

As in the Total Stock Report, the two tables generated by the New Stock Report function of CIMS, show new stock additions in each year of the simulation and the market share of that stock purchased in that simulation year. This NEWMAR spreadsheet header, like the others, displays specific information concerning scenario and economic criteria. Figure 3.3 displays the layout of a typical New Stock Report.

Figure 3.3 ISTUM's New Stock Report

### 45.1.4 Levelized Costs Report

The LEVEL report lists each technology and contains tables displaying two categories of information:

1) The first set of columns include data used to generate the levelized costs. The first numeric column displays capital costs (CC) of each technology, followed by its annualized CC ( $CC \times CC$  recovery factor, found in the sixth column). The third column shows the sum of annualized CC and costs of operations and maintenance (excluding fuel and emissions costs) for each technology. This, divided by the material output found in column 4 generates the lifecycle cost per unit output in column 5. Column six indicates whether any non-cost parameter was applied to the capital costs ( $CC \times$  non-cost parameter).

2) The second set of columns, each headed by a simulation year, presents the levelized cost of that technology in that year. The levelized cost is the sum of the per-unit cost (column four) and the annual fuel (and emission) costs. (Figure 3.4)

The lower portion of this table provides information concerning retrofits applied in that simulation. Each set of lead and retrofit technologies, displayed in separate groups, show the lead technology first followed by the technology(ies) that can be retrofit to it. Note that the capital cost of the lead technology is always 0 since it is considered a sunk cost. The capital costs associated with the retrofit technologies represents the cost of retrofit purchase and installation.

Figure 3.4 ISTUM's Levelized Cost Report

### 45.1.5 Node Competition Report

There is a separate node competition report for every sector in every region in every year. They are title (region)NC(year)J, e.g. for all sectors in Alberta the 2005 report is called "ABNC2005J.xls". The region codes are: BC, AB, SK, MB, ON, PQ, ATL (for Atlantic).

The node competition report presents the results of a run as a comparison between BAU and POLICY in terms of fuel use, services, and emissions. The key difference between this report

and the last four is that it organized by technology competition. As there are several fuels, services, and emissions in CIMS, these reports tend to be very large. Figure 6 on the next page provides a sample node competition report for the baseload electricity production competition in Alberta in 2005. The shown figure gives the difference in stock, fuel use, and emissions, while further to the right on the spreadsheet will be the BAU and policy absolute values. Stock is valued in the output unit of the sector in question, energy in GJ, and emissions in tonnes.

Figure 6: Sample node competition report – base load electricity production Alberta 2005

## 45.2 End user reports

Reports of energy use, financials, intensities, travel, etc. at end user level

For example, household energy spending, average travel, and costs

MACC Curve like in EDF report, MACC 2.0

## 45.3 LMDI Decomposition

Can this be done appropriately at the sector level or only at region/national?

Decomposition of results (to also use in policy attribution)

Include automatic waterfall chart?

Factors:

- Activity
- $E_{tot} / \text{Activity}$  (efficiency)
- $Eff / E_{tot}$  (fuel switch)
- $GHG_{tot} / Eff$  (fuel decarbonisation)
- $GHG_{net} / GHG_{tot}$  (CCS?)

## 45.4 Attributing emissions reductions to policies

Shapley val

## 46 Post-Processing and Reporting

TODO: Describe the post-processing workflow that turns raw CIMS outputs into the figures and tables typically presented in a study. Likely includes: pivoting and aggregating output CSVs, computing derived metrics (emissions intensities, levelized costs, sector totals), comparing scenarios against a baseline, building standard chart sets (energy demand by sector, technology stock evolution, emissions trajectories). If any scripts in the repo support this, link to them and show example usage.

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## A Acronyms

TODO: Single source-of-truth list of acronyms used throughout the book. Build this by walking the existing chapters and pulling out abbreviations on first use — CIMS, ISTUM, LCC, FIC, DIC, DCC, GCC, AEEI, CRF, BS, NS, MS, NMS, PR, CEUD, NIR, NRCan, ECCC, IEA, MVR (mechanical vapour recompression), COP (coefficient of performance), and so on. Use a Pandoc definition list (the same `.eq-vars`-style markup used for equation variables) so the list renders as a clean two-column glossary.

## B File Format Reference

TODO: Column-by-column specification of every file format the model reads or writes. At minimum: model\_inputs CSV (the 24-column Branch/Type/Region/Sector/Service/Technology/Parameter/Component structure), region\_primary and region\_aggregation\_\* mapping CSVs, sector\_map CSV, energy\_map, and energy\_conversions CSVs, nir\_to\_cims\_map CSV. Note the “navigate by typing” banner on the first line of every model\_inputs CSV (which is why all read code uses `skiprows=1`). Document the expected datatypes and any required vs. optional

## C Equation Summary

TODO: A quick-reference table listing every numbered equation in the methodology chapters with its label, name, and one-line description. Sortable by label (eq-stock\_retirement\_existing, eq-market\_share\_new, etc.). Lets readers locate an equation by name without scanning the methodology chapters. Could also be auto-generated from the qmd source by parsing `$$...$$ {#eq-...}` blocks.

## D Version History

TODO: A chronologically-organized log of substantive changes to the model and the documentation. Start with the current “version” and seed it with the recent reorganization (per-chapter bibliographies, equation reformatting, dynamic region-sector table, etc.). When a new model release is cut, add a section documenting what’s new, what’s deprecated, and any migration steps needed for inputs or scenarios from previous versions.